



OneATS

User Manual

Confidential

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株式会社 東光高岳
TAKAOKA TOKO CO., LTD.



OneATS user manual

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Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

Comment

Notes

Warnings

Content

Chapter	Content
A.	System overview
B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- OneATS SmartHIS

20/Feb/2025

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Software Version	Date of Issue	Change Description
V4.2.1.0	May 30, 2024	
v4.2.14.0	Aug,20,2024	- Update Report Viewer, Data Link support with interval month ("mo") - Use OneATS.cfg, OneATS.lic - Fix bug register archive in OAArea - Add method QueryLog for Mudic project - Change default interval time of each archive to 30 days - Add config to reject quality from UA Server (DataServer) - Add mode Previous_2 in RetrivalMode - Add property GroupName in OAAAlarm - Add filter alarm with GroupId - Update string filter when query Alarm, support condition filter, ex: *HL01*Active* *HL02* - Change data type of Digital Set point from Int8 to Int32 - Add 2 options of readProcessed UA funtion: + Start: to return previous value + End: to return next value - Update Report Viewer - Update ODBC Driver - Fixed archie empty path - ArchiveDB log file size limit (<1MB * 5 files) - Fixed interval mo, year in GetCalculateData function

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G. OneATS SmartHIS

OneATS SmartHIS is a component of the OneATS product suite, organized in a *Client-Server structure*. *OneATS SmartHIS* performs tasks including receiving data from the data source, storing data for **the OneATS system**, and being the server that provides the stored data to the data mining application (client).

G.1. OneATS SmartHIS

G.1.1. OneATS SmartHIS Structure

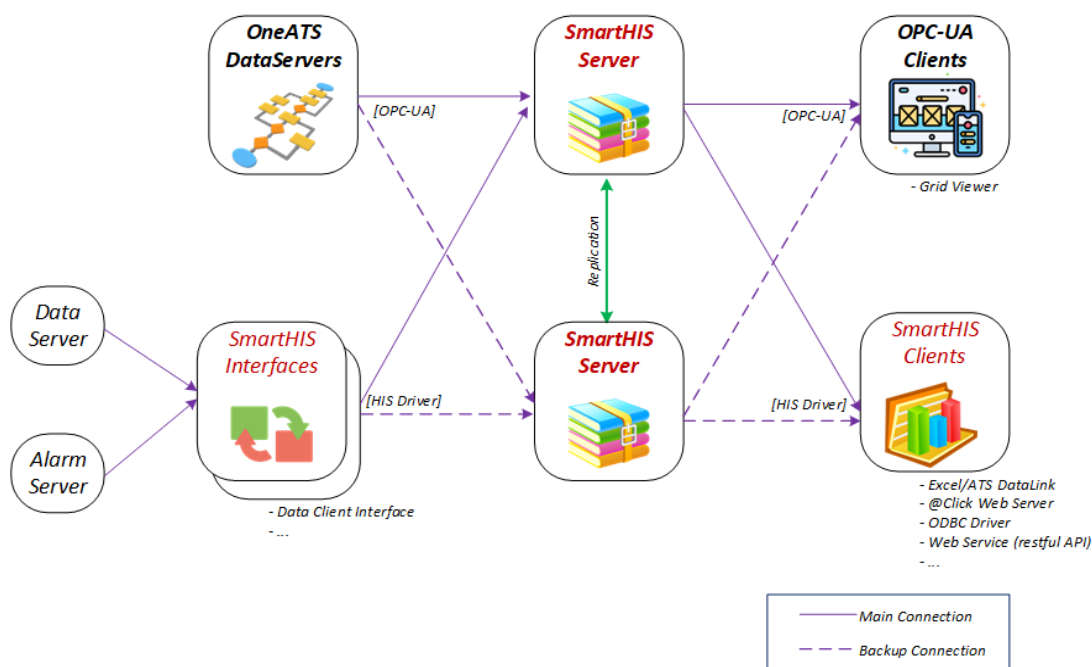


Figure G-1. OneATS SmartHIS package structure

- The **SmartHIS** package center is a **SmartHIS Server** that performs the following services:
 - Data compression and storage.
 - Supports calculation and storage of calculation results from data points
 - Respond to data query commands from clients.

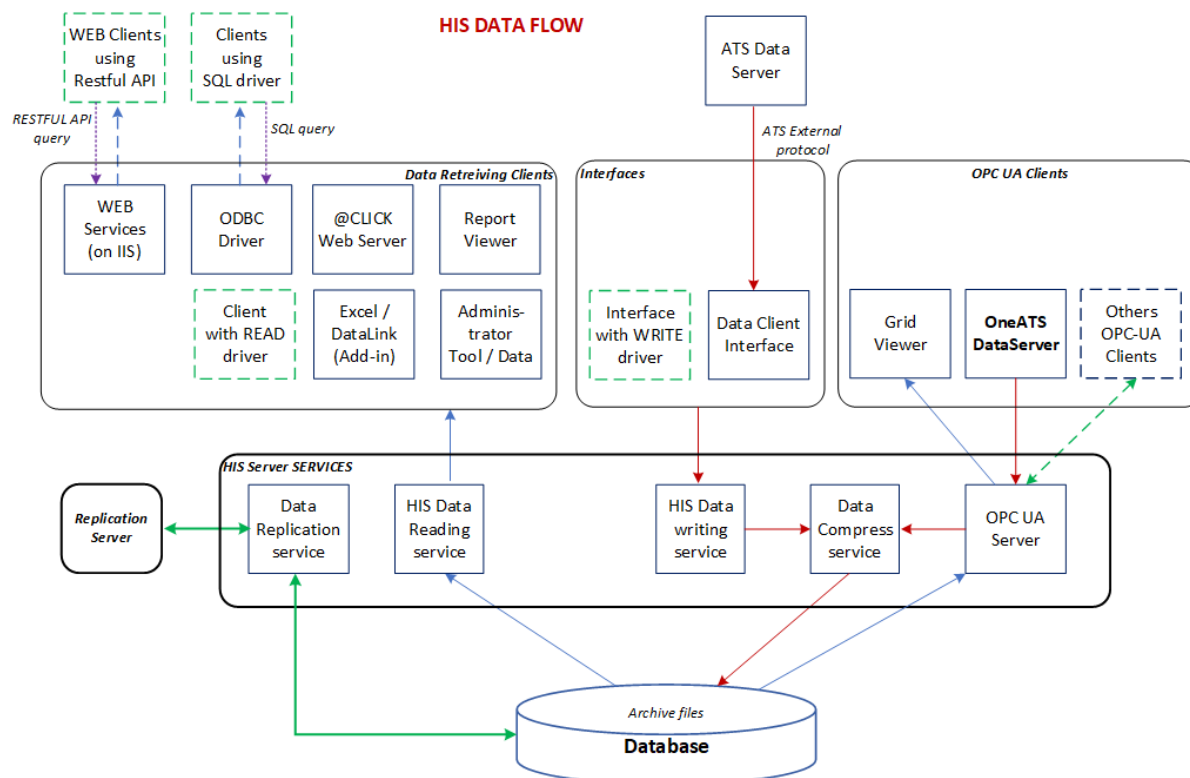
SmartHIS Server supports hot-hot redundancy mode with the mechanism of replication servers that are automatically synchronized in terms of data and configuration.

SmartHIS Server is configured with data lists including OneATS DataModel integration and other standalone data lists

- The Data Source provides data to **the SmartHIS Server** through the **Interface**, or OPC-UA Client. In the OneATS system, **OneATS DataServer** is the Client that provides data to the SmartHIS Server via the OPC-UA protocol. Data Source includes Data (value/timestamp/quality) Server & Alarm Server. The DataSource (Interface, OPC-UA Client) supports redundancy (main & backup) connections to SmartHIS Servers.
- Clients exploit data from SmartHIS Server through OPC-UA or SmartHIS driver, including basic applications in the OneATS system (GridViewer,

Excel/Addin ATS DataLink, @Click Web Server), protocol interfaces (ODBC Driver, Restful API web service).
The client supports redundancy (main & backup) connections to the SmartHIS Server.

G.1.2. Components in OneATS SmartHIS



FigureG-2. Components in OneATS SmartHIS package

Ingredient	Describe
OneATS SmartHIS Servers	Functions: • Writing (receiving data) services from HIS Interfaces • OPC-UA Server (read & write data) • Processing & storing data value newsletters and alarm messages • Supports calculation and storage of calculation results from data points • Read (return data) services for HIS clients • Data synchronization and configuration (Replication) with other HIS Servers in the redundancy system structure
Database	SmartHIS Server stores and retrieves data from NoSQL databases, called Archives.

His Data Retrieving Client: Applications that exploit (read) data	
Admin Tool / TAB Data	The tool to check/view data on SmartHIS Server. In addition, this is also one of the SmartHIS Server configuration tools
Excel/ DataLink Add-in	Excel Add-in, Library of SmartHIS Server → Excel Data Retrieval Functions
ODBC Driver	Windows ODBC Driver for HIS Server, which provides a SQL syntax data retrieval interface for clients that integrate SQL drivers.
WEB Service	IIS-based Web Services provide Restful API data retrieval interfaces for Web Clients that support Restful APIs
Report Viewer	The report miner is independent of Microsoft Excel installed on the operating computer. In particular, the reports are designed from Excel/ DataLink Add-in on the environment-computer design is different from the operating computer.
@Click Web Server	Web Server Module in OneATS System
Other Client	<i>User-trained applications on demand based on the SmartHIS Driver library</i>
Interfaces: Protocol ports that transmit (write) data to the HIS Server	
Data Client Interface	Interface (Client) according to the External protocol (ATS). <i>(for connection to @Station system version 2.0)</i>
Other Interface	<i>User-trained applications on demand based on the SmartHIS Driver library</i>
OPC-UA Clients: The OPC-UA Client exchanges data with	
Grid Viewer	The HMI application module in the OneATS system, acts as an OPC-UA Client to retrieve (read) data from SmartHIS Server (OPC-UA Server) to display graphical interfaces, draw charts

OneATS DataServer	The real-time data processing module in the OneATS system, acts as an OPC-UA Client that writes data to the SmartHIS Server (OPC-UA Server).
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G.1.3. Copyright Technical Policy

OneATS technically manages the copyright of the components in the OneATS SmartHIS module through the license of the SmartHIS Server, calculated according to the maximum number of data points.

There are two methods of license registration: License soft-key and License Dongle-Key. The license registration method is specified by the respective software version.

When no license is installed, SmartHIS Server automatically runs with **a demo license** with the following limitations:

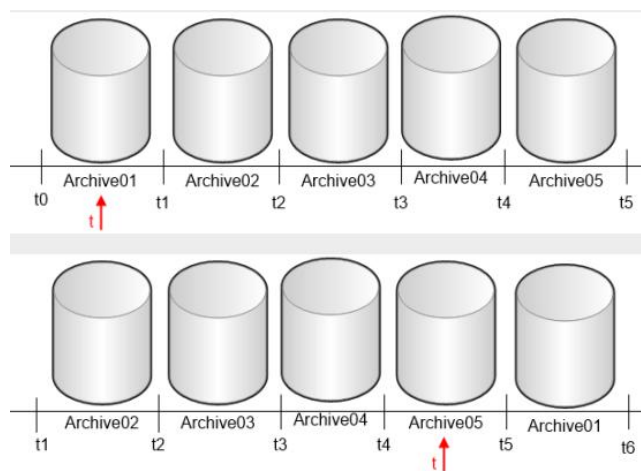
- Maximum number of data points: Unlimited
- Maximum HIS Server running time: 10h. (After 10h, HIS Server service will automatically stop)

G.1.4. Ability to store data on SmartHIS Server

Instead of using conventional periodic data storage algorithms, SmartHIS Server uses high-performance data compression and storage algorithms to store and store only all significant (meaningful) data variations on the data characteristic. Therefore, the parameters of data storage capacity depend on the data characteristics of each project and have higher storage performance and quality than periodic data storage solutions. The parameter table below summarizes the basic descriptions:

Total Data Points	Depends on license.
Storage Capacity / 1 Data Point / 1 Time Unit	Depends on the volatility of the data.
Maximum data retention time	By hard drive capacity / (total storage capacity / 1 unit of time)
Minimum amount of time between 2 data storage points	It depends on the design with each data point. Configurable up to 0s (ms size).

G.1.5. Data Storage Mechanism on SmartHIS Server



The data stored by the SmartHIS Server into a file on the hard drive is called an Archive. Each Archive stores time-stamped data that is within a fixed storage period (start time – end time) of the Archive.

Archives are designed to have sequential timelines and not have any overlapping time periods. SmartHIS Server only writes 1 value to a single archive at a time.

Archives have 2 statuses: Online and Offline. The Archive Online is managed by SmartHIS Server, which is ready for data storage and retrieval. In contrast, Offline Archives are not managed by SmartHIS Server. The Offline Archive is created by the process of backing up data (automatically or manually) in the system. In order to exploit the data, Offline Archives need to be added to the SmartHIS Server to switch to the Online state.

- The entire online data storage cycle is a sequential list of archives. SmartHIS Server is active on the Archive according to the time condition (Archive End time > SmartHIS Server > Archive Start time).
- When the last Archive in the list runs out (SmartHIS Server > Archive End time), the oldest Archive is automatically transferred to the Offline state and moved to the **Backup Offline folder** (DefaultPathBackupDB configuration parameter).
- The standard design recommends that the Backup Offline folder be located on another hard drive with plug-and-play capability, from which this offline data is automatically backed up and the hard drive can be easily replaced when the offline data storage space is exhausted.

At the same time as the offline process of the oldest Archive, SmartHIS Server automatically creates a new Archive with the Start time being the End time of the last Archive in the list, the storage period is equal to the length of the storage time is the same as the oldest Archive that has just been moved to the Offline state to continue the data storage process.

In the event that the hard drive space is full during the data storage process, the SmartHIS Server stops the new data storage process.

In case the total capacity of Archives exceeds the configured parameter (**MaxTotalSizeOfArchive**), SmartHIS Server will automatically delete unused Archives (in the future) from the list of Online Archives.

The data is stored until it is deleted by the administrator. Deleting a data point does not affect the data that has been stored.

SmartHIS Server also supports a tool that helps users manually back up any data in the list of Archive Online.

SmartHIS combines device monitoring (hard drive capacity) from other modules in the OneATS system to issue a full hard drive space alert, not only for the area where past data is stored, but for all drives in the system.

G.1.6. Data synchronization in redundancy mode

SmartHIS Server supports redundancy mode with replication service function: synchronize data and configuration from the source → server to the destination server. Unlimited number of Destination Servers. The system supports the configuration of 2-way communication (anti-loop message), in a relationship between 2 servers, each server can be the source server and the destination server at the same time.

In case the nodes are all online, the replication service automatically synchronizes the configuration and the data stored on the nodes.

In the case of the destination node going offline (network problems, power problems, etc.), when that node is restored, changes in the configuration and stored data of SmartHIS during the time the node is offline are automatically restored by the replication service that synchronizes the data from the source node.

G.1.7. Data Characteristics on SmartHIS Server

G.1.7.1 The data stored on the SmartHIS Server:

Alarm/Event Message with OPC-UA standard attributes.

Attribute	Describe
Message Type	Message Type: Alarm or Event
Time	Alarm/Event time (device time)- 1ms resolution

Receive Time	Time when ... Server receive Alarm/Event Message-1ms resolution
Severity	Alarm/Event Severity
Operator	Operator name
Source Name	Alarm/Event Source (point key name)
Message	Alarm/Event Message
Value	Alarm/Event Data Value

Data points with attributes

Attribute	Describe
Value	Value
Timestamp	Time Label- 1ms Resolution
Quality	according to OPC-UA standard

G.1.7.2 Point classes

Classic	Data points receive data newsletters from data sources
Equation	Data points are calculated (+, -, *, /, ...) from other data points at the same time
Totalizer	Data points are calculated and aggregated from data bulletins of 1 data point in a period of time (min, max, average, count, ...)

G.1.7.3 Data types

Data Type	Describe	Value Range
Boolean		true/false; 0 / 1
Digital	Digital Set Style	Defined in Digital Set
Int8	8-bit signed integer	-128 → 127
Int16	16-bit signed integers	-32,768 → 32,767
Int32	32-bit signed integers	-2,147,483,648 → 2,147,483,647
Int64	64-bit signed integer	-9,223,372,036,854,775,808 → 9,223,372,036,854,775,807
UInt8	8-bit positive integer	0 → 255
UInt16	16-bit positive integer	0 → 65,535
UInt32	32-bit positive integer	0 → 4,294,967,295

uint64	64-bit positive integer	→ 0 18,446,744,073,709,551,615
Float	32-bit floating-point number real	$1,5 \cdot 10^{-45} \rightarrow 3,4 \cdot 10^{38}$
Double	64-bit floating-point number real	$5,0 \cdot 10^{-345} \rightarrow 1,7 \cdot 10^{308}$
String	Character Strings	
DateTime	Time Type	

G.1.7.4 Data Point Storage Configuration Attributes

General	Archive	System	
Server Name	127.0.0.1	Area Name	OAArea
Point Name	B01.XCBR1.BlkLogic	Data Source	OA
Description	Show XCBR Logic Interlock Panel	Point Class	Classic
Point Type	Boolean	Digital Set	System

Attribute	Describe
General - General attributes of a data point	
Area Name	Area Name
Point Name	Point Name -Rule: - The first character must be a number or letter: a-z, A-Z, 0-9 - Does not contain control characters (downline, tabs, ...) - Does not contain the following special characters: * ' ? ; { } [] ~ \ ` ' “ ” - Distinguishing Upper/Lower
Point Description	Point Description
Point Type	Point data types (int, double, string, ...) Reference G.1.7.3 Data types
Data Source	Data source name. - Data imported from the Data Model (Database) is automatically assigned Data Source = OA
Point Class	Data Type (Classic, Equation, Totalizer) Reference G.1.7.2 Point classes
Digital Set	Digital Set Style - Applies when Data Type = Digital Set - Choose from a list of defined Digital Sets

General | Archive | System

Archiving
☒ True ☐ False

Zero Deviation
☐ Step

Exception
 Deviation
 Min. Time Day Hour Min Sec
 Max. Time
 Unexpected change
 Multiples of Deviation
 Time Estimate (s)

Compression
 Deviation
 Min. Time Day Hour Min Sec
 Max. Time

Parameter	Describe
Archive - Data point data storage parameters	
Archiving	Data saved to Archive True: the data is saved to the Archive False: the data is not saved to the Archive but still exists on the Snapshot
Step	Fixed data exploitation (query command results) data/data interpolation results: The server provides data to the client according to the hierarchical characteristics (equivalent to the previous mode interpolation method), independent of the interpolation type parameter in the client's query command. Applies to data with step characteristics. Notes: This parameter simplifies data query commands from the Client. In principle, the server supports simultaneous query commands for multiple data points according to conditions and 1 data interpolation method. Points with different types of data interpolation must use different data query commands. The step parameter allows in the same command to query data multiple data points simultaneously, combining points with cascading characteristics with points with other types of data interpolation characteristics. This is of great significance on clients such as excel, when the data input is in the form of a cell range, with the operation of dragging and selecting a batch of cells
Zero Deviation	Zero Zone Data Noise Threshold - Reference G.1.8.2.4 Algorithm at point 0
Exception/ Deviation	Exception - Deadband - Reference G.1.8.2.2 Exception Filter Algorithm
Exception/ Min. Time	Exception – Min Time - Reference G.1.8.2.2 Exception Filter Algorithm
Exception/ Max. Time	Exception – Max Time - Reference G.1.8.2.2 Exception Filter Algorithm

Parameter	Describe
Exception/ Unexpected change/ Multiple of Deviation	- The latter value changes compared to the previous value with a greater number of times than this value, it can be considered as a noise value and is ignored.
Exception/ Unexpected change/ Time estimate	- If the change in data between the latter and the previous value exceeds the Multiples of Deviation and occurs over a period of time less than this value, it is considered a noise and does not save the value to HIS. - Unit: second(s)
Compression/ <i>Deviation</i>	Compression - Standard Deviation - Reference G.1.8.2.3 LIM3 Algorithm
Compression / <i>Min. Time</i>	Compression – Min Time - Reference G.1.8.2.3 LIM3 Algorithm
Compression / <i>Max. Time</i>	Compression – Max Time - Reference G.1.8.2.3 LIM3 Algorithm

General | Archive | System |

Point Id

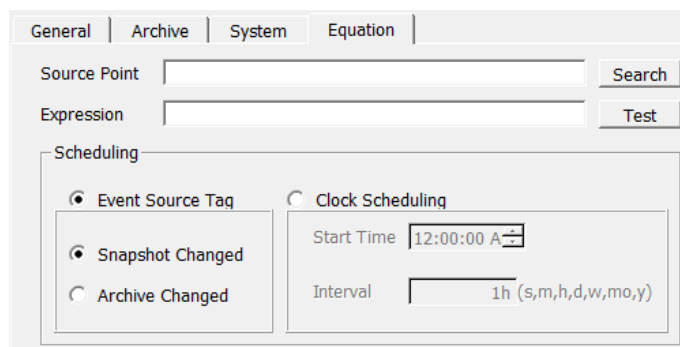
Creator

Changer

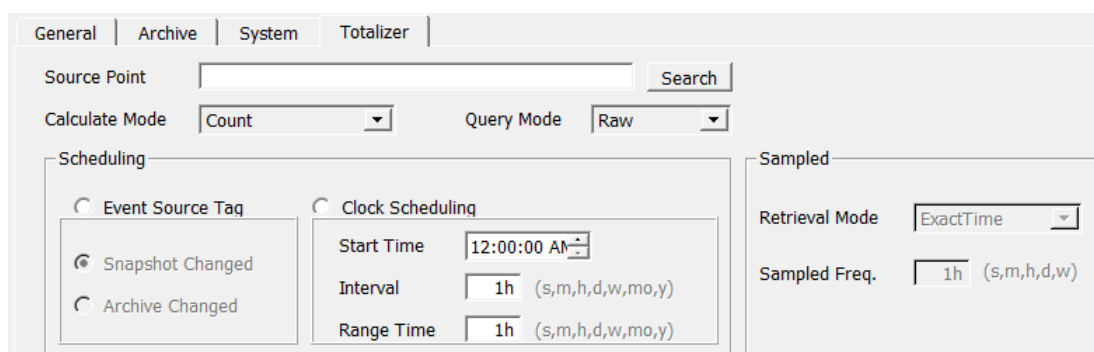
Creator Time

Last Change Time

Parameter	Describe
System - The system (software) attributes of the Data Point	
Point ID	The ID of the point in the HIS Server, which is generated by the system itself.
Creator	User create (new/import mode Create) out point
Changer	User performs the operation to change the last point attribute (save / import with mode Edit)
Creator Time	When to create a point
Last Change Time	The last time a change was made to the point attribute



Parameter	Describe
Equation - The data calculation function properties of the Point Equation	
Source Point	Data points provide trigger events that update <i>the point equation value</i> - It only makes sense to select Scheduling = Event Source Tag
Expression	Execution expressions, point fields, etc.; Point.v; Point.q; Point.t corresponds to the value, quality, and timestamp of the Point. - Reference syntax G.10. Appendix - Expression Syntax
Scheduling	Event Source Tag: <i>Point Equation</i> is updated when <i>the Source Tag</i> changes its value Clock Scheduling: <i>Point Equation</i> is updated periodically
Snapshot Changed	Trigger Event when the Source Tag changes on the Snapshot. (when Scheduling = Event Source Tag)
Archive Changed	Trigger Event when the Source Tag changes on the Archive. (when Scheduling = Event Source Tag)
Start Time & Interval	Frequency of data updates. (when Scheduling = Clock Scheduling)



Parameter	Describe
Totalizer Computational function (statistical) properties of point totalizer	
Source Point	Raw data points to perform statistics

Parameter	Describe
Calculate Mode	• Count: Counts the number of stored values • Total: Calculate the sum of the stored values • Minimum: Find the smallest value • Maximum: Find the Greatest Value • Average: Calculate the average value • Subtraction: The difference between the following value and the previous immediate value • Total_Abs: Calculate the total absolute value • Minimum_Abs: Find the smallest absolute value • Maximum_Abs: Find the Greatest Absolute Value • Average_Abs: Calculate Average Absolute Value • Subtraction_Abs: Calculate the absolute value of the sign between the latter value and the previous immediate value
Query Mode	• Raw: Math all values stored in the Archive • Sampled: Calculated by frequency
Scheduling	• Event Source Tag: <i>Point Equation</i> is updated when <i>the Source Tag</i> changes its value • Clock Scheduling: <i>Point Equation</i> is updated periodically
Snapshot Changed	Trigger Event when the Source Tag changes on the Snapshot. (when Scheduling = Event Source Tag)
Archive Changed	Trigger Event when the Source Tag changes on the Archive. (when Scheduling = Event Source Tag)
Start Time & Interval	Frequency of data updates. (when Scheduling = Clock Scheduling)
Range Time	The period to be calculated (when Scheduling = Clock Scheduling)
Retrieval	Calculation mode: • ExactTime • Interpolate • Previous • Next • Nearest • Previous_2 (when Query Mode = Sampled)
Sampled Freq.	Calculation frequency (when Query Mode = Sampled)

G.1.8. Data Processing with OneATS SmartHIS Server

G.1.8.1 Compute data on the server

In order to meet the specific computing needs of stored data that are not normally met by Data Sources, SmartHIS Server supports the ability to compute with:

- Data point totalizer to meet the needs of aggregating data over time series (e.g., calculating max load capacity in a day).
- Equation-style data points to meet the needs of aggregating data from multiple different data points at the same timestamp.

The calculation points are updated with the value/quality/time-stamp periodically or when there is a change in the input data in the calculation formula depending on the data configuration.

Types of calculations provided:

- The arithmetic of addition, subtraction, multiplication, and division with real numbers, integer values, and natural numbers.
- Absolute value.
- Squared, 3rd power, high power.
- Check the conditions ($>$, $>=$, $=$, $<=$, $<$, $!=$, $<>$).
- Logical operations (or ($|$ |)), and ($&$ &)).
- Structured conditional math: IF (THEN, ELSE).
- Trigonometric functions.
- Maximum, minimum, summary, and average calculations.

Structure of computational expressions using Lua language5.3. Refer to the calculation functions supported by SmartHIS Server in detail at [G.10. Appendix – Expression syntax](#)

G.1.8.2 Data storage

G.1.8.2.1 Overview

OneATS SmartHIS Server is designed with the following objectives in mind:

- Ensure data integrity. Store all meaningful data points (significant fluctuations, irregularities).
- High-performance data compression:
 - Rough removal of values that are considered system interference
 - Eliminate values that do not make much sense in storage but still maintain the integrity of the data, increasing storage capacity.
- The data stored in the HIS is persisted and retrieved until it is deleted by the administrator. Deleting data points/tags in the Data Model does not affect the stored data.

To meet those goals, SmartHIS Server processes data with algorithms

- All Alarm Message newsletters are archived.
- For data value bulletins, a high-performance compression algorithm is used including a 2-step compression technique that combines Exception Filter and Slim3.

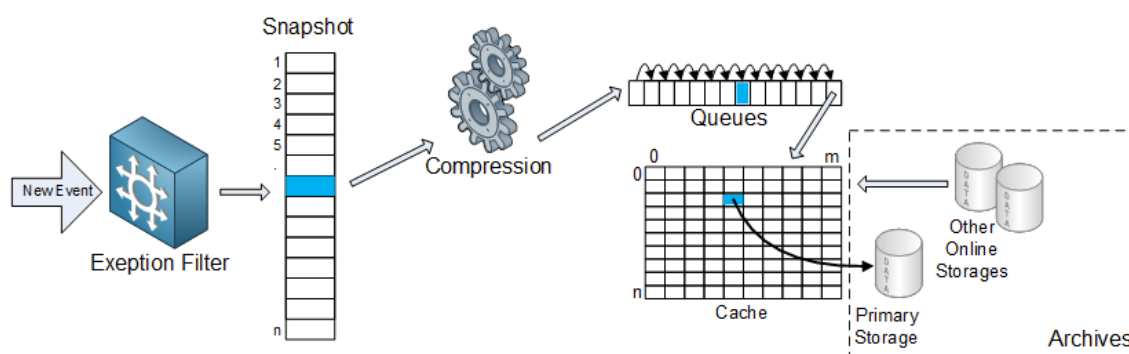


Figure G-3. Data Flow

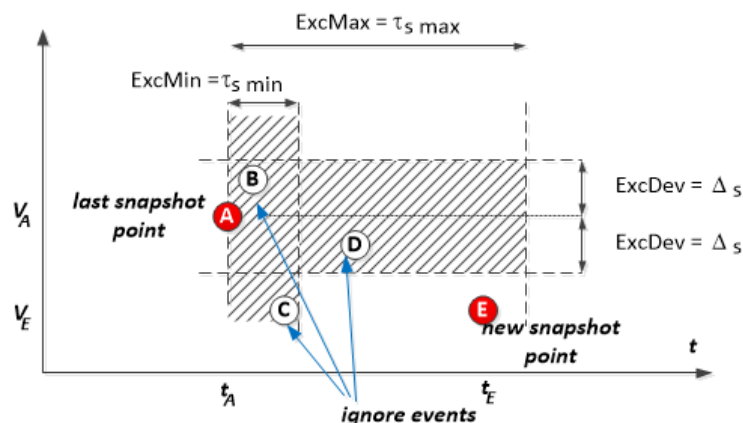
- **Filter Exception:**
 - The data were evaluated using the exclusion test method.
 - In this process, each data point is compared to the previous data and excluded if the variation is negligible.
 - It is a method of coarsely removing values that are considered to be system noise. The excluded data will be included in the Snapshot area.
- **Snapshot:**
 - Snapshot contains all of the tag's most recent data
 - When a new data point arrives, it becomes a snapshot
 - The tag's previous data point will be included in the calculation performed by the Slim3 Compression method
- **Compression:**
 - When a data point is included in a snapshot, the previous data is checked by **Slim3 Compression** to see if it is important for storage. If it is considered critical, the data is sent to the Event Queue for storage, otherwise it will be discarded.
 - This method removes data from values that don't make much sense in storage (e.g., points between the ends of a line) while preserving the integrity of the data.
- **Queues:**
 - The queue acts as a buffer for data after processing by **Slim3 Compression**, waiting to be stored.
 - This element neutralizes the difference between the two processes of **Slim3 Compression** data compression and storing the data to a hard disk.
- **Cache:**

- Cache is a cache used to minimize the frequency of access to the hard disk, it will automatically push data down the hard disk when the cache is full or when a specified time threshold is exceeded, helping to speed up data storage.
- **Archive:**
 - These are data storage files that reside on a hard drive
 - Each Archive contains only data with a specified time stamp that falls between the start and end times of that Archive.
 - If a data is written to the Archive with a timestamp that is not in any of the Archives' time periods, it will be discarded.

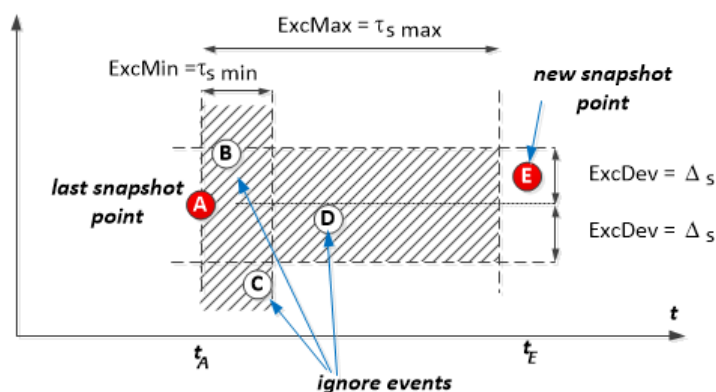
The data storage parameter configuration needs to be designed:

- Matching the natural characteristics of each data
- Suitable for data storage/exchange parameters of pre-SmarHIS Server data collection, e.g. deadband parameter (report by exception) in IED/IEC61850 Server or IED/IEC60870-4-104 Slave

G.1.8.2.2 Exception Filter Algorithm



FigureG-4. Exception Filter Algorithm (1)



FigureG-5. Exception Filter Algorithm (2)

This process is characterized by an exclusion zone (a striped area on the image), which means that the noisy or fluctuating data area does not matter. Compared to the existing data point on the snapshot (A), new data points (B, C, D) that have the same Quality group and are in the exclusion zone will be discarded. Conversely, a new data point (E) with the same Quality group and not in the exclusion zone will be updated to the snapshot, and the existing data point (A) will be transferred to the Slim3 Compression data compression calculations.

The exclusion zone is set by the set of parameters:

Symbol	Configuration Parameters	Meaning
$\tau_s \min$	ExcMin	(Exception Min): time-deadband is the period of time when all data fluctuations during that period are negligible
Δ_s	ExcDev	(Exception Deviation): value-deadband A value range in which all data fluctuations within that value range are negligible
$\tau_s \max$	ExcMax	(Exception Max): time max Override The amount of time that data after that period needs to be stored to reduce the deviation of the stored data characteristics from the actual data characteristics

A value (E) is saved to the snapshot if it is outside the exclusion zone:

$$\left\{ \begin{array}{l} \tau_s \min < \tau_E - \tau_A < \tau_s \max \\ |V_E - V_A| > \Delta_s \\ Q_A, Q_E \text{ are the same quality} \end{array} \right.$$

or

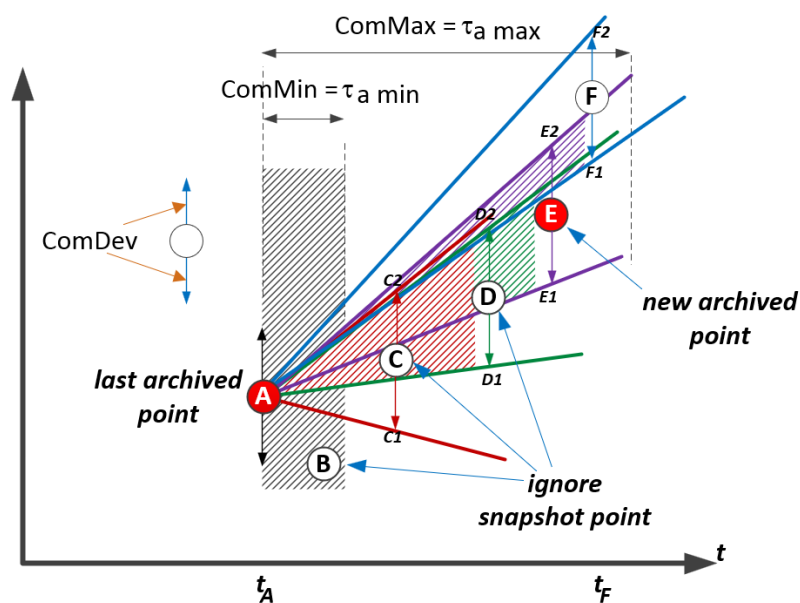
$$\left\{ \begin{array}{l} \tau_E - \tau_A \geq \tau_s \max \\ Q_A, Q_E \text{ are the same quality} \end{array} \right.$$

In which:

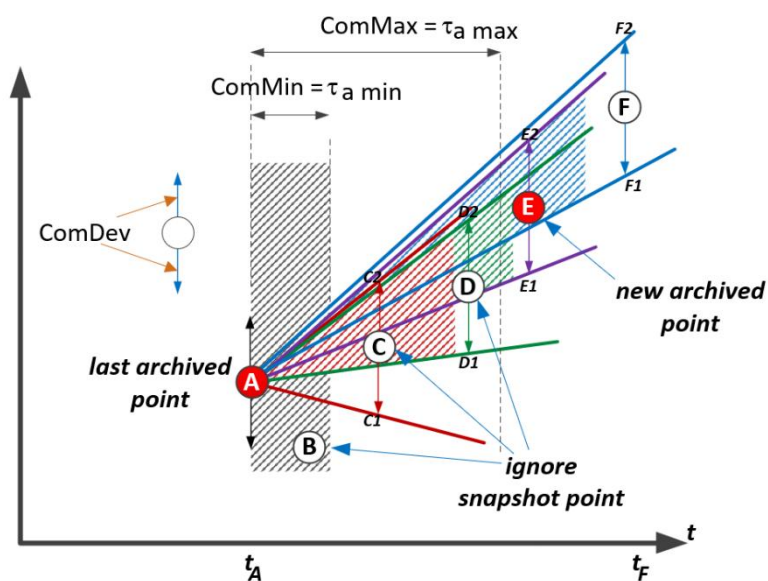
τ_A	Time-stamp of A	τ_E	Time-stamp of E
V_A	Value of A	V_E	The value of E
Q_A	Quality of A	Q_E	Quality of E

Data processing exceptions (quality changes, data fluctuations at point 0, etc.) are described in the 0, G.1.8.2.5, ...

G.1.8.2.3 SLIM3 Algorithm



FigureG-6. SLIM3 Algorithm (1)



FigureG-7. SLIM3 Algorithm (2)

This process is characterized by an exclusion zone (striped area in the image), which signifies the region where data points can be linearly interpolated with an error within a configurable limit based on the data characteristics.

In FigureG-6:

- Data points evaluated (B, C, D, E) within the same Quality group are sequentially added to the snapshot.
- Point B, when added to the snapshot, falls within the temporal exclusion zone ($\tau_B - \tau_A < \tau_a \text{ min}$) and is therefore discarded.
- When C and D are added to the snapshot, the exclusion zone becomes the angle D_1AD_2 . Point C falls within this zone and satisfies $\tau_a \text{ min} < \tau_C - \tau_A < \tau_a \text{ max}$, so it is discarded.
- When E is added to the snapshot, the exclusion zone becomes the angle E_1AE_2 . Point D falls within this zone and satisfies $\tau_a \text{ min} < \tau_D - \tau_A < \tau_a \text{ max}$, so it is discarded.
- When F is added to the snapshot, the exclusion zone becomes the angle F_1AE_2 . Point E does not fall within this zone and satisfies $\tau_a \text{ min} < \tau_E - \tau_A < \tau_a \text{ max}$, so it is stored in the archive.

In FigureG-7:

- Points B, C, and D are discarded due to the same conditions as in Figure G-6.
- Point E falls within the exclusion zone but has a time interval of $\tau_E - \tau_A > \tau_a \text{ max}$, so it is still stored in the archive.

The exclusion zone is set by the set of parameters:

Symbol	Configuration Parameters	Meaning
$\tau_a \text{ min}$	ComMin	(Compression Min): time-deadband is the period of time when all data fluctuations during that period are negligible
Δ_a	ComDev	(Compression Deviation): Rate of Change-deadband is the limit error interval of data interpolation between 2 stored data points
$\tau_a \text{ max}$	ComMax	(Compression Max): time max Override The amount of time that data after that period needs to be stored to reduce the deviation of the stored data characteristics from the actual data characteristics

A value (E) in a snapshot is saved to Archived if that value (E) is outside the striped area:

$$\begin{cases} \tau_a \min < \tau_E - \tau_A < \tau_a \max \\ (E) \text{ is outside the striped area} \\ Q_A, Q_E \text{ are the same quality} \end{cases}$$

or

$$\begin{cases} \tau_E - \tau_A \geq \tau_a \max \\ Q_A, Q_E \text{ are the same quality} \end{cases}$$

τ_A	Time-stamp of A	τ_E	Time-stamp of E
Q_A	Quality of A	Q_E	Quality of E

Data processing exceptions (quality changes, data fluctuations at point 0, etc.) are described in the G.1.8.2.4, G.1.8.2.5, ...

The Slim3 data exclusion and compression test method does not apply to the following data types:

- Digital
- Boolean
- String
- DateTime

G.1.8.2.4 Algorithm at point 0

The absolute 0 score is a special point because of the error and interference effect of the measurement (measurement algorithms, measuring devices, communication errors, ...) at 0. The error at point 0 may differ from other measurement value ranges. In fact, there are many signals where a value of 0 is interpreted as $c(0 \pm \Delta)$ with Δ being the allowable margin of error at point 0.

SmartHIS supports this characteristic of the data with the **Zero Vision configuration parameter**. Any data value that falls within the range ($\pm$ **Zero Devision**) is calculated and stored by SmartHIS with a value of 0.

With so many signals, the zero point also has special significance including reflecting changes in the state of the device. For example, "Inverter run" → String Current >0; "Inverter stop" → String Current =0. Therefore, a value of 0 is meaningful to reflect a significant change in the actual process and is not suitable for removal from data storage algorithms. SmartHIS stores all data fluctuations from a value of $\neq 0$ → to a value of 0, as well as all data fluctuations from a value = 0 → to a value of $\neq 0$.

G.1.8.2.5 Algorithms with quality change events

All changes in the quality of the data are abnormal changes and are updated by SmartHIS in the snapshot and stored in the Archive without considering the conditions of time and value.

G.1.8.2.6 Other Treatments

Future Data Processing

In theory, the data is stored from a real-time process system, so the timestamp of the data cannot come from the future. The data with the future time stamp is the result of a faulty data collection process, or a system failure to synchronize the data for the collected data. These data do not guarantee time accuracy and are removed from storage by SmartHIS Server.

SmartHIS Server evaluates future ability by comparing the data time label with the SmartHIS Server time (operating system time). In practice, due to the possibility of deviations in the time synchronization systems at the SmartHIS Server and at the data collection stages, the SmartHIS Server allows for future time errors that the system accepts data with the **FutureMinute** configuration parameter.

Initialization Values & First Data

Right at the time of point creation, SmartHIS Server will store the first data for that point with the value "**Pt Created**" with a time stamp. This value reflects when the system starts generating and is ready to store data for that point. Data queries before the initialization time will not make sense.

The first data that the SmartHIS Server receives after the point is initialized is not excluded by "Future Data" will be updated to the snap-shot and stored in the Archive.

G.1.8.3 Data Mining

G.1.8.3.1 Data query commands

SmartHIS Server & Client supports a list of data query commands:

Function	Describe
Current Value	Query the latest value (snap-shot) of the point
Archive Value	Query point/expression data at a specified time in the archive
Compressed Data(Raw Data)	Query all stored values of a point for a specified period of time
Sampled Data	Query point/expression data over a period of time with a defined sampling frequency
Timed Data	Query data with any number of defined timelines
Calculated Data	Statistical calculation of historical data over a specified period of time
Condition Data	Calculation according to the conditions
Time Filtered	Calculate the total time that an expression is satisfied.
Get Alarm	Alarm/Event Statistics for a Period of Time

Detailed list of data query commands and parameters are referred to the corresponding documentation sections of client applications (Admin Tool-Data, Excel-ATSDData Link, Web Service, ...)

G.1.8.3.2 Data query modes

In contrast to data storage that uses data compression algorithms, data mining is the "decompression" activity – restoring the compressed data to meet the data query request. The data recovery method must be selected in accordance with the applied data compression storage algorithm (parameters). These two processes must generally conform to the real data characteristics (data characteristics before storage).

Querying data at a timestamp is processed as follows:

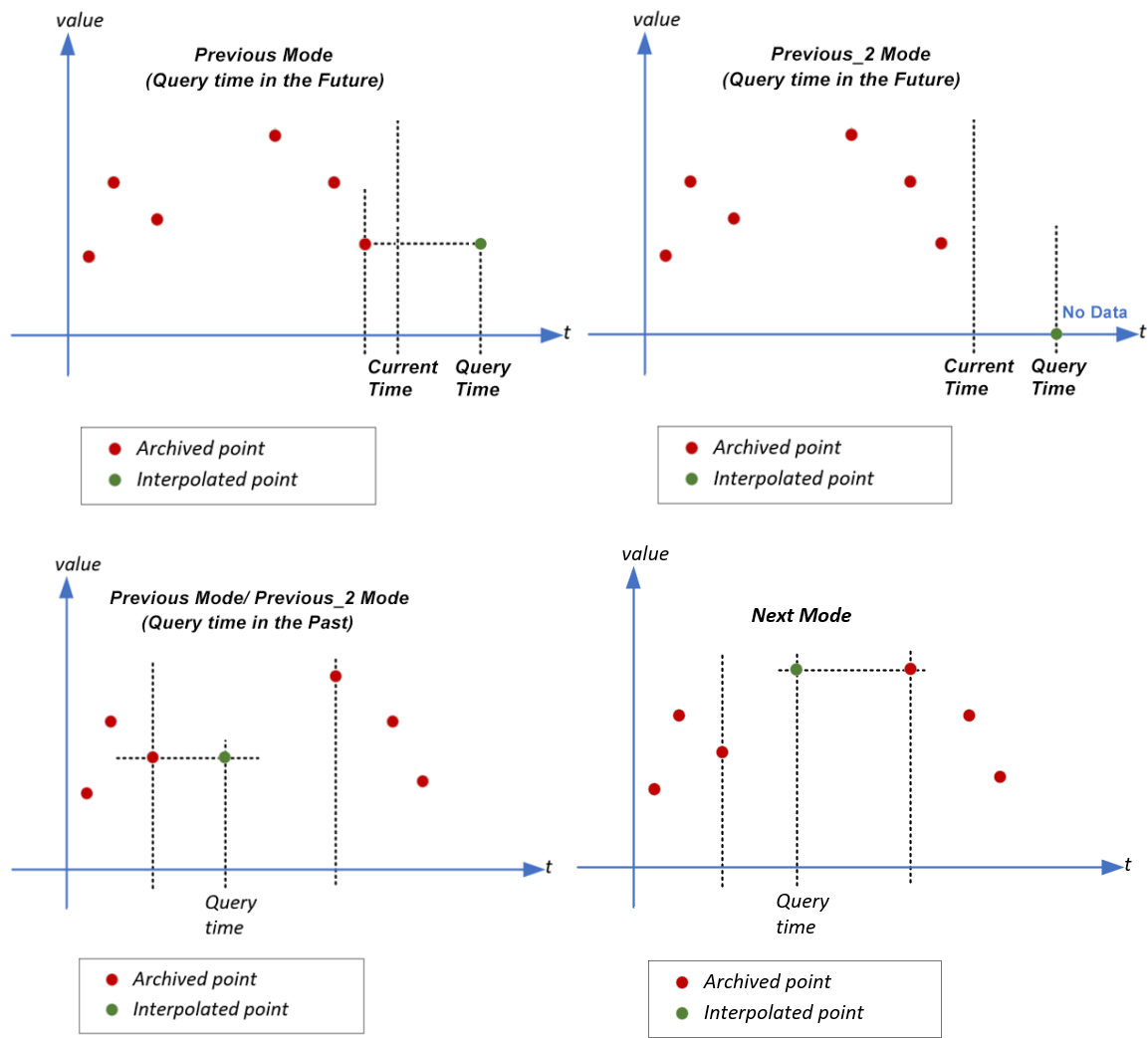
- If at the timestamp to be queried there is stored data. The stored information (value/quality/time-stamp) will be returned to the query request.
- If there is no data stored at the query timestamp, the information (value/quality/time-stamp) will be calculated from the data stored around (before and after) the query timestamp, depending on the data query mode selected by the client

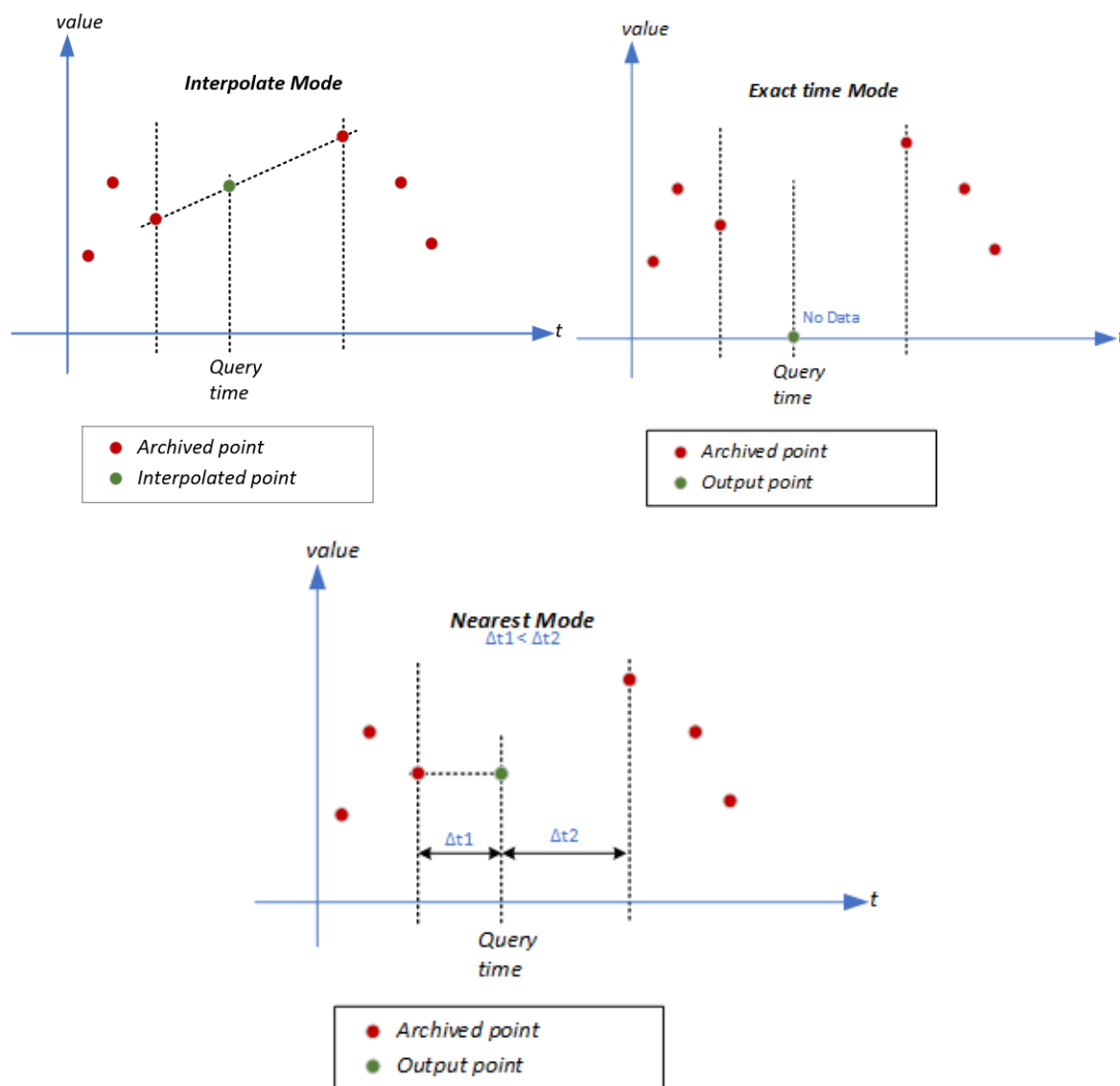
Data query modes:

Previous	The data has the same value and quality as the data that was stored at the time-stamp immediately preceding the time-stamp being queried
Previous_2	The data has the same value and quality as the data that was stored at the time-stamp immediately preceding the time-stamp being queried. Querying data at a future time (now + futureTime) will result in a NoData response
Next	The data has the same value and quality as the data that has been stored at the time-stamp immediately after the time-stamp is queried.
Interpolate	The data is of the same quality as the data that has been stored at the time-stamp immediately preceding and following the time-stamp being queried. If the data immediately before and after has a different quality, the returned data has a quality of Uncertain The valuable data is linearly interpolated from 02 data that have been stored at the time-stamp immediately before and after the time-stamp is queried
Exact time	The data has the same value and quality as the data that has been stored at the time-stamp is the same as the time-stamp being queried. If no data at the time-stamp is queried, the returned data is No Data

Nearest	The data has the same value and quality as the data that has been stored at the time-stamp closest to the time-stamp being queried
---------	--

However, with data with a ladder value characteristic (configuration parameter Step = true), the data is always interpolated in Previous mode without depending on the client's query request.





FigureG-8. Data query modes

G.1.8.3.3 Calculation expressions in data query commands

SmartHIS Server also supports expressions in data query statements. When it receives requests to query data according to the expression, the Server will execute:

- Query the component data points in the expression.
- Calculate the values of the expression according to the frequency of the data in the query command.
- Return the calculated data to the Client application.

Refer to the calculation functions supported by SmartHIS Server in detail at [G.10 Appendix – Expression Syntax](#)

G.1.8.3.4 Exceptions

The data query functions that return the usual results include the value – time label – quality label of the data. In special cases, the results returned are the exceptions:

Value	Describe
<i>Exceptions</i>	
Pt Created (121)	is the data returned at the time the point is generated in the SmartHIS Server
No Data (122)	Data returned when no data point is found that matches the data query mode
Data Error (123)	Data is returned when data is found but fails due to storage.

G.1.9. Configuration and Data Privacy Policy

SmartHIS uses User policies to secure configurations and data. Clients use the User information that has been configured to connect to the HIS Server, perform configuration operations, and read/write data according to the user's permissions provided.

G.1.9.1 Configuration Security Policy and Activity Monitoring

SmartHIS uses a user classification policy to secure the configuration. Users are divided into 03 levels:

- Admin
- Operator
- User

Delegated levels to HIS Server configurations:

Feature	Decentralization	Admin	Operator	User
Software Upgrade				
License	Activate License	O	X	X
Configuration				
Account Configuration	Edit User's own information	O	O	O
	Create, delete, edit other users	O	X	X
Operation Configuration	View information Archives	O	X	X
	Create, delete, edit, register (restore) Archives	O	X	X
	View Replication Configuration	O	O	O

	Configuring Replication	O	O	X
Area Configuration	View Area Configuration	O	O	O
	Create, delete, edit Areas	O	O	X
	Add, remove (decentralize) Group Users for Areas	O	X	X
Digital Dataset Configuration	View Digital Dataset Configuration	O	O	O
	Create, delete, and edit Digital Datasets	O	O	X
Point Configuration	View Point Configuration	(A)	(A)	(A)
	Create, delete, edit points	(A)	(A)	X
Operation Management				
Operation Management	Log Viewer	O	O	O
	Network Statistic	O	O	O

(A) : according to Area decentralization

O : YES

X : FULL

G.1.9.2 Data Privacy Policy

SmartHIS uses a user/Area matrix decentralization policy to secure data.

- Each data is categorized under one of the Areas.
- Each User can be configured to belong to different User Groups, or not belong to any User Group. The User inherits all the permissions of the User Group with the Data Areas.
- Each area is configured with a list of User Groups that are allowed to read/write data.

Feature	Decentralization	Admin	Operator	User
Data Read/Write				
Data Point Read/Write	Read the value (Snapshot/Archive)	(A)	(A)	(A)
	Write values (Snapshot, Archive)	(A)	(A)	X
	Delete Recorded Data	X	X	X
Alarm Read/Write	Read the Alarm newsletter	O	O	O
	Alarm Logging	O	O	X
	Delete Recorded Data	X	X	X
Data Update	Calculation	(A)	(A)	(A)



Backup & Restore Data				
Data Point		(A)	(A)	(A)
Alarms		O	O	O

(A) : according to Area decentralization
O : YES
X :FULL

G.2. Installing the SmartHIS package

G.2.1. App Environment Settings

G.2.1.1 Install system libraries

OneATS SmartHIS version on the Windows operating system requires some of the application environment libraries (.NET, Visual C++, ... libraries) to be pre-installed.

G.2.1.2 Install Excel for the DataLink Add-in component

In order to implement the Excel Add-in Datalink component installation in the SmartHIS installer, Excel software needs to be installed with the "Office Shared Tools" & "Office Tools" libraries (automatic installations include these components).

SmartHIS version	Compatible Excel Versions
4.x and later versions	Excel 2007 and later versions

Refer to the corresponding version of Excel installation documents from the Microsoft manufacturer.

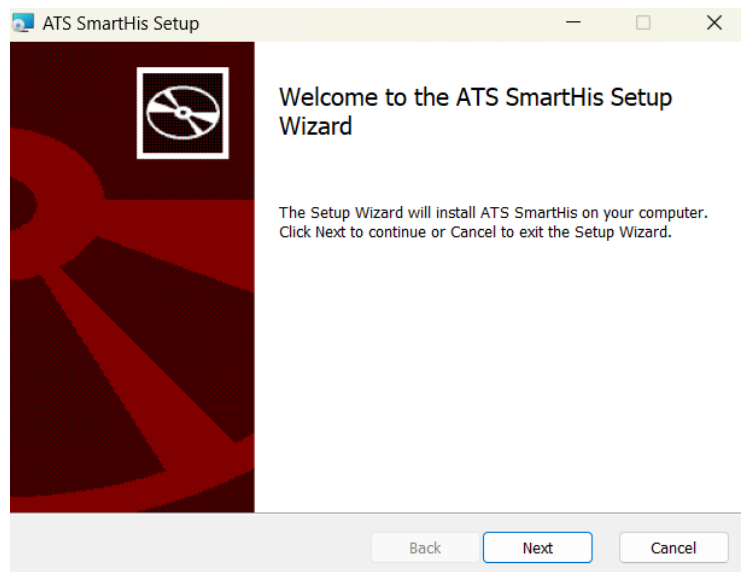
G.2.1.3 Installing Internet Information Services (IIS) for the WEB Service component

To deploy the WEB Service component in the SmartHIS installer, the IIS window feature needs to be installed/enabled.

Refer to the IIS installation documents according to the corresponding operating system version from the Microsoft manufacturer.

G.2.2. Installing ATS SmartHIS Components

Run the ATS SmartHIS Setup.msi **installation file**.

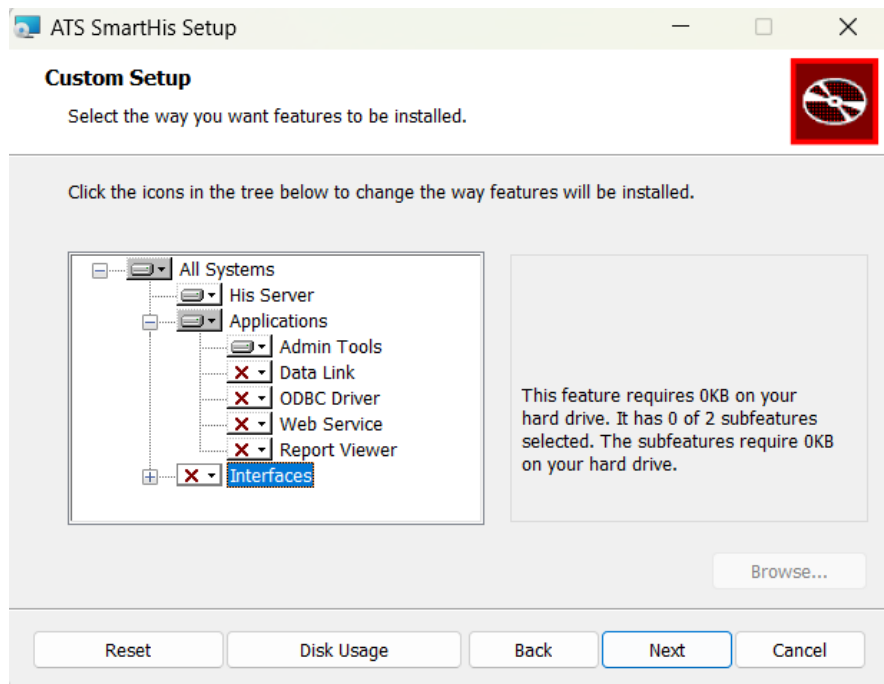


FigureG-9. Install ATS SmartHIS Components

Select **Next** to continue:

In the **All Systems** section, users can choose to install additional components according to their needs.

Select **Next** to continue:

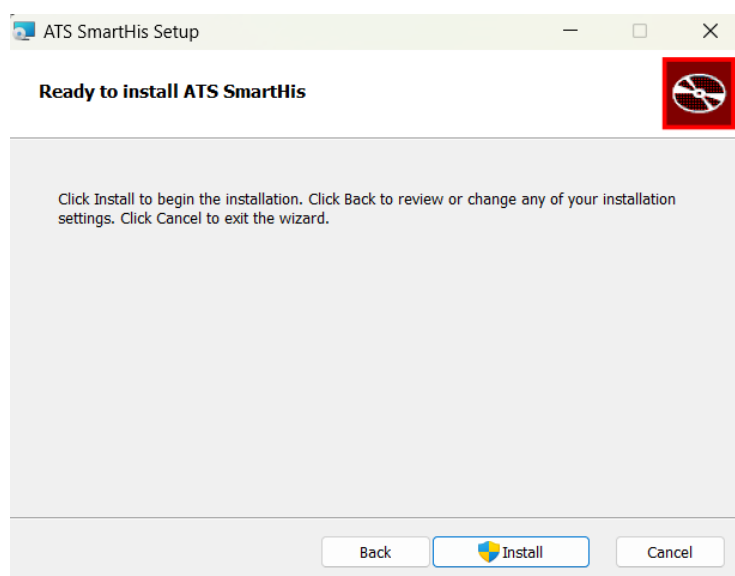


FigureG-10. Install ATS SmartHIS Components

Ingredient	Describe
His Server	Data Storage Server
Applications: Data mining applications	

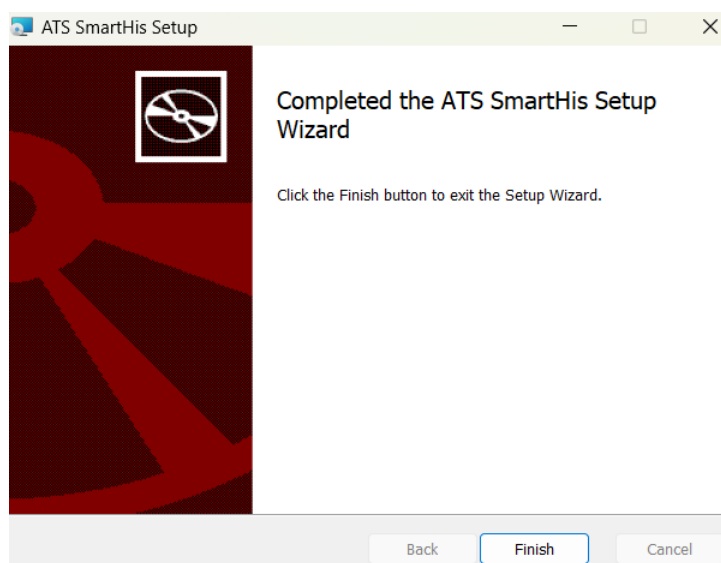
Admin Tools	SmartHIS Administrator – a tool to configure and check HIS Server
Data Link	Excel Add-in, Library of HIS Server → Excel Data Retrieval Functions
ODBC Driver	Windows ODBC Driver for HIS Server
WEB Service	REST API Web Service (running on IIS).
Report Viewer	*.xlsx file report viewer application independent of Microsoft Excel
Interfaces: Protocol ports that transmit data to the HIS Server	
Data Client	DataClient Protocol-Based Interface (ATS)
OPC UA	OPC UA Protocol

Select **Install** and wait for the installation to take place:



FigureG-11. Install ATS SmartHIS Components

Select **Finish** to finish the installation process:



FigureG-12. Finish the installation process

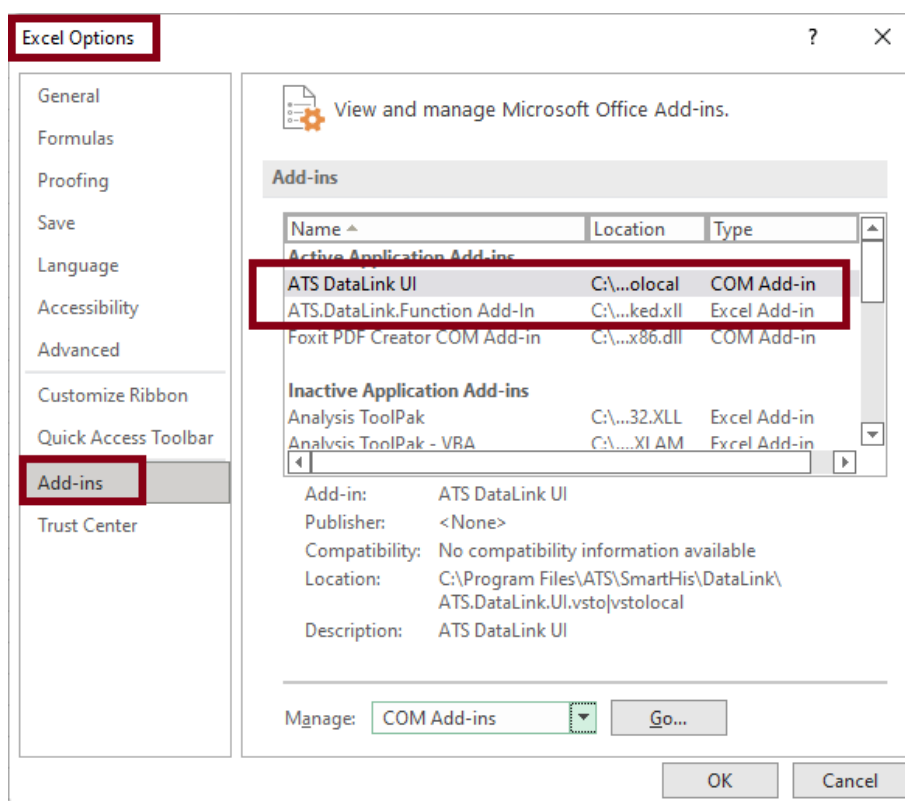
G.2.3. Install and check the HIS Server license

Refer to the [ChapterH.OneMasterTool documentation](#)

G.2.4. Integrate DataLink Add-in into Excel

DataLink Excel Add-in is an integrated library into excel that includes data retrieval functions and configuration support interfaces.

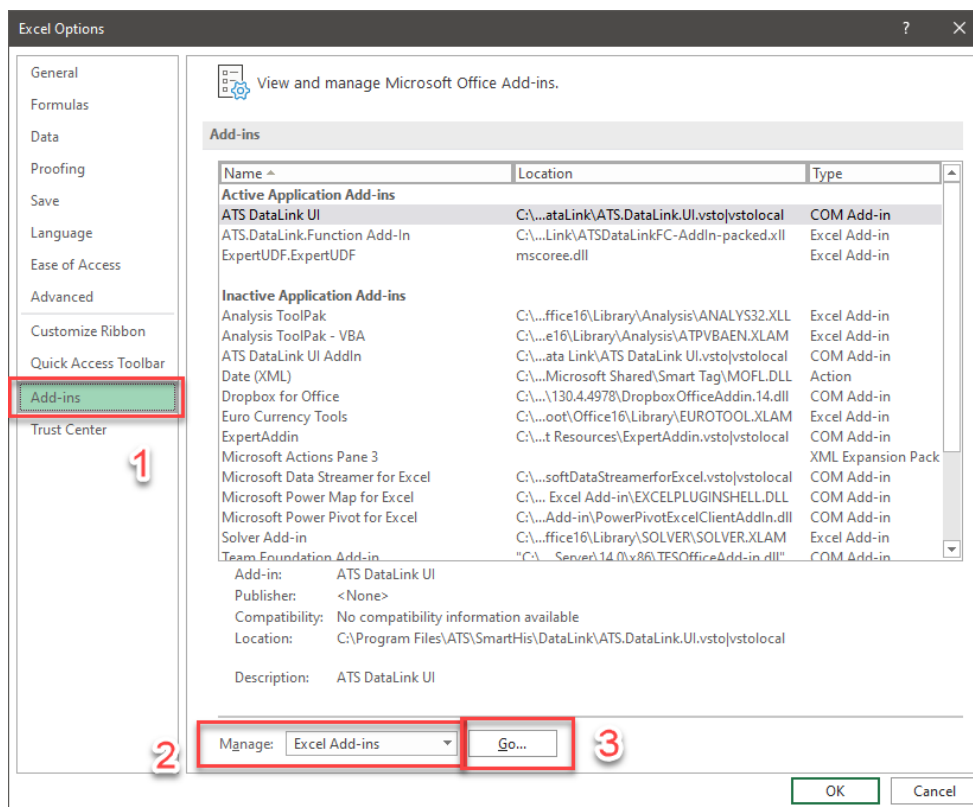
Add-ins	Describe
ATS DataLink UI	Configuration Support InterfaceDisplays the ATS tab on the Menu bar
ATS DataLink Function Addin	Library of Data Retrieval Functions



FigureG-13. Integrate DataLink Add-in into Excel

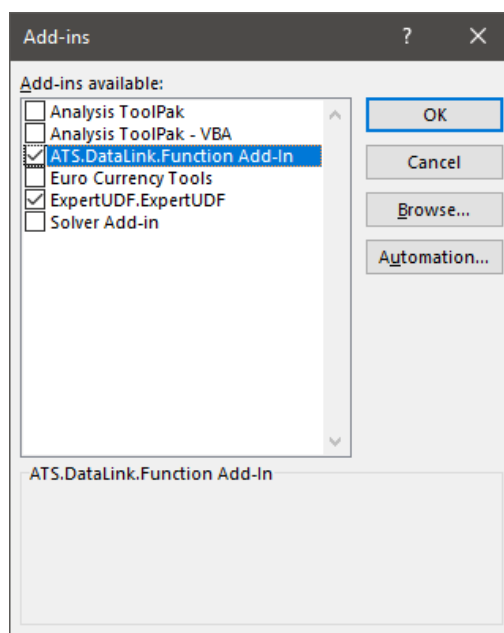
Once the Excel & SmartHIS/DataLink add-in has been installed, check that the add-in has been automatically integrated into Excel.

In case Excel is installed after SmartHIS, or the operating system is not properly decentralized, resulting in the DataLink add-in not being integrated into Excel, proceed with manual integration.



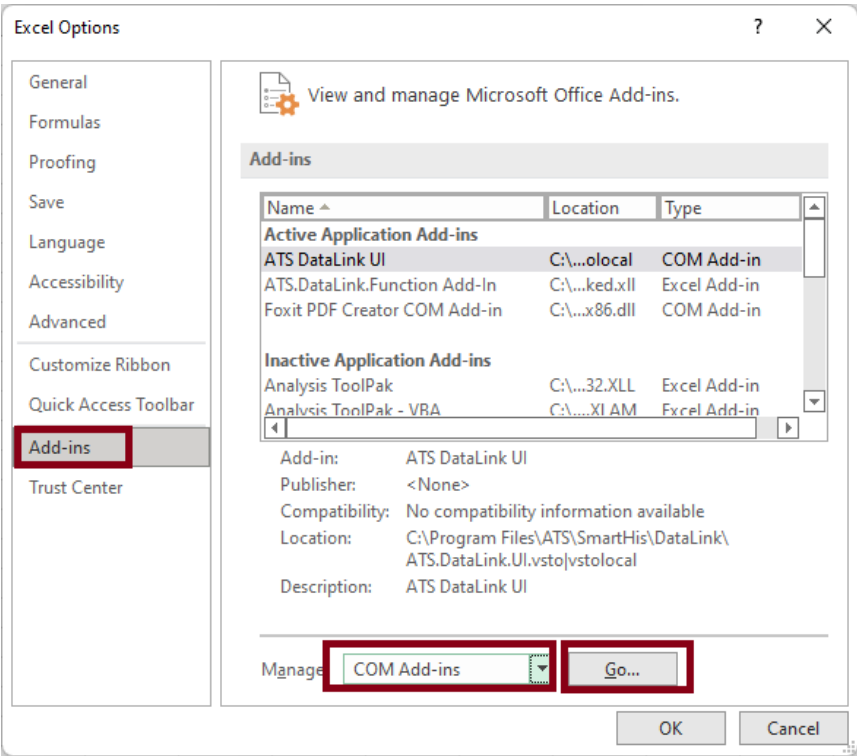
FigureG-14. Configuration Interface Add-in Function in Excel Option

File → Option → Add-ins → Manager Excel Add-ins → Go



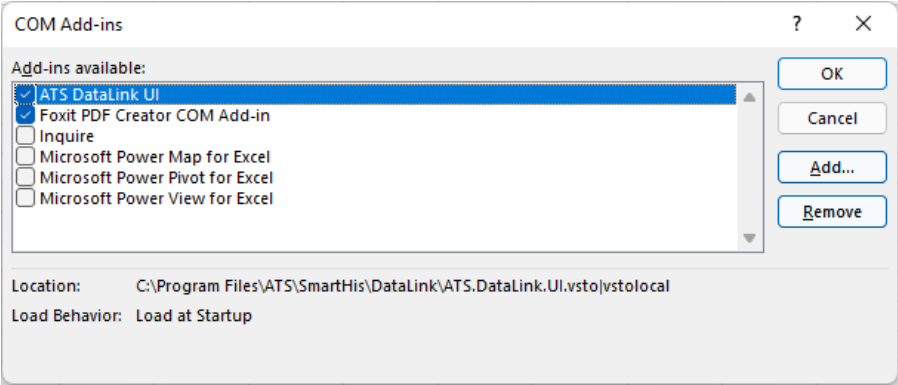
FigureG-15. Choose to install ATS Add-ins. DataLink.Function

ATS Add-ins → dialog. DataLink.Function Add-in → OK



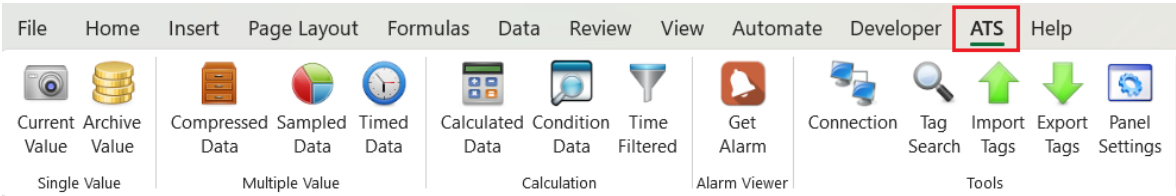
FigureG-16. Configuration Interface Add-in UI in Excel Option

Excel Option → Add-ins → Manager COM Add-ins → Go



FigureG-17. Choose to install ATS Add-ins. DataLink.UI

The ATS **menu** will be added on the Excel menu:

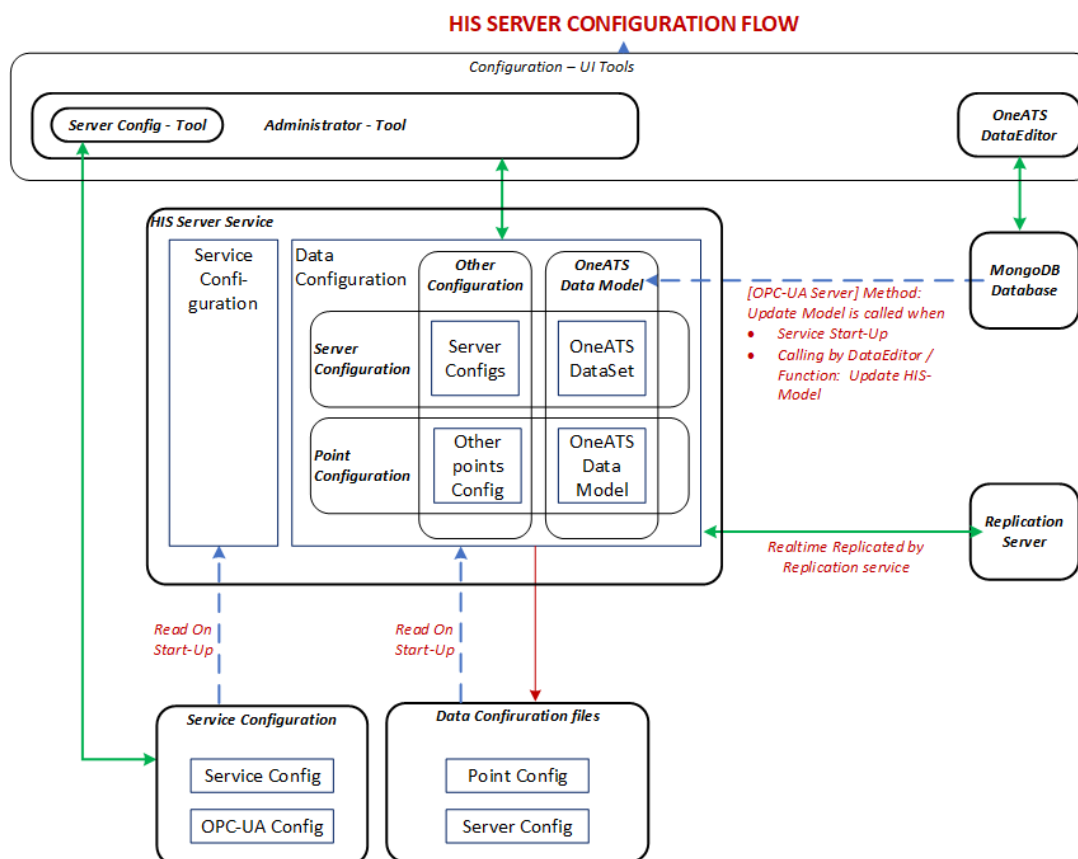


FigureG-18. Supported features at ATS Data Link Add-ins

In this ATS menu, there will be features that allow HIS data to be mined directly from Excel.

G.3. SmartHIS Server

G.3.1. SmartHIS Server Data Configuration



FigureG-19. SmartHIS Server Configuration Diagram

G.3.1.1 Data Storage Design

G.3.1.1.1 Design storage capacity

The total storage time should be designed so that the total storage capacity does not exceed the designed hard drive capacity, or conversely, the total hard drive capacity that needs to be designed to be sufficient for the data storage time. Although the storage capacity depends on the fluctuation characteristics of the data, the average storage capacity / 1 period is a parameter calculated after a period of stable system operation or reference from the equivalent system, thereby building a suitable design.

The total storage capacity is divided into Archives. The storage period for an archive affects the capacity of that archive. The minimum storage period per **Archive** is 1 hour, the maximum is 1440 hours.

In addition, the total number of Archives (number of files) affects the speed of data retrieval in a data retrieval command. Typical data retrieval needs (time periods) should be considered during the design process.

For a smooth data storage process, it is recommended to configure:

- Maximum capacity parameter for Archives (**MaxTotalSizeOfArchive**) < (hard drive capacity designed for data storage – 02 x average capacity for 1 Archive)
- Balance between 2 factors of retrieval speed and maximum use of available storage to design the capacity (time) for 1 Archive.

Configure the Archive with the Administrator tool. Refer to [G.4.6.1 Archive Manager for details](#).

G.3.1.1.2 Design data permissions (User & Area).

SmartHIS uses a privacy policy and decentralization according to the User & Data Area. Refer to the detailed description at G.1.9. These policies need to be designed during the implementation of SmartHIS.

The basis of these designs depends on the analysis of the needs of sharing and configuring data in the system.

Configure the User with the Administrator tool. Refer to detail [G.4.5. Account](#)

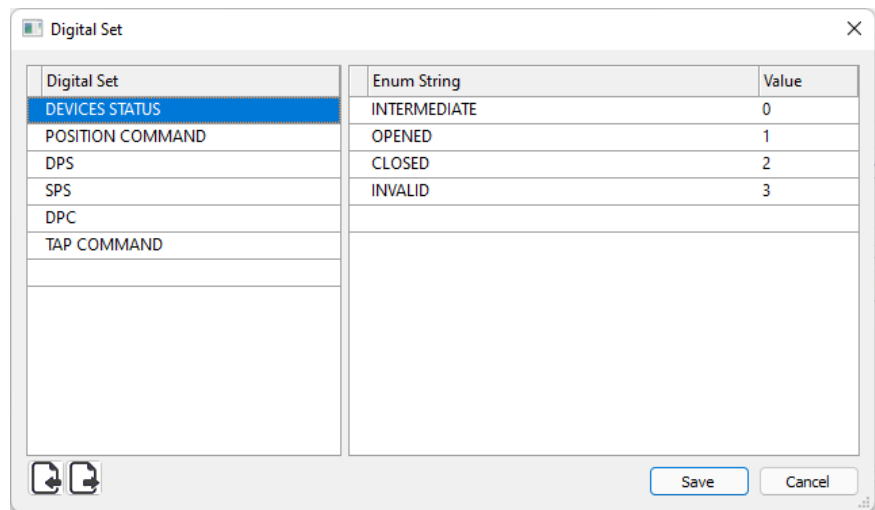
Configure the (Data) Area with the Administrator tool. Refer to [G.4.3.1 Area Manager for details](#).

G.3.1.2 SmartHIS Server Configuration

Refer to the **ChapterH.OneMasterTool** documentation

OneATS SmartHIS can operate as a standalone system with ***the SmartHIS Administrator configuration tool***

- The general configuration of OneATS SmartHIS Server operation is configured by the SmartHIS Administrator tools. Refer to G.4 for details . [SmartHIS Administrator](#)
- The data according to the OneATS DataModel is configured by the OneATS Data Editor tool.
 - Digital Dataset Configuration

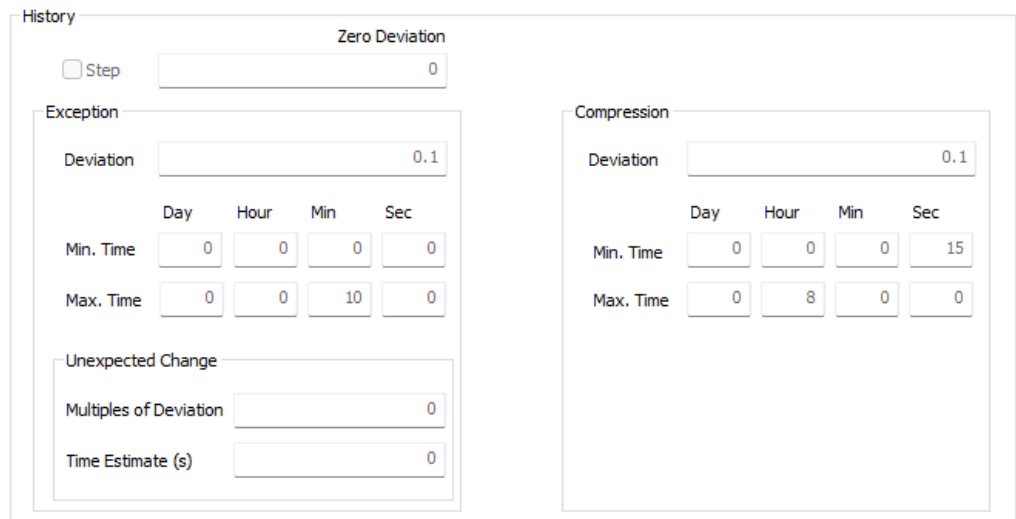


The 'Digital Set' window shows a list of digital sets on the left and a table of enum strings on the right. The 'DEVICES STATUS' set is selected. The table lists three enum strings: INTERMEDIATE (0), OPENED (1), and CLOSED (2). There is also an 'INVALID' entry with a value of 3. The window has 'Save' and 'Cancel' buttons at the bottom right.

Enum String	Value
INTERMEDIATE	0
OPENED	1
CLOSED	2
INVALID	3

FigureG-20. Digital Dataset Configuration

○ Data Point Configuration / History Tab



The 'History' tab of the 'Data Point Configuration' window shows settings for 'Zero Deviation' and 'Compression'. The 'Zero Deviation' section includes a 'Step' checkbox, a 'Deviation' field (0.1), and time fields for 'Min. Time' and 'Max. Time'. The 'Compression' section includes a 'Deviation' field (0.1) and time fields for 'Min. Time' and 'Max. Time'. There is also an 'Unexpected Change' section with 'Multiples of Deviation' and 'Time Estimate (s)' fields.

Field	Day	Hour	Min	Sec
Min. Time	0	0	0	0
Max. Time	0	0	10	0

Field	Day	Hour	Min	Sec
Min. Time	0	0	0	15
Max. Time	0	8	0	0

FigureG-21. Data Point Configuration

Refer to the data configurations in detail in the OneATS DataServer documentation.

- Other data that does not follow the OneATS DataModel, including points and datasets, is configured by the SmartHIS Administrator. Refer to item. G.4

G.3.1.3 Configuration update method

SmartHIS Server configurations are stored by configuration files. The configuration files are loaded when the service starts.

OneATS DataModel data profiles are updated synchronously when:

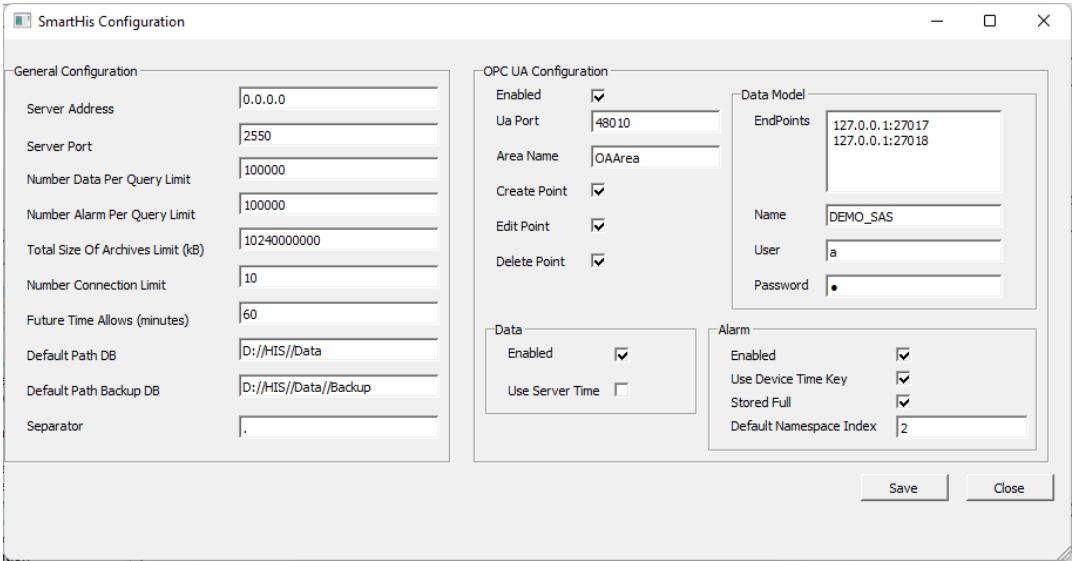
- Start the SmartHIS Server service.
- Run the data synchronization function from the **OneATSDataEditor/His/Update Data Model** tool.

The SmartHIS Administrator tool allows for interactive configuration updates online.

G.3.2. Configure the HIS Server runtime deployment

G.3.2.1 Configure OneATS HIS Server service

- **Method 1:** Refer to [G.3.1.2 SmartHIS Server Configuration](#)
- **Option 2:** With SmartHis Administrator tool: Menu Tools/ Server Config or select the Configuration  icon ®

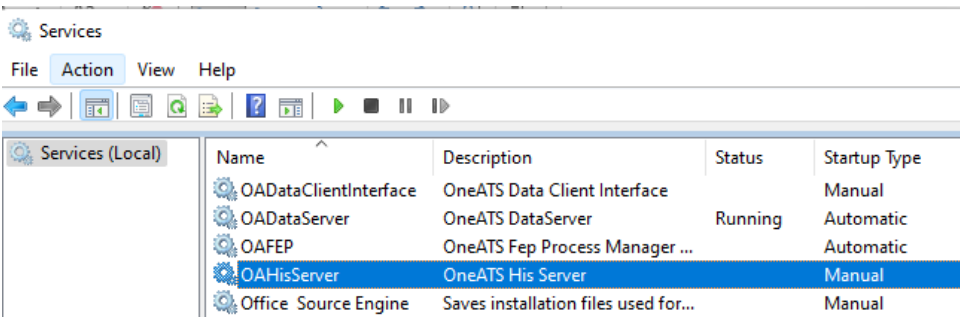


FigureG-22. Configure OneATS HIS Server service

G.3.2.2 Start OneATS HIS Server service with Windows

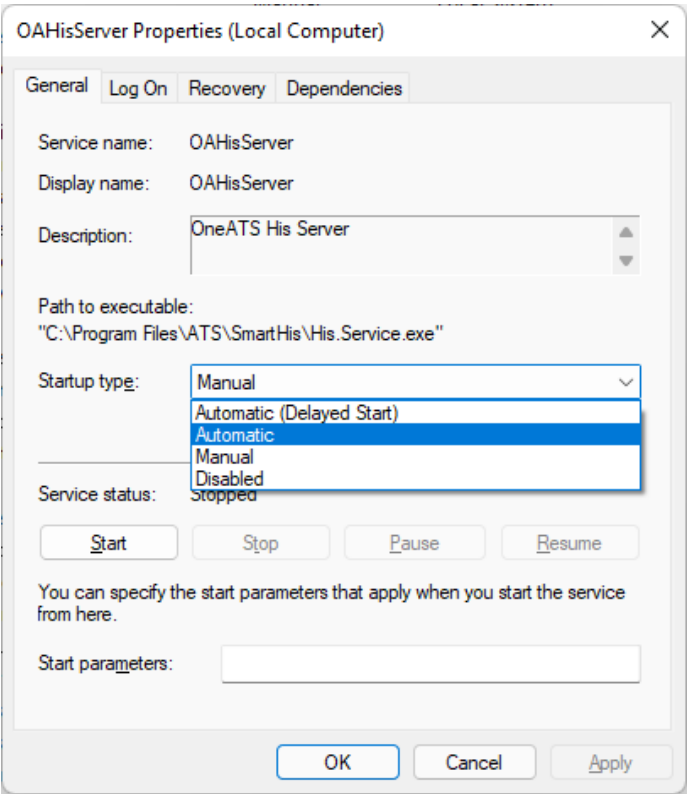
To start OneATS HIS Server service with Windows:

- Run the Window Services app



FigureG-23. Start OneATS HIS Server service with Windows

- Setting up OAHIServer service: Start-up Type = Automatic



FigureG-24. Start OneATS HIS Server service with Windows

G.3.3. Backup/Restore SmartHIS Server Configuration

Backup and restore SmartHIS Server configuration by copy/paste configuration files. Refer to the location of the configuration file at [G.3.1.2 SmartHIS Server Configuration](#)

Depending on the configuration, SmartHIS Server Service can automatically synchronize data from (MongoDB) OneATS DataModel at startup

G.3.4. SmartHIS Server Data Backup/Restore

G.3.4.1 Automatic data back-up

Refer to the automatic data back-up mechanism at G.1.5 – Data storage mechanism

G.3.4.2 Manual data back-up

Archive Backup	
Admin Tool/Operation/ Archive Manager /Backup	Archive Backup - Refer to G.4.6.1.5 Backup Archive
	In addition, Archive Offline files (Unregistered or Unregistered) can be manually backed up with

	copy/paste operations on the operating system environment
Data Backup	
Admin Tool/Operation/ Backup & Restore /Backup	Back up arbitrary data for a customized period of time - Refer to G.4.6.5 Backup and Restore Data

G.3.4.3 Data Recovery

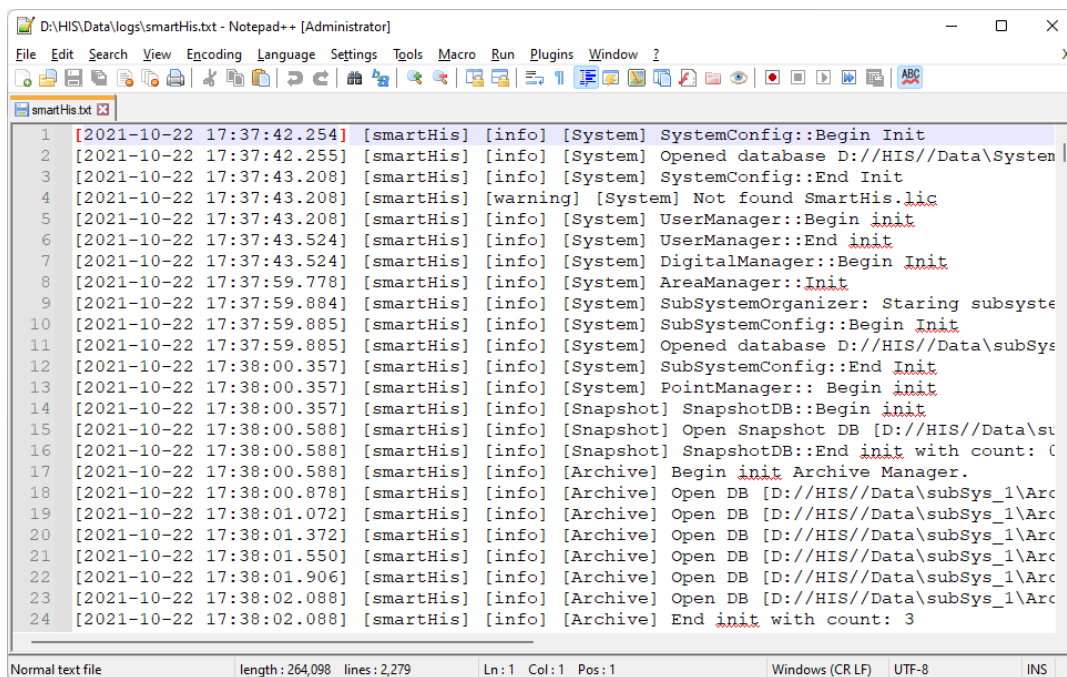
Restore Archive	
Admin Tool / Operation / Archive Manager / Register	Restore Archive - Refer to G.4.6.1.6 Register Archive
Data Recovery	
Admin Tool/Operation/ Backup & Restore /Restore	Restore arbitrary data during the customized period that has been backed up - Refer to G.4.6.5 Backup and Restore Data

G.3.5. Monitoring HIS Server Log – runtime

File smartHis.txt	Log the operation process (start, stop, license, download configuration files, ...) of OneATS HIS Server service
Admin Tool / Operation / Log Viewer	Log activities/Errors in data storage, processing, and query operations (snap-shot, Archive, ...) on OneATS HIS Server
Admin Tool / Operation / Network Statistic	Statistical Monitoring of Data Exchange

G.3.5.1 HIS Server service activity log

The logs of the OneATS HIS Server service are stored in file **smartHis.txt** in the configured folder (DefaultPathDB) in the OneATS HISServer configuration file.



```

1 [2021-10-22 17:37:42.254] [smartHis] [info] [System] SystemConfig::Begin Init
2 [2021-10-22 17:37:42.255] [smartHis] [info] [System] Opened database D://HIS//Data\System
3 [2021-10-22 17:37:43.208] [smartHis] [info] [System] SystemConfig::End Init
4 [2021-10-22 17:37:43.208] [smartHis] [warning] [System] Not found SmartHis.lic
5 [2021-10-22 17:37:43.208] [smartHis] [info] [System] UserManager::Begin init
6 [2021-10-22 17:37:43.524] [smartHis] [info] [System] UserManager::End init
7 [2021-10-22 17:37:43.524] [smartHis] [info] [System] DigitalManager::Begin Init
8 [2021-10-22 17:37:59.778] [smartHis] [info] [System] AreaManager::Init
9 [2021-10-22 17:37:59.884] [smartHis] [info] [System] SubSystemOrganizer: Staring subsystem
10 [2021-10-22 17:37:59.885] [smartHis] [info] [System] SubSystemConfig::Begin Init
11 [2021-10-22 17:37:59.885] [smartHis] [info] [System] Opened database D://HIS//Data\subSys
12 [2021-10-22 17:38:00.357] [smartHis] [info] [System] SubSystemConfig::End Init
13 [2021-10-22 17:38:00.357] [smartHis] [info] [System] PointManager:: Begin init
14 [2021-10-22 17:38:00.357] [smartHis] [info] [Snapshot] SnapshotDB::Begin init
15 [2021-10-22 17:38:00.588] [smartHis] [info] [Snapshot] Open Snapshot DB [D://HIS//Data\su
16 [2021-10-22 17:38:00.588] [smartHis] [info] [Snapshot] SnapshotDB::End init with count: 0
17 [2021-10-22 17:38:00.588] [smartHis] [info] [Archive] Begin init Archive Manager.
18 [2021-10-22 17:38:00.878] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
19 [2021-10-22 17:38:01.072] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
20 [2021-10-22 17:38:01.372] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
21 [2021-10-22 17:38:01.550] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
22 [2021-10-22 17:38:01.906] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
23 [2021-10-22 17:38:02.088] [smartHis] [info] [Archive] Open DB [D://HIS//Data\subSys_1\Arc
24 [2021-10-22 17:38:02.088] [smartHis] [info] [Archive] End init with count: 3

```

FigureG-25. OneATS HIS Server application log file

Basic Information:

- Load configuration file **serverConfig.json** complete & configuration file error if any
- License error (if any)
- Connecting to Database / project name
- Redundancy status
- ...

G.3.5.2 Log activities / Errors in data storage, processing, query

Reference [G.4.6.2 Log Viewer](#)

G.3.5.3 Data Exchange Statistics

Reference [G.4.6.3 Network Statistics](#)

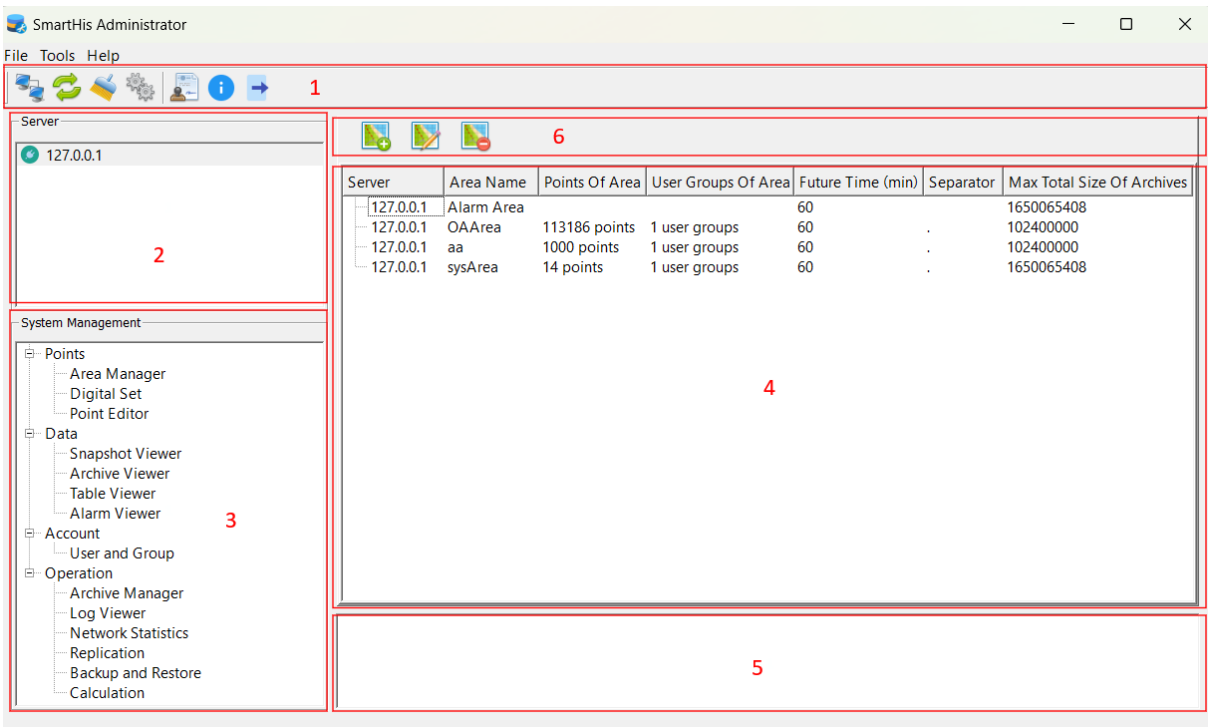
G.3.6. Monitor operational data on SmartHIS Server

Reference [G.4.4 Data – data monitoring](#)

G.4. SmartHIS Administrator

SmartHIS Administrator is a tool for managing, configuring, and checking the status and data on OneATS SmartHIS Server.

G.4.1. SmartHIS Administrator interface overview

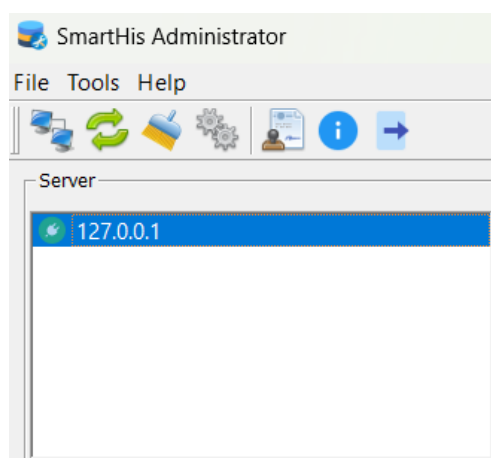


FigureG-26. SmartHIS Administrator Interface

[1]	App Menu Bar		
	Function	Symbol	Describe
	File/Exit		Close the app window
	Tools/Connections		HIS Server Connections Management List
	Tools/Refresh		Refresh the displayed information (update monitoring values, configuration attributes, etc.)
	Tools/Clear Logs		Delete log messages on the display [5]
	Tools/Server Config		Call the Server Configuration Interface

	Help/License		License Management Interface for Server
	Help/About		Admin Tool App Info
	—		
[2]	Server List – List of HIS Server connections		
[3]	System Management – Catalogue of management, configuration, and testing of HIS Server		
	Points	Manage and configure data points	
	Data	Querying alarms/data	
	Account	Account Management	
	Operation	Configure, exploit, and manage HIS Server operations	
	—		
[4]	Configuration Properties & Data Window		
[5]	Activity log window on Administrator		
[6]	Toolbar New/Edit/Delete/...		

G.4.2. Manage and connect HIS Servers

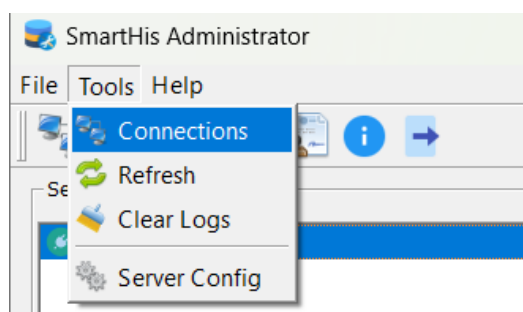


The list of Admin **Tool servers** is contained in the [adminConfig.xml](#) file located in the C:\ProgramData\ATS\SmartHis folder

G.4.2.1 Manage the list of HIS Server connections

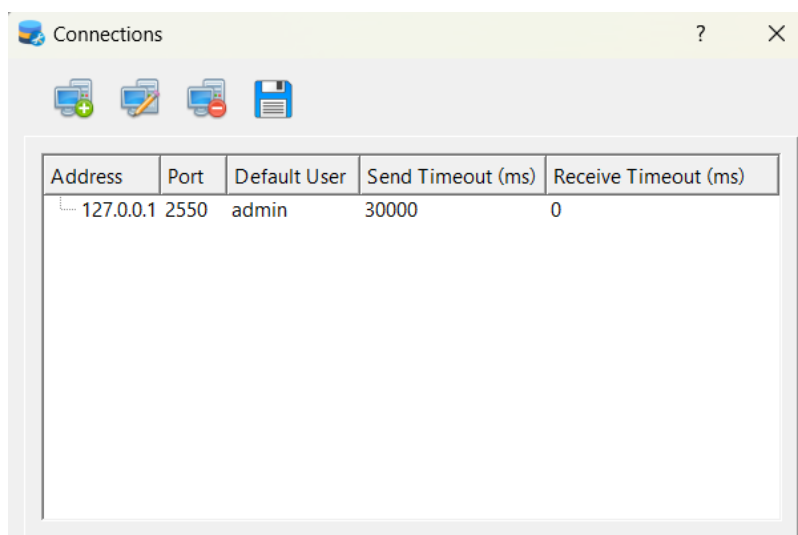
G.4.2.1.1 See the list of HIS Server connections

Menu **Tools** → **Connections** or click 



FigureG-27. Connections Menu

The list of HIS Server connections is displayed:



FigureG-28. List of connections to HIS Server

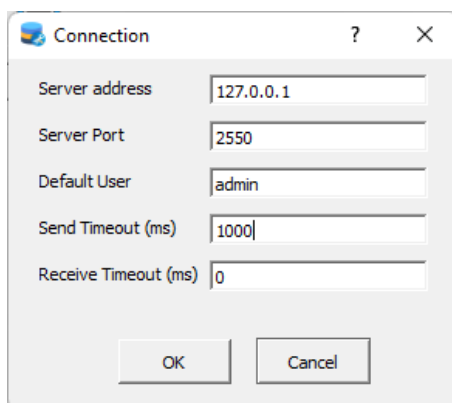
HIS Server connection information:

Address	IP of HIS Server
Port	His Server Administration Port
Default User	User defaults on the HIS Server login window
Send Timeout (ms)	Timeout (ms) query command from Admin Tool
Receive Timeout (ms)	Time (ms) timeout to receive results (data) from Admin Tool

G.4.2.1.2 Add a new HIS Server connection

From the list of HIS Server connections: Click the **Add** button 

Enter the HIS Server connection information



A dialog box titled "Connection" with a question mark icon and a close button. It contains six input fields: "Server address" with "127.0.0.1", "Server Port" with "2550", "Default User" with "admin", "Send Timeout (ms)" with "1000", and "Receive Timeout (ms)" with "0". At the bottom are "OK" and "Cancel" buttons.

FigureG-29. HIS Server connection information

Click the **Save** button  to save the calibrated information to the configuration file (*adminConfig.xml*).

G.4.2.1.3 Edit HIS Server Connection

From the HIS Server Connection List window

- → Click to select the HIS Server to edit → Click the **Edit** button 
- → Or Double-Click on the HIS Server connection to be edited

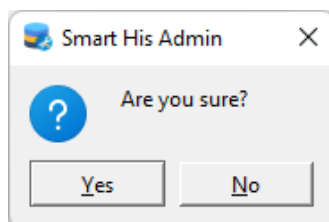
Enter the HIS Server connection information

Click the **Save** button  to save the calibrated information to the configuration file (*adminConfig.xml*).

G.4.2.1.4 Clearing the HIS Server Connection

From the list of HIS Server connections window, Click the **Delete** button 

Click Yes to confirm (No to ignore).



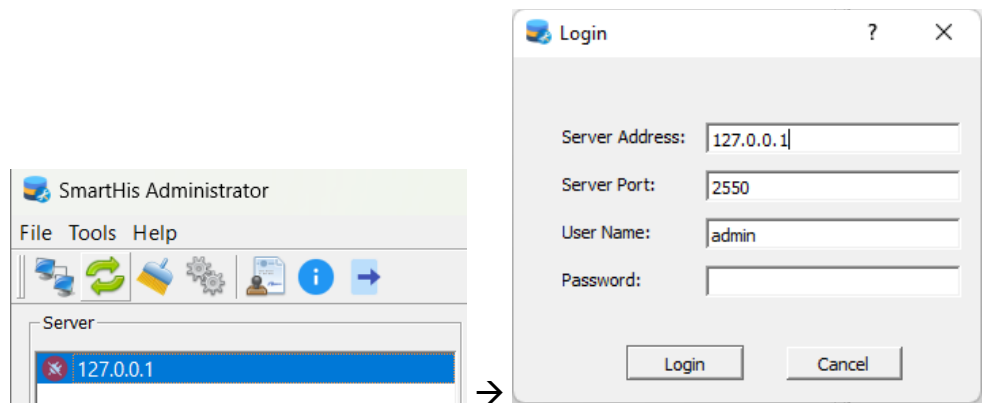
A dialog box titled "Smart His Admin" with a close button. It contains a question mark icon and the text "Are you sure?". At the bottom are "Yes" and "No" buttons.

Click the **Save** button  to save the calibrated information to the configuration file (*adminConfig.xml*).

G.4.2.2 Connecting/disconnecting to the HIS Server

G.4.2.2.1 HIS Server Connection

Double-click server not yet connected () in the list



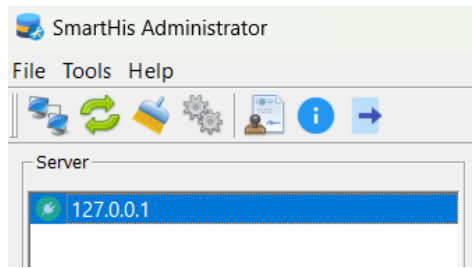
FigureG-30. HIS Server Login Window

ServerAddress	IP of HIS Server
ServerPort	His Server Administration Port
User Name / Password	User/Password Log in to HIS Server

Default User: admin
Default Password: admin

Click **login** to connect.

The Server address icon turns blue indicating that the connection was successful.



G.4.2.2.2 *Disconnect the HIS Server*

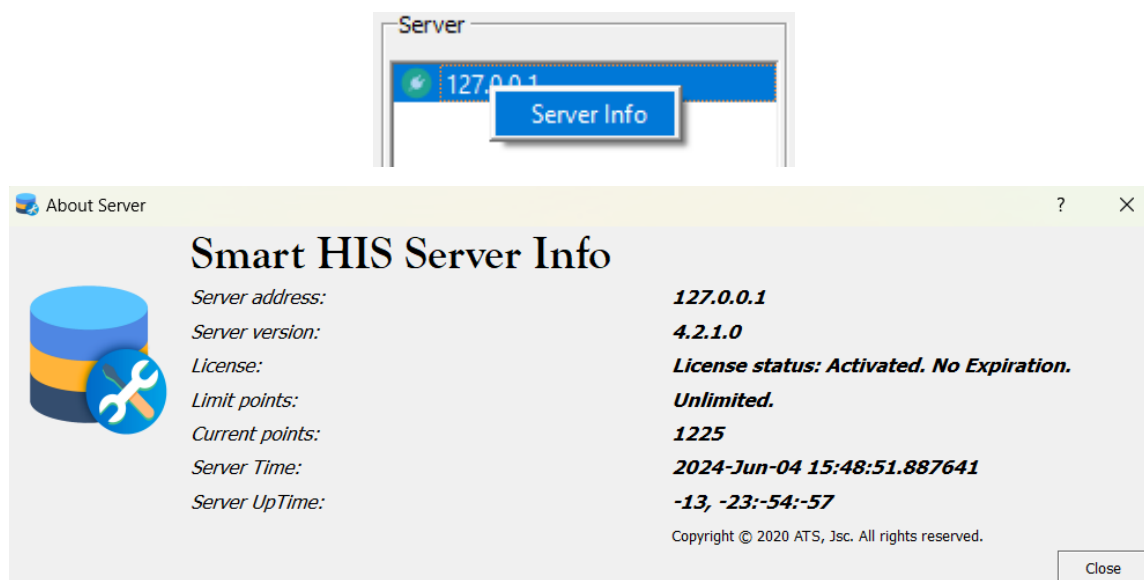
Double-click on the **connecting server** () to disconnect that server.

The Server address icon turns red indicating that the connection has been disconnected.

G.4.2.3 *View HIS Server information*

Connect to the HIS Server.

Right-click on HIS Server in the list of Servers to view the information.



FigureG-31. HIS Server Information

Information displayed:

Server Address	IP Address of HIS Server
Server version	HIS Server version
License	License Status
Limit points	Number of points per license
Current Point	Number of points being exploited
Server Time	HIS Server Time

G.4.3. Points – manage the configuration of datapoints

Menu Points provides a tool to manage the configuration of the *system's* data points:

Area Manager	Data Partition Management
Digital Dataset	Manage datasets
Point Editor	Point Configuration

G.4.3.1 Area Manager

Data Area is a data partition that serves the decentralization of data manipulation in the system. Each User Group only has the right to manipulate (read/write) data in the configured data areas.

Area Manager provides tools for managing data areas.

Area partition design generally follows the regulations on control decentralization/data governance decentralization.

G.4.3.1.1 *List of Areas*

Menu **System Management** → **Points / Area Manager**

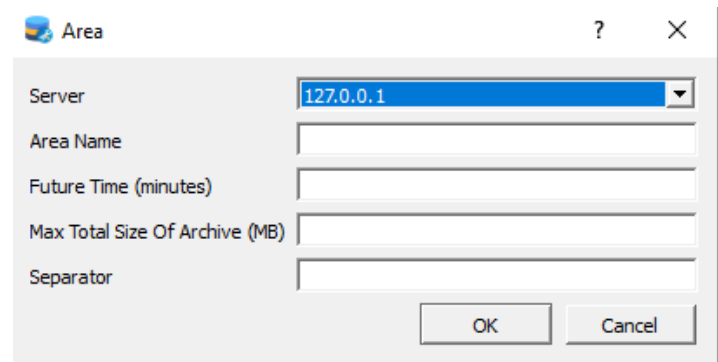
Server	Area Name	Points Of Area	User Groups Of Area
127.0.0.1	OAArea	31179 points	1 user groups
127.0.0.1	sysArea	14 points	0 user groups

Area Information:

Server	HIS Server contains Area
Area Name	Name of Area
Points of Area	Total points in Area
User Groups of Area	Total Area User Groups

G.4.3.1.2 *Add Area*

Select the **Add icon** 



FigureG-32. Add an area for the server

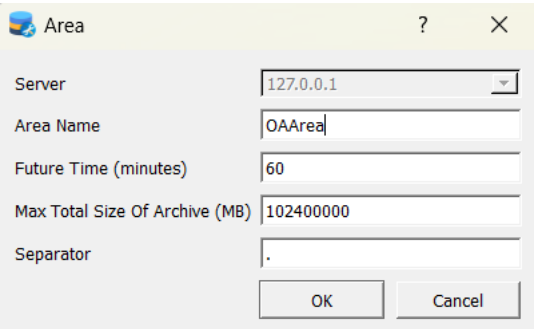
Enter the information for Area:

Server	HIS Server in the list of HIS Server connections
Area Name	Name for area
Future Time (minutes)	The period of time ahead of a data point compared to the current time allows the data to still be saved to His Server
Max Total Size Of Archive (MB)	Largest Archive Capacity

Separator	Character separator of layers in the name of the data point (Supports displaying data structures on application interfaces in SmartHIS system)
-----------	---

G.4.3.1.3 Edit Area

Select the **Edit**  → to customize the Area's configuration parameters.



The 'Area' dialog box contains the following fields:

Server	127.0.0.1
Area Name	OAArea
Future Time (minutes)	60
Max Total Size Of Archive (MB)	102400000
Separator	.

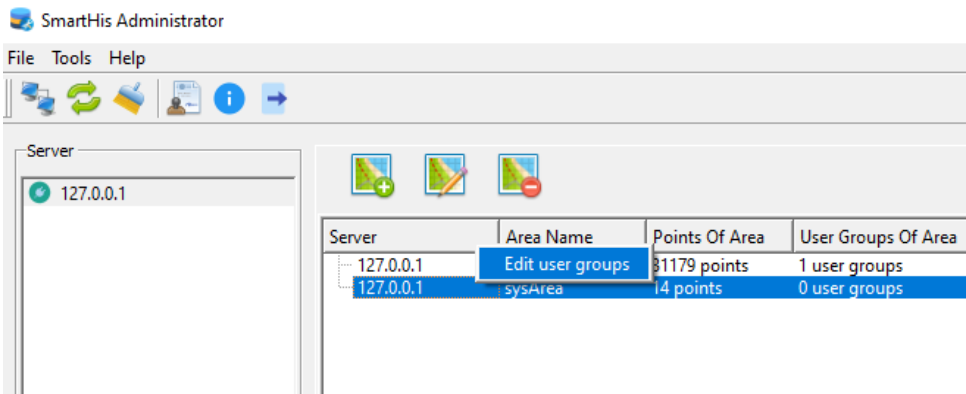
Buttons: OK, Cancel

FigureG-33. Edit area

G.4.3.1.4 Clear Area

Select **the Area Name** to be deleted →, select the **Delete icon** 

G.4.3.1.5 Delegate user group permissions to Area



The SmartHis Administrator interface shows a list of servers on the left and a table of areas on the right.

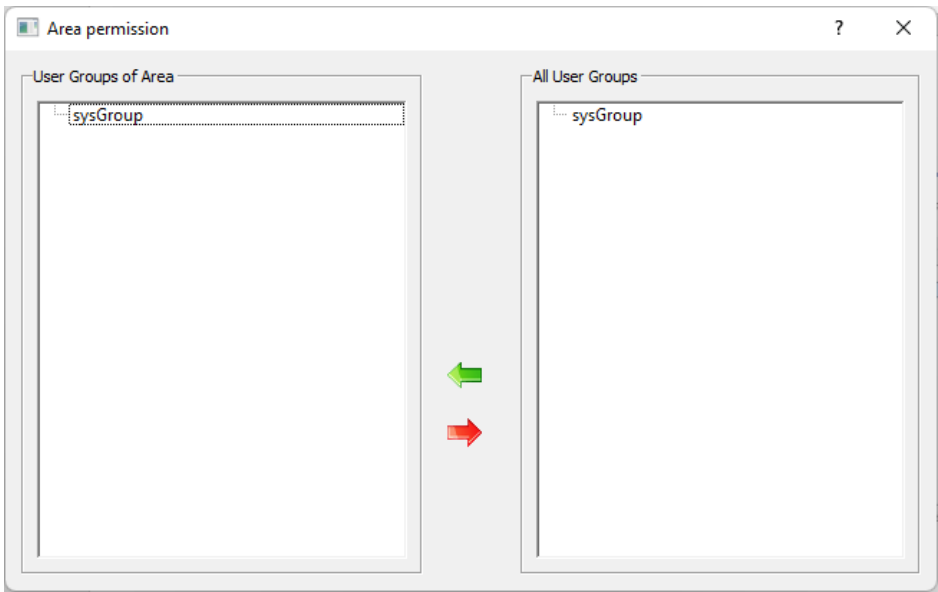
Servers:

- 127.0.0.1

Areas Table:

Server	Area Name	Points Of Area	User Groups Of Area
127.0.0.1	Edit user groups	1179 points	1 user groups
127.0.0.1	sysArea	14 points	0 user groups

Right Click the name **of the corresponding** ® **Αρφα Edit user groups**

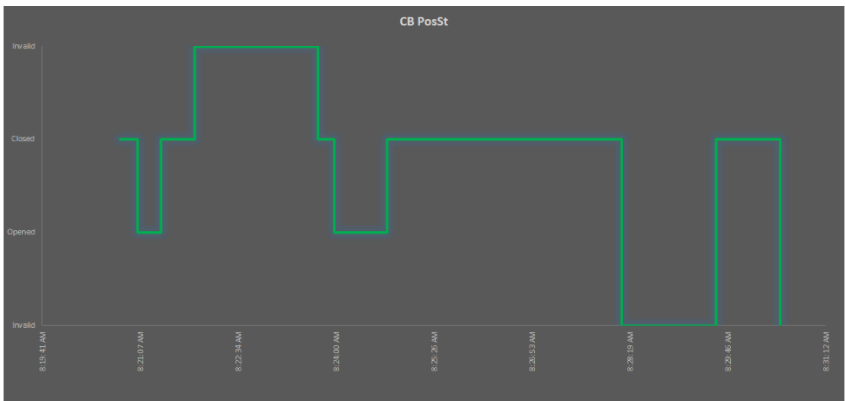


FigureG-34. Edit a user group in Area Permission

- Configure **User Groups** for Area:

User Groups of Area	User groups have permission to manipulate the Area
All User Groups	All User groups are declared on the software
Add User Group 	Add a User Group selected in the All User Groups → list to the User Groups of Area list
Remove User Group 	Delete a User Group selected in the User Groups of Area list

G.4.3.2 Digital Set



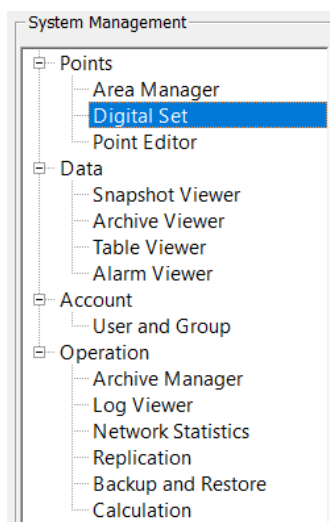
A digital set is a set of values (enums) defined for set-type data points. Digital Sets are enum-type in nature, consisting of a table of definitions of "names" and their corresponding numeric values.

Digital Set means display. The data that is actually stored and processed in the system is the numerical values (raw), the resulting data displayed on the clients is the digital set values (descriptive values).

Example:

Data Point: Cutter Status 171	
Digital Set Type: STATUS_2BIT	
Digital set value	Numerical Values
INTERMEDIATE	0
OPEN	1
CLOSE	2
INVALID	3

G.4.3.2.1 List of Digital Sets



System Management → Points / Digital Set Menu:

Server	Digital Set Name	State Number	State Name
127.0.0.1	System		
	DEVICES STATUS	0	INTERMEDIATE
		1	OPENED
		2	CLOSED
		3	INVALID
	POSITION COM...		
	DPS		
	SPS		
	DPC		
	TAP COMMAND	0	RESET CMD
		1	LOWER CMD EXECUTED
		2	RAISE CMD EXECUTED
		11	LOWER REQ - HMI
		12	RAISE REQ - HMI
		21	LOWER REQ - A0
		22	RAISE REQ - A0
		31	LOWER REQ - A1
		32	RAISE REQ - A1

FigureG-35. Manage digital sets

Digital Set Information:

Server	HIS Server contains a Digital Set
Digital Name	Name of Digital Set
State Number	The raw value of the data
State Name	Digital set of data

Ways to configure a Digital Set:

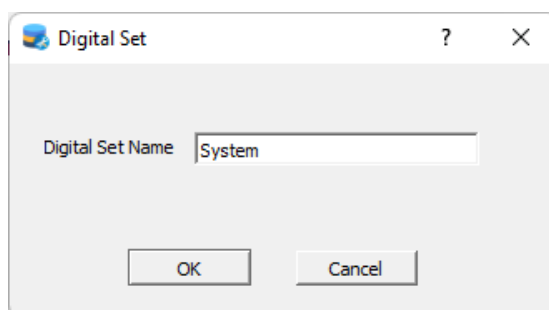
- App Default Digital Set: System
- Digital Sets are automatically updated according to the database (MongoDB). These data are configured according to *the OneATS Data Editor tool*
- Digital Set created by the user with *SmartHIS Administrator*

G.4.3.2.2 Add a new Digital Set

Select server → click , or Right-Click server → New Digital Set



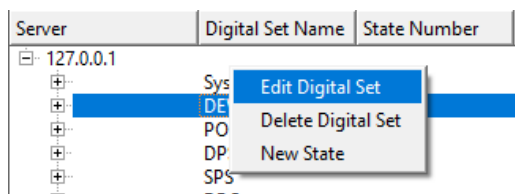
Fill in the name Digital Set → **OK**



FigureG-36. Add a new digital set

G.4.3.2.3 Edit Digital Set

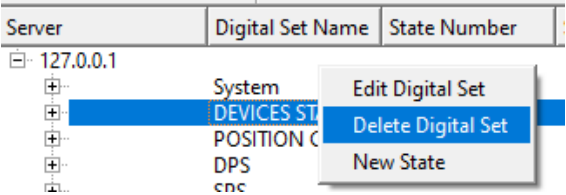
Select Edit Digital Set → , or Right-Click Digital Set → Edit Digital Set



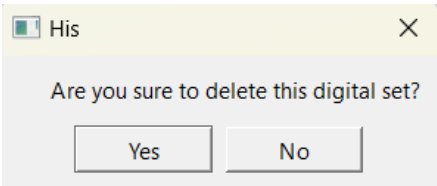
Edit Digital Set → **Name OK**

G.4.3.2.4 Deleting a Digital Set

Select Delete Digital Set → , or Right-Click Digital Set → Delete Digital Set



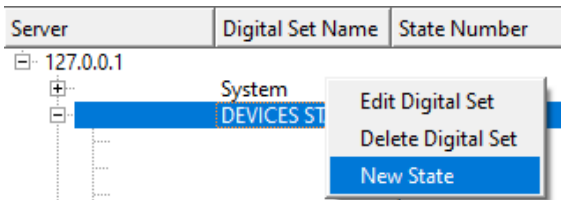
Confirm **YES** to Clear Digital Set



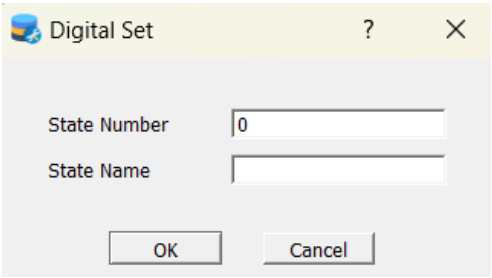
FigureG-37. Delete digital set

G.4.3.2.5 Adding a State in a Digital Set

Select Digital Set → click **Add State** , or Right-Click Digital Set → New State



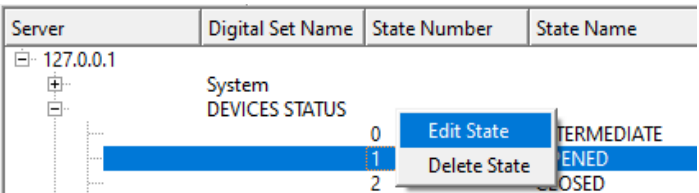
Enter the **State Number & State Name** → values Click **OK**



FigureG-38. Adding a State in a Digital Set

G.4.3.2.6 Edit the State Name in a Digital Set

Select State → **Edit State** , or Right-Click State → Edit State



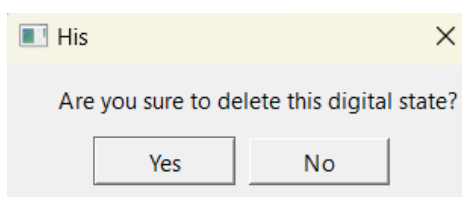
Edit **State Name** → **OK**

G.4.3.2.7 Deleting a State in a Digital Set

Select State → **Delete State** , or Right-Click State → **Delete State**

Server	Digital Set Name	State Number	State Name	
127.0.0.1	System DEVICES STATUS			
		0	INTER	Edit State
		1	OPEN	Delete State
		2	CLOSE	
		3	INVALID	

Confirm **YES** to delete State in Digital Set



FigureG-39. Deleting a State in a Digital Set

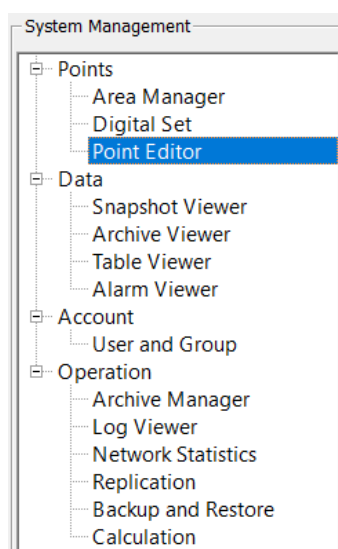
G.4.3.3 Point Editor

Point Editor is a tool that configures datapoints to achieve the following objectives:



- Add, edit, delete data points
- Import/Export data point

Access the Point Editor tool: Menu **System Management** → **Points / Point Editor**



G.4.3.3.1 Data point search

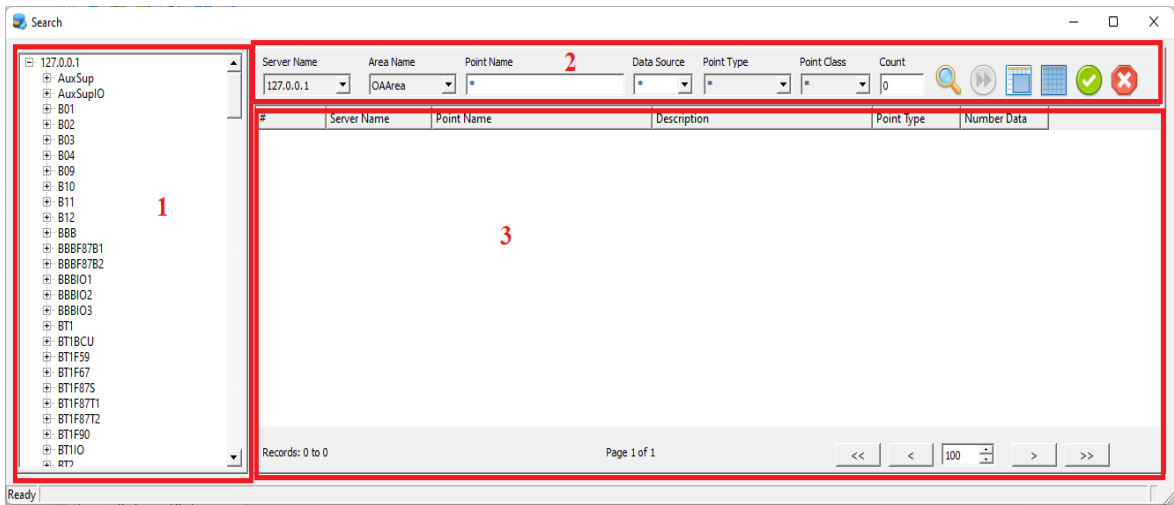
Data-point search is the basic operation to Edit/Delete/Export point.

Summary of operation cycles:

- → Enter search conditions
- → Show search results
- → Select points among the points that satisfy the search results
- → (Apply) List of selected → points for next operations (Edit/Delete/Export point)

Specific operations:

Click **Search**  from the **Point/Point Editor** → toolbar:



FigureG-40. Search point tool interface

Region	Describe
[1]	Data Model Tree → supports filtering by point name
[2]	Filter conditions
[3]	List of points, displayed in page style

Filter conditions:

Server Name	Area Name	Point Name	Data Source	Point Type	Point Class	Count
127.0.0.1	OAArea	B01*	*	*	*	0

Server Name	Server Address - Select from a list of HIS Server connections
Area Name	Data Area - Select from a list of Data Areas
Point Name	Syntax to search for text in point name. The exact type of search for text content. - Selectable from the Data Model Tree on the left - The character "*": represents any character. Example:

	- Search syntax: B0*. XCBR1.Blk → Results: B01.XCBR1.Blk, B011.XCBR1.Blk, B03.XCBR1.Blk
Data Source	DataSource attribute of data - Choose from a list of data sources - "*" character: search from any data source
Point Type	Data value type attribute - Choose from a list of data types - "*" character: search for any type of data
Point Class	Data type (type) attribute - Choose from a list of data types - "*" character: search for any type of data
Count	The amount of data you want to display. - A value of 0 will show all

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

Summary of information displayed in search results:

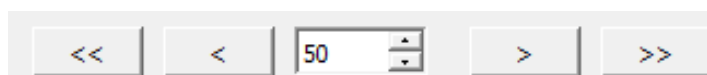
Records shown: 0 to 50

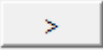
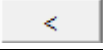
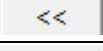
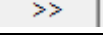
Page 1 of 2

Found points: 57

Records shown	List of the order of the records displayed
Page ... of	Page order / total number of pages results
Found points	Total search result points

Customization tools that display the results page:



Tool	Symbol	Describe
Number result / page		Number of points / 1 page displayed
Next page		Show the next points page
Previous page		Display the previous results page (points)
First Page		Display the first page of results (points)
Last Page		Display the final points page

Tools on the toolbar / point list:

Tool	Symbol	Describe
Search		Search for points that satisfy the filter conditions

Continue		Continue to show more results due to being limited by the amount of data you want to show
Select Page		Select all points on the page that is being displayed
Select All		Select all result points (all pages)
Multi-select		On the list of points, support the operation of selecting points - Click: single selection - CTRL+Click: multi-selection - SHIFT+Click: continuous selection
SATISFACTORY		Apply the selected points to the Search result. The results will be displayed on the Data Editor window interface
Cancel		Cancel a search

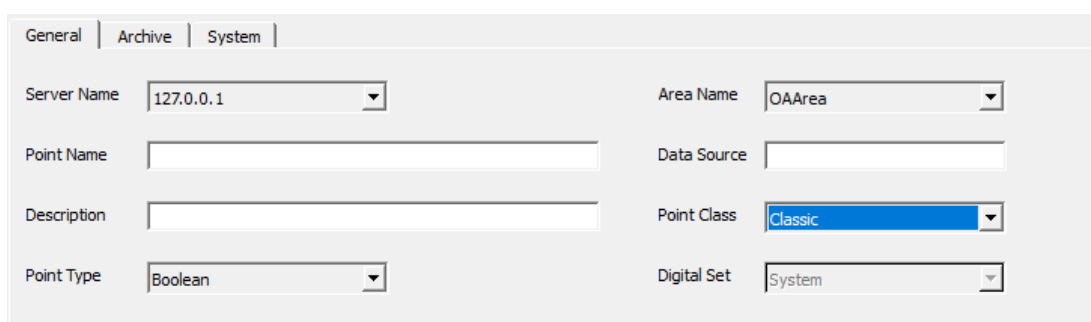
G.4.3.3.2 Clear search results

Click **Clear**  from the **Point/Point Editor** → to clear the list of searched points.

G.4.3.3.3 Add Data point

Click **New**  from the **Point/Point Editor** toolbar.

General Tab:

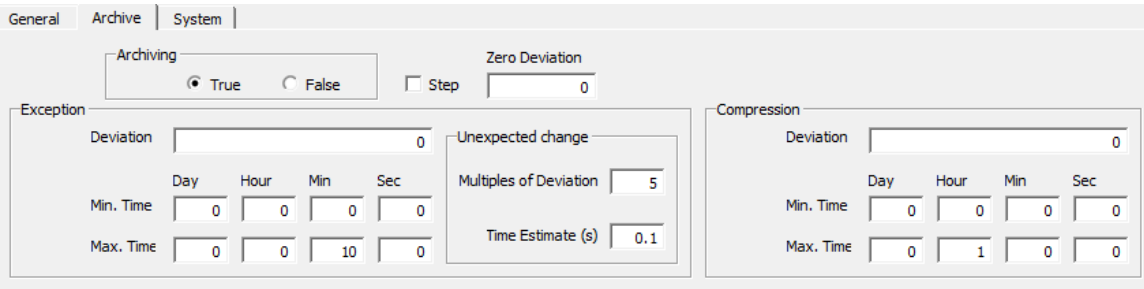


FigureG-41. Added a new point – Tab General

Parameter	Describe
<i>General tab : General attributes of the data point</i>	
Server Name	The server address I want to add points to

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

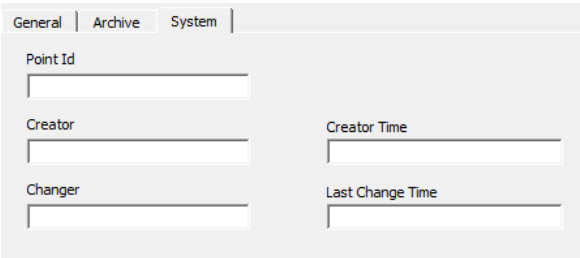
Archive Tab:



FigureG-42. Tab Archive

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

System Tab:

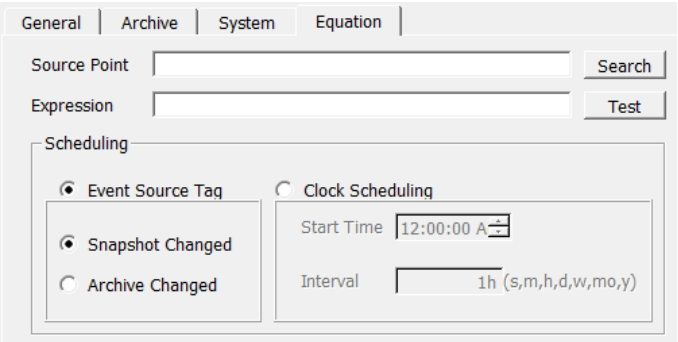


FigureG-43. Tab System

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

Equation Tab:

Appears when selecting Point Class as Equation:



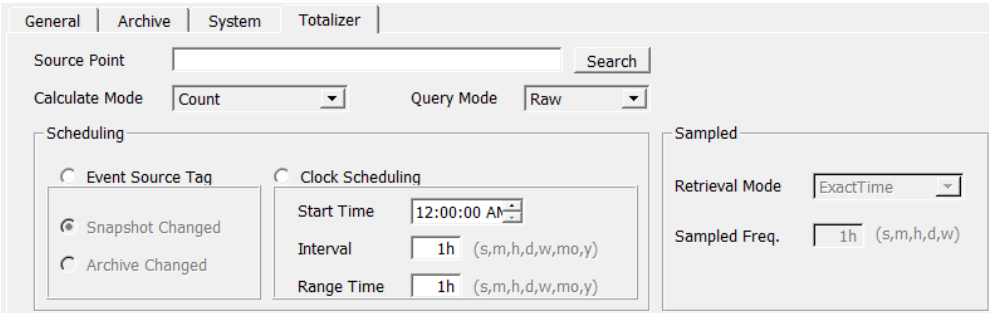
FigureG-44. Tab Equation

Parameter	Describe
Equation Tab: Data Point Compute Function Properties	
Test	Check if the expression is valid
Search	Support to select points from Data Model

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

Totalizer Tab:

Appears when selecting Point Class as Totalizer (related to statistics)



FigureG-45. Tab Totalizer

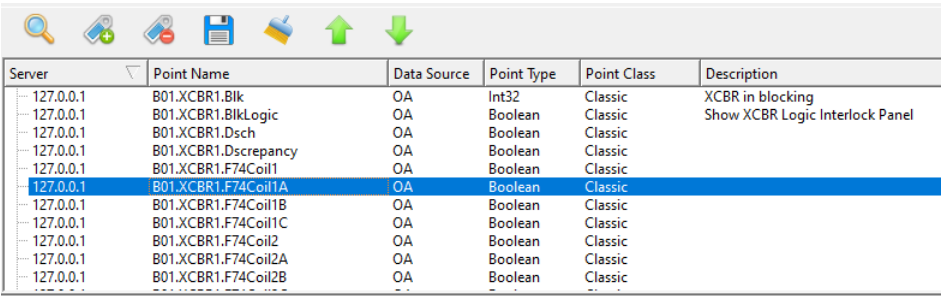
Parameter	Describe
Totalizer Tab: Compute (statistical) data function properties in the Archive	
Search	Support to select points from Data Model

Meaning of Data Point Attributes Reference [G.1.7.4 Data Point Storage Configuration Attributes](#)

G.4.3.3.4 Deleting a Data Point

Click **Search**  from the Point/Point Editor → toolbar to perform a points search. Refer [to G.4.3.3.1 Data point search](#)

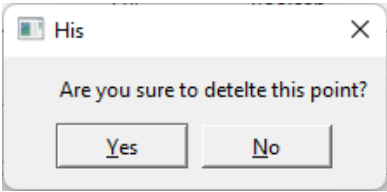
Select **a point** from the list of search results.



Server	Point Name	Data Source	Point Type	Point Class	Description
127.0.0.1	B01.XCBR1.Blk	OA	Int32	Classic	XCBR in blocking
127.0.0.1	B01.XCBR1.BlkLogic	OA	Boolean	Classic	Show XCBR Logic Interlock Panel
127.0.0.1	B01.XCBR1.Dsch	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.Dscrepancy	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1A	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1B	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1C	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2A	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2B	OA	Boolean	Classic	

Click **Delete**  from the Point/Point Editor toolbar.

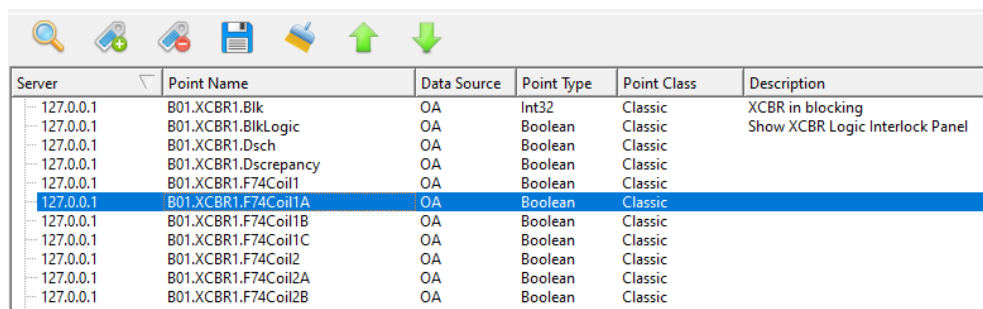
Confirm the data deletion operation



G.4.3.3.5 Edit Data Points

Click **Search**  from the Point/Point Editor → toolbar to perform a points search. Refer [to G.4.3.3.1 Data point search](#)

Select **a point** from the list of search results.



Server	Point Name	Data Source	Point Type	Point Class	Description
127.0.0.1	B01.XCBR1.Blk	OA	Int32	Classic	XCBR in blocking
127.0.0.1	B01.XCBR1.BlkLogic	OA	Boolean	Classic	Show XCBR Logic Interlock Panel
127.0.0.1	B01.XCBR1.Dsch	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.Dscrepancy	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1A	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1B	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil1C	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2A	OA	Boolean	Classic	
127.0.0.1	B01.XCBR1.F74Coil2B	OA	Boolean	Classic	

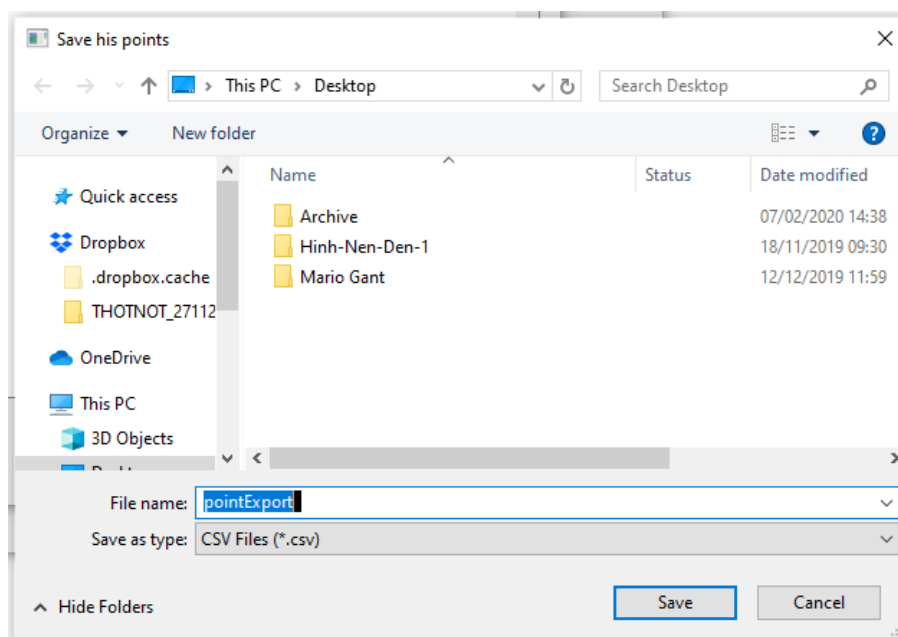
Edit the Point properties on the General/Archive/System/Equation/Totalizer Tabs. Refer to the table of Attribute Tabs in [G.4.3.3.3 Adding Data points](#)

Click **Save**  from the Point/Point Editor → toolbar to update the Data Point attribute edits.

G.4.3.3.6 Export Data point

Click **Export**  from the Point/Point Editor → toolbar to → perform a search for points. Refer [to G.4.3.3.1 Data point search](#)

Click **OK** → in the export data file storage window.

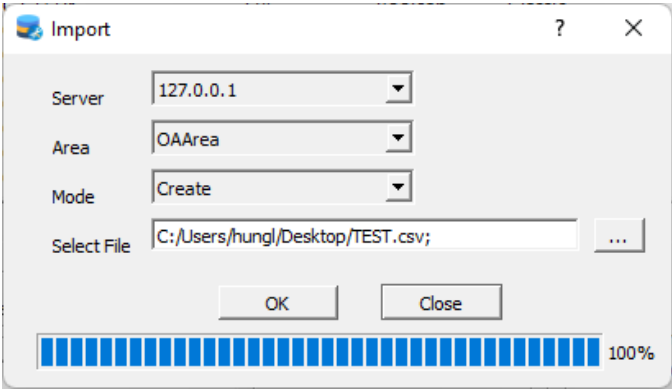


FigureG-46. Save point to file

Select the location where the file → is stored, Click **Save**.

G.4.3.3.7 Import Data point

Click **Import**  from the Point/Point Editor → toolbar in the Import Data interface.

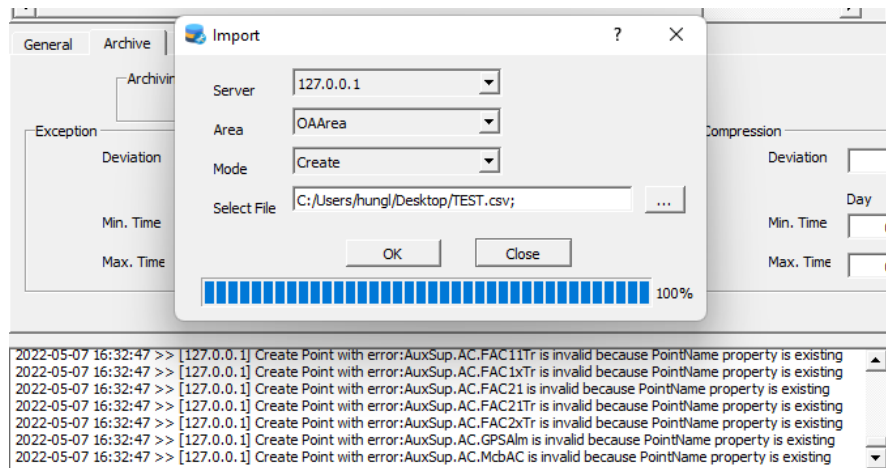


FigureG-47. Import point from file

Enter the Import file data parameters.

Parameter	Describe	
Server	Destination Server Address for Data Import	
Fashion	Select Data Import Mode	
	Create	▪ If the point already exists → ignore it ▪ If the point does not exist → create a new one
	Edit	▪ If the point already exists → update the configuration attribute ▪ If the point does not exist → ignore it
	Create or Edit	▪ If the point already exists → update the configuration attribute ▪ If the point does not exist → create a new one
	Delete	▪ If a point already exists → delete ▪ If the point does not exist → ignore it
Select File	Select the path to the file to Import	
Column %	Showing the Import process	

The Warning/Error Messge during the Import process is shown on the Log window area of the application.



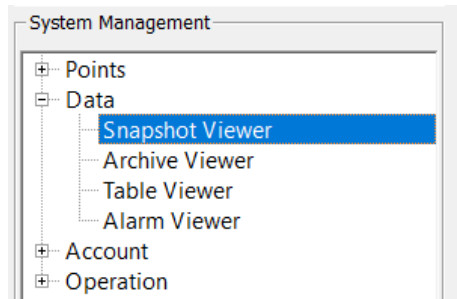
G.4.4. Data – data monitoring

The Data menu provides tools for monitoring data.

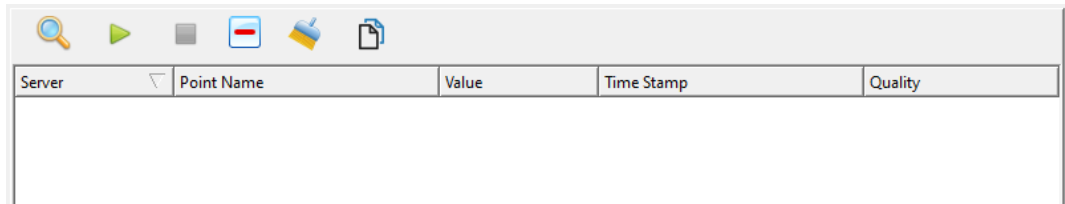
Snapshot Viewer	Mine data stored on a Snap-shot of 1 datapoint
Archive Viewer	Exploit data stored on the Archive of 1 datapoint
Table Viewer	Exploit data stored on the Archive of multiple datapoints
Alarm Viewer	Exploit Alarm/Event

G.4.4.1 Snapshot Viewer

Monitoring data stored on Snap-shot



System Management → Data/Snapshot Viewer → Menu Snap shot Viewer interface.



Click **Search**  from the **Snap-shot Viewer** → toolbar to search for points. Refer to [G.4.3.3.1 Searching for Data points](#). The value of the selected points at the time of Click **OK**  is displayed in the data monitoring panel.

Server	Point Name	Description	Value	Time Stamp	Quality
127.0.0.1	E1_16.V110.BAY131.MMXU1.Amax	MAX OF 3 PHASE CURRENT (A)	Pt Created	2023-Jul-21 16:57:43.884517	Point_Created
127.0.0.1	E1_16.V110.BAY131.MMXU1.AphsA	PHASE A CURRENT (A)	Pt Created	2023-Jul-21 16:57:45.828350	Point_Created
127.0.0.1	E1_16.V110.BAY131.MMXU1.AphsB	PHASE B CURRENT (A)	Pt Created	2023-Jul-21 16:57:43.884517	Point_Created
127.0.0.1	E1_16.V110.BAY131.MMXU1.AphsC	PHASE C CURRENT (A)	Pt Created	2023-Jul-21 16:57:45.828350	Point_Created
127.0.0.1	E1_16.V110.BAY131.MMXU1.Vlin	MAX OF 3 LINE VOLTAGE (kV)	Pt Created	2023-Jul-21 16:57:43.885514	Point_Created

FigureG-48. Snapshot data

Monitoring information:

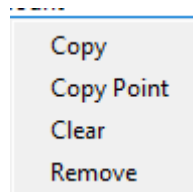
Server	Server contains data points
Point Name	Point Name
Description	Point Description
Value	Point value
Time-stamp	Time Labels
Quality	Data quality

Monitored Data List Calibration Tools:

- Toolbar:



- Right-Click Menu on Selected Points



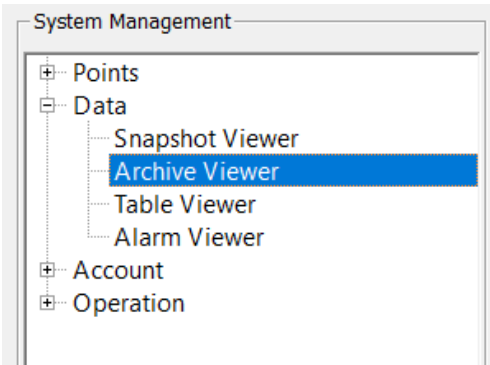
Tool	Symbol	Describe
Search		Search for points that satisfy the filter conditions
Start		Online Monitoring (Continuous Update) Value
Stop		Stop updating online data values
Remove		Remove a selected point (row) from the data monitoring list
Clear		Remove all points (rows) from the data monitoring list
Copy		Copy the display information of the selected points (rows) to *.csv format (data fields separated by ";") → Serving the operations of copying the displayed information to a file
Copy Point		Copy the point name of the selected points (rows)

		→ Serving the operations of copying the point list to a file
--	--	--

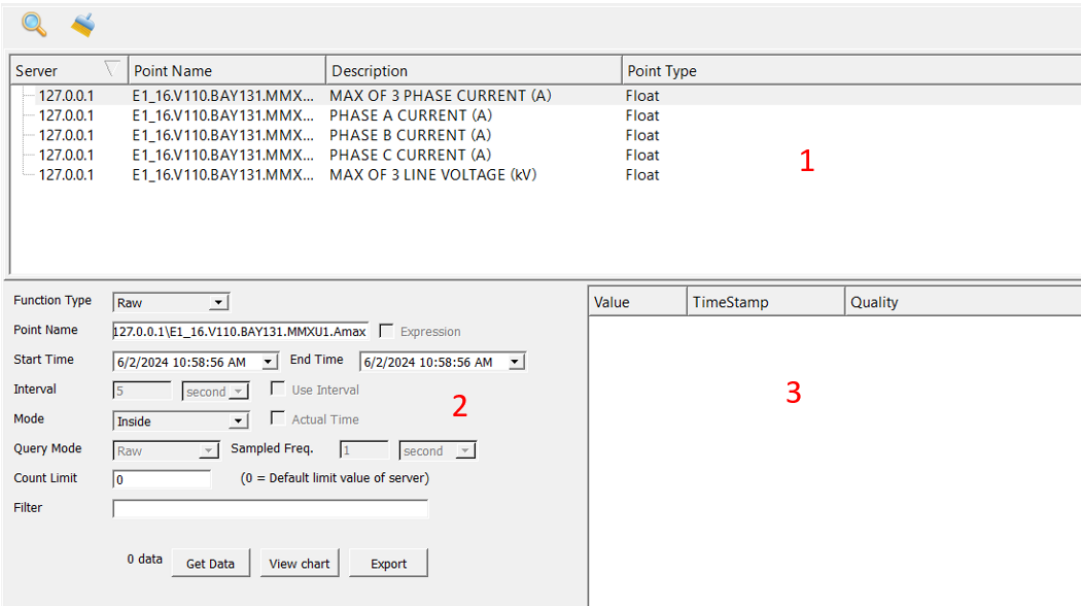
G.4.4.2 Archive Viewer

Archive Data Monitoring & Logging Tool

G.4.4.2.1 Monitor data on the Archive



System Management → Data → Archive Viewer



FigureG-49. Data archive

1	Point List
2	Configure the Data Query Command
3	Queried data results

Click  **Search** from the **Archive Viewer** → toolbar to search for points. Refer to [G.4.3.3.1 Data point search](#)

Select the point to monitor the data Information → is automatically assigned to the **Point Name** in the query statement.

Configure data query commands.

Function Type Data Query Function		
<i>Query function</i>	<i>Describe</i>	<i>Configuration parameters on the interface</i>
Archive	Query point/expression data at a specified time in the Archive	- Point Name / Expression- TimeStamp- Mode
Raw	Query all values that have been saved for a specified period of time	- Point Name- StartTime- EndTime- Mode- Count Limit- Filter
Sampled	Query data over a period of time with a defined sampling frequency	- Point Name /Expression- StartTime- EndTime- Interval- Mode- Count Limit- Filter
Calculated	Statistical calculation of data stored over a specified period of time (Count, Total, Min, Max...)	- Point Name- StartTime- EndTime- Interval- Mode- Query Mode- Sampled Frequency- Count Limit- Filter
Time Filter	Calculate the total time that an expression is satisfied.	- Expression- StartTime- EndTime- Query Mode- Count Limit

Condition	Conditional calculation function	- Point Name- StartTime- EndTime- Condition Start- Condition Stop- Expression Start- Expression Stop- Offset Time Start- Offset Time Stop- Count Limit- Only Good- Filter
------------------	----------------------------------	---

Data list calibration tools:



Tool	Symbol	Describe
Search		Search for points that satisfy the filter conditions
Clear		Remove all points (rows) from the data monitoring list

Data Visualization and Query Tools

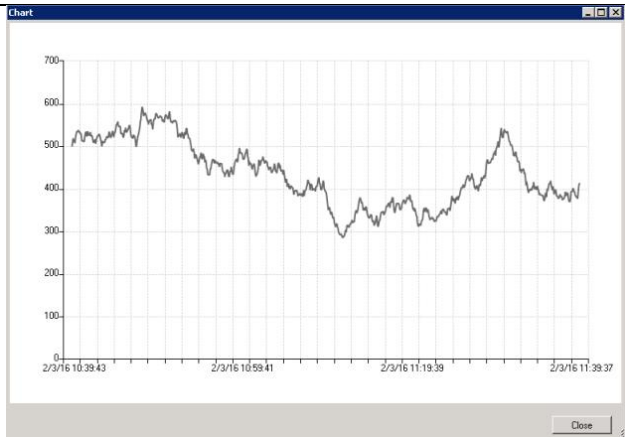
19 data in 4ms

Get Data

View chart

Export

Tool	Symbol	Describe																																							
<i>Get Data</i>		Send data query commands. The result of the data query command will be displayed on the results table <table border="1"> <thead> <tr> <th>Value</th><th>TimeStamp</th><th>Quality</th></tr> </thead> <tbody> <tr><td>Pt Created</td><td>03-Feb-2016 10:07:00.694</td><td>No_Value</td></tr> <tr><td>0</td><td>03-Feb-2016 10:16:02.000</td><td>Bad</td></tr> <tr><td>335.072</td><td>03-Feb-2016 10:28:53.000</td><td>Good</td></tr> <tr><td>355.0139</td><td>03-Feb-2016 10:30:23.000</td><td>Good</td></tr> <tr><td>352.7063</td><td>03-Feb-2016 10:30:27.000</td><td>Good</td></tr> <tr><td>344.3048</td><td>03-Feb-2016 10:30:31.000</td><td>Good</td></tr> <tr><td>344.4897</td><td>03-Feb-2016 10:30:35.000</td><td>Good</td></tr> <tr><td>344.8091</td><td>03-Feb-2016 10:30:39.000</td><td>Good</td></tr> <tr><td>341.9639</td><td>03-Feb-2016 10:30:43.000</td><td>Good</td></tr> <tr><td>338.1541</td><td>03-Feb-2016 10:30:48.000</td><td>Good</td></tr> <tr><td>343.4848</td><td>03-Feb-2016 10:30:52.000</td><td>Good</td></tr> <tr><td>354.0379</td><td>03-Feb-2016 10:30:56.000</td><td>Good</td></tr> </tbody> </table>	Value	TimeStamp	Quality	Pt Created	03-Feb-2016 10:07:00.694	No_Value	0	03-Feb-2016 10:16:02.000	Bad	335.072	03-Feb-2016 10:28:53.000	Good	355.0139	03-Feb-2016 10:30:23.000	Good	352.7063	03-Feb-2016 10:30:27.000	Good	344.3048	03-Feb-2016 10:30:31.000	Good	344.4897	03-Feb-2016 10:30:35.000	Good	344.8091	03-Feb-2016 10:30:39.000	Good	341.9639	03-Feb-2016 10:30:43.000	Good	338.1541	03-Feb-2016 10:30:48.000	Good	343.4848	03-Feb-2016 10:30:52.000	Good	354.0379	03-Feb-2016 10:30:56.000	Good
Value	TimeStamp	Quality																																							
Pt Created	03-Feb-2016 10:07:00.694	No_Value																																							
0	03-Feb-2016 10:16:02.000	Bad																																							
335.072	03-Feb-2016 10:28:53.000	Good																																							
355.0139	03-Feb-2016 10:30:23.000	Good																																							
352.7063	03-Feb-2016 10:30:27.000	Good																																							
344.3048	03-Feb-2016 10:30:31.000	Good																																							
344.4897	03-Feb-2016 10:30:35.000	Good																																							
344.8091	03-Feb-2016 10:30:39.000	Good																																							
341.9639	03-Feb-2016 10:30:43.000	Good																																							
338.1541	03-Feb-2016 10:30:48.000	Good																																							
343.4848	03-Feb-2016 10:30:52.000	Good																																							
354.0379	03-Feb-2016 10:30:56.000	Good																																							
<i>View chart</i>		View data graphs																																							

		
Export		Export results to *.csv file

G.4.4.2.2 Manually Write Data to Archive

Not available

G.4.4.2.3 Archive Viewer – function: Archive

Function Type: **Archive**

Point Name: ☐ Expression

TimeStamp: End Time:

Interval: ☐ Use Interval

Mode: **ExactTime** ☐ Actual Time

Query Mode: **Raw** Sampled Freq.

Count Limit: (0 = Default limit value of server)

Filter:

Describe:

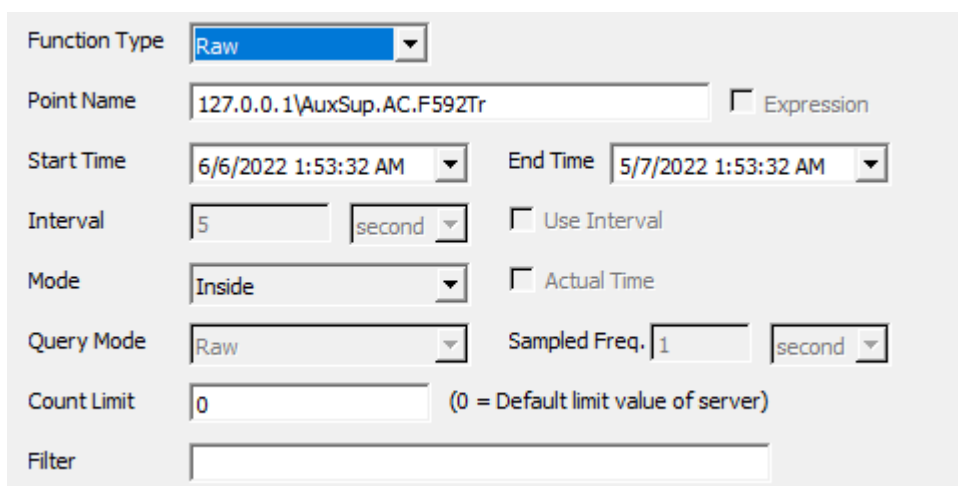
Query data 01 point/expression at a specified time

Configuration parameters

Parameter	Input Type	Describe
Function Type: Archive		
Expression Mode	True/False Selection	- False: query Point data - True: query expression data
Point Name	Select from the list of points	Data Query Point - Makes sense only when Expression Mode = False

Expression	Choose from a list of points and input strings	Data query expressions - Meaning only when Expression Mode = True - Expression syntax: reference G.10. Appendix - Expression Syntax
Timestamp	Input/Selection Dialog	Data query time. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Mode	Select from the list	Data Interpolation Methods for Points - Makes sense only when Expression Mode = False. With Expression Mode = True, the data points in the expression are queried with Interpolate interpolation mode - Modes: • Exact Time • Interpolate • Next • Previous • Nearest • Previous_2 Reference to the meaning of data interpolation modes in G.1.8.3.2 Data query modes

G.4.4.2.4 Archive Viewer - function: Raw



Function Type: **Raw**

Point Name: **127.0.0.1\AuxSup.AC.F592Tr** ☐ Expression

Start Time: **6/6/2022 1:53:32 AM** End Time: **5/7/2022 1:53:32 AM**

Interval: **5** **second** ☐ Use Interval

Mode: **Inside** ☐ Actual Time

Query Mode: **Raw** Sampled Freq. **1** **second**

Count Limit: **0** (0 = Default limit value of server)

Filter:

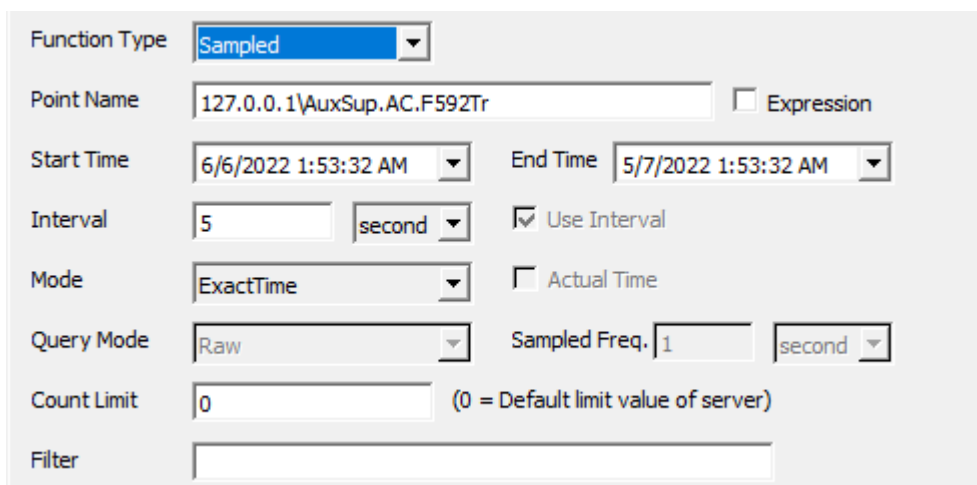
Describe:

- The function takes all the values that have been saved for a specified period of time
- Unwanted values can be filtered with filter expressions.

Configuration parameters

Parameter	Input Type	Describe
Function Type: Raw		
Point Name	Select from the list of points	Data Query Point
Start Time End Time	Input/Selection Dialog	Start Time: The time when the data query starts. End Time: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Mode	Select from the list	Data acquisition method 2 start and end timelines ▪ 0: In side ▪ 1: Out side ▪ 2: Interpolate
Count Limit	Input	Limit the number of results returned by the server (parameter with optimal performance meaning)
Filters	Input	(option) Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax

G.4.4.2.5 Archive Viewer - function: Sampled



Describe:

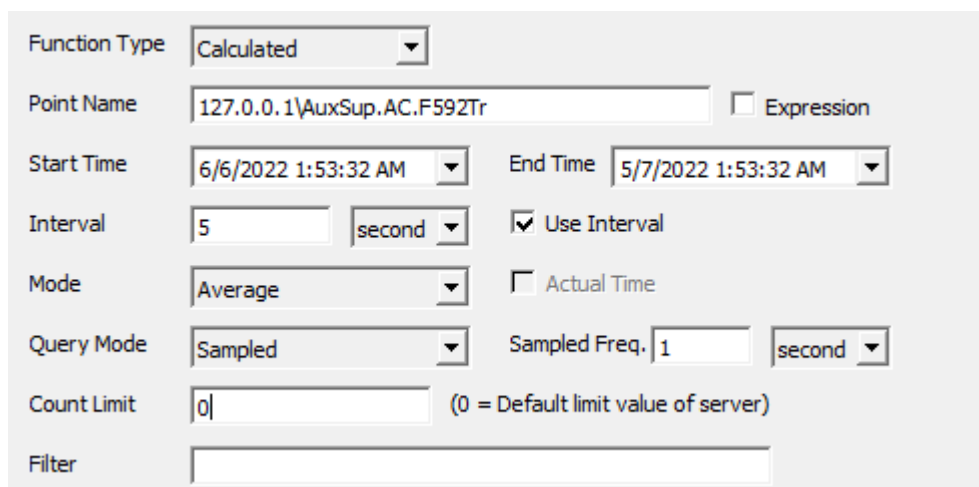
- The function retrieves data over a period of time with a specified sampling frequency
- Data query of 1 point/expression
- Unwanted values can be filtered with filter expressions

Configuration parameters

Parameter	Input Type	Describe
Function Type: Sample		
Expression Mode	True/False Selection	▪ False: query Point data ▪ True: query expression data
Point Name	Select from the list of points	Data Query Point - Makes sense only when Expression Mode = False
Expression	Choose from a list of points and input strings	Data query expressions - Meaning only when Expression Mode = True - Expression syntax: reference G.10. Appendix - Expression Syntax
Start Time End Time	Input/Selection Dialog	Start Time: The time when the data query starts. End Time: The end time of the data query. Time format by operating system.

		Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Interval	Input	Data query frequency
Interval Unit	Select from the list	Units for Interval parameter • second • minute • hour • day • week
Mode	Select from the list	Data Interpolation Methods for Points - Makes sense only when Expression Mode = False. With Expression Mode = True, the data points in the expression are queried with Interpolate interpolation mode - Modes: • Exact Time • Interpolate • Next • Previous • Nearest Reference to the meaning of data interpolation modes in G.1.8.3.2 Data query modes
Count Limit	Input	Limit the number of results returned by the server (parameter with optimal performance meaning)
Filters	Input	<i>(option)</i> Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax

G.4.4.2.6 Archive Viewer - function: Calculated



Describe:

- The function calculates the statistics (min, max, ...) of the data over a specified period of time.
- Query the data of 1 or more points.
- It is possible to calculate according to fixed frequency milestones during the selected time period.
- Unwanted values can be filtered with filter expressions.

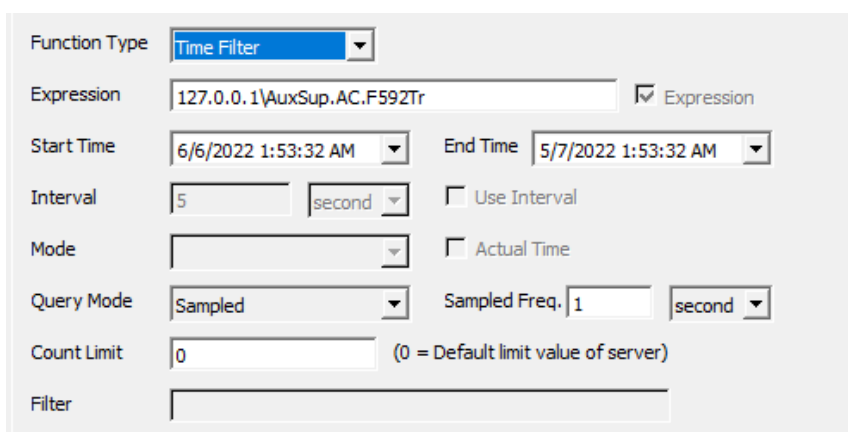
Configuration parameters

Parameter	Input Type	Describe
Function Type: Calculated		
Expression Mode	True/False Selection	▪ False: query Point data ▪ True: query expression data
Point Name	Select from the list of points	Data Query Point - Makes sense only when Expression Mode = False
Expression	Choose from a list of points and input strings	Data query expressions - Meaning only when Expression Mode = True - Expression syntax: reference G.10. Appendix - Expression Syntax
Start Time End Time	Input/Selection Dialog	Start Time: The time when the data query starts. End Time: The end time of the data query. Time format by operating system.

		Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Use Interval	True/False Selection	Statistical calculations according to time intervals
Interval	Input	Statistical calculation time sample interval - Makes sense only when Use Interval = True
Interval Unit	Select from the list	Units for Interval parameter • second • minute • hour • day • week
Mode	Select from the list	Calculation method: • Count • Total • Minimum • Maximum • Average • Subtraction • Minimum_Abs • Maximum_Abs • Average_Abs • Subtraction_Abs Notes: • Mode Subtraction & Subtraction_Abs uses the sampled data function (Query Mode = Sampled) to get the interpolation data at the timelines, subtract the value of the preceding milestone to give the calculation result, the time is taken as the time of the previous milestone.
Query Mode	Select from the list	Data query type to perform statistical calculations • 0: Raw: queries stored values

		1: Sampled: query data at a specified frequency (sampleFreq) with the Interpolate interpolation type
Sample Freq	Input	Data query frequency - Makes sense only when Query Mode = Sampled
Sample Freq Unit	Select from the list	Unit for Sample Freq Parameter - second - minute - hour - day - week
Count Limit	Input	Limit the number of results returned by the server (parameter with optimal performance meaning)
Filters	Input	<i>(option)</i> Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax

G.4.4.2.7 Archive Viewer - function: Time Filter



Function Type: Time Filter

Expression: 127.0.0.1\AuxSup.AC.F592Tr ☒ Expression

Start Time: 6/6/2022 1:53:32 AM End Time: 5/7/2022 1:53:32 AM

Interval: 5 second ☐ Use Interval

Mode: ☐ Actual Time

Query Mode: Sampled Sampled Freq.: 1 second

Count Limit: 0 (0 = Default limit value of server)

Filter:

Describe:

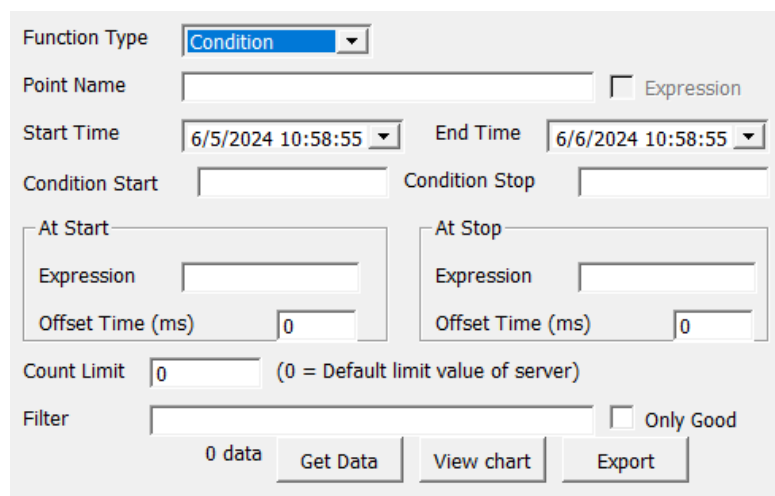
- The function is the sum of the lengths of the time periods that the expression satisfies.

Configuration parameters

Parameter	Input Type	Describe
-----------	------------	----------

<i>Function Type: Time Filtered</i>		
Expression	Choose from a list of points and input strings	Data query expressions - Expression syntax: reference G.10. Appendix - Expression Syntax
Start Time End Time	Input/Selection Dialog	Start Time: The time when the data query starts. End Time: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Query Mode	Select from the list	Data query type to perform statistical calculations • 0: Raw: queries stored values • 1: Sampled: query data at a specified frequency (sampleFreq) with the Interpolate interpolation type
Sample Freq	Input	Data query frequency - Makes sense only when Query Mode = Sampled
Sample Freq Unit	Select from the list	Unit for Sample Freq Parameter • second • minute • hour • day • week
Count Limit	Input	Limit the number of results returned by the server (a parameter that means optimal performance)

G.4.4.2.8 Archive Viewer - function: Condition



Describe:

- The function calculates according to conditions.

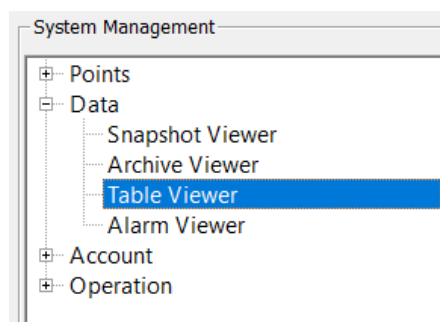
Configuration parameters

Parameter	Input Type	Describe
Function Type: Condition		
Point Name	Select from the list of points	Data Query Point
Start Time End Time	Input/Selection Dialog	Start Time: The time when the data query starts. End Time: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Condition Start Condition Stop	Input/String Format	Calculation start/end conditions
Expression Start Expression Stop	Input/String Format	The expression is calculated at the start/end time
Offset Time Start Offset Time Stop	Input/Int Format	Timeline deviation from the start/end time for calculation

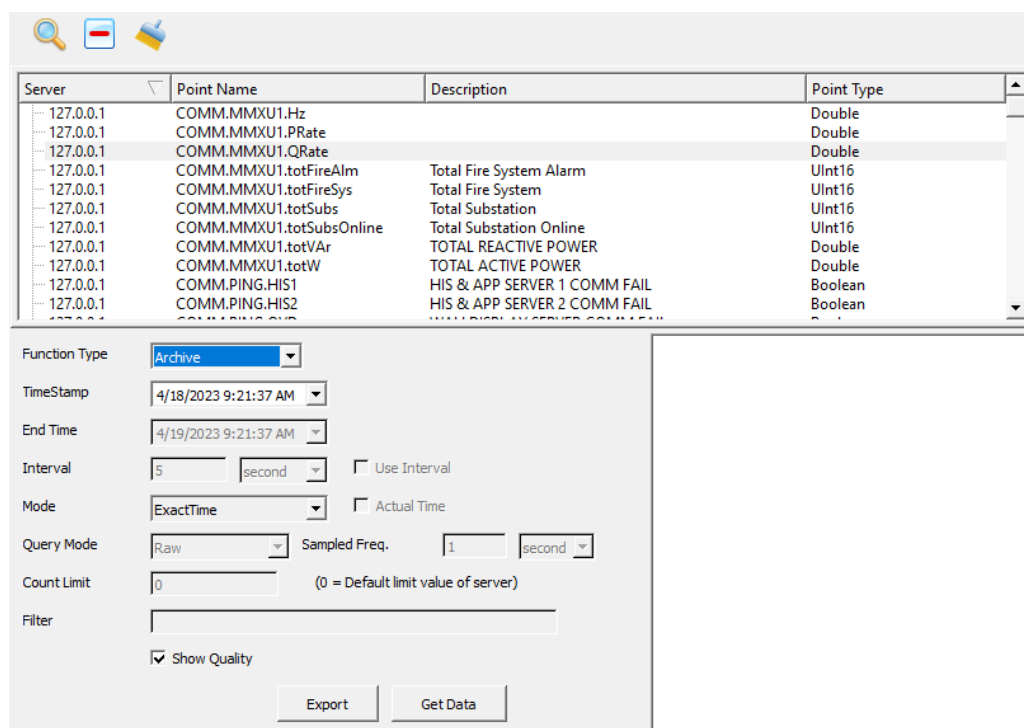
Count Limit	Input	Limit the number of results returned by the server (parameter with optimal performance meaning)
Filters	Input	<i>(option)</i> Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax
Only Good	True/False Selection	- True: Only fetch data with good quality - False: Displays all data

G.4.4.3 Table Viewer

Multi-datapoint monitoring tool stored on Archive



System Management → Data / Table Viewer → Menu Alarm Viewer Interface.



Server	Point Name	Description	Point Type
127.0.0.1	COMM.MMXU1.Hz		Double
127.0.0.1	COMM.MMXU1.PRate		Double
127.0.0.1	COMM.MMXU1.QRate		Double
127.0.0.1	COMM.MMXU1.totFireAlm	Total Fire System Alarm	UInt16
127.0.0.1	COMM.MMXU1.totFireSys	Total Fire System	UInt16
127.0.0.1	COMM.MMXU1.totSubs	Total Substation	UInt16
127.0.0.1	COMM.MMXU1.totSubsOnline	Total Substation Online	UInt16
127.0.0.1	COMM.MMXU1.totVAr	TOTAL REACTIVE POWER	Double
127.0.0.1	COMM.MMXU1.totW	TOTAL ACTIVE POWER	Double
127.0.0.1	COMM.PING.HIS1	HIS & APP SERVER 1 COMM FAIL	Boolean
127.0.0.1	COMM.PING.HIS2	HIS & APP SERVER 2 COMM FAIL	Boolean

Function Type: **Archive**

TimeStamp: 4/18/2023 9:21:37 AM

End Time: 4/19/2023 9:21:37 AM

Interval: 5 second ☐ Use Interval

Mode: ExactTime ☐ Actual Time

Query Mode: Raw Sampled Freq. 1 second

Count Limit: 0 (0 = Default limit value of server)

Filter:

☒ Show Quality

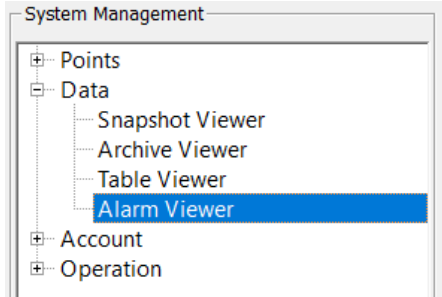
Export Get Data

FigureG-50. Data Archive - Table Viewer

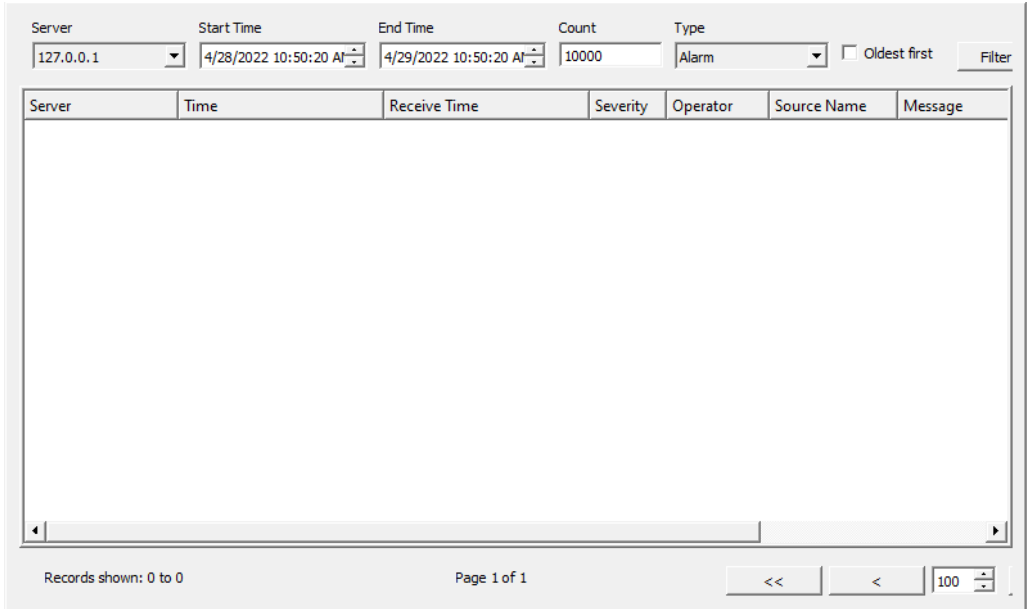
The functions are similar to the Archive Viewer.

G.4.4.4 Alarm Viewer

Alarm Monitoring



System Management → Data / Alarm Viewer → Alarm Viewer Interface



FigureG-51. Alarm Data

Data query configuration options:

Server	Start Time	End Time	Count	Type
127.0.0.1	4/28/2022 10:50:20 AM	4/29/2022 10:50:20 AM	10000	Alarm
Server	Alarm message data storage server			
Start Time	When to start a query			
End Time	When the query ends			
Count	Limit the number of results you can receive			
Type	Message Type: - Alarms - Event			

Configure the display arrangement (in order from top to bottom) by the **Time column** on the results table: ☐ Oldest first

- Oldest First = True: timed display from old → to new
- Oldest First = False: timed display from the old new →

Send a → command that displays the results of the stored Alarms/Events.

Results are displayed in a page style.

Alarm newsletter information:

Attribute	Describe
Server	Server hosting alarm/event message
Time	Alarm/Event Time (device time)
Receive Time	Time when ... Server receive Alarm/Event Message
Severity	Alarm/Event Severity
Operator	Alarm/Event Operator
Group	Alarm/Event Category ID
Source Name	Alarm/Event Source point key name
Message	Alarm/Event Message, support condition filter, ex: *HL01*Active* *HL02*
Value	Alarm/Event Value

Refer to [G.1.7.1 The data stored on the SmartHIS Server](#) – Alarm Message about the meaning of the Alarm attributes.

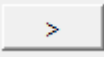
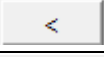
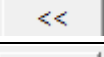
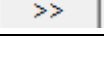
Aggregate information about the information displayed in search results.

Records shown: 0 to 50 Page 1 of 2

Records shown	List of the order of the records displayed
Page ... of	Page order / total number of pages results
Found points	Total results (alarm message)

Customization tools that display the results page

Tool	Symbol	Describe
Number result / page	50	Number of points / 1 page displayed

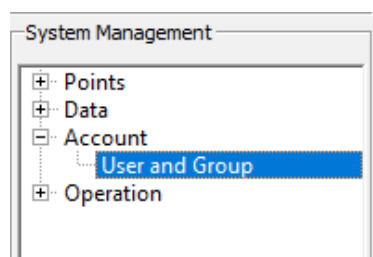
Next page		Show the next points page
Previous page		Display the previous results page (points)
First Page		Display the first page of results (points)
Last Page		Display the final points page

G.4.5. Account

The Account menu provides tools for managing and configuring the list of users and user groups. Decentralization policy for reference user types G.1.9 – Privacy Policy. User Groups are used to decentralize data access by Data Area. Consult [G.4.3.1.5 Assign user group permissions to Area](#)

G.4.5.1 List of Users & User Groups

Go to the System Management → Account / User and Group Menu.



→ User & Group Menu interface.

User				Group User			
Server	User Name	Full Name	Role	Server	Group Name	Group Description	Users o
127.0.0.1	admin	Administrator	Admin	127.0.0.1	sysGroup	System Group	1 users

The User.

Server	Server Name (Address)
UserName	Username
Full Name	Full Name
Password	Login Password
Role	User Type: • Admin • Operator • User

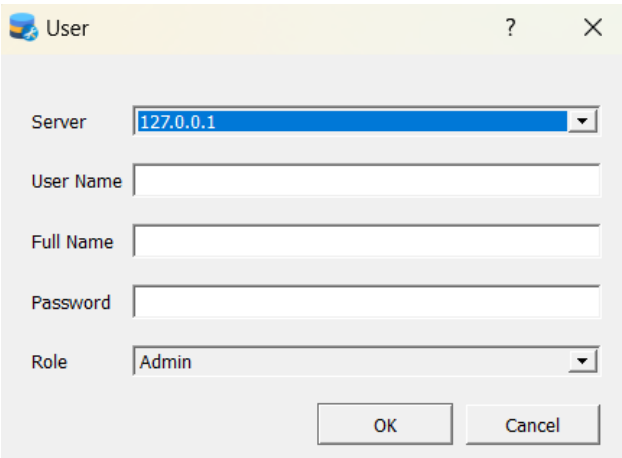
User Group parameters.

Server	Server Name (Address)
Group Name	Group Name
Group Description	Group Description

G.4.5.2 Add User

Log in with the User with the **Admin role**.

Select the **Add User tool** 



FigureG-52. Create and delegate user permissions

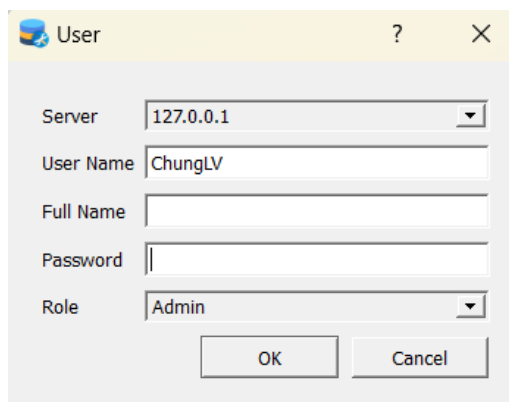
Enter the information of the User parameters.

Server	Server Name (Address) - Choose from a list of connected servers
UserName	Username
Full Name	Full Name
Password	Login Password
Role	User Type: • Admin • Operator • User

G.4.5.3 Edit User Information

Log in with **Admin** privileges if you need to edit other users. Type **Operator/User** only allows editing of the User itself .

Select **the User** to be edited →, select the **Edit User tool**, **Edit the**  → User's information



The 'User' dialog box contains the following fields:

- Server: 127.0.0.1 (dropdown)
- User Name: ChungLV (text input)
- Full Name: (empty text input)
- Password: (empty password input)
- Role: Admin (dropdown)
- Buttons: OK, Cancel

FigureG-53. Edit user information

Note: Editing of User Admin usernames and permissions is not allowed .

G.4.5.4 Delete User

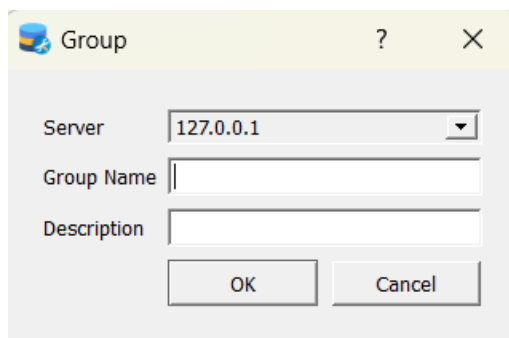
Log in with the User with the **Admin role**.

Select **the User** to be deleted →, select the **Delete User tool**. 

G.4.5.5 Add a User Group

Log in with the User with the **Admin role**.

Select the **Add Group tool** 



The 'Group' dialog box contains the following fields:

- Server: 127.0.0.1 (dropdown)
- Group Name: (empty text input)
- Description: (empty text input)
- Buttons: OK, Cancel

FigureG-54. Add a user group

Enter the information of the User Group parameters.

Server	Server Name (Address)
Group Name	Group Name
Group Description	Group Description

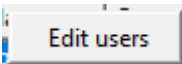
G.4.5.6 Edit User Group Information

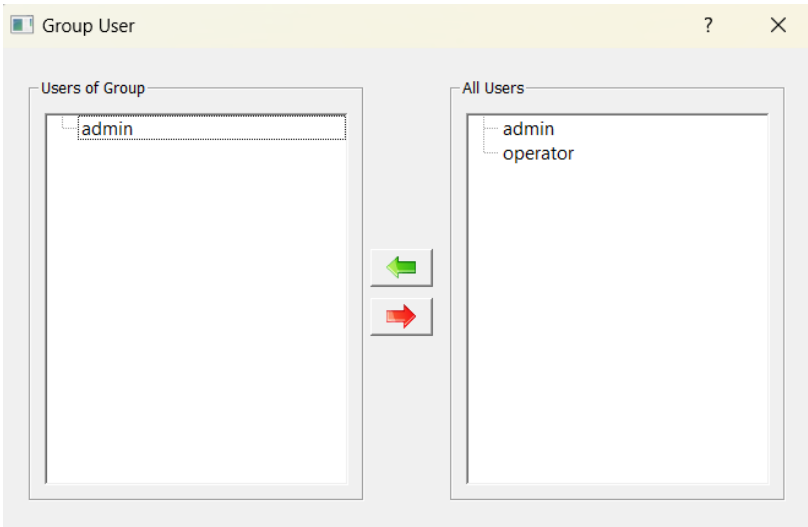
Log in with the User with the **Admin role**.

Select **the User Group** to edit → Select the Edit **Group**  tool → Edit the information of the User Group.

G.4.5.7 Editing the List of Users in a User Group

Log in with the User with the **Admin role**.

Select the User Group to edit → Right Click: **Edit Users** 



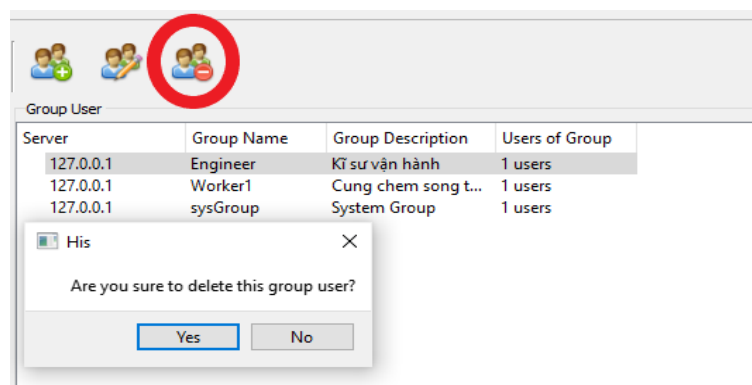
FigureG-55. Edit a user group

All Users	List of Users in the system
Users of Group	List of users in the group
Add User 	Add a User selected in the All User → list to the Users of Group list
Remove User 	Delete a User Selected in the Users of Group List

G.4.5.8 Delete a User Group

Log in with the User with the **Admin role**.

Select **the User Group** to be deleted →, select the **Delete Group** tool .



FigureG-56. Delete a User Group

G.4.6. Operation

Menu Operation provides tools to manage the operation of the SmartHIS Server.

Archive Manager	Managing Archives
Log Viewer	Monitoring Activity Log Newsletters
Network Statistic	Statistics on data exchange
Replication	Replication Operations Management (Data Sync)
Backup and Restore	Back-up and data recovery tools
Calculation	Tool to update values for Equation/Totalizer points - Serves in cases where the source data points are updated with additional values due to the data synchronization process / data supplementation

G.4.6.1 Archive Manager

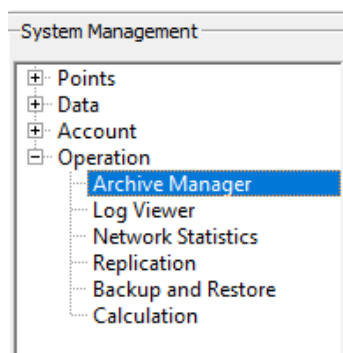
Manage Archives



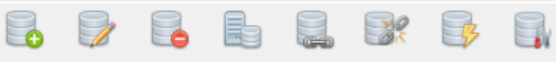
- Add/edit/delete
- Backup
- Registration / Cancellation
- Validate / Repair

G.4.6.1.1 List of Archives

Access the System **Management** → **Operation / Archive Manager Menu**



List of Archives.

				
127.0.0.1 Alarm Area OAArea sysArea	Server	Area	Archive Path	Start Time
	127.0.0.1	OAArea	D://HIS//Data\subSys_2\Archive02	2022-A1
	127.0.0.1	OAArea	D://HIS//Data\subSys_2\Archive03	2022-A1
	127.0.0.1	OAArea	D://HIS//Data\subSys_2\Archive01	2022-A1

FigureG-57. List of archives

Archive Information.

Attribute	Describe
Server	Archive Management Server
Area	Archive Data Management Partition: + Alarm Area (Alarm event storage) + OAArea (OAArea data model storage) + sysArea (System event storage)
Archive Path	Folder containing Archive
Start Time End Time	Archive start and end times. - Each Archive can define a different retention period.- The minimum duration of an Archive is 1 hour- The maximum duration of an Archive is 1440 hours.- The Archives do not overlap in terms of retention intervals
Enabled	Archive Activation Status - True/False values
State	Archive Data Available Status - Has data: Archive - Empty: Archive does not have data yet
Size	File size on the Archive's hard drive

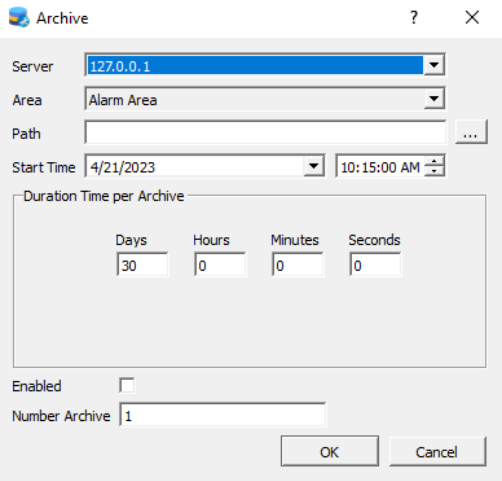
G.4.6.1.2 Add new Archives

Click **Add**



Enter the parameters for the Archive.

Attribute	Describe
Server	Archive Management Server - Choose from a list of connected servers
Area	Archive Data Management Partition - Select from a list of Data Areas
Path	Folders containing Archives - Supports selection from Window Explorer with button 
Start Time	The start time of the archive of the first Archive. (The start time of the next Archive is calculated automatically, immediately adjacent to the previous Archive)
Duration Time per Archive	Retention period for each Archive. - The minimum duration of an Archive is 1 hour - The maximum duration of an Archive is 1440 hours.
Enabled	Archive Activation Status - True/False values - If not selected, His data will not be saved to the Archive for the duration of the Archive
Number Archive	Number of Archives to Add



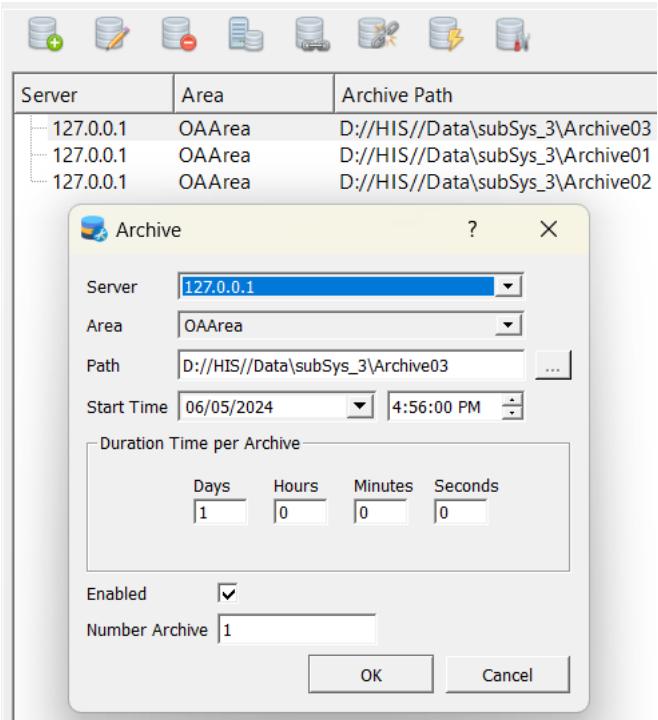
The 'Archive' dialog box contains the following fields and controls:

- Server:** A dropdown menu with '127.0.0.1' selected.
- Area:** A dropdown menu with 'Alarm Area' selected.
- Path:** A text field with a browse button ('...').
- Start Time:** A date and time picker showing '4/21/2023' and '10:15:00 AM'.
- Duration Time per Archive:** A section with four input fields: 'Days' (30), 'Hours' (0), 'Minutes' (0), and 'Seconds' (0).
- Enabled:** A checkbox that is currently unchecked.
- Number Archive:** A text field with the value '1'.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

FigureG-58. Add a new archive

G.4.6.1.3 Edit Archive

Select Archive to edit → Click  Edit or Double-Click on Archive.
Edit Archive Information



The figure shows a table of existing archives and an 'Archive' dialog box for editing one of them.

Server	Area	Archive Path
127.0.0.1	OAArea	D://HIS//Data\subSys_3\Archive03
127.0.0.1	OAArea	D://HIS//Data\subSys_3\Archive01
127.0.0.1	OAArea	D://HIS//Data\subSys_3\Archive02

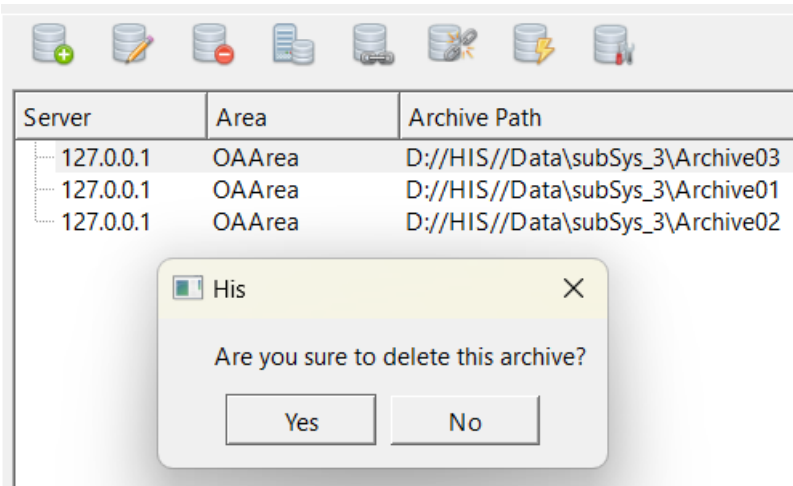
The 'Archive' dialog box (shown in the foreground) has the following values:

- Server:** 127.0.0.1
- Area:** OAArea
- Path:** D://HIS//Data\subSys_3\Archive03
- Start Time:** 06/05/2024 4:56:00 PM
- Duration Time per Archive:** Days: 1, Hours: 0, Minutes: 0, Seconds: 0
- Enabled:** Checked
- Number Archive:** 1

FigureG-59. Edit archive

G.4.6.1.4 Delete Archive

Select Archive to delete → Click **Delete**  → to confirm.



FigureG-60. Delete archive

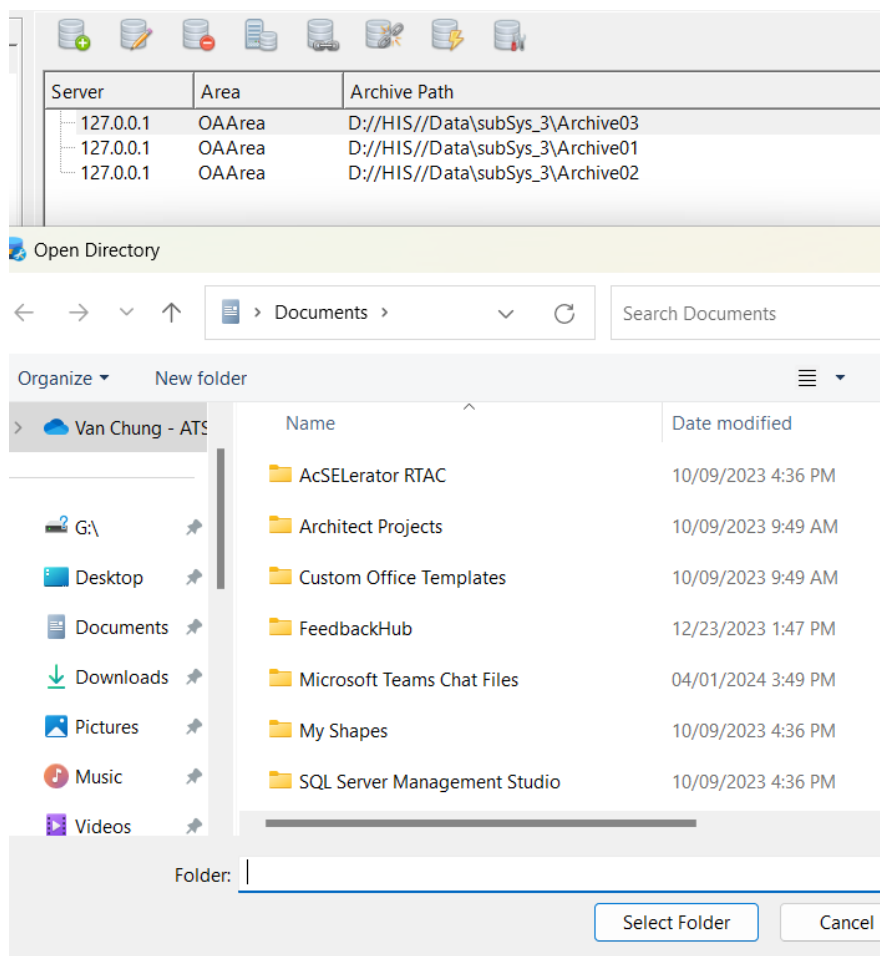
Archive Deletion involves deleting the Archive file on the hard drive.

G.4.6.1.5 Backup Archive

Select Archive to Backup → Click **Backup**



Select the Back-up folder.



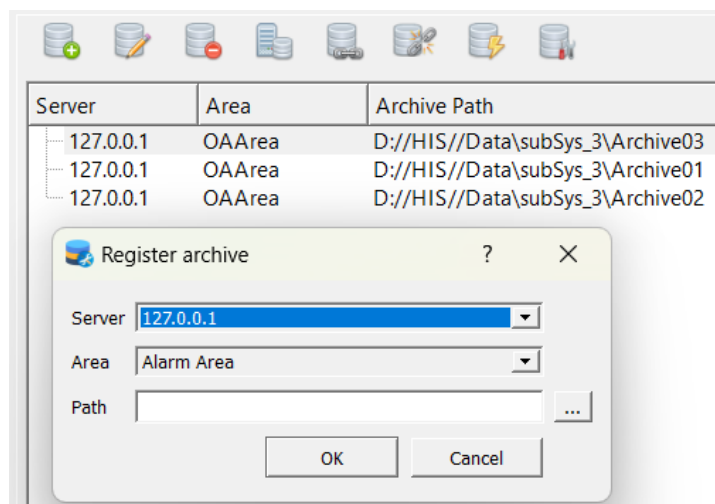
FigureG-61. Archive

G.4.6.1.6 Register Archive

Register Archive is a correction of a list of archives, which can be used in data recovery operations. In the event that the Archives for some reason (e.g. the expiration of the Server's storage time limit) are stored offline and are not included in the list of Online Archives managed by SmartHIS Server, resulting in the inability to store and query data. Accordingly, when it is necessary to store/query data from these Archives, it is necessary to perform the Register Archive operation. Registered Archives will return to the list of Online Archives managed by SmartHIS Server.

In addition, from the Windows operating system environment, it is not possible to copy/delete archive files when managed by SmartHIS Server.

- Click **Register**  → The Archive registration interface appears Select the Archive you want to Register.
- Select the information related to the desired Archive.



FigureG-62. Subscribe to the archive

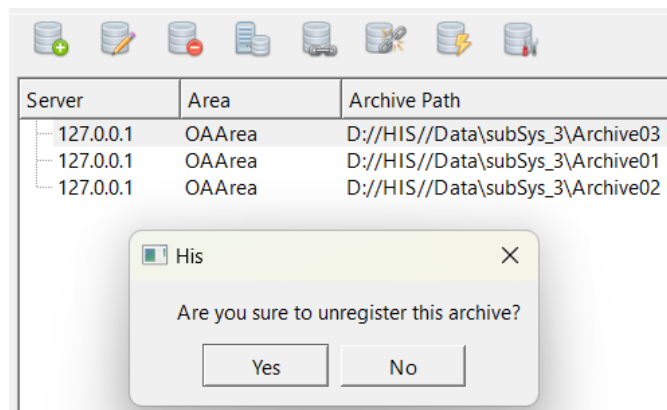
Attribute	Describe
<i>Server</i>	Archive Management Server - Choose from a list of connected servers
<i>Area</i>	Archive Data Management Partition - Select from a list of Data Areas
<i>Path</i>	Folders containing Archives - Supports selection from Window Explorer with button 

G.4.6.1.7 Unregister Archive

Unregister Archive is an operation to edit the list of Archives. Archive after Unregister is no longer on the list of Online Archives managed by SmartHIS Server, resulting in the inability to store and query data to this Archive.

Archive files still exist on the hard disk, and copy/delete operations can be performed from the Windows operating system environment.

- Select Archive to Unregister → Click **Unregister** 

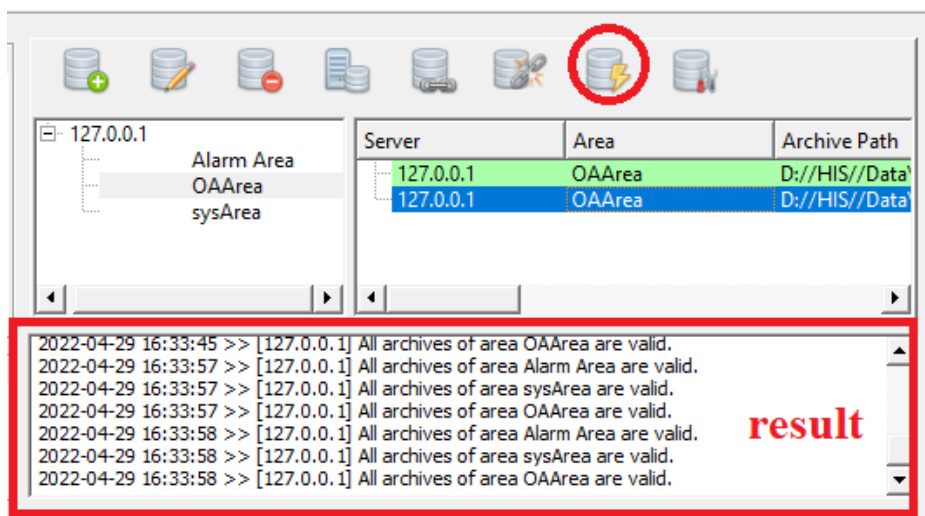


FigureG-63. Unsubscribe archive

G.4.6.1.8 Validate Archive

Check the validity of the Archive.

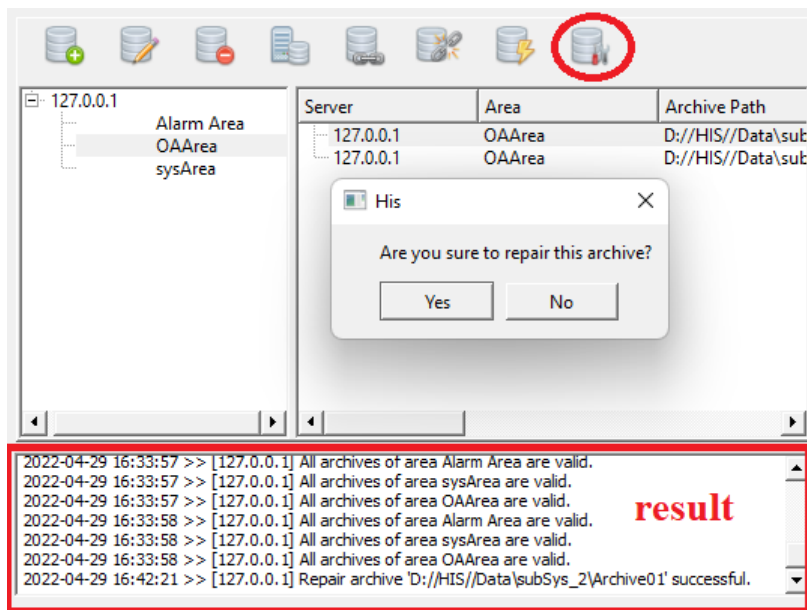
- Click **Validate** .
- The Validate **result** is displayed on the Activity Log window of the Admin Tool.
 - Valid: Valid archive
 - Invalid: Invalid archive



FigureG-64. Validate archive

G.4.6.1.9 Repair Archive

In case the Archive fails to lose the stored information of the points in the archive
→, use the Repair Archive tool to correct it.



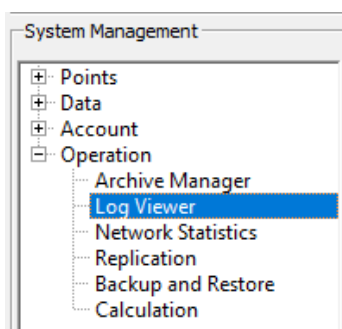
FigureG-65. Repair archive

- Select Archive → Click **Repair**  → Confirm.
- The Repair **result** is displayed on the Activity Log window of the Admin Tool.
 - Successful: repair successfully
 - Unsuccessful: repair failed

G.4.6.2 Log Viewer

Provides information that is logged back during the operation of the SmartHIS Server.

- Access the System Management → Operation / Log Viewer Menu.



- Log Viewer Display

Server: 127.0.0.1 Start Time: 4/1/2022 10:50:20 AM End Time: 4/29/2022 10:50:20 AM Group: All Level: All Message: 63065 logs Get Auto receive

Server	Time Stamp	Group	Level	Message
127.0.0.1	2022-04-05 18:00:30.711000	System	Info	SystemConfig::Begin Init
127.0.0.1	2022-04-05 18:00:30.770000	System	Info	Opened database D:/HIS/Data/SystemConfig.db successfully.
127.0.0.1	2022-04-05 18:00:30.771000	System	Info	SystemConfig::End Init
127.0.0.1	2022-04-05 18:00:31.203000	System	Info	License key is valid with limit points 100000 and expiration date at 9999-Jan-01 00:00:00.000000
127.0.0.1	2022-04-05 18:00:31.203000	System	Info	UserManager::Begin init
127.0.0.1	2022-04-05 18:00:31.203000	System	Info	UserManager::End init
127.0.0.1	2022-04-05 18:00:31.203000	System	Info	DigitalManager::Begin Init
127.0.0.1	2022-04-05 18:00:31.204000	System	Info	AreaManager::Begin Init
127.0.0.1	2022-04-05 18:00:31.204000	System	Info	SubSystemOrganizer: Staring subsystem sysArea
127.0.0.1	2022-04-05 18:00:31.204000	System	Info	SubSystemConfig::Begin Init
127.0.0.1	2022-04-05 18:00:31.205000	System	Info	Opened database D:/HIS/Data/subSys_1\SubSystemConfig.db successfully.
127.0.0.1	2022-04-05 18:00:31.205000	System	Info	SubSystemConfig::End Init
127.0.0.1	2022-04-05 18:00:31.205000	System	Info	PointManager:: Begin init
127.0.0.1	2022-04-05 18:00:31.206000	System	Info	PointManager:: End init
127.0.0.1	2022-04-05 18:00:31.206000	Snapshot	Info	SnapshotDB::Begin init
127.0.0.1	2022-04-05 18:00:31.206000	Snapshot	Info	Open Snapshot DB [D:/HIS/Data/subSys_1\Snapshot] with status: OK
127.0.0.1	2022-04-05 18:00:31.206000	Snapshot	Info	SnapshotDB::End init with count: 14
127.0.0.1	2022-04-05 18:00:31.206000	Archive	Info	ArchiveManager:: Begin init.
127.0.0.1	2022-04-05 18:00:31.206000	Archive	Info	D:/HIS/Data/subSys_1\Archive03; 2022-Mar-30 05:01:00; 2022-Mar-31 05:01:00
127.0.0.1	2022-04-05 18:00:31.206000	Archive	Info	Open DB [D:/HIS/Data/subSys_1\Archive03\Info] with status: OK
127.0.0.1	2022-04-05 18:00:31.206000	Archive	Info	Open DB [D:/HIS/Data/subSys_1\Archive03\Data] with status: OK
127.0.0.1	2022-04-05 18:00:31.206000	Archive	Info	D:/HIS/Data/subSys_1\Archive01; 2022-Mar-31 05:01:00; 2022-Apr-01 05:01:00

Records shown: 0 to 100 Page 1 of 631 << < 100 > >>

FigureG-66. Log information

- Data query configuration options:

Server: 127.0.0.1 Start Time: 4/1/2022 10:50:20 AM End Time: 4/29/2022 10:50:20 AM Group: All Level: All Message: Get Auto receive

Server	Alarm message data storage server
Start Time	When to start a query
End Time	When the query ends
Group	Newsletter Type
Level	Severity
Message	Log Content

- Custom only shows new logs from the time of monitoring ☐ Auto receive
- Send query → command Displays the results of log messages. Results are displayed in a page style. The information of the Log newsletter.

Attribute	Describe
Server	Server
Time Stamp	Timing of the Bulletin Log
Group	Log Newsletter Classification
	System - General system log newsletters
	Archive - Log newsletters related to the Archive
	Snap-shot - Log newsletters related to Snap-shots

	Alarms	- Alarm-related log newsletters
	Data	- Data-related log newsletters
	Network	- Log newsletters related to network connections
	—	
Level	Severity	
	Info	-Information
	Debug	- Activity Log
	Warning	-Alert
	Error	-Fault
	—	
Message	Log content	

- Aggregate information about the information displayed in search results.

Records shown: 0 to 50

Page 1 of 2

Message

63065 logs

Get

Records shown	List of the order of the records displayed
Page ... of	Page order / total number of pages results
Log	Total results (log message)

- Customization tools that display the results page

<<

<

50

>

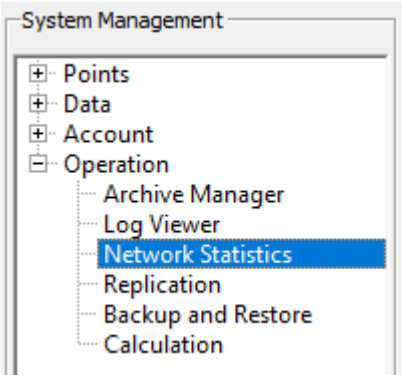
>>

Tool	Symbol	Describe
Number result / page	50	Number of points / 1 page displayed
Next page	>	Show the next points page
Previous page	<	Display the previous results page (points)
First Page	<<	Display the first page of results (points)
Last Page	>>	Display the final points page

G.4.6.3 Network Statistics

The Network Statistic menu provides statistical information on data exchange, → supporting analysis of the operational performance of the SmartHIS Server → to optimize the configuration.

- Go to the System Management → Operation / Network Statistics Menu.



- Network Statistic Interface

General		Errors							
Server	Data	Alarm	File	Queue Snapshot	Queue Archive	Snapshot (data/s)	Archive (data/s)	Alarm (data/s)	
127.0.0.1	37	0	0	0	0	3	2	0	

Server	Client End Point	Type	Server Sent	Server Received
127.0.0.1	127.0.0.1:64955	ATS.Admin.Tools	4	5

FigureG-67. Data Exchange Statistics

Monitoring parameters:

Parameter	Describe
General / Data Statistics	
Server	IP address of HIS Server
Data	Number of Data Points on HIS Server
Alarms	Number of Alarm/Event messages on HIS Server
File	-

Parameter	Describe
Queue Snap-shot	The number of data points (data value/ quality/ timestamp) waiting to be written to the snap-shot
Queue Archive	The number of data points (data value/ quality/ timestamp) waiting to be written to the archive
Snapshot (data/s)	Speed of writing data to snap-shot
Archive (data/s)	Speed of writing data (value/quality/timestamp) to Archive
Alarm (data/s)	Alarm data recording speed to Archive

Parameter	Describe
General / Connection Statistic	
Server	IP address of HIS Server
Client End Point	Client's Address (IP:Port)
Type	Client Type (AdminTool, DataLink, WEB Service, ...)
Server Sent	Sent HIS Server Data Capacity
Server Received	HIS Server Data Capacity Received

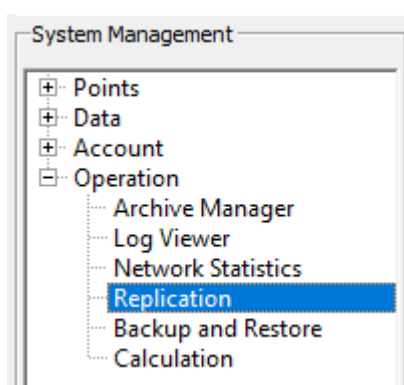
Parameter	Describe
Error / Data Statistic	
Server	IP address of HIS Server
Time too Old	- The amount of data not recorded to the archive because the time-stamp is older than the start-time of the oldest archive
Time too New	- The amount of data not written to the archive due to having a time-satmp newer than the end-time of the latest archive
Alarm not found	- The number of alarms is not recorded in the archive because the time-stamp is not in any of the SmartHIS archives
Data Commit	- The amount of data that has been stored
Alarm Commit	- Number of alarms that have been stored
Data Cache	- The amount of data that is in SmartHIS's cache
Cache Alarms	- The number of alarms that are in SmartHIS's cache
Other (data)	- The amount of data that cannot be written to the archive due to other reasons
Other (alarm)	- The number of alarms that cannot be recorded in the archive due to other reasons
File Size	-

G.4.6.4 Replication

The Replication menu provides a redundant data configuration tool – synchronously stores data from the source → server to the destination server.

G.4.6.4.1 List of Replications

Access the System Management → Operation/Replication Menu.



The Replication interface is displayed.

Source Server	Target Server	Enable	Status	Source Sent	Source Rece
127.0.0.1:2551	192.168.234.220:2551	Enabled	Disconnected	0 bytes	0 bytes

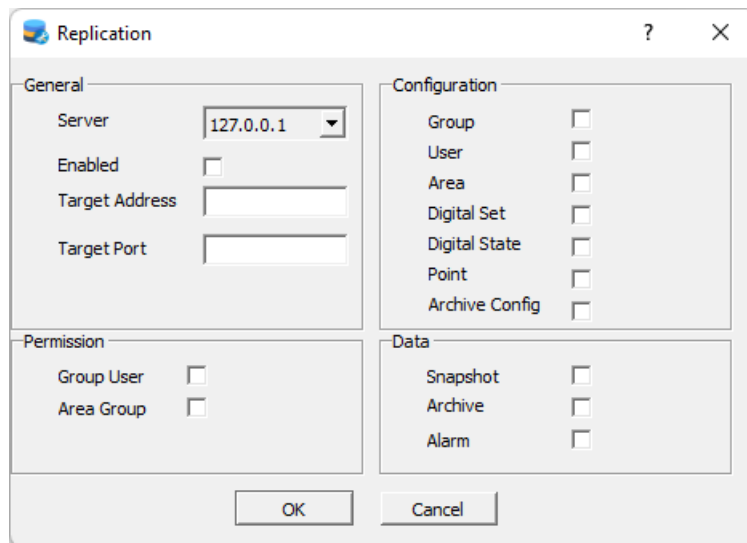
FigureG-68. List of replications

Replication monitoring information.

Source Server	Address (IP:Port) Server containing source data
Target Server	Address (IP:Port) The server will receive data from the source
Enable	Replication Activation Status
Status	Source Server Connection Status – Target Server ▪ Disconnected: lost connection ▪ Connected
Source Sent	Capacity (bytes) sent from the Source Server → Target Server
Source Receive	Bytes received from Target Server → Source Server

G.4.6.4.2 Add a new Replication (Connection)

Click **Add** 



The image shows a 'Replication' dialog box with the following sections:

- General:**
 - Server: 127.0.0.1 (dropdown)
 - Enabled: ☐
 - Target Address:
 - Target Port:
- Configuration:**
 - Group: ☐
 - User: ☐
 - Area: ☐
 - Digital Set: ☐
 - Digital State: ☐
 - Point: ☐
 - Archive Config: ☐
- Permission:**
 - Group User: ☐
 - Area Group: ☐
- Data:**
 - Snapshot: ☐
 - Archive: ☐
 - Alarm: ☐

Buttons: OK, Cancel

FigureG-69. Configuring replication

Configure the Replication information.

<i>Parameter</i>	<i>Describe</i>
General – General information about Replication Connection	
Server	Server Name Contains Source Data
Target Server Address / Port	The server address will receive data from the source
Enable	Replication Activation Status
Permission	
Group User	-True: Allows synchronous configuration of user permissions in groups -False: Do not allow synchronization of the configuration of users in groups
Area Group	-True: Allows synchronous configuration of decentralization of user groups in different areas -False: Do not allow synchronization of the configuration of decentralization of user groups in areas
Configuration – Synchronized configuration information	
Group	-True: Allows synchronization of group users

<i>Parameter</i>	<i>Describe</i>
	-False: Do not allow group user synchronization
User	-True: Allows user synchronization -False: Do not allow user synchronization
Area	-True: Allows area synchronization -False: Do not allow area synchronization
Digital Set	-True: Allows digital set synchronization -False: Do not allow digital set synchronization
Digital State	-True: Allows digital state synchronization -False: Do not allow digital state synchronization
Point	-True: Allows point synchronization -False: Point synchronization is not allowed
Archive Config	-True: Allows archive synchronization -False: Do not allow archive synchronization
Data – Synchronized data	
Snap shot	-True: Allows syncing of snap-shot data -False: Snap-shot data synchronization is not allowed
Archive	-True: Allows synchronization of archive data -False: Archive data synchronization is not allowed
Alarms	-True: Allows alarm synchronization -False: Alarm synchronization is not allowed

G.4.6.4.3 Edit Replication (Connection)

Select Replication → Click **Edit**



Edit the parameters of Replication.

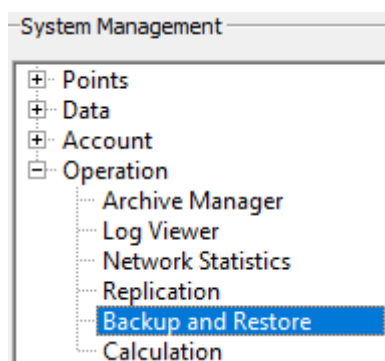
G.4.6.4.4 Delete Replication (Connection)

Select Replication → Click **Delete**  → **Confirm**.

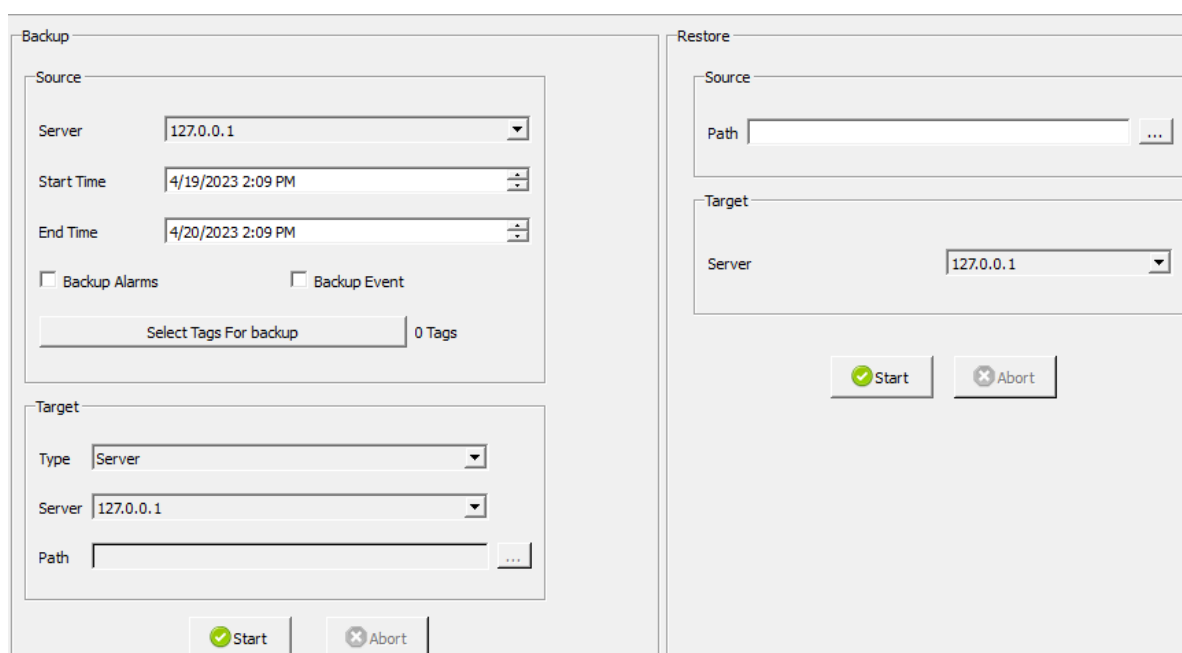
G.4.6.5 Backup and Restore Data

The Backup & Restore menu provides tools for manually backing up and restoring data. The data points and the time period to be backed up are arbitrary, independent of the Archive.

- Go to the System Management → Operation/Backup & Restore Menu.

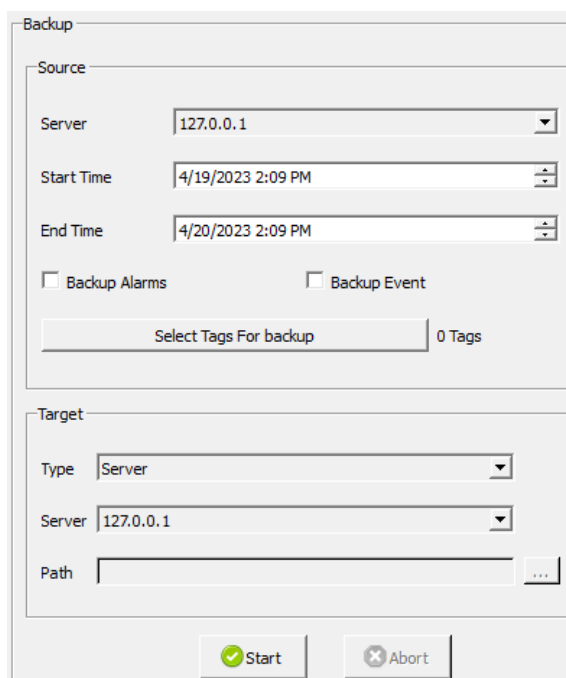


- Backup & Restore interface.



FigureG-70. System backup and recovery

G.4.6.5.1 Backup Data



- **Source** - Configure the selection of time periods and data to be backed up

<i>Parameter</i>	<i>Describe</i>
Server	Server Name Contains Source Data
Start Time End Time	Time period for data back-up
Backup Alarm Backup Events	Option to backup Alarm/Event message data
Select Tag for Backup	Select the points (Tags) that need to be backed up - Refer to the configuration operations at G.4.3.3.1 Searching for Data points

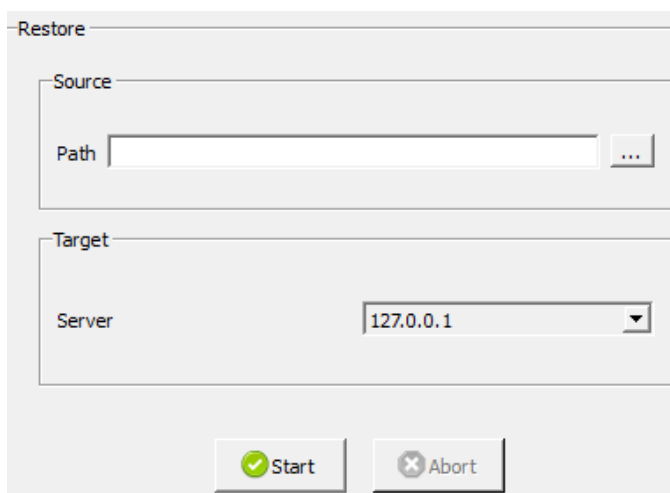
- **Target** - Configure the selection of the location where the data is backed.

<i>Parameter</i>	<i>Describe</i>
Type	▪ Server: stores data to SmartHIS Server ▪ File: stores data into a file
Server	The name of the server that receives the stored data (only means when Type = Server)
Path	The location of the file where the data is stored (meaning only when Type = File)

- Tools.

<i>Function</i>	<i>Describe</i>
Start	Run the data back-up process
Abort	Cancel the data back-up process

G.4.6.5.2 Data Recovery



- **Source** - Configure the selection of the recovery data source.

<i>Parameter</i>	<i>Describe</i>
Path	Data storage file location

- **Target** - Configure the server selection to receive the recovered data

<i>Parameter</i>	<i>Describe</i>
Server	Name of the server receiving the recovery data

- Tools



<i>Function</i>	<i>Describe</i>
Start	Run the data restore process
Abort	Cancel the data restore process

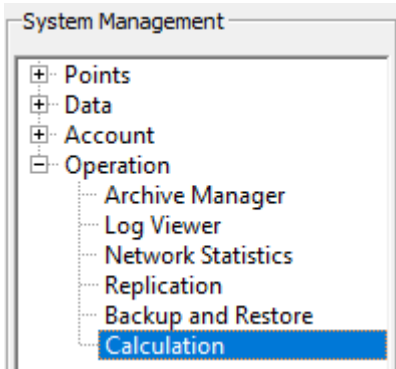
Restore Data will restore all data points, alarms/event messages stored in the data file.

The data will be automatically distributed to the Archives according to time and Data Area.

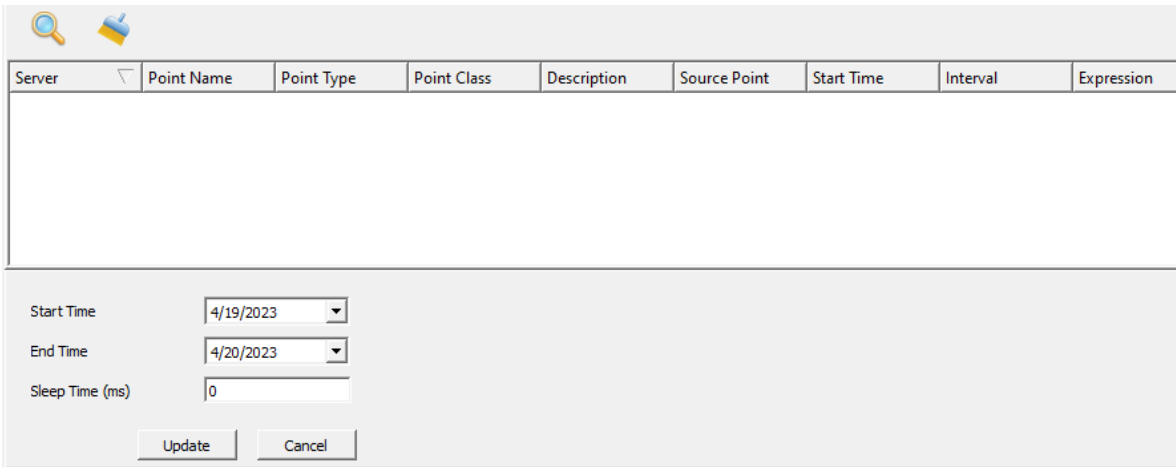
G.4.6.6 Calculation

The Calculation menu provides a tool to update the values of the calculation data points (Equation / Totalizer) in case the source data (Source Point) is updated (as a consequence of the process of synchronizing additional data, restoring Data, ...)

- Access the System Management → Operation / Calculation Menu



- Calculation Interface



FigureG-71. Calculation

Calculated data list calibration tools:

Tool	Symbol	Describe
Search		Perform a points search. Refer to G.4.3.3.1 Searching for Data points . The value of the selected points at the time of Click OK  is displayed in the data list table
Clear		Remove all points (rows) from the data list

Equation / Totalizer data point configuration information:

Server	Server contains data points
--------	-----------------------------

Point Name	Data point name
Point Type	Data Point Type
Point Class	Data Point Classes
Description	Descriptive name of the data point
Source Point	Source Data Name
Start Time	Expression Calculation Start Time
Interval	Expression calculation cycle
Expression	Expressions are performed

Refer to data attributes in [G.1.7.4 Data Point Storage Configuration Attributes](#)

Configure the interval and frequency of data updates.

Start Time	04/28/2022
End Time	04/29/2022
Sleep Time (ms)	0

<i>Parameter</i>	<i>Describe</i>
Start Time	Data update interval
End Time	
Step Time (ms)	Frequency of data updates - Unit: ms

Click **Update**  to update the data

G.4.7. Apply for a HIS Servers license

- Menu Help /License 
- Reference [G.2.3. Install and check the HIS Server license](#)

G.4.8. Configuring SmartHIS Server Runtime Service

- Menu Tool /Server Config 
- Reference [G.3.2 HIS Server Runtime Deployment Configuration](#)

G.5. SmartHIS - ATS DataLink

SmartHIS - ATS DataLink provides an add-in via Excel environment to help mine HIS Server data in a flexible and convenient way.

FileHomeInsertPage LayoutFormulasDataReviewViewAutomateDeveloperATSHelp

Current Value

Single Value

Archive Value

Compressed Data

Sampled Data

Timed Data

Calculated Data

Condition Data

Time Filtered

Get Alarm

Alarm Viewer

Connection

Tag Search

Import Tags

Export Tags

Panel Settings

Tools

G64

<

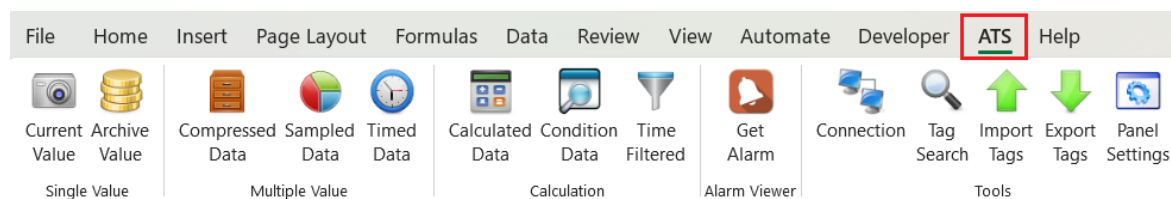
FigureG-72. Operation report (excel file) extracting data from HIS Server

Add-ins offer:

- Functions retrieve data from the Server & add-on functions.
- The interface (Ribbon Menu) supports the configuration of data retrieval functions.
- Interface (Ribbon Menu) supporting tools (Connection, ...).

G.5.1. Interface overview

ATS Data Link on Microsoft Excel 2007 or later is designed to use the ribbon menu.



FigureG-73. Menu Interface Functions on ATS Datalink

The Data Query Functions interface provides a tool to support the configuration of formulas for cells in the form of **formula arrays** (formulas placed in pairs of {}).

Operations on the interface:

- When the user clicks on each button on the ATS ribbon, task panels corresponding to each button will appear, each task panel creates each corresponding function.
- The task panels are movable and expandable.
- When the task panel is opened, users can fill in or change the content of the parameters on the panel. Depending on the properties, the input interface may include:
 - Choose from a list of parameter values
 - Import content
 - Fill in the address of the corresponding cell containing the value to be entered
- Confirmation of data entry:
 - → Click OK → to save and close the panel
 - → Click Apply → to save but do not close the panel.
 - → Click X → to close the panel and not save.

G.5.2. Data query functions

G.5.2.1 List of Data Query Functions

<i>Query function</i>	<i>Icon</i>	<i>Code</i>	<i>Describe</i>
Current Value		ATSCurrVal	Query the last value (snap-shot) of 1 or more points
Archive Value		ATSArchVal ATSArchValEx	Query data at a specified time ATSArchVal: Query data of 1 or more points ATSArchValEx: Querying data of 1 or more expressions Data interpolation method options
Compressed Data (Raw Data)		ATSCompressValues	Query all the values of 1 or more points that have been saved in a specified period of time Unwanted values can be filtered with the filter parameter
Sampled Data		ATSSampleValues ATSSampleValuesEx	Query data over a period of time with a defined sampling frequency

			ATSSampleValues: Query data of 1 or more points ATSSampleValuesEx: Query data of 1 or more expressions Wide selection of data interpolation methods Unwanted values can be filtered with the filter parameter
Timed Data		ATSTimeValues ATSTimeValuesEx	Query past data with multiple defined timelines ATSTimeValues: Query data of 1 or more points ATSTimeValuesEx: Query data of 1 or more expressions Wide selection of data interpolation methods
Calculated Data		ATSCalculatedValues	Statistical calculation of historical data over a specified period of 1 or more points It is possible to calculate and make statistics according to the specified frequency
Condition Data		ATSConditionValues	Conditional calculation function
Time Filtered		ATSTimeFiltered	Calculate the total time that an expression is satisfied
Get Alarm		ATSAalarmFiltered	Alarm/Event Statistics for a Period of Time
MaxValue		@ATSMMaxValue	Get the largest value of the selected region
MaxIndex		@ATSMMaxIndex	Get the position (ordinal number) of the largest value in the selected region

MinValue		@ATSMinValue	Get the smallest value of the selected region
MinIndex		@ATSMinIndex	Get the position (ordinal number) of the smallest value in the selected region

G.5.2.2 Data query results

The data query functions that return the usual results include the value – time label – quality label of the data.

In exceptional cases, the results returned are exceptions. Reference [G.1.8.3.4 Exceptions](#)

The data query results are displayed at the Cell (Range) arrays depending on the data display configuration parameter.

Parameter	Data Type	Configuration values
<i>Parameters that display results in Excel</i>		
Calculate Now	bool	Select to execute a query when there is an event that changes any value in the Parameter Cells of the function ▪ True: the function that takes the past value will be executed. ▪ False: the function that takes the past value will not be executed
Output Cell	Range	The first Cell address in the Cell array (range) contains the result returned by the data query function.

OutCode	Int	How to display return data ▪ 0: Column ▪ 1: Row ▪ 2: Column_TimeStamp ▪ 3: Row_TimeStamp ▪ 4: Column_Quality ▪ 5:Row_Quality ▪ 6:Column_TimeStamp_Quality ▪ 7:Row_TimeStamp_Quality The Show parameter value is compiled from a set of options: Calculate Now FALSE Output Cell Sheet1!\$D\$17 Show Column ☐ Time Stamp ☐ Quality ☒ Show value not good <table><tr><td>Show</td><td> The Cell array type (range) contains the data results ▪ Row: The return data for each tag or expression is displayed on 1 line. ▪ Column: The return data for each tag or expression displayed on 1 column </td></tr><tr><td>Time stamp</td><td> The data returned for each tag will include the corresponding time label line/column of the data. Value: True/False </td></tr><tr><td>Quality</td><td> The data returned for each tag will include the corresponding quality label line/column of the data. Value: True/False </td></tr><tr><td>Show Value not Good</td><td> ▪ False: Data with a Not Good quality label will display a data label instead of a data value on the data value line/column. </td></tr></table>	Show	The Cell array type (range) contains the data results ▪ Row: The return data for each tag or expression is displayed on 1 line. ▪ Column: The return data for each tag or expression displayed on 1 column	Time stamp	The data returned for each tag will include the corresponding time label line/column of the data. Value: True/False	Quality	The data returned for each tag will include the corresponding quality label line/column of the data. Value: True/False	Show Value not Good	▪ False: Data with a Not Good quality label will display a data label instead of a data value on the data value line/column.
Show	The Cell array type (range) contains the data results ▪ Row: The return data for each tag or expression is displayed on 1 line. ▪ Column: The return data for each tag or expression displayed on 1 column									
Time stamp	The data returned for each tag will include the corresponding time label line/column of the data. Value: True/False									
Quality	The data returned for each tag will include the corresponding quality label line/column of the data. Value: True/False									
Show Value not Good	▪ False: Data with a Not Good quality label will display a data label instead of a data value on the data value line/column.									

		▪ True: Data labeled Not Good will show the data value on the data value line/column.
--	--	--

G.5.2.3 Current Value - ATSCurrVal

Describe:

- Query the latest value (Snapshot) of the point
- Query data of 1 or more points

Jaw syntax

ATSCurrVal(int serverType, string address, int port, string userName, string password, string tagNames, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:

Current Value

Server Type

SmartHisServer

Address

Sheet1!\$H\$3

Port

Sheet1!\$H\$5

User Name

Sheet1!\$H\$7

Password

Tagname(s)

Sheet1!\$H\$11

Calculate Now

FALSE

Output Cell

Sheet1!\$D\$17

Show

Column

☐ Time Stamp

☐ Quality

☒ Show value not good

OK

Apply

The configuration parameters display the result on Excel:

Parameter	Input Type								
Parameters that display results in Excel									
Calculate Now	Select (TRUE/FALSE) / Cell Address								
Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs: <table><tr><td>Show</td><td>Row/Column Selection</td></tr><tr><td>Time stamp</td><td>True/False Selection</td></tr><tr><td>Quality</td><td>True/False Selection</td></tr><tr><td>Show Value not Good</td><td>True/False Selection</td></tr></table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
tagNames	String	Input String / Address Range (multi Cells)	List of point-names to query data.

G.5.2.4 Point Archive Value – ATSArchVal

Describe:

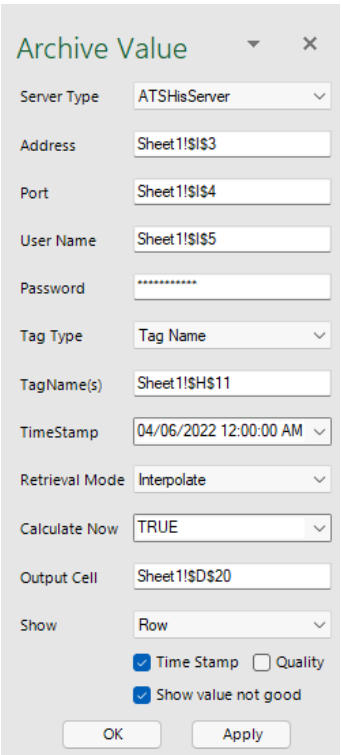
- Function to retrieve point data values stored on Archives
- Query data of 1 or more points
- Data interpolation mode option

Jaw syntax

```
ATSArchVal(int serverType, string address, int port, string userName,
string password, string tagNames, string timeStamp, int
retrievalMode, int outCode, bool calculateNow)
```

Configuration Interface:

- Icon: 
- Configuration Interface:



Archive Value

Server Type: ATSHisServer

Address: Sheet1!\$I\$3

Port: Sheet1!\$I\$4

User Name: Sheet1!\$I\$5

Password: *****

Tag Type: Tag Name

TagName(s): Sheet1!\$H\$11

TimeStamp: 04/06/2022 12:00:00 AM

Retrieval Mode: Interpolate

Calculate Now: TRUE

Output Cell: Sheet1!\$D\$20

Show: Row

☒ Time Stamp ☐ Quality

☒ Show value not good

OK Apply

The configuration parameters display the result on Excel:

Parameter	Input Type								
Parameters that display results in Excel									
Calculate Now	Select (TRUE/FALSE) / Cell Address								
Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs:<table><tr><td>Show</td><td>Row/Column Selection</td></tr><tr><td>Time stamp</td><td>True/False Selection</td></tr><tr><td>Quality</td><td>True/False Selection</td></tr><tr><td>Show Value not Good</td><td>True/False Selection</td></tr></table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Meaningful reference of the parameters that display the results at [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
Point/Expression data function type	
Tag Type	=Tag Name: to use the ATSArchVal function to query point data

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
tagNames	String	Input String / Address Range (multi Cells)	List of data query points.
Timestamp	String	Input String / Selection	Data query time.

		Dialog / Cell Address. -Support for excel functions DATE-TIME	Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
retrievalMode	Int	Select from the list	Data interpolation method: • 0: Auto • 1: Exact Time • 2: Interpolate • 3: Previous • 4: Next Reference to the meaning of data interpolation modes in G.1.8.3.2 Data query modes

G.5.2.5 Expression Archive Value – ATSArchValEx

Describe:

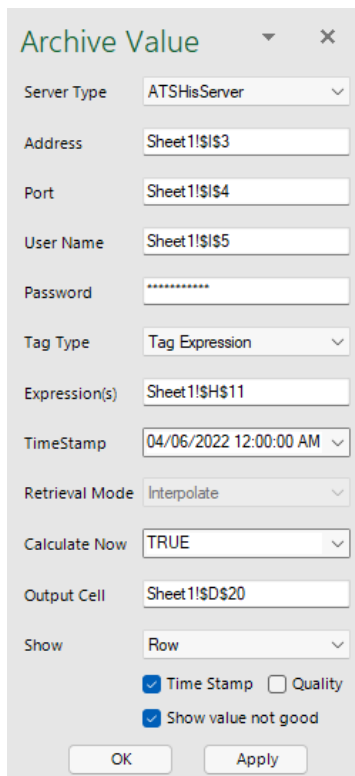
- The function takes the data value (point) stored in the Archives and calculates the corresponding expression value
- Querying data of 1 or more expressions
- Points in the expression are queried with Interpolate interpolation mode

Jaw syntax

```
ATSArchValEx(int serverType, string address, int port, string
userName, string password, string expression, string timeStamp, int
outCode, bool calculateNow)
```

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type								
<i>Parameters that display results in Excel</i>									
Calculate Now	Select (TRUE/FALSE) / Cell Address								
Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs: <table border="1"> <tr> <td>Show</td><td>Row/Column Selection</td></tr> <tr> <td>Time stamp</td><td>True/False Selection</td></tr> <tr> <td>Quality</td><td>True/False Selection</td></tr> <tr> <td>Show Value not Good</td><td>True/False Selection</td></tr> </table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
<i>Point/Expression data function type</i>	
Tag Type	=Expression Tag: to use the ATSArchValEx function to query the expression data

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
Expressions	String	Input String / Address Range (multi Cells)	A list of expressions that query the data. Expression syntax: reference G.10. Appendix - Expression Syntax
Timestamp	String	Input String/Cell Address. -Support for excel functions DATE-TIME	Data query time. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)

G.5.2.6 Compressed Data – ATSCompressValues

Describe:

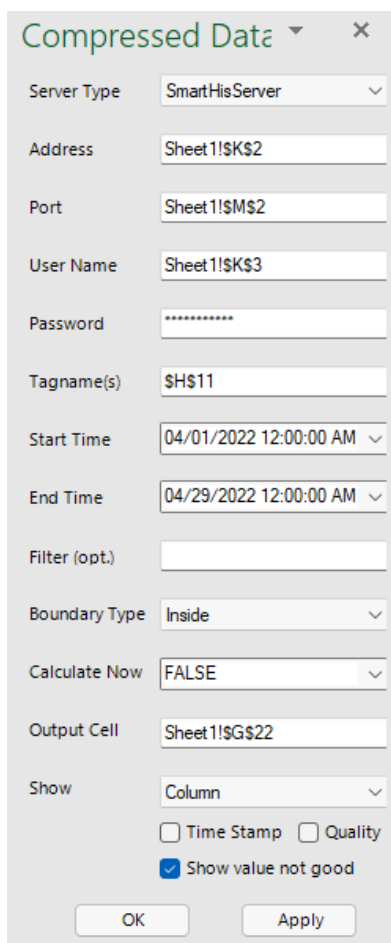
- The function takes all the values that have been saved for a specified period of time
- Query data of 1 or more points
- Unwanted values can be filtered with filter expressions.

Jaw syntax

ATSCompressValues(int serverType, string address, int port, string userName, string password, string tagNames, string startTime, string endTime, string filter, int boundaryType, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type								
<i>Parameters that display results in Excel</i>									
Calculate Now	Select (TRUE/FALSE) / Cell Address								
Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs: <table> <tr> <td>Show</td><td>Row/Column Selection</td></tr> <tr> <td>Time stamp</td><td>True/False Selection</td></tr> <tr> <td>Quality</td><td>True/False Selection</td></tr> <tr> <td>Show Value not Good</td><td>True/False Selection</td></tr> </table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

Parameter	Data Type	Input type on the interface	Describe
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port of HIS Server
userName	String	Input String / Cell Address	User Name

password	String	Input String / Cell Address	password
tagNames	String	Input String / Address Range (multi Cells)	List of data query points.
startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
Filter	String	Input String / Cell Address	(option) Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax
boundaryType	Int	Select from the list	Data acquisition method 2 start and end timelines ▪ 0: In side ▪ 1: Out side ▪ 2: Interpolate ▪ 3: Auto

G.5.2.7 Point Sampled Data – ATSSampleValues

Describe:

- The function retrieves data over a period of time with a specified sampling frequency
- Query data of 1 or more points
- Data interpolation mode option
- Unwanted values can be filtered with filter expressions.

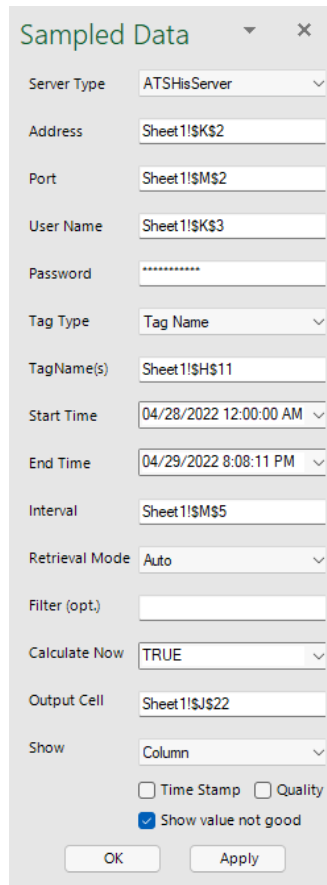
Jaw syntax

```
ATSSampleValues(int serverType, string address, int port, string
userName, string password, string tagNames, string startTime, string
```

endTime, string interval, int retrievalMode, string filter, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type	
<i>Parameters that display results in Excel</i>		
Calculate Now	Select (TRUE/FALSE) / Cell Address	
Output Cell	Cell Address	
outCode	The type of data result display. Choose from a set of inputs:	
	Show	Row/Column Selection

	Time stamp	True/False Selection
	Quality	True/False Selection
	Show Value not Good	True/False Selection

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
<i>Point/Expression data function type</i>	
Tag Type	=Tag Name: to use the ATSSampleValues function to query point data

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password

tagNames	String	Input String / Address Range (multi Cells)	List of data query points.																					
startTime endTime	String	Input String/Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)																					
Interval	String	Input String/Cell Address.	Data query frequency Syntax: value + unit symbol <table><tr><th>Symbol</th><th>Unit</th><th>Example</th></tr><tr><td>s</td><td>second</td><td>5s</td></tr><tr><td>m</td><td>minute</td><td>2m</td></tr><tr><td>h</td><td>hour</td><td>1h</td></tr><tr><td>d</td><td>day</td><td>1d</td></tr><tr><td>w</td><td>week</td><td>1W</td></tr><tr><td>Mo</td><td>month</td><td>1 mo</td></tr></table>	Symbol	Unit	Example	s	second	5s	m	minute	2m	h	hour	1h	d	day	1d	w	week	1W	Mo	month	1 mo
Symbol	Unit	Example																						
s	second	5s																						
m	minute	2m																						
h	hour	1h																						
d	day	1d																						
w	week	1W																						
Mo	month	1 mo																						
retrievalMode	Int	Select from the list	Data interpolation method: 0: Auto 1: Exact Time 2: Interpolate 3: Previous 4: Next Reference to the meaning of data interpolation modes in G.1.8.3.2 Data query modes																					
Filter	String	Input String/Cell Address.	<i>(option)</i> Filter result condition expression.																					

			Filter expression syntax: reference G.10. Appendix - Expression Syntax
--	--	--	--

G.5.2.8 Expression Sampled Data – ATSSampleValuesEx

Describe:

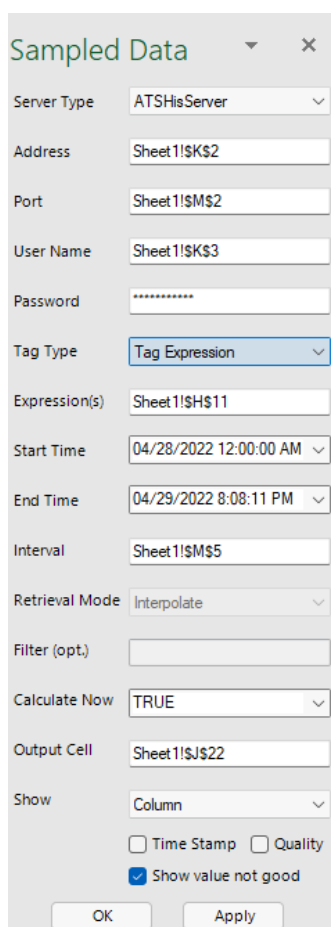
- The function retrieves data over a period of time with a specified sampling frequency
- Query data of 1 or more expressions
- Points in the expression are queried with Interpolate interpolation mode

Jaw syntax

ATSSampleValuesEx(int serverType, string address, int port, string userName, string password, string expression, string startTime, string endTime, string interval, string filter, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type								
<i>Parameters that display results in Excel</i>									
Calculate Now	Select (TRUE/FALSE) / Cell Address								
Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs: <table> <tr> <td>Show</td><td>Row/Column Selection</td></tr> <tr> <td>Time stamp</td><td>True/False Selection</td></tr> <tr> <td>Quality</td><td>True/False Selection</td></tr> <tr> <td>Show Value not Good</td><td>True/False Selection</td></tr> </table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
<i>Point/Expression data function type</i>	
Tag Type	=Expression Tag: to use the ATSSampleValuesEx function to query the expression data

Data query parameters

Parameter	Data Type	Input type on the interface	Describe
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	With SmartHIS Server: SmartHIS Server's IP/DNS address

			In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool																		
Port	Int	Input String / Cell Address	Data Port của HIS Server																		
userName	String	Input String / Cell Address	User Name																		
password	String	Input String / Cell Address	password																		
Expressions	String	Input String / Address Range (multi Cells)	A list of expressions that query the data. Expression syntax: reference G.10. Appendix - Expression Syntax																		
startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)																		
Interval	String	Input String/Cell Address.	Data query frequency Syntax: value + unit symbol <table><tr><th>Symbol</th><th>Unit</th><th>Example</th></tr><tr><td>s</td><td>second</td><td>5s</td></tr><tr><td>m</td><td>minute</td><td>2m</td></tr><tr><td>h</td><td>hour</td><td>1h</td></tr><tr><td>d</td><td>day</td><td>1d</td></tr><tr><td>w</td><td>week</td><td>1W</td></tr></table>	Symbol	Unit	Example	s	second	5s	m	minute	2m	h	hour	1h	d	day	1d	w	week	1W
Symbol	Unit	Example																			
s	second	5s																			
m	minute	2m																			
h	hour	1h																			
d	day	1d																			
w	week	1W																			

G.5.2.9 Point Time Data – ATSTimeValues

Describe:

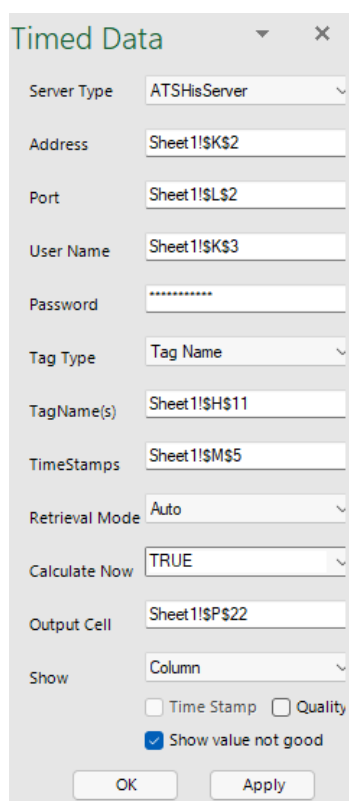
- Function retrieves data with any specified timelines
- Query data of 1 or more points
- Data interpolation mode option

Jaw syntax

```
ATSTimeValues(int serverType, string address, int port, string
userName, string password, string tagNames, object timeStamps, int
retrievalMode, int outCode, bool calculateNow)
```

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type
<i>Parameters that display results in Excel</i>	
Calculate Now	Select (TRUE/FALSE) / Cell Address

Output Cell	Cell Address								
outCode	The type of data result display. Choose from a set of inputs: <table> <tr> <td>Show</td><td>Row/Column Selection</td></tr> <tr> <td>Time stamp</td><td>True/False Selection</td></tr> <tr> <td>Quality</td><td>True/False Selection</td></tr> <tr> <td>Show Value not Good</td><td>True/False Selection</td></tr> </table>	Show	Row/Column Selection	Time stamp	True/False Selection	Quality	True/False Selection	Show Value not Good	True/False Selection
Show	Row/Column Selection								
Time stamp	True/False Selection								
Quality	True/False Selection								
Show Value not Good	True/False Selection								

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
<i>Point/Expression data function type</i>	
Tag Type	=Tag Name: to use the ATSTimeValues function to query point data

Data query parameters

Parameter	Data Type	Input type on the interface	Describe
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server

userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
tagNames	String	Input String / Address Range (multi Cells)	List of data query points.
Timestamps	Object	Input String / Selection Dialog / Address Range (multi- Cells) -Support for excel functions DATE-TIME	List of data retrieval times Gathering-type timelines Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
retrievalMode	Int	Select from the list	Data interpolation method: • 0: Auto • 1: Exact Time • 2: Interpolate • 3: Previous • 4: Next Reference to the meaning of data interpolation modes in G.1.8.3.2 Data query modes

G.5.2.10 Expression Time Data – ATSTimeValuesEx

Describe:

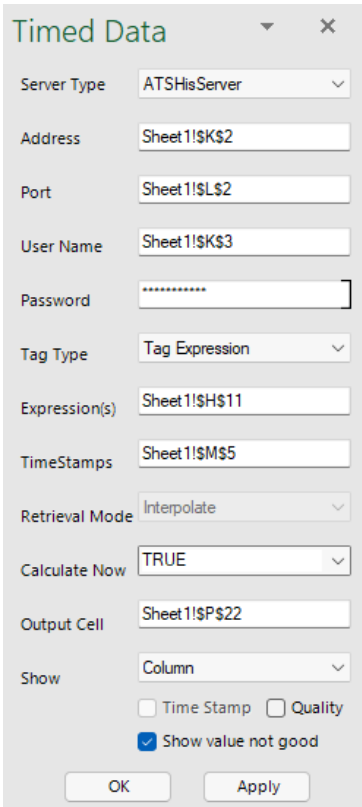
- Function retrieves data with any specified timelines
- Query data of 1 or more expressions
- Points in the expression are queried with Interpolate interpolation mode

Jaw syntax

ATSTimeValuesEx(int serverType, string address, int port, string userName, string password, string expression, object timeStamps, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:



The 'Timed Data' configuration dialog box includes the following fields and options:

- Server Type: ATSHisServer (dropdown)
- Address: Sheet1!\$K\$2 (text box)
- Port: Sheet1!\$L\$2 (text box)
- User Name: Sheet1!\$K\$3 (text box)
- Password: (password field)
- Tag Type: Tag Expression (dropdown)
- Expression(s): Sheet1!\$H\$11 (text box)
- TimeStamps: Sheet1!\$M\$5 (text box)
- Retrieval Mode: Interpolate (dropdown)
- Calculate Now: TRUE (dropdown)
- Output Cell: Sheet1!\$P\$22 (text box)
- Show: Column (dropdown)
- Time Stamp: ☐
- Quality: ☐
- Show value not good: ☒
- Buttons: OK, Apply

The configuration parameters display the result on Excel:

Parameter	Input Type
<i>Parameters that display results in Excel</i>	
Calculate Now	Select (TRUE/FALSE) / Cell Address
Output Cell	Cell Address

outCode	The type of data result display. Choose from a set of inputs:	
	Show	Row/Column Selection
	Time stamp	True/False Selection
	Quality	True/False Selection
	Show Value not Good	True/False Selection

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Point/Expression data query function type parameter:

Parameter	Describe
<i>Point/Expression data function type</i>	
Tag Type	=Expression Tag: to use the ATSTimeValuesEx function to query the data point

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server

userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
Expressions	String	Input String / Address Range (multi Cells)	A list of expressions that query the data. Expression syntax: reference G.10. Appendix - Expression Syntax
Timestamps	Object	Input String / Selection Dialog / Address Range (multi-Cells) -Support for excel functions DATE-TIME	List of data retrieval times Gathering-type timelines Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)

G.5.2.11 Calculated Data – ATSCalculatedValues

Describe:

- Statistical calculation function (min, max, ...) data over a specified period of time
- Query data of 1 or more points
- It is possible to calculate according to fixed frequency milestones during the selected time period.
- Unwanted values can be filtered with filter expressions.

Jaw syntax

```
ATSCalculatedValues(int serverType, string address, int port, string
userName, string password, string tagNames, string startTime, string
endTime, string interval, string filter, int calculateMode, int outCode,
bool calculateNow, string sampleFrequency, int sampleType)
```

Configuration Interface:

- Icon: 
- Configuration Interface:

Calculated Data

Server Type

ATSHisServer

Address

Port

User Name

Password

Tag Type

Tag Name

TagName(s)

Start Time

End Time

Interval (opt.)

Filter (opt.)

Sample Type

Compressed

Sampling Freq.

Mod

Count

Calculate Now

TRUE

Output Cell

Show

Column

☐ Time Stamp

☐ Quality

☒ Show value not good

OK

Apply

The configuration parameters display the result on Excel:

Parameter	Input Type
<i>Parameters that display results in Excel</i>	
Calculate Now	Select (TRUE/FALSE) / Cell Address
Output Cell	Cell Address

outCode	The type of data result display. Choose from a set of inputs:	
	Show	Row/Column Selection
	Time stamp	True/False Selection
	Quality	True/False Selection
	Show Value not Good	True/False Selection

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port của HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password

tagNames	String	Input String / Address Range (multi Cells)	List of data query points.																		
startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)																		
Interval	String	Input String/Cell Address.	<i>(option)</i> Statistical calculation frequency Syntax: value + unit symbol <table><tr><th>Symbol</th><th>Unit</th><th>Example</th></tr><tr><td>s</td><td>second</td><td>5s</td></tr><tr><td>m</td><td>minute</td><td>2m</td></tr><tr><td>h</td><td>hour</td><td>1h</td></tr><tr><td>d</td><td>day</td><td>1d</td></tr><tr><td>w</td><td>week</td><td>1W</td></tr></table>	Symbol	Unit	Example	s	second	5s	m	minute	2m	h	hour	1h	d	day	1d	w	week	1W
Symbol	Unit	Example																			
s	second	5s																			
m	minute	2m																			
h	hour	1h																			
d	day	1d																			
w	week	1W																			
Filter	String	Input String / Cell Address	<i>(option)</i> Filter result condition expression. Filter expression syntax: reference G.10. Appendix - Expression Syntax																		
sampleType	Int	Select from the list	Data query type to perform statistical calculations 0: Compressed: queries stored values 1: Interpolated: query data at a specified frequency																		

			(sampleFrequency) with the Interpolate interpolation type																		
sampleFrequency	String	Input String / Cell Address	The frequency of data queries to perform statistical calculations. Makes sense only when sampleType = Interpolated Syntax: value + unit symbol <table><thead><tr><th>Symbol</th><th>Unit</th><th>Example</th></tr></thead><tbody><tr><td>s</td><td>second</td><td>5s</td></tr><tr><td>m</td><td>minute</td><td>2m</td></tr><tr><td>h</td><td>hour</td><td>1h</td></tr><tr><td>d</td><td>day</td><td>1d</td></tr><tr><td>w</td><td>week</td><td>1W</td></tr></tbody></table>	Symbol	Unit	Example	s	second	5s	m	minute	2m	h	hour	1h	d	day	1d	w	week	1W
Symbol	Unit	Example																			
s	second	5s																			
m	minute	2m																			
h	hour	1h																			
d	day	1d																			
w	week	1W																			
calculationMode	Int	Select from the list	Calculation method: • 0: Count • 1: Total • 2: Minimum • 3: Maximum • 4: Average • 5:Minimum_Abs • 6:Maximum_Abs • 7:Average_Abs																		

G.5.2.12 Condition Data – ATSConditionValues

Describe:

- Conditional calculation function
- Query data of 1 or more points
- It is possible to calculate according to fixed frequency milestones during the selected time period.
- Unwanted values can be filtered with filter expressions.

Jaw syntax

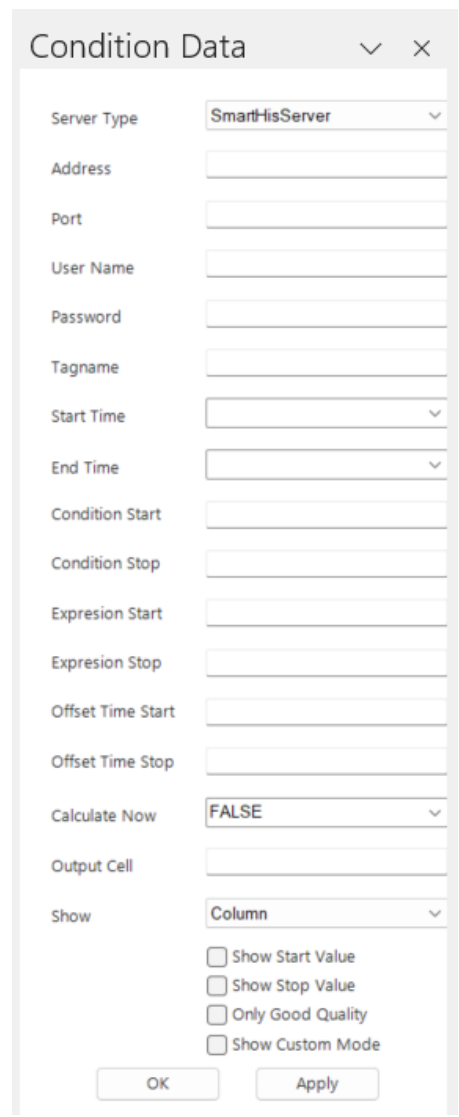
```
ATSConditionValues(string address, int port, string userName, string password, string tag, string startTime, string endTime, string conditionStart,
```

string conditionStop, string expressionAtStart, string expressionAtStop, int
offsetTimeAtStart, int offsetTimeAtStop, bool onlyGoodQuality, bool
showStartValue, bool showStopValue, int outCode, bool calculateNow)

Configuration Interface:



- Icon:
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type
<i>Parameters that display results in Excel</i>	
Calculate Now	Select (TRUE/FALSE) / Cell Address

Output Cell	Cell Address										
outCode	The type of data result display. Choose from a set of inputs: <table> <tr> <td>Show</td><td>Row/Column Selection</td></tr> <tr> <td>Show Start Value</td><td>True/False Selection</td></tr> <tr> <td>Show Stop Value</td><td>True/False Selection</td></tr> <tr> <td>Only Good Quality</td><td>True/False Selection</td></tr> <tr> <td>Show Custom Mode</td><td></td></tr> </table>	Show	Row/Column Selection	Show Start Value	True/False Selection	Show Stop Value	True/False Selection	Only Good Quality	True/False Selection	Show Custom Mode	
Show	Row/Column Selection										
Show Start Value	True/False Selection										
Show Stop Value	True/False Selection										
Only Good Quality	True/False Selection										
Show Custom Mode											

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

Parameter	Data Type	Input type on the interface	Describe
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port of HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password

Tag	String	Input String / Address Range (multi Cells)	List of data query points.
startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
conditonStart conditonStop	String	Input String / Cell Address	conditonStart: The condition for starting the data query. conditonStop: The condition for the end of the data query.
expressionStart expressionStop	String	Input String / Cell Address	expressionStart: The expression is calculated at the start time. expressionStop: The expression computed at the end time.
offsetTimeAtStart offsetTimeAtStop	Int	Input String / Cell Address	offsetTimeStart: The timeline that deviates from the start time to calculate the full expression. offsetTimeStop: The timeline that deviates from the end time to calculate the full expression.

G.5.2.13 Time Filtered – ATSTimeFiltered

Describe:

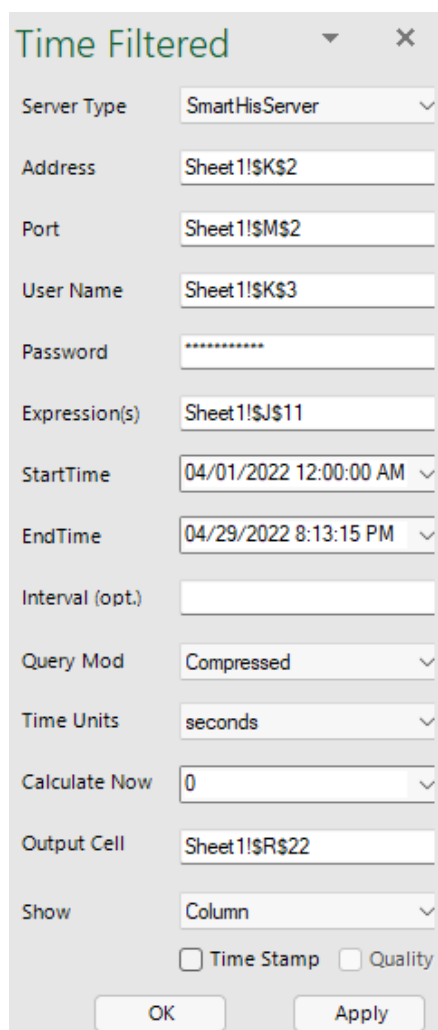
- The function of the sum of the lengths of time that the expression satisfies
- Query data of 1 or more expressions
- It is possible to calculate according to fixed frequency milestones during the selected time period.

Jaw syntax

ATSTimeFiltered(int serverType, string address, int port, string userName, string password, string expression, string startTime, string endTime, string interval, int typeUnit, int outCode, bool calculateNow)

Configuration Interface:

- Icon: 
- Configuration Interface:



The configuration parameters display the result on Excel:

Parameter	Input Type
<i>Parameters that display results in Excel</i>	
Calculate Now	Select (TRUE/FALSE) / Cell Address
Output Cell	Cell Address

outCode	The type of data result display. Choose from a set of inputs:	
	Show	Row/Column Selection
	Time stamp	True/False Selection
	Quality	True/False Selection
	Show Value not Good	True/False Selection

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

<i>Parameter</i>	<i>Data Type</i>	<i>Input type on the interface</i>	<i>Describe</i>
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version) 1: PI Server 2: SmartHis Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port of HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
Expressions	String	Input String / Address Range (multi Cells)	A list of expressions that query the data. Expression syntax: reference G.10. Appendix - Expression Syntax

startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)																		
Interval	String	Input String/Cell Address.	<i>(option)</i> Statistical calculation frequency Syntax: value + unit symbol <table><tr><th>Symbol</th><th>Unit</th><th>Example</th></tr><tr><td>s</td><td>second</td><td>5s</td></tr><tr><td>m</td><td>minute</td><td>2m</td></tr><tr><td>h</td><td>hour</td><td>1h</td></tr><tr><td>d</td><td>day</td><td>1d</td></tr><tr><td>w</td><td>week</td><td>1W</td></tr></table>	Symbol	Unit	Example	s	second	5s	m	minute	2m	h	hour	1h	d	day	1d	w	week	1W
Symbol	Unit	Example																			
s	second	5s																			
m	minute	2m																			
h	hour	1h																			
d	day	1d																			
w	week	1W																			
timeUnit	Int	Select from the list	Units of time for returned results 0: second 1: minute 2: hour 3: Thick																		
queryMode	Int	Select from the list	Query Type 0: Compressed 1: Interpolated																		

G.5.2.14 Get Alarm – ATSArmFiltered

Describe:

- Alarm/Event Statistics Function for a Period of Time

Jaw syntax

ATSArmFiltered(string address, int port, string userName, string password, int alarmType, DateTime startTime, DateTime endTime, int fromPriority, int toPriority, string operatorName, string sourceName, string message, string value, bool calculateNow, string col1, string col2, string col3, string col4, string col5, string col6, string col7, string col8, string col9, string col10, string col11, string col12, string col13)

Configuration Interface:

- Icon: 
- Configuration Interface:

Alarm Filtered

Server Type

SmartHisServer

Address

Config!\$B\$7

Port

Config!\$C\$7

User Name

Config!\$D\$7

Password

Alarm Type

Alarm

Start Time

Config!\$B\$9

End Time

Config!\$B\$10

Priority From

0

to

1000

Operator

*

Source Name

*

Message

*

Value

*

Calculate Now

Config!\$B\$1

Output Cell

Sheet2!\$A\$1

Show Alarm

Select	Attributes
☒	Device Time
☒	Server Time
☒	Name
☒	Source
☒	Acknowledgement
☒	Status
☒	Type
☒	Condition Name

The configuration parameters display the result on Excel:

Parameter	Input Type																										
<i>Parameters that display results in Excel</i>																											
Calculate Now	Select (TRUE/FALSE) / Cell Address																										
Output Cell	Cell Address																										
outCode	The type of data result display. Choose from a set of inputs: <table> <tr> <td>Device Time</td><td>True/False Selection</td></tr> <tr> <td>Server Time</td><td>True/False Selection</td></tr> <tr> <td>Name</td><td>True/False Selection</td></tr> <tr> <td>Source</td><td>True/False Selection</td></tr> <tr> <td>Acknowledgement</td><td>True/False Selection</td></tr> <tr> <td>Status</td><td>True/False Selection</td></tr> <tr> <td>Type</td><td>True/False Selection</td></tr> <tr> <td>Condition Name</td><td>True/False Selection</td></tr> <tr> <td>Serverity</td><td>True/False Selection</td></tr> <tr> <td>Operator</td><td>True/False Selection</td></tr> <tr> <td>Message</td><td>True/False Selection</td></tr> <tr> <td>Value</td><td>True/False Selection</td></tr> <tr> <td>Categories</td><td>True/False Selection</td></tr> </table>	Device Time	True/False Selection	Server Time	True/False Selection	Name	True/False Selection	Source	True/False Selection	Acknowledgement	True/False Selection	Status	True/False Selection	Type	True/False Selection	Condition Name	True/False Selection	Serverity	True/False Selection	Operator	True/False Selection	Message	True/False Selection	Value	True/False Selection	Categories	True/False Selection
Device Time	True/False Selection																										
Server Time	True/False Selection																										
Name	True/False Selection																										
Source	True/False Selection																										
Acknowledgement	True/False Selection																										
Status	True/False Selection																										
Type	True/False Selection																										
Condition Name	True/False Selection																										
Serverity	True/False Selection																										
Operator	True/False Selection																										
Message	True/False Selection																										
Value	True/False Selection																										
Categories	True/False Selection																										

Reference to the meaning of the parameters displayed in [G.5.2.2 Data query results](#)

Data query parameters

Parameter	Data Type	Input type on the interface	Describe
serverType	Int	Select from the list	Server Type: 0: ATS HIS Server (old version)

			1: PI Server 2: <i>SmartHis</i> Server
address	String	Input String / Cell Address	- With SmartHIS Server: SmartHIS Server's IP/DNS address - In addition, with PI Server / ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Int	Input String / Cell Address	Data Port of HIS Server
userName	String	Input String / Cell Address	User Name
password	String	Input String / Cell Address	password
alarmType	Int	Select from the list	Alarm or event query type: 0: Alarm 1: Event
startTime endTime	String	Input String / Selection Dialog / Cell Address. -Support for excel functions DATE-TIME	startTime: The time when the data query starts. endTime: The end time of the data query. Time format by operating system. Example: 11/30/2014 12:00:00 (with the time format of the operating system mm/dd/yyyy HH:mm:ss)
fromPriority toPriority	Int	Enter a value	The Priority value of the alarm/event variable.
operatorName	String	Input String / Cell Address	Operator's name.
sourceName	String	Input String / Cell Address	The name tag (source) occurred alarm/event.
Message	String	Input String / Cell Address	Message of the alarm/event variable.

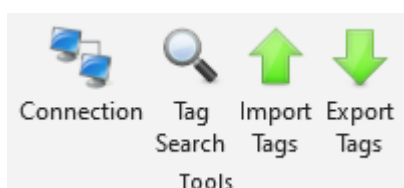
Value	String	Input String / Cell Address	The value of the alarm/event variable.
--------------	--------	--------------------------------	--

G.5.3. Add-on functions

<i>Code</i>	<i>Describe</i>
ATSMinIndex	<i>Describe</i> -Return the order of the Cell with the Min-Abs value in the selected range. Remove cells that do not contain numeric values from the calculation results. -Cell order is calculated from the 0 mark <i>Syntax</i> ATSMinIndex(range) <i>Example:</i> ATSMinIndex(S3:S7)
ATSMinValue	<i>Describe</i> -Returns the value of the Cell with the Min-Abs value in the selected range. Remove cells that do not contain numeric values from the calculation results. <i>Syntax</i> ATSMinValue(range) <i>Example:</i> ATSMinValue(S3:S7)
ATSMaxIndex	<i>Describe</i> -Return the order of the Cell with the Max-Abs value in the selected range. Remove cells that do not contain numeric values from the calculation results. -Cell order is calculated from the 0 mark <i>Syntax</i> ATSMaxIndex(range) <i>Example:</i> ATSMaxIndex(S3:S7)
ATSMaxValue	<i>Describe</i>

	-Return the value of the Cell with the Max-Abs value in 1 selected range for calculation. Remove cells that do not contain numeric values from the calculation results. Syntax ATSMaxValue(range) Example: ATSMaxValue(S3:S7)
--	--

G.5.4. Add-on tools



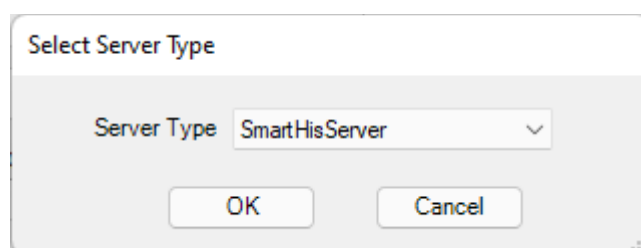
Tool	Symbol	Describe
Connection		Data Connection Test Tool to SmartHIS Server
Tag Search		<i>(The tool is only supported for OSI-PI Server)</i>
Import Tags		<i>(The tool is only supported for OSI-PI Server)</i>
Export Tag		<i>(The tool is only supported for OSI-PI Server)</i>

G.5.4.1 Connection-

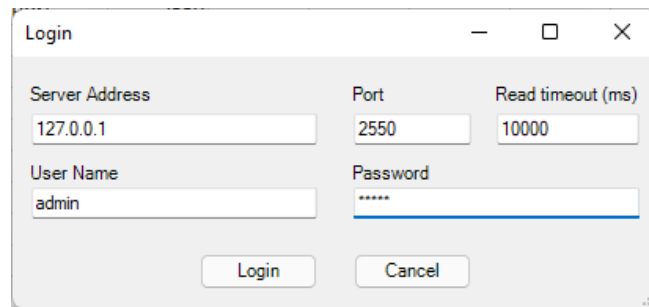


Data Connection Test Tool to HIS Server

- Click Connection  → Select Server Type: **SmartHIS Server**



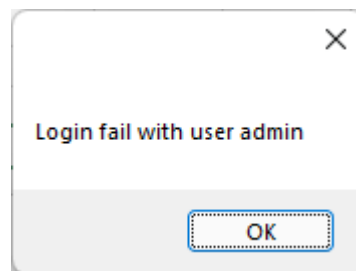
- Enter Log in *παραμετερος*®



A screenshot of a 'Login' dialog box. It contains three input fields at the top: 'Server Address' with the value '127.0.0.1', 'Port' with the value '2550', and 'Read timeout (ms)' with the value '10000'. Below these are two more input fields: 'User Name' with the value 'admin' and 'Password' with masked characters '*****'. At the bottom are two buttons: 'Login' and 'Cancel'.

Log-in Parameters Meaning Reference at [G.4.2.1.1 Viewing the List of HIS Server Connections](#)

- Test Results:
 - If Log-in Fails → Show Message



- If Log-in is successful → , hide the operation window.

G.6. SmartHIS - ATSReport Viewer

SmartHIS – ATS Report Viewer provides a tool to view reports designed on MS Excel – ATS Datalink in an operating system environment that does not have MS Excel installed. Report Viewer is applied in centralized management systems, in which 01-02 engineering computers equipped with MS Excel are responsible for designing report templates, then distributing them to operating station computers not equipped with MS Excel. The operator at the station computers only performs the task of viewing, downloading, and printing reports with the ATS Report Viewer. The work of designing, exploiting and analyzing data, and editing reports is carried out centrally at the Engineering computer by a full-time engineer.

G.6.1. ATS Report Viewer Interface

The ATS Report Viewer application works as a Desktop Application.

ATS Report Viewer interface.

ATS Report Viewer

FileToolsCommunicationsConfigurationsUser AuthorizationHelp

Root

Trung tam van hanh

D01

E1.1 Bao cao thong so van hanh phia 110kV

E1.1 Bao cao thong so van hanh phia 22kV

E1.1 Bao cao thong so van hanh phia 35kV

E1.16 Bao cao thong so van hanh phia 110kV

E1.16 Bao cao thong so van hanh phia 22kV

E1.17 Bao cao thong so van hanh phia 110kV

E1.17 Bao cao thong so van hanh phia 22kV

E1.24 Bao cao thong so van hanh phia 110kV

E1.24 Bao cao thong so van hanh phia 22kV

E1.36 Bao cao thong so van hanh phia 110kV

E1.36 Bao cao thong so van hanh phia 22kV

E1.36 Bao cao thong so van hanh phia 35kV

E1.41 Bao cao thong so van hanh phia 110kV

E1.41 Bao cao thong so van hanh phia 22kV

E1.41 Bao cao thong so van hanh phia 35kV

E1.42 Bao cao thong so van hanh phia 110kV

E1.42 Bao cao thong so van hanh phia 22kV

E1.49 Bao cao thong so van hanh phia 110kV

E1.49 Bao cao thong so van hanh phia 22kV

D02

D03

D04

D05

D06

D07

D08

Realtime Data

From:01/02/20245:48:48 PMTo:01/02/20245:48:48 PM

E1.1 Bao cao thong so van hanh phia 110kV (Daily Rep...

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	CÔNG TY LƯỚI ĐIỆN CAO THÉ TP HÀ NỘI																
2	*****																
3	E1.1 TRẠM BIẾN ÁP 110kV ĐỒNG ANH																
4	BÁO CÁO VẬN HÀNH THÔNG SỐ NGĂN LỘ 110kV																
5																	
6																	
7																	
8	Giờ	C11		C12		C19		I31				I32					
9		Uab	f	Uab	f	Uab	f	Uab	Ib	P	Q	Uab	Ib	P	Q	Uab	
10	0:00	-	-	-	-	-	-	-	85.95	16.72	0.97	-	67.92	13.74	1.65	-	
11	0:30	-	-	-	-	-	-	-	96.15	19.94	1.88	-	60.78	12.19	0.06	-	
12	1:00	-	-	-	-	-	-	-	114.71	22.90	3.05	-	60.59	12.33	-0.82	-	
13	1:30	-	-	-	-	-	-	-	80.38	15.59	0.15	-	58.30	12.01	-0.84	-	
14	2:00	-	-	-	-	-	-	-	86.85	17.84	2.54	-	57.36	11.88	-0.67	-	
15	2:30	-	-	-	-	-	-	-	75.56	15.61	3.07	-	55.45	11.35	-0.50	-	
16	3:00	-	-	-	-	-	-	-	59.39	12.84	-1.25	-	58.80	11.65	-0.34	-	
17	3:30	-	-	-	-	-	-	-	69.41	14.62	-1.85	-	57.87	11.76	-0.19	-	
18	4:00	-	-	-	-	-	-	-	73.78	15.28	-1.57	-	78.92	15.22	2.49	-	
19	4:30	-	-	-	-	-	-	-	75.06	14.97	-0.25	-	93.90	16.68	2.59	-	
20	5:00	-	-	-	-	-	-	-	78.76	16.56	0.14	-	65.60	13.43	-0.51	-	
21	5:30	-	-	-	-	-	-	-	70.13	13.85	1.32	-	72.18	14.64	-0.40	-	
22	6:00	-	-	-	-	-	-	-	83.33	16.79	-0.69	-	100.69	19.48	2.55	-	
23	6:30	-	-	-	-	-	-	-	105.59	21.04	1.47	-	102.08	20.91	0.72	-	
24	7:00	-	-	-	-	-	-	-	106.72	20.55	1.18	-	108.81	22.04	1.00	-	

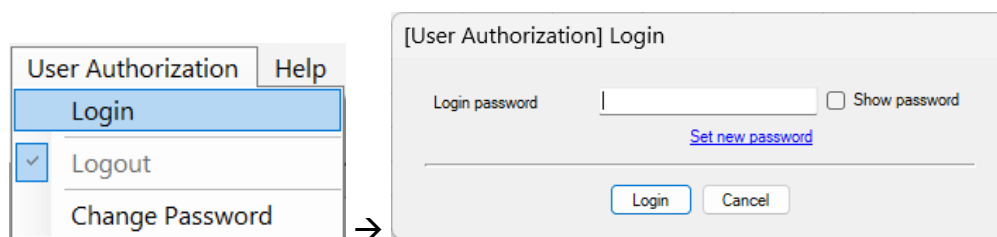
FigureG-74. Report Viewer Interface

G.6.2. Configuring the ATS Report Viewer

G.6.2.1 Build report forms

Users must log in to the system to perform configuration and report form editing tasks.

Navigate to the Menu bar → User Authorization → Login.



FigureG-75. Login ATS Report Viewer

Enter the password to log in to the system (default password: At\$Jsc_123).

To log out of the system, navigate to:

Menu → User Authorization → Logout

To change the login password, navigate to:

Menu → User Authorization → Change Password

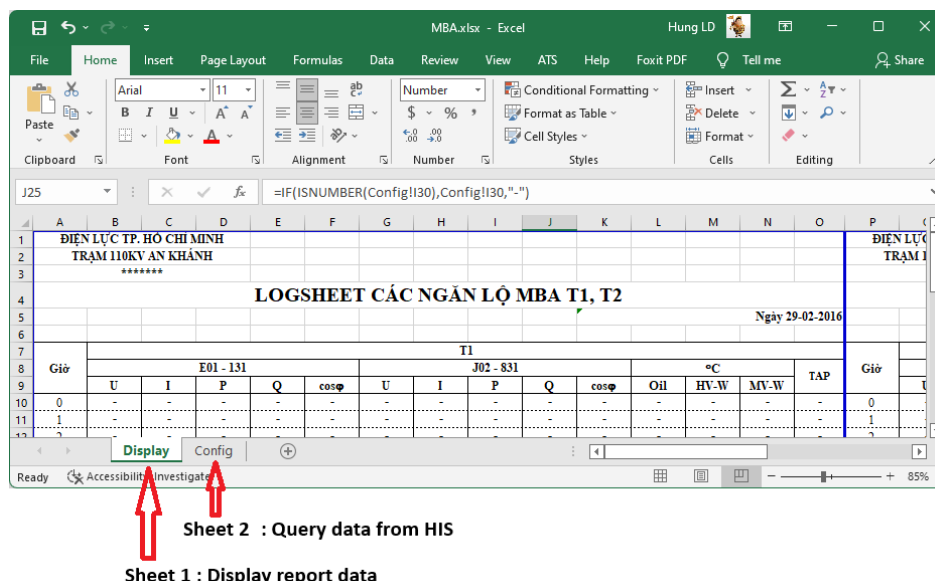
G.6.2.2 Build report forms

Report forms are built on Microsoft Excel with the ATS Data Link tool. Reference data query functions [G.5. SmartHis - ATS DataLink](#)

Design rules:

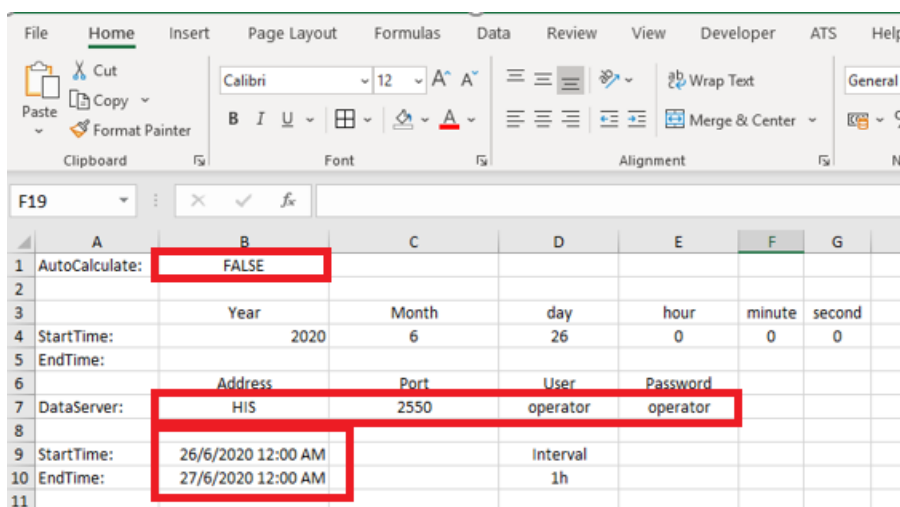
- The report template does not limit the number of sheets. However, in accordance with the following principles:
 - The interface for displaying report results is designed on the 1st sheet in the list of worksheets.
 - The data query function is designed on the second sheet in the list of worksheets.

- ATS Report Viewer will display the 1st sheet on the application interface
- ATS Report Viewer will update the data query function parameter to sheet 2, at the location of the fixed cells as described below



FigureG-76. Report Form Interface

- The Cell address contains the data query parameter for the report on the data query sheet (2nd sheet)



FigureG-77. Report Form Configuration

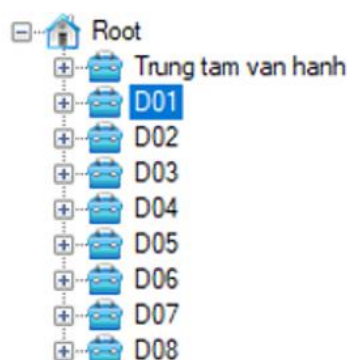
Cell Address	Parameter
B2	Calculate Now
B7	Server Address
C7	Server Port
D7	User Name
E7	Password
B9	Base Start time

Cell Address	Parameter
B10	Base Stop time

When loading data on a report, ATS Viewer automatically performs tasks

- Update cell value **B2** = TRUE
- Update the values of cells **B7, C7, D7, E7** according to the connection parameters declared on the application
- With periodic reports (Daily, Weekly, ...), update the **Start Time parameter** selected by the user from the application interface into cell **B9**.
- With interval reports, update the **Start Time parameter** selected by the user from the application interface into cell **B9** and the **Stop Time** parameter selected by the user from the application interface into cell **B10**.

G.6.2.3 Configure a report category



FigureG-78. Report category structure

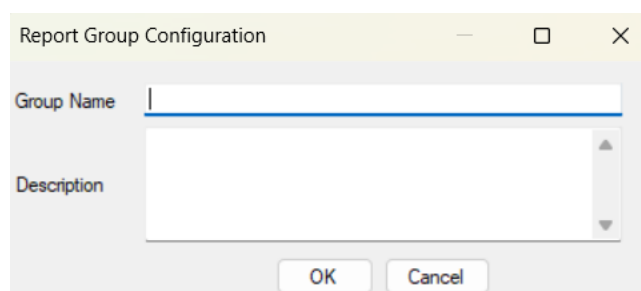
Reports are organized in the form of trees.

- You can add new, edit, or delete report groups.
- Reports can be added, edited, or deleted.

G.6.2.3.1 Add Group Report

Menu Tool → Add Group or Right-Click on Root → Add Group.

Enter the configuration parameters:



Report Group Configuration

Group Name:

Description:

OK Cancel

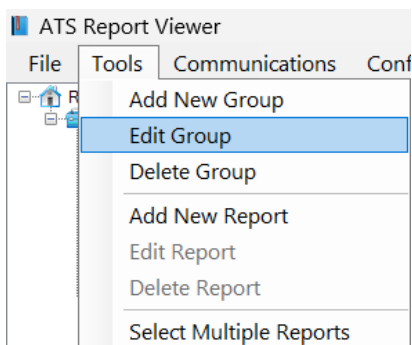
FigureG-79. Report Group Configuration

- Group Name: the name of the report group
- Description: description of the report group

Select the **File/Save** menu or press **ctrl + S** to save.

G.6.2.3.2 Edit a Group Report

Left Click Select Group Report → Menu Tool/**Edit Group** or Right-Click → **Edit Group**.



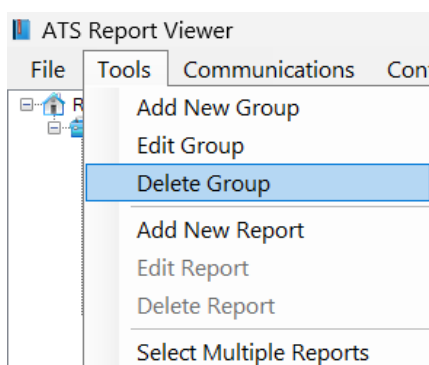
FigureG-80. Edit Report Group Menu

Enter the group configuration parameter.

Select the **File/Save** menu or press **ctrl + S** to save.

G.6.2.3.3 Delete a report group

Left Click Select Group Report → Menu Tool/**Delete Group** or Right-Click → **Delete Group**.



FigureG-81. Delete Report Group Menu

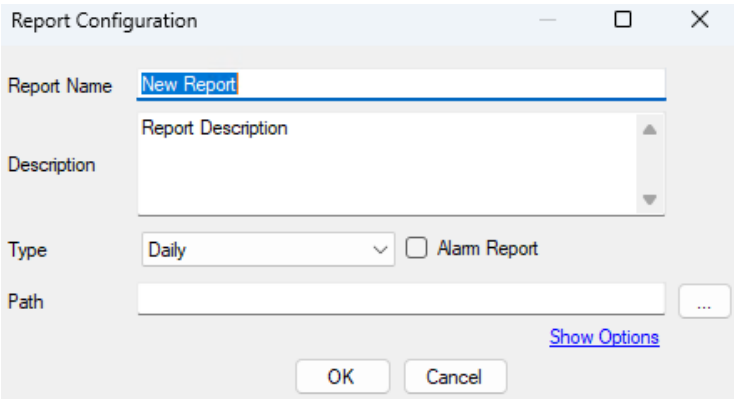
Confirm the operation.

Select the **File/Save** menu or press **Ctrl + S** to save.

G.6.2.3.4 Add a report

Menu Tool → **Add Report** or Right-Click on Group → **Add Report**.

Enter the configuration parameters:



FigureG-82. New Report Configuration

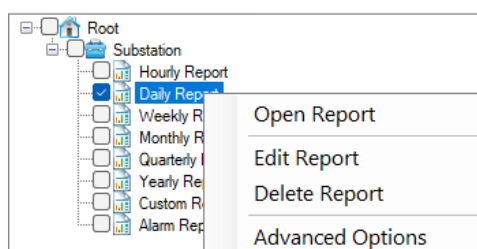
Report Name	Report Name	
Description	Report Description	
Type	Report Type	
	Hourly	Hourly reports. - Timeline: 00' 00"
	Daily	Daily reports. - Timeline: 0h 00' 00"
	Weekly	Weekly reports. - Timeline: 0h 00'00" every Monday
	Monthly	Monthly reports. - Timeline: 0h 00' 00" day 01
	Quarterly	Monthly reports. - Timeline: 0h 00'00" on the 1st day of the first month of the quarter
	Yearly	Annual report. - Timeline: 0h 00' 00" on 01/01
	Custom	Report within the optional time period (Start Time- Stop Time)

Alarm Report	➤ Checked: the report only uses data that are alarm/event signals ➤ Unchecked: the report only uses measurement data.
Path	Link to the report template file (excel file *.xlsx)
Show Options	Show Advanced Options (Refer to G.6.2.6)

Select the File/Save menu or press ctrl + S to save.

G.6.2.3.5 Edit a report configuration

Left Click to select the Report → Menu Tool/**Edit Report** or Right-Click on the → **Edit Report** report.



FigureG-83. Edit Report Configuration

Enter the report configuration parameter.

Select **the File/Save** menu or press **ctrl + S** to save.

G.6.2.3.6 Delete a report

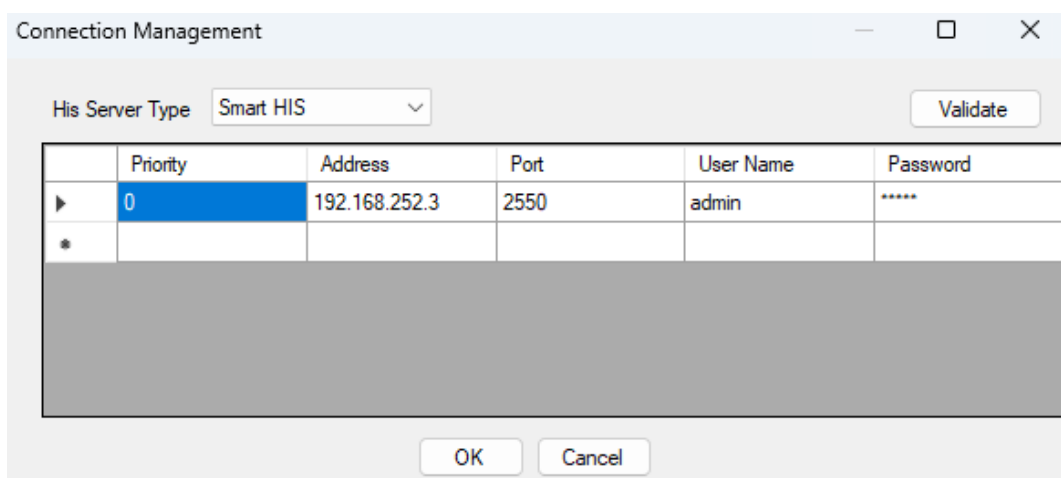
Left Click to select Menu → Report Tool/**Delete Report** or Right-Click® **Delete Report**.

Confirm the operation.

Select **the File/Save** menu or press **Ctrl + S** to save.

G.6.2.4 DataServer Configuration

From the Utilities/ **Connections** menu:



FigureG-84. List of DataServers

The HIS Server servers are arranged in the order of priority of the Priority flag from small to large. The program automatically forwards the connection to the server with the smallest Priority flag that is successfully connected

- HIS Server Type

SmartHIS	Apply with OneATS SmartHIS Server
ATS HIS	Applies to ATS HIS Server (old version)

- Connection parameters

Priority	Connection Priority
Address	▪ With SmartHIS Server: SmartHIS Server's IP/DNS address ▪ In addition, with ATS HIS Server (old version): Connection Name to Server is declared on the Admin Tool
Port	Data Port of HIS Server
User	User Name
Password	password

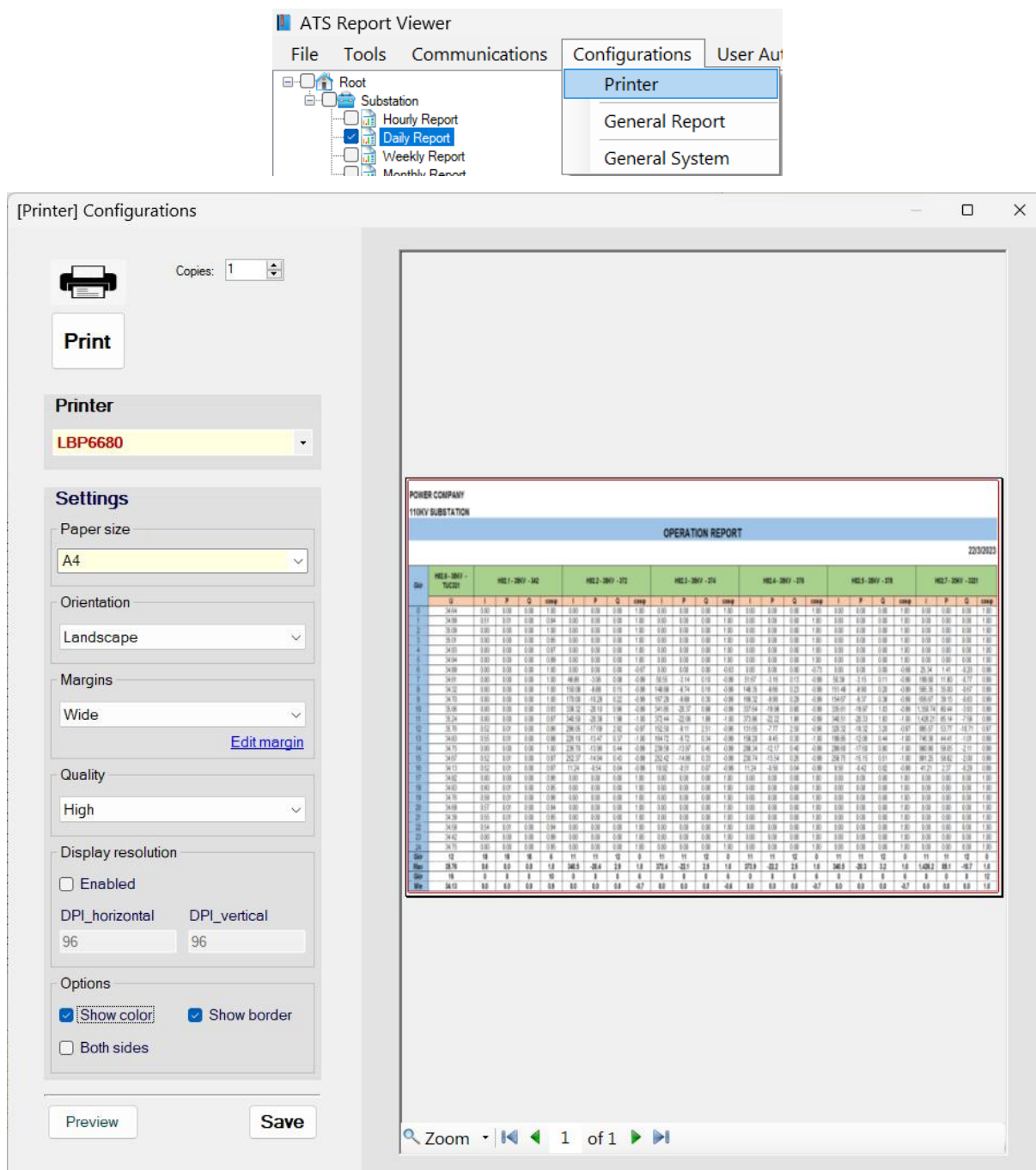
- DataServer list calibration operations
 - Enter the Server Address parameter in the bottom line (marked with *) to add a new connection to the HIS Server

- Double-Click to calibrate connection parameters
- Delete Connection:
 - Left-Click Connection Right-Click → Remove

Select the **File/Save** menu or press **ctrl + S** to save.

G.6.2.5 Configuring the connection to the printer

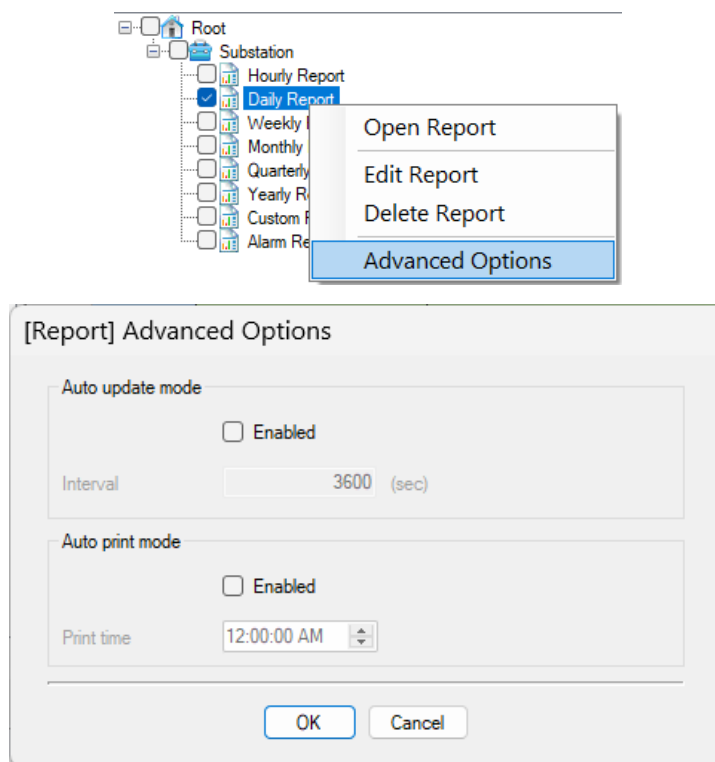
From **Configurations/Printer**:



FigureG-85. Printer Connection Configuration

G.6.2.6 Configure automatic data updates/print daily reports

Right Click on the report select **Advanced Options**



FigureG-86. Configure automatic data updates/print daily reports

Enable	Enable/Disable Features
Interval	Frequency of updating (querying) of report data - Units of seconds.
Print time	When to print the report

Select the **File/Save** menu or press **Ctrl + S** to save.

G.6.3. Report retrieval

G.6.3.1 Query Realtime Data reports

From the report category, Double Click on the report form or Right Click on the report, select **Open**:

Realtime Data

From: 01/02/2024 12:00:00 AM To: 01/03/2024 12:00:00 AM

E1.1 Báo cáo thông số vận hành phía 110kV (Daily Report)

Read

Export Pdf

Export Excel

Print

CÔNG TY LƯỚI ĐIỆN CAO THÉ TP HÀ NỘI

E1.1 TRẠM BIẾN ÁP 110kV ĐỒNG ANH

BÁO CÁO VẬN HÀNH THÔNG SỐ NGÂN LỘ 110kV

Ngày 02 tháng 01 năm 2024

Giờ	C11		C12		C19		131			132			133			171		
	Uab	f	Uab	f	Uab	f	Uab	Ib	P	Q	Uab	Ib	P	Q	Uab	Ib	P	
0:00	-	-	-	-	-	-	-	35.00	4.65	-5.86	-	36.51	7.31	-2.27	-	23.96	4.31	-1.47
0:30	-	-	-	-	-	-	-	40.77	6.05	-5.28	-	35.59	6.94	-2.24	-	22.77	4.08	-1.43
1:00	-	-	-	-	-	-	-	40.01	6.38	-5.60	-	35.33	6.69	-2.19	-	19.20	4.14	-1.39
1:30	-	-	-	-	-	-	-	37.53	4.19	-5.99	-	33.45	6.64	-2.14	-	20.66	4.20	-1.36
2:00	-	-	-	-	-	-	-	39.78	6.14	-5.55	-	35.09	6.74	-2.09	-	21.61	4.77	-1.32

FigureG-87. Realtime Data report query interface

Click **Export Pdf** to → export the report to a ***.pdf file**

Click **Export Excel** → to export the report to a ***.xlsx file**

Click **Print** → Print Report

G.6.3.2 Historical Data Report Query

From the report category, Double Click on the report form or Right Click on the report, select **Open**

Select the Historical Data query mode

Select when to query the report (From)

Click **Read** → to query the data and display it on the interface

Historical Data

Realtime Data

Historical Data

From:

12/08/2023

12:00:00 AM

Historical Data

From: 01/01/2024 12:00:00 AM To: 01/02/2024 12:00:00 AM

E1.1 Báo cáo thông số vận hành phía 110kV (Daily Report)

Read

Export Pdf

Export Excel

Print

CÔNG TY LƯỚI ĐIỆN CAO THÉ TP HÀ NỘI

E1.1 TRẠM BIẾN ÁP 110kV ĐỒNG ANH

BÁO CÁO VẬN HÀNH THÔNG SỐ NGÂN LỘ 110kV

Ngày 01 tháng 01 năm 2024

Giờ	C11		C12		C19		131			132			133			171					
	Uab	f	Uab	f	Uab	f	Uab	Ib	P	Q	Uab	Ib	P	Q	Uab	Ib	P				
0:00	-	-	-	-	-	-	-	67.14	13.72	-0.88	-	49.95	9.83	-2.03	-	50.51	6.06	-7.71	-	109.64	17.07
0:30	-	-	-	-	-	-	-	63.03	12.95	-1.04	-	63.68	13.24	1.32	-	50.24	5.65	-7.65	-	107.35	16.65
1:00	-	-	-	-	-	-	-	63.92	12.32	-0.95	-	83.26	16.97	-0.06	-	45.82	5.41	-7.59	-	107.68	16.60
1:30	-	-	-	-	-	-	-	55.96	11.65	-1.49	-	72.40	15.65	0.54	-	47.25	5.18	-8.05	-	98.00	14.32
2:00	-	-	-	-	-	-	-	60.74	13.07	-1.56	-	79.25	16.58	0.43	-	46.10	5.51	-8.13	-	96.91	12.52

FigureG-88. Historical Data report query interface

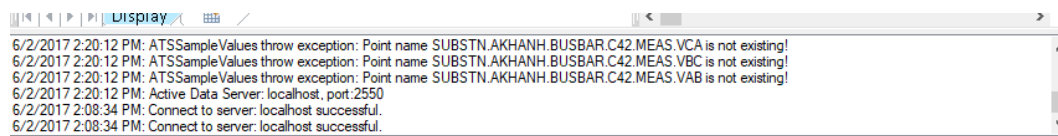
Click **Export Pdf** to → export the report to a ***.pdf file**

Click **Export Excel** → to export the report to a ***.xlsx file**

Click **Print** → Print Report

G.6.4. Log Viewer

Display application activity logs



FigureG-89. Program log viewing interface

Basic log messes

- Connection status to Server (Connection to Server AAA successful/unsuccessful, ...)
- Data query error (Point is not exist, Timeout, ...)

G.6.5. Backup/Restore ATS Report Viewer Configuration

Backup and restore SmartHIS Server configuration by copy/paste configuration files.

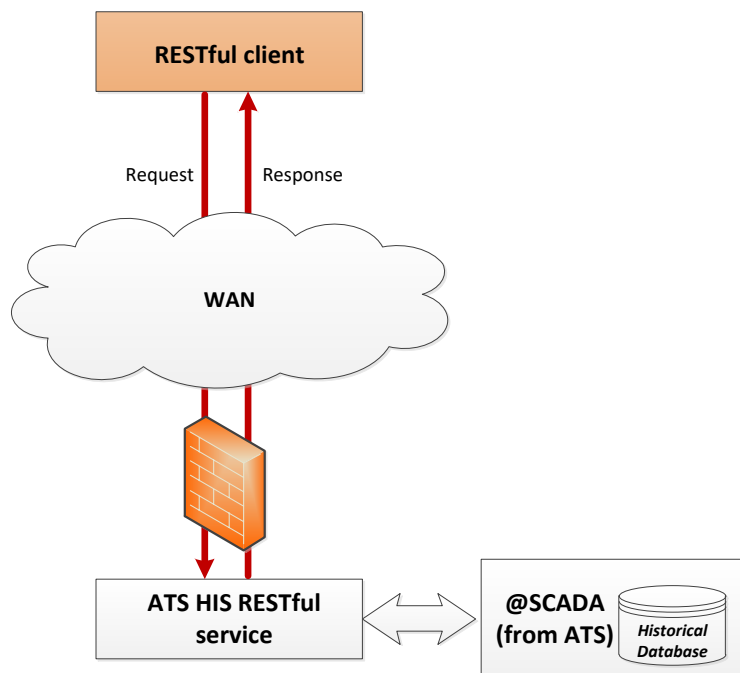
Configuration	Configuration file
Report forms (*.xlsx)	*.xlsx file folders are user-selectable. - It is recommended to archive C:\ProgramData\ATS\SmartHis\ReportViewer\Reports
Server Connections	C:\ProgramData\ATS\SmartHis\ReportViewer\Servers.xml
List of reports	C:\ProgramData\ATS\SmartHis\ReportViewer\Data.xml
General Configuration	C:\ProgramData\ATS\SmartHis\ReportViewer\AppSettings.json
Printing Configuration	C:\ProgramData\ATS\SmartHis\ReportViewer\PrinterSettings.json

G.7. SmartHIS - WEB Service/API

G.7.1. Overview

Web Service **ATS HIS RESTful** is a REST server, supporting 2 GET and POST commands in JSON format. UTF-8 data code.

ATS HIS RESTful connects to the SmartHIS Server database.



FigureG-90. Connection between ATS HIS RESTful and SmartHIS Server

G.7.2. Web Service Configuration

Summary table of configurations

Configuration	Configuration file	Tool
IIS Web Service	—	IIS Manager Tool (Windows built-in)
SmartHIS Server Connection	C:\Program Files\ATS\SmartHis\WebService\ <i>appsettings.json</i>	Text Editor (Notepad, ...)
Log folder	C:\Program Files\ATS\SmartHis\WebService\ <i>NLog.config</i>	Text Editor (Notepad, ...)

Configuration	Configuration file	Tool
https	_ Certificate	—

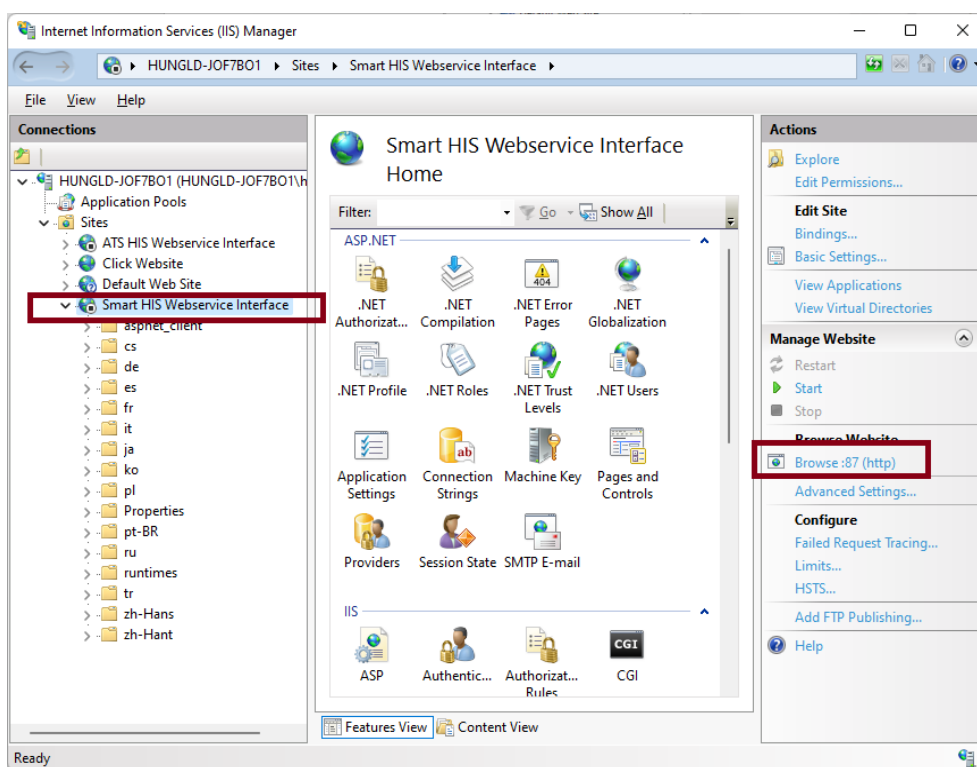
Perform Start/Stop/Restart web Service with IIS Manager

G.7.2.1 Configuring IIS Web Service

Web Service *ATS HIS RESTful* is deployed on the IIS platform as *SmartHIS Webservice Interface*.

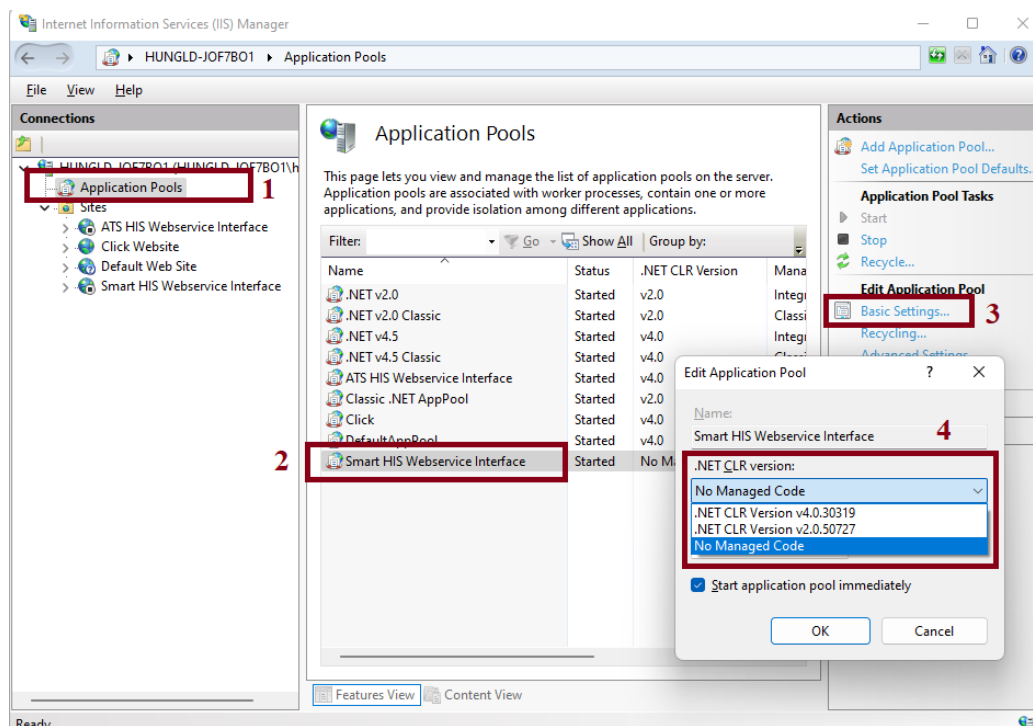
Access IIS Manager to configure Webservice

- Connection Port: (default port 87)



FigureG-91. IIS Web Service Configuration

- Other Web Service Parameters:
 - .NET CLR Version: = No Managed Code



FigureG-92. IIS Web Service Configuration

G.7.2.2 Configuring SmartHIS Server Connection

WEB Service supports 1 list of SmartHIS Servers. The WEB Service will in turn and automatically transfer the connection to the SmartHIS Server with a good connection.

Configure SmartHIS Server Connection at the configuration file

C:\Program Files\ATS\SmartHis\WebService\ **appsettings.json**



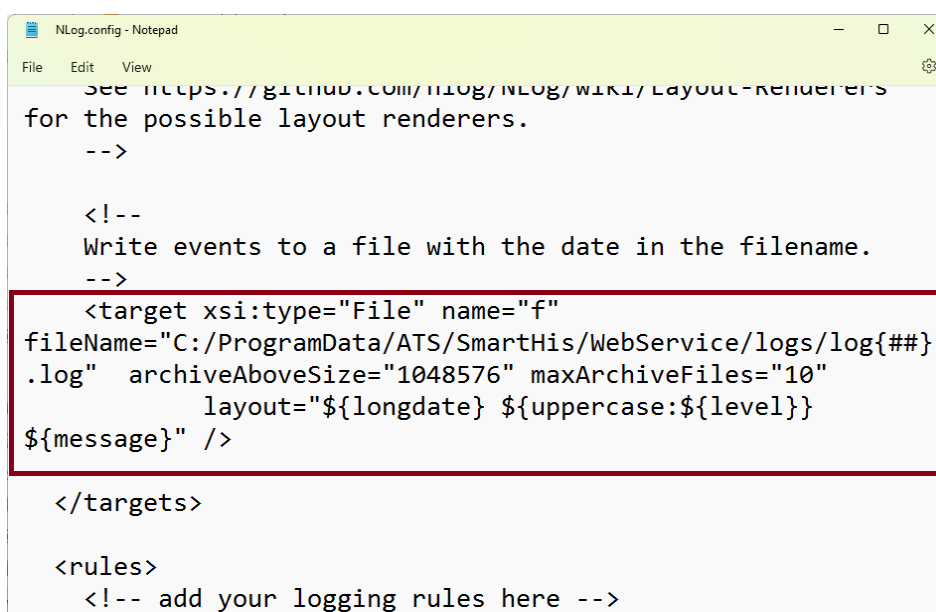
Configure the SmartHIS Server list in the SmartHisServers tab, including the following parameters:

ServerName	SmartHIS Server IP Address
Port	SmartHIS Server data port

G.7.2.3 Configuring the Log Directory for WEB Service

Configure SmartHIS Server Connection at the configuration file

C:\Program Files\ATS\SmartHis\WebService\ **NLog.config**



```

NLog.config - Notepad
File Edit View
See https://github.com/NLog/NLog/wiki/Layout-Renderers
for the possible layout renderers.
-->

<!--
Write events to a file with the date in the filename.
-->

<target xsi:type="File" name="f"
fileName="C:/ProgramData/ATS/SmartHis/WebService/logs/log{##}
.log" archiveAboveSize="1048576" maxArchiveFiles="10"
layout="${longdate} ${uppercase:${level}}
${message}" />

</targets>

<rules>
<!-- add your logging rules here -->

```

Configure the Log folder in **the target tag**, including the basic parameters:

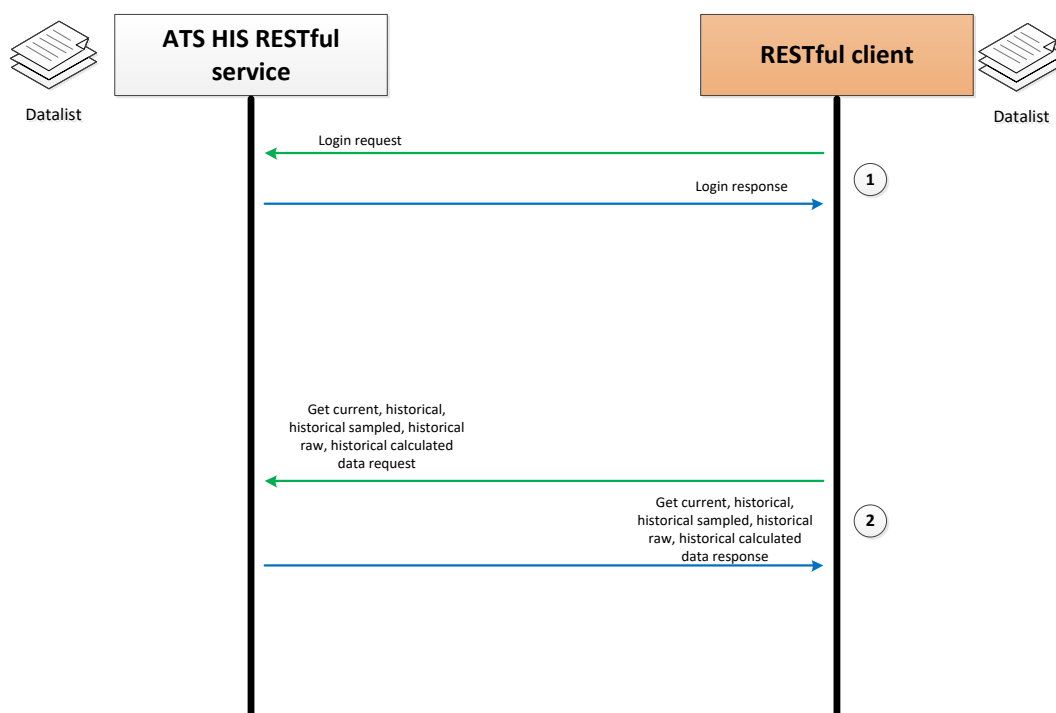
Filename	Folder, file Log.Default C:/ProgramData/ATS/SmartHis/WebService/logs/log{##}.log
archiveAboveSize	Maximum size 1 file (bytes)
maxArchiveFiles	Number of Log Files

G.7.2.4 https configuration

...

G.7.3. Data process

Data communication procedure using ATS HIS RESTful is as follows:



FigureG-93. Data Exchange Process

G.7.4. Error handling

[1] In the Login request (**1**), if username or password is incorrect, the server will respond status 400 (Bad request).

[2] In the requests (**2**), if the token is invalid or no token, the server will respond status 401 (Unauthorized). If it has an error on the server while query, this error will set into err property in the response message.

G.7.5. Data format

Web Service **ATS HIS RESTful** uses JSON format.

G.7.6. Data methods

G.7.6.1 Login method

HTTP method	POST	
URL	<i>http://[your address]/users/login</i>	
HTTP header	Content/Type: application/json	
INPUT	{ "username": "your username", "password": "your password"	*username: The name to access. *password: The password to access.

	}	
OUTPUT	<pre>{ "username": "your username", "fullname": "your full name", "enabled": true, "token": "your token id" }</pre>	*username: The name of the account is accessing. *fullname: The full name of the account is accessing. *enabled: The state of the account *token: The unique id for each session, which created by server when the RESTful client login successfully to the RESTful server. The RESTful client will use this token to query data after that.

G.7.6.2 Get current data method

HTTP method	GET/POST	
URL	<i>http://[your address]/smarthisapi/current</i>	
HTTP header	Content/Type: application/json Authorization: + type: Bearer token + token: "your token id"	
INPUT	<pre>{ "point_names": ["IVT1. MEAS. PhsA", "IVT1. MEAS. PhsB"], "expression_mode": false }</pre>	*point_names: The array of point name or expression of point name in the database of the SCADA. *expression_mode: If this value equal true, point_names is an array of expression. If this value equal false, point_names is an array of point name. *Note: Support the input is the expression (see syntax in Appendix 1)
OUTPUT	<pre>[{</pre>	*point_name: The name of the point or expression of point name.

	<pre> "point_name": "IVT1. MEAS. PhsA", "v": "123", "q": 192, "t": 1438011255230, "type":10, "err": "" }, { "point_name": "IVT1. MEAS. PhsB", "v": "456", "q": 192, "t":1438011255230, "type":10, "err": "" }] </pre>	*v: The value of tag as a string or a number depend on the data type *q: The quality of value: 4: Good 0: Bad 5:Good_Override 8: Uncertain 10: No Value *t: The timestamp, in milliseconds, that data was published to endpoint since the UNIX epoch. *err: The error of processing query data on the server, empty means no error on the server while query data.
--	--	--

G.7.6.3 Get historical data method

HTTP method	GET/POST	
URL	<i>http://[your address]/smarthisapi/historical</i>	
HTTP header	Content/Type: application/json Authorization: + type: Bearer token + token: "your token id"	
INPUT	<pre> { "timestamp": 1438011255230, "expression_mode": false, "point_names": ["IVT1. MEAS. PhsA", "IVT1. MEAS. PhsB"] } </pre>	*timestamp: The timestamp, in milliseconds, to query data since the UNIX epoch. *point_names: The array of point name or expression of point name in the database of the SCADA.

	<pre>], "retrieval_mode": 1 } </pre>	*expression_mode: If this value equal true, point_names is an array of expression. If this value equal false, point_names is an array of point name. *retrieval_mode: The method to retrieval historical data: 1: ExactTime 2: Interpolated 3: Previous 4: Next 5: Nearest *Note: Support the input is the expression (see syntax in Appendix 1)
OUTPUT	<pre> [{ "point_name": "IVT1. MEAS. PhsA", "v": "123", "q": 192, "t": 1438011255230, "type": 10, "err": "" }, { "point_name": "IVT1. MEAS. PhsB", "v": "456", "q": 192, "t": 1438011255230, "type": 10, "err": "" }] </pre>	*point_name: The name of the point or expression of point name. *v: The value of tag as a string or a number depend on the data type *q: The quality of value: 4: Good 0: Bad 5: Good_Override 8: Uncertain 10: No Value *t: The timestamp, in milliseconds, that data was published to endpoint since the UNIX epoch. *type: The data type of value: 0: Boolean, 1: Digital

	]	2: Int8 3: Int16 4: Int32 5: Int64 6: UInt8 7: UInt16, 8: UInt32 9: UInt64 10: Float 11: Double 12: String 13: DateTime *err: The error of processing query data on the server, empty means no error on the server while query data.
--	---	--

G.7.6.4 Get historical sampled data method

HTTP method	GET/POST	
URL	<i>http://[your address]/smarthisapi/historical/sampled</i>	
HTTP header	Content/Type: application/json Authorization: + type: Bearer token + token: "your token id"	
INPUT	<pre>{ "start_time": 1438011255230, "end_time": 1438011355230, "interval": "1h", "expression_mode": false, "point_name": "IVT1. MEAS. PhsA " "retrieval_mode": 1 }</pre>	*start_time: The start time, in milliseconds, to query data since the UNIX epoch. *end_time: The end, in milliseconds, to query data since the UNIX epoch. *interval: The interval time with format: [xxx][yyy] +xxx: The number +yyy: The character: s(second), m(minute), h(hour)

		+examples: 10s, 30m, 1h *point_name: The name of the point or expression of point name. *expression_mode: If this value equal true, point_names is an array of expression. If this value equal false, point_names is an array of point name. *retrieval_mode: The method to retrieval historical data: 1: ExactTime 2: Interpolated 3: Previous 4: Next 5: Nearest *Note: Support the input is the expression (see syntax in Appendix 1)
OUTPUT	<pre>[{ "point_name": "IVT1. MEAS. PhsA", "v": "123", "q": 192, "t": 1438011255230, "type":10, "err": "" }, { "point_name": "IVT1. MEAS. PhsA", "v": "456", "q": 192,</pre>	*point_name: The name of the point or expression of point name. *v: The value of tag as a string or a number depend on the data type *q: The quality of value: 4: Good 0: Bad 5:Good_Override 8: Uncertain 10: No Value *t: The timestamp, in milliseconds, that data was published to endpoint since the UNIX epoch.

	<pre> "t":1438011355230, "type":10, "err":"" }] </pre>	*type: The data type of value: 0: Boolean, 1: Digital 2: Int8 3: Int16 4: Int32 5: Int64 6: UInt8 7: UInt16, 8: UInt32 9: UInt64 10: Float 11: Double 12: String 13: DateTime *err: The error of processing query data on the server, empty means no error on the server while query data.
--	---	--

G.7.6.5 Get historical raw data method

HTTP method	GET/POST	
URL	<i>http://[your address]/smarthisapi/historical/raw</i>	
HTTP header	Content-Type: application/json Authorization: + type: Bearer token + token: "your token id"	
INPUT	<pre> { "start_time": 1438011255230, "end_time": 1438011355230, "point_name": "IVT1. MEAS. PhsA" } </pre>	*start_time: The start time, in milliseconds, to query data since the UNIX epoch. *end_time: The end, in milliseconds, to query data since the UNIX epoch. *point_name: The name of the point.

		*Note: DON'T support the input is the expression.
OUTPUT	<pre>[{ "point_name": "IVT1. MEAS. PhsA", "v": "123", "q": 192, "t": 1438011255230, "type":10, "err": "" }, { "point_name": "IVT1. MEAS. PhsA", "v": "456", "q": 192, "t":1438011355230, "type":10, "err": "" }]</pre>	*point_name: The name of the point *v: The value of tag as a string or a number depend on the data type *q: The quality of value: 4: Good 0: Bad 5:Good_Override 8: Uncertain 10: No Value *t: The timestamp, in milliseconds, that data was published to endpoint since the UNIX epoch. *type: The data type of value: 0: Boolean, 1: Digital 2: Int8 3: Int16 4: Int32 5: Int64 6: UInt8 7: UInt16, 8: UInt32 9: UInt64 10: Float 11: Double 12: String 13: DateTime

		*err: The error of processing query data on the server, empty means no error on the server while query data.
--	--	---

G.7.6.6 Get historical calculated data method

HTTP method	GET/POST	
URL	<i>http://[your address]/smarthisapi/historical/calculated</i>	
HTTP header	Content/Type: application/json Authorization: + type: Bearer token + token: "your token id"	
INPUT	<pre>{ "start_time": 1438011255230, "end_time": 1438011255230, "interval": "1h", "calculate_mode":0, "query_mode":0, "query_sample": "1m", "point_name": "IVT1. MEAS. PhsA" }</pre>	*start_time: The start time, in milliseconds, to query data since the UNIX epoch. *end_time: The end, in milliseconds, to query data since the UNIX epoch. *interval: The interval time with format: [xxx][yyy] +xxx: The number +yyy: The character: s(second), m(minute), h(hour) +examples: 10s, 30m, 1h *calculate_mode: The calculate mode, include 0: Total 1: Minimum 2: Maximum 3: Average 4: Minimum_Abs 5:Maximum_Abs 6: Average_Abs *query_mode: The query mode of data to calculate, include 0: Raw data

		1: Sampled data *query_sample: The interval time to query sample data, it was used when query_mode is Sampled data. *point_name: The name of the point. *Note: If the expression contains more a point name, must use query_mode=1.
OUTPUT	<pre>[{ "point_name": "IVT1. MEAS. PhsA", "v": "123", "q": 192, "t": 1438011255230, "type": 10, "err": "" }, { "point_name": "IVT1. MEAS. PhsB", "v": "456", "q": 192, "t": 1438011255230, "type": 10, "err": "" }]</pre>	*point_name: The name of the point *v: The value of tag as a string or a number depend on the data type *q: The quality of value, true means good quality, false means bad quality *t: The timestamp, in milliseconds, that data was published to endpoint since the UNIX epoch. *type: The data type of value: 0: Boolean, 1: Digital 2: Int8 3: Int16 4: Int32 5: Int64 6: UInt8 7: UInt16, 8: UInt32 9: UInt64 10: Float 11: Double

		12: String 13: DateTime *err: The error of processing query data on the server, empty means no error on the server while query data.
--	--	---

G.7.7. Test Web Service

You can use the Postman tool (<https://www.postman.com/>) to test WEB Services. Refer to the tool documentation from the manufacturer.

G.8. SmartHIS - ODBC Driver

G.8.1. ODBC Driver Overview

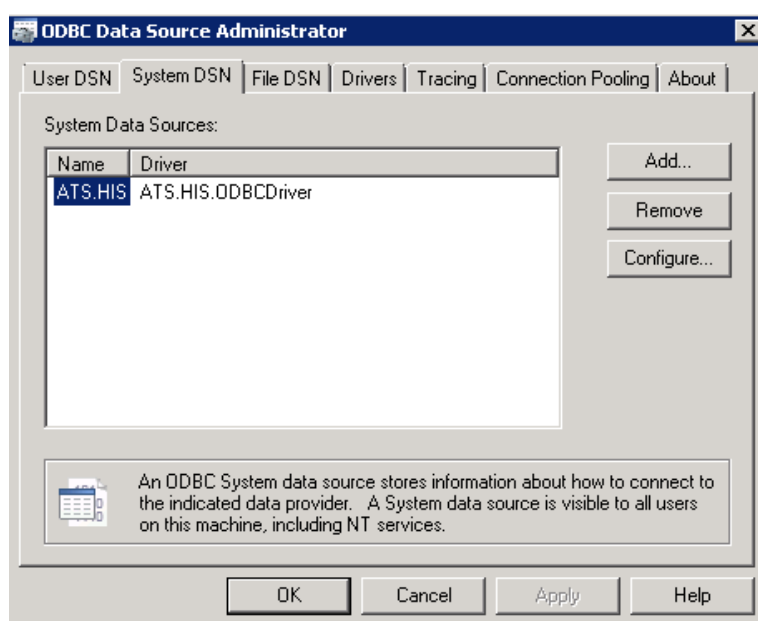
An ODBC Driver (32-bit) on Windows Operating System provides tools to access historical data from SmartHIS Server, with using SQL statements.

SmartHIS ODBC Driver helps users use standard communication easily to exploit historical data.

SmartHIS ODBC Driver can be installed on any computer on your LAN or WAN can connect to SmartHIS Server.

G.8.2. ODBC Driver Configuration

After installing SmartHIS ODBC Driver, driver name will appear in list table of Windows's ODBC Driver.



FigureG-94: SmartHIS ODBC Driver in Windows

G.8.3. Data query functions

G.8.3.1 List of Data Table

Data table is provided in SmartHIS ODBC Driver includes:

Points: Contains a list of all available data in SmartHIS Server connecting.

Snapshot: Contains Snapshot data of all data points.

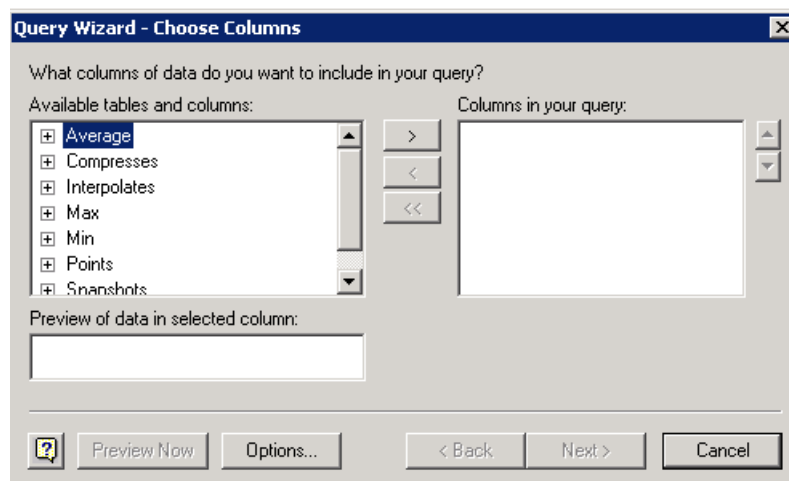
Interpolates: Contains interpolation data of data points.

Compresses: Contains stored data of data points.

Max: Contains maximum data of data points.

Min: Contains minimum data of data points.

Average: Contains average data of data points.



FigureG-95: Data tables in SmartHIS ODBC Driver

G.8.3.2 Points Table

Table contains a list of all data points in SmartHIS Server connecting, includes properties of corresponding data points to columns in table:

TagName, Description, Archiving, ExcDev, ExcMin, ExcMax, CompDev, commin, CompMax, PointType, DigitalSet, PointClass, PointAccess, PointUser, PointGroup, DataAccess, DataUser, Datagroup, DataSource, SourceTag, Expression, CalculateMode, Scheduling, StartTime, Interval, Creator, CreationTime, Changer, LastChangeTime.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' or 'like' (eg: TagName = 'sine' or TagName like 'si%').

PointType: Use operator '=' (eg: PointType = 'Float').

PointClass: Use operator '=' (eg: PointClass = 'Classic').

DataSource: Use operator '=' (eg: DataSource = 'OPC').

G.8.3.3 Snapshots Table

Table contains snapshot data of all data points, includes corresponding properties to columns in table:

TagName, DataSource, DataType, Value, Timestamp, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' or 'like' (eg: TagName = 'sine' or TagName like 'si%').

DataSource: Use operator '=' (eg: DataSource = 'OPC').

DataType: Use operator '=' (eg: DataType = 'Float').

G.8.3.4 Interpolates Table

Table contains interpolates data of data points, includes corresponding properties to columns in table:

TagName, Value, Timestamp, Interval, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' (eg: TagName = 'sine').

Timestamp: Use operators '>' or '<' (eg: TimeStamp> '7/15/2016').

Interval: Use operator '=' (eg: Interval = '1h').

G.8.3.5 Compresses Table

Table contains compresses data of data points, includes corresponding properties to columns in table:

TagName, Value, Timestamp, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' (eg: TagName = 'sine').

Timestamp: Use operators '>' or '<' (eg: TimeStamp> '7/15/2016').

G.8.3.6 Max Table

Table contains maximum data of data points, includes corresponding properties to columns in table:

TagName, Value, Timestamp, Interval, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' (eg: TagName = 'sine').

Timestamp: Use operators '>' or '<' (eg: TimeStamp> '7/15/2016').

Interval: Use operator '=' (eg: Interval = '1h').

G.8.3.7 Min Table

Table contains minimum data of data points, includes corresponding properties to columns in table:

TagName, Value, Timestamp, Interval, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' (eg: TagName = 'sine').

Timestamp: Use operators '>' or '<' (eg: TimeStamp> '7/15/2016').

Interval: Use operator '=' (eg: Interval = '1h').

G.8.3.8 Average Table

Table contains average data of data points, includes corresponding properties to columns in table:

TagName, Value, Timestamp, Interval, Quality.

Users can filter (using Where statement) on columns with following operators:

TagName: Use operator '=' (eg: TagName = 'sine').

Timestamp: Use operators '>' or '<' (eg: TimeStamp> '7/15/2016').

Interval: Use operator '=' (eg: Interval = '1h').

G.8.4. Test ODBC Driver

G.9. SmartHIS – Data Client Interface

Interface for connecting data to ATS DataServer software (old generation software version @Station 2.0)

G.10. SmartHIS – @Click Web Server (Web Portal)

@Click Web Server is an application that allows users to retrieve data in HIS Server through Web services.

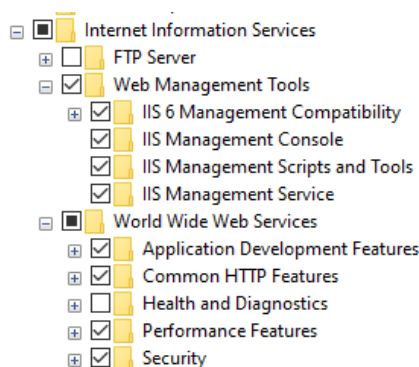
G.10.1. Install the app

Installation Steps:

- Installing Internet Information Service (IIS)
- SQL Settings: Where to Store and Manage Databases
- @Click: App Interface

G.10.1.1 Installing Internet Information Service (IIS)

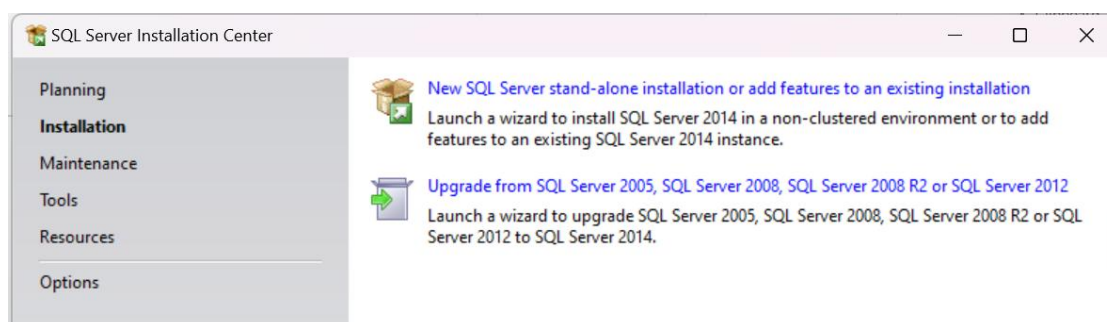
Enable IIS features, go to **Control Panel > Programs and Features > Turn Windows features on or off > Internet Information Services**, check **Web Management Tools** and **World Wide Web Services**.



FigureG-96: Enable Internet Information Services

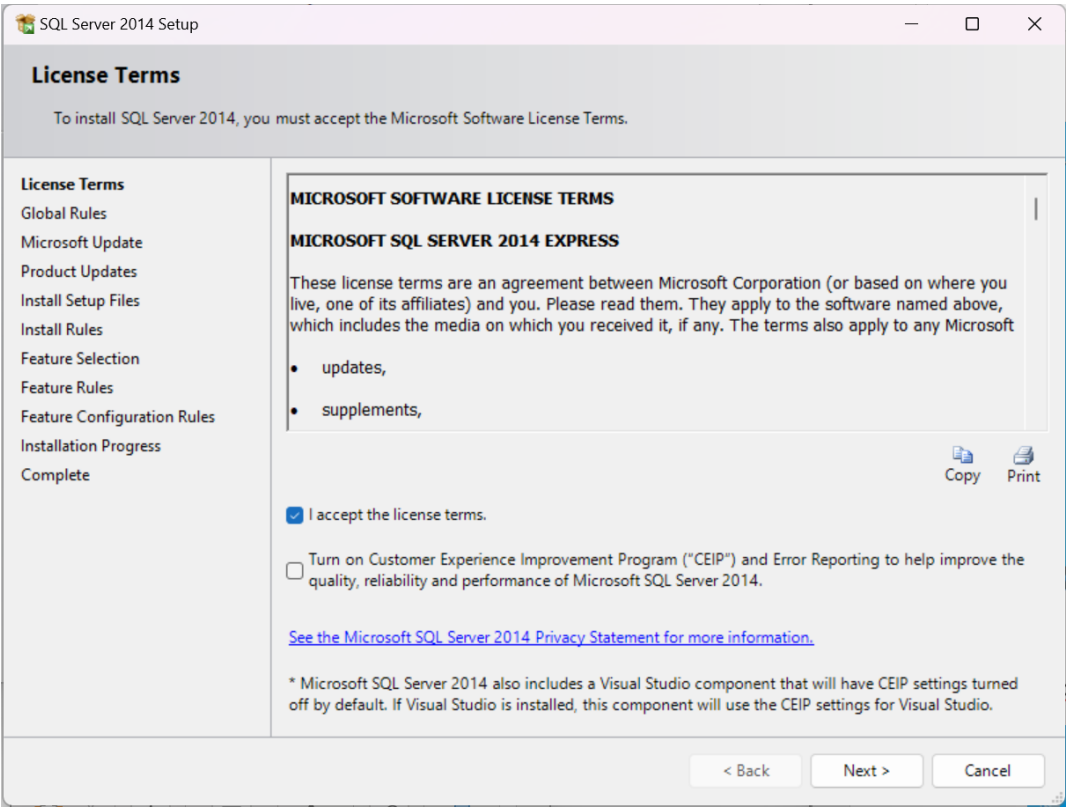
G.10.1.2 Install SQL2014

Run the SQL2014 installation file with Administrator permissions, select **New SQL Server stand-alone installation or add features to an existing installation**



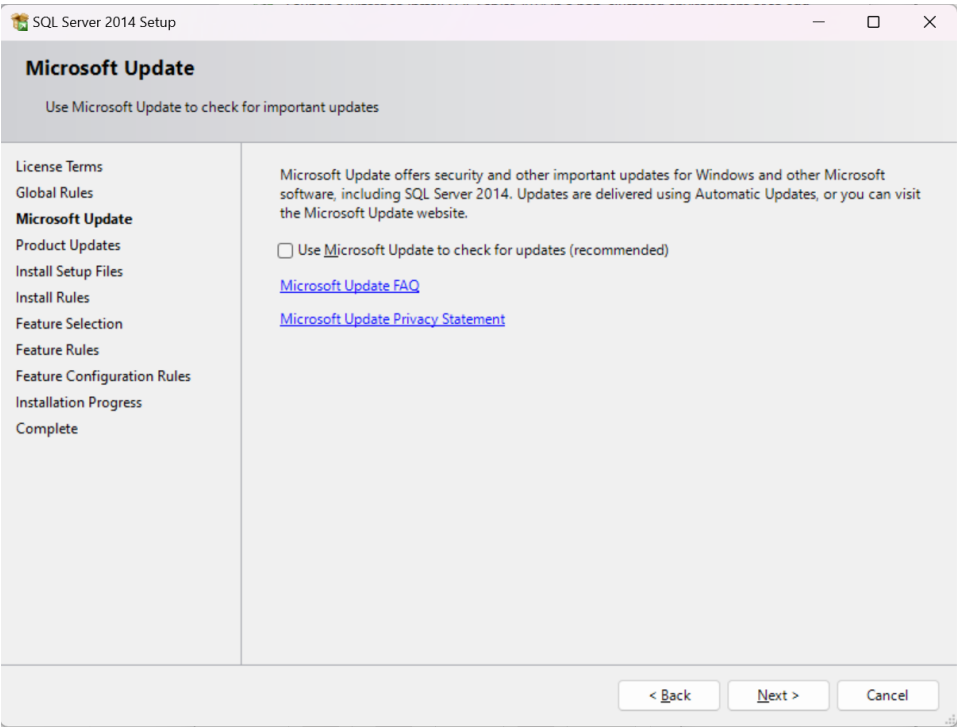
FigureG-97: New SQL or Upgrade SQL option interface

Click I accept the license, select **Next** to continue



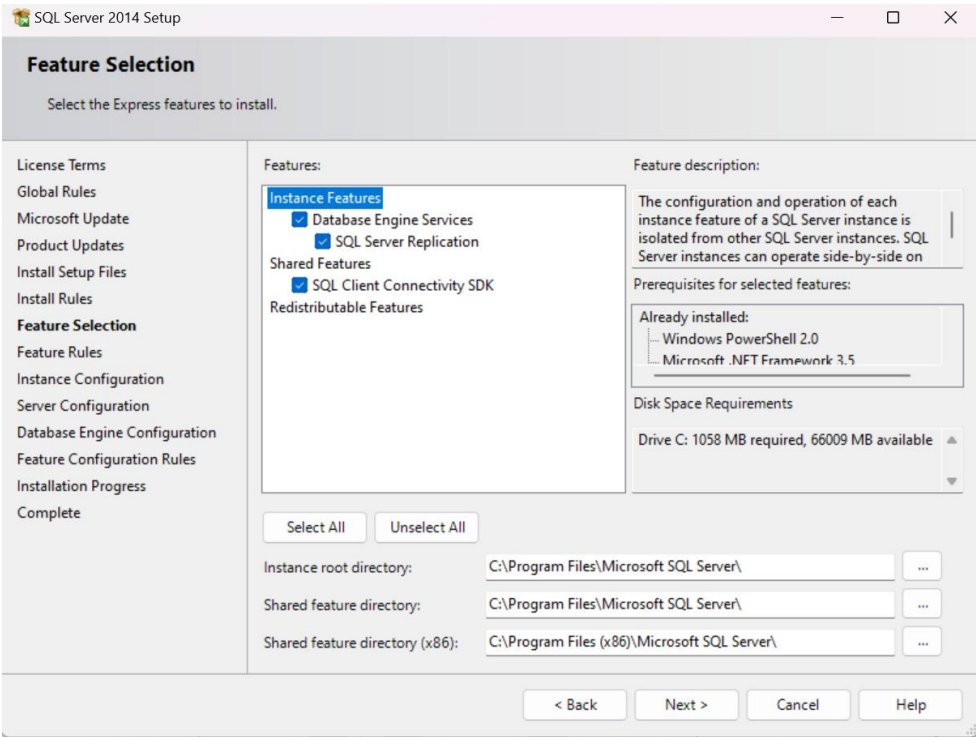
FigureG-98: License Terms Interface

Select **Use Microsoft Update** to check for updates, select **Next** to continue



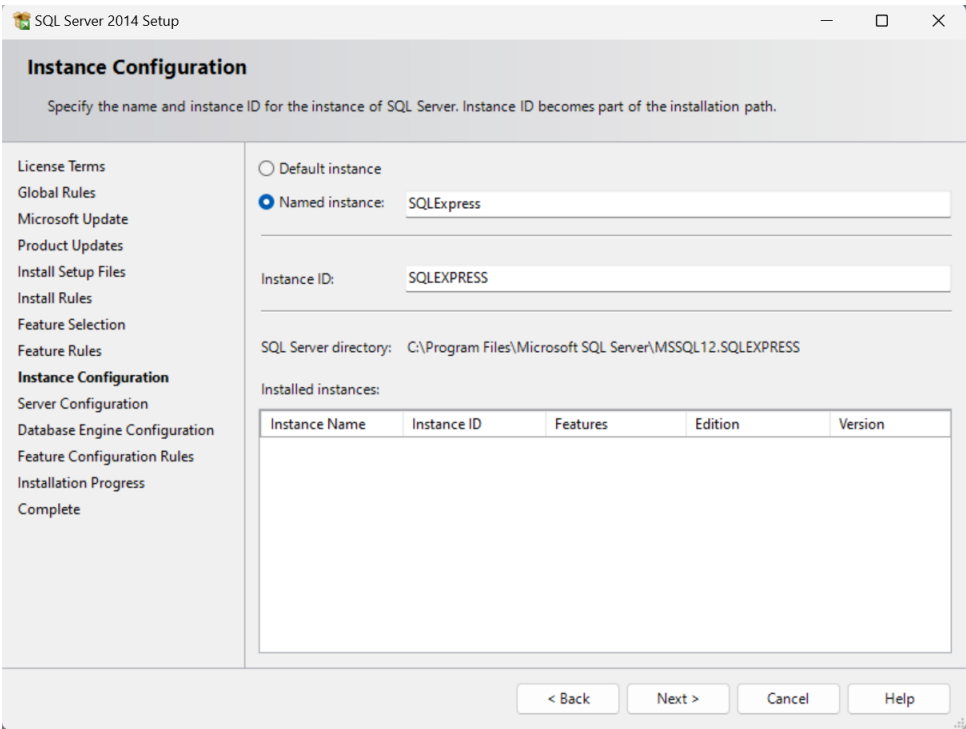
FigureG-99: Microsoft Update Configuration Interface

Select **Select All** to install all Features, select **Next** to continue



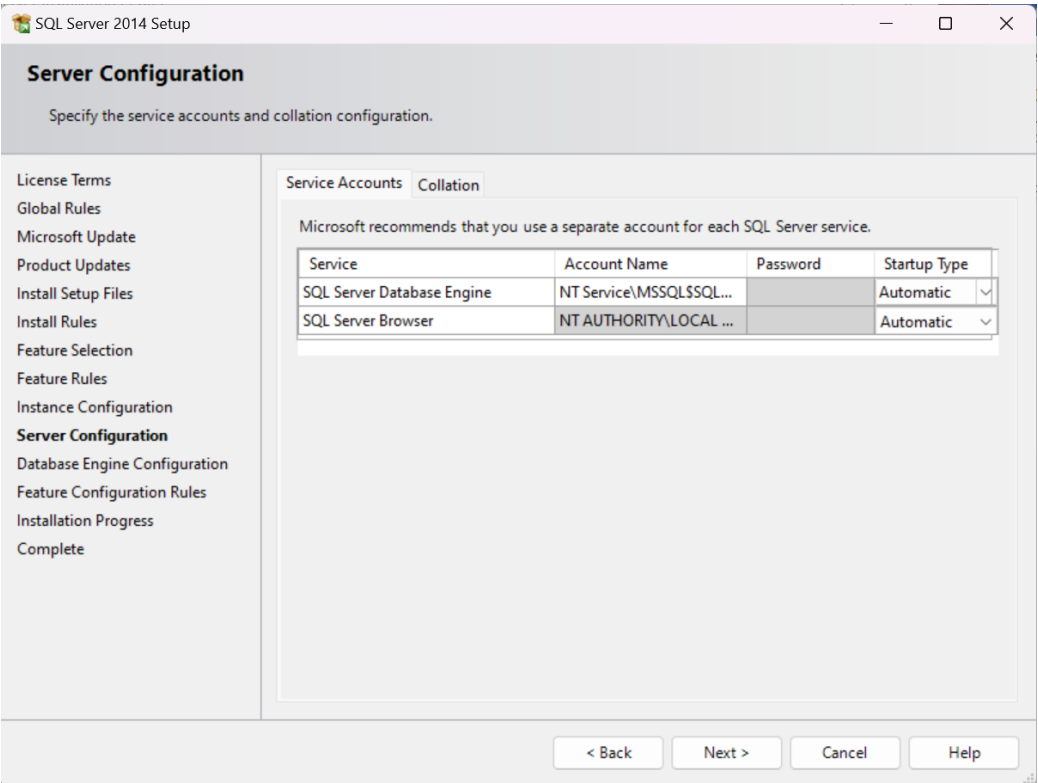
FigureG-100: Optional Interface Features

Configure the **Instance name** as shown, select **Next** to continue.



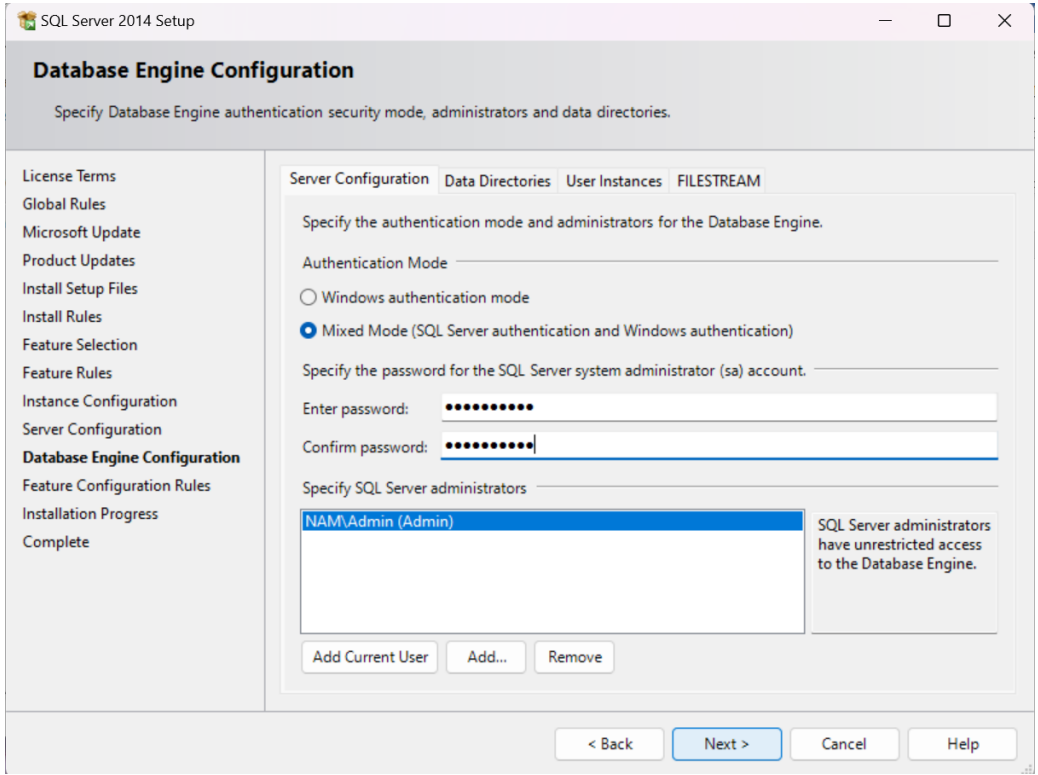
FigureG-101: Instance Configuration Interface

In **Startup Type**, select **Automatic**, select **Next** to continue.



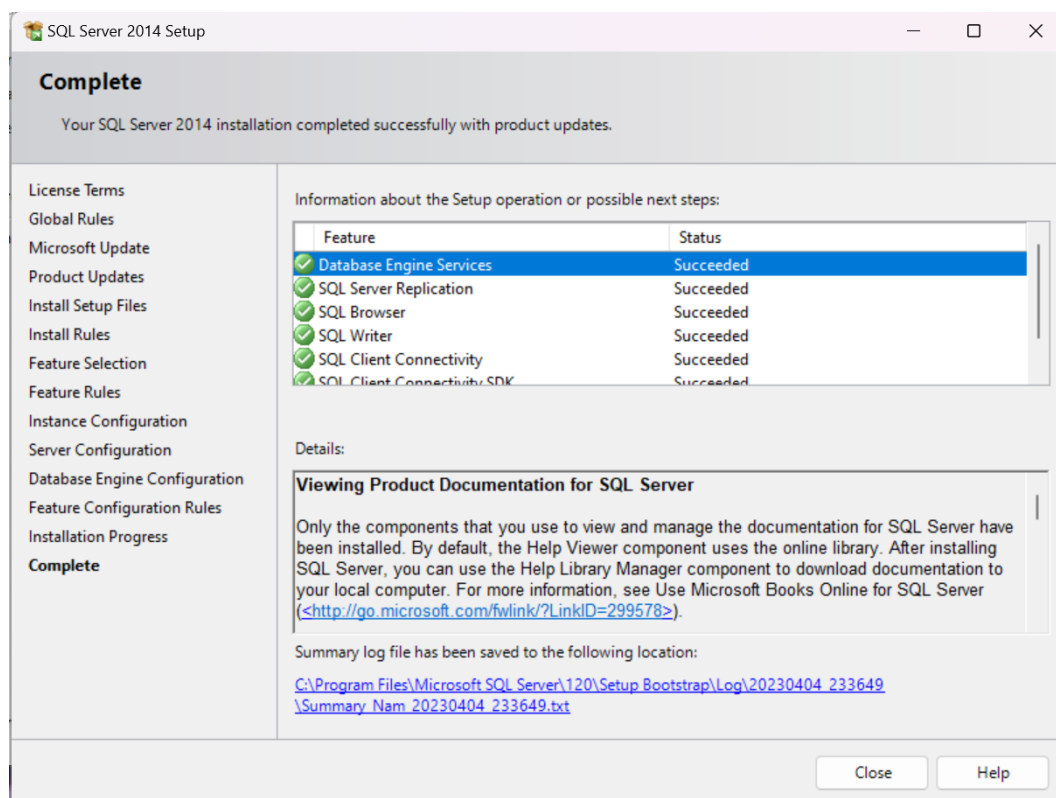
FigureG-102: Server Configuration Interface

Set the password at the user's discretion, select **Next** to continue



FigureG-103: Database Engine Configuration Interface

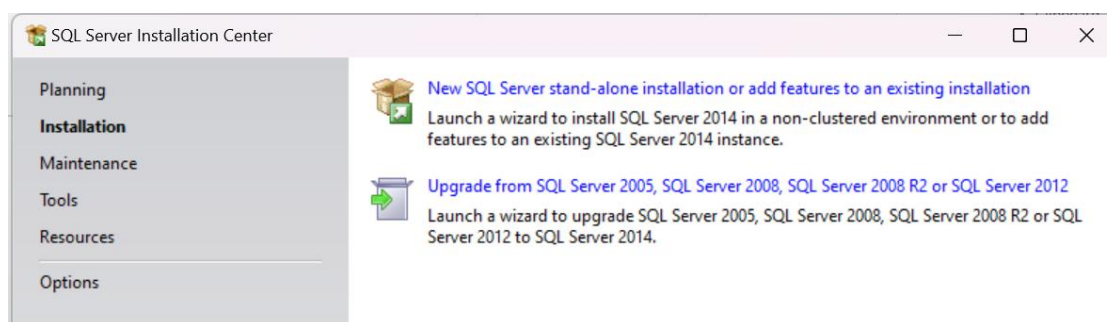
Select **Close** to finish the installation process



FigureG-104: Interface ends the installation process

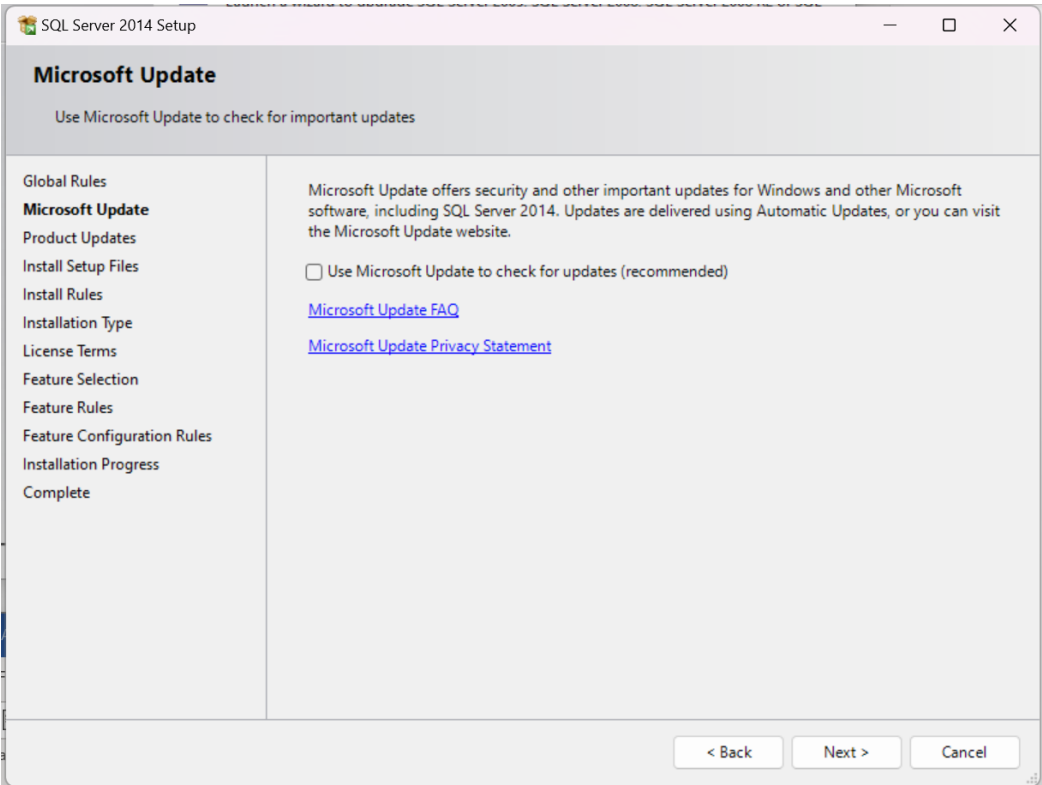
G.10.1.3 Install SQL Management Studio

Run the SQL Management Studio installation file with Administrator permissions, select **New SQL Server stand-alone installation** or add features to an existing installation



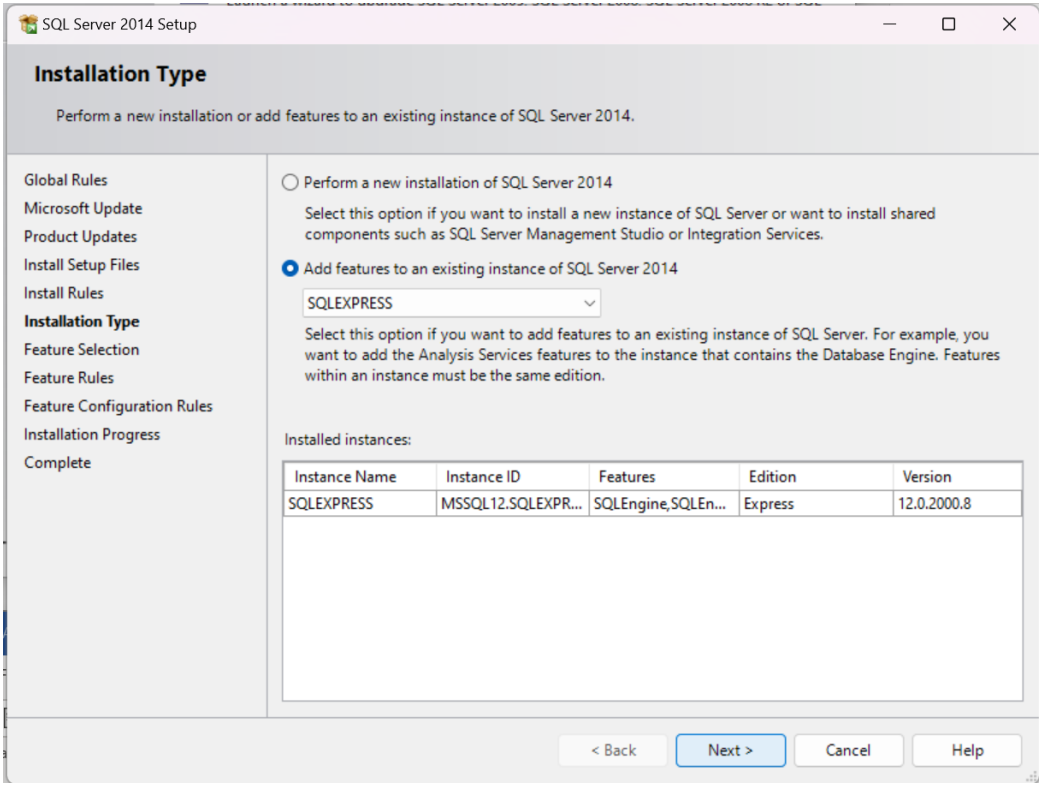
FigureG-105: New SQL or Upgrade SQL option interface

Select **Use Microsoft Update to check for updates**, select **Next** to continue



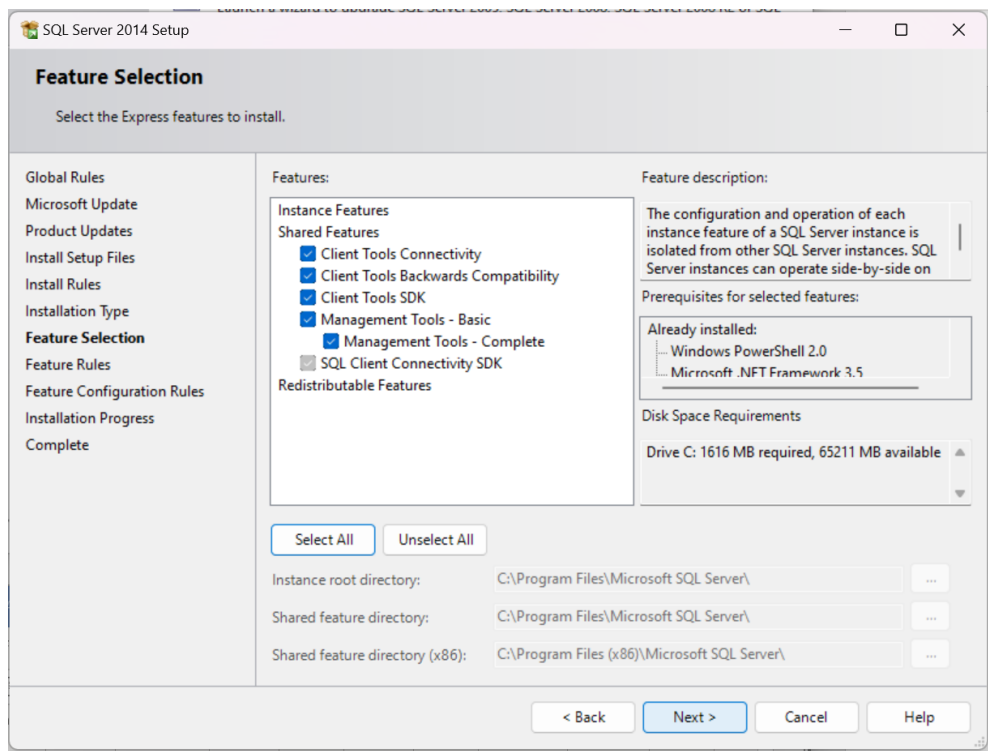
FigureG-106: Microsoft Update Configuration Interface

Configure the **Installation Type** as shown, select **Next** to continue



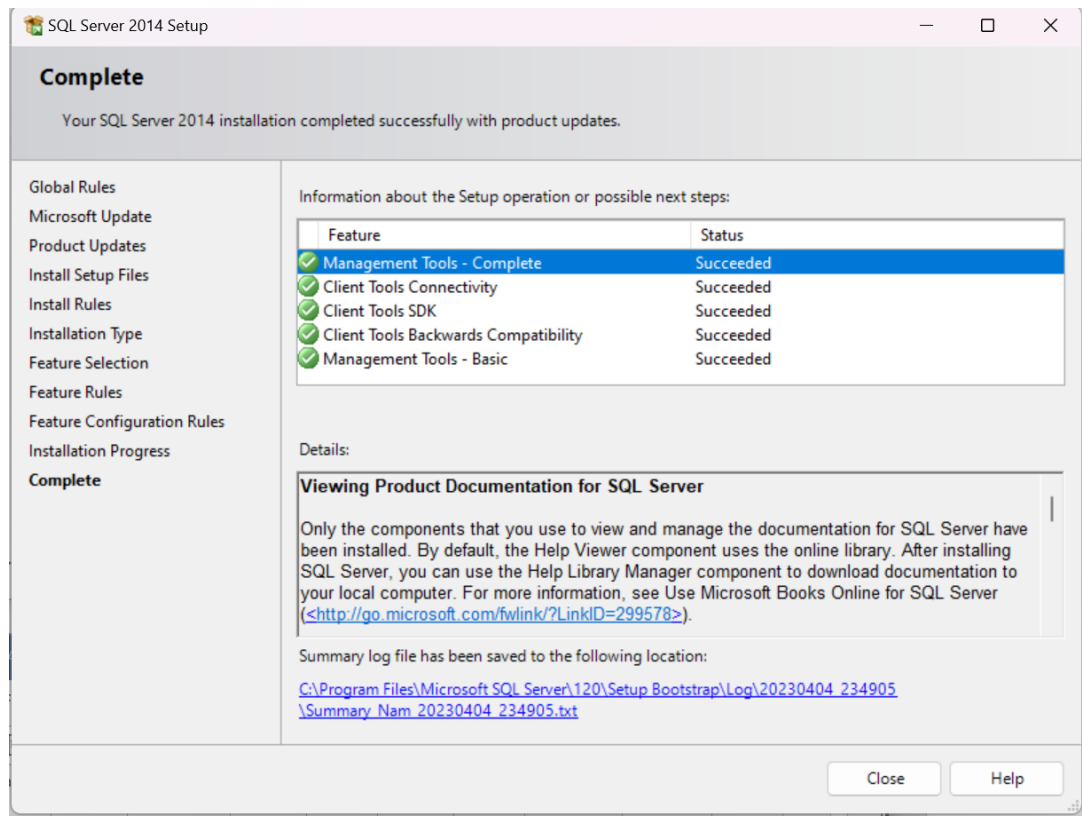
FigureG-107: Installation Type Configuration Interface

Select **Select All** to install all Features, then select **Next** to continue



FigureG-108: Optional Interface Features

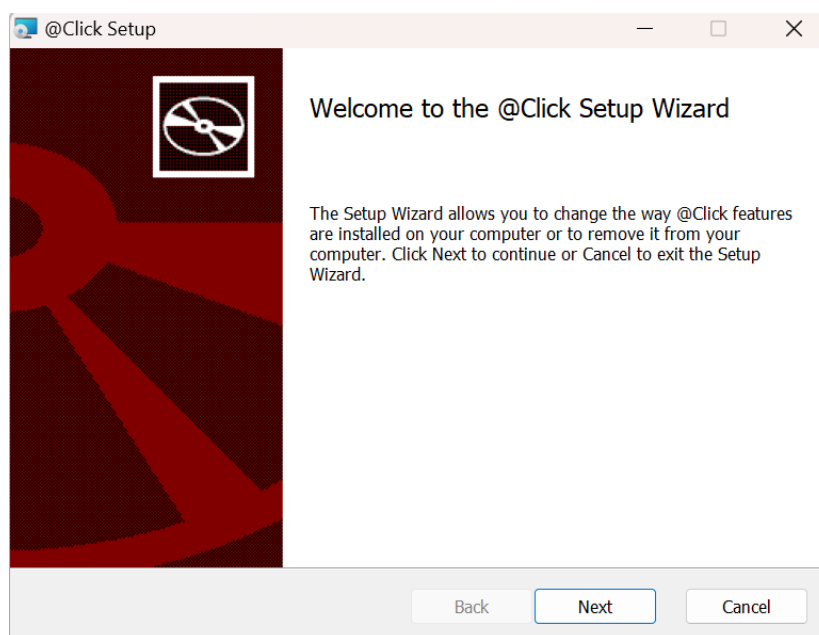
Select **Close** to finish the installation process



FigureG-109: Interface ends the installation process

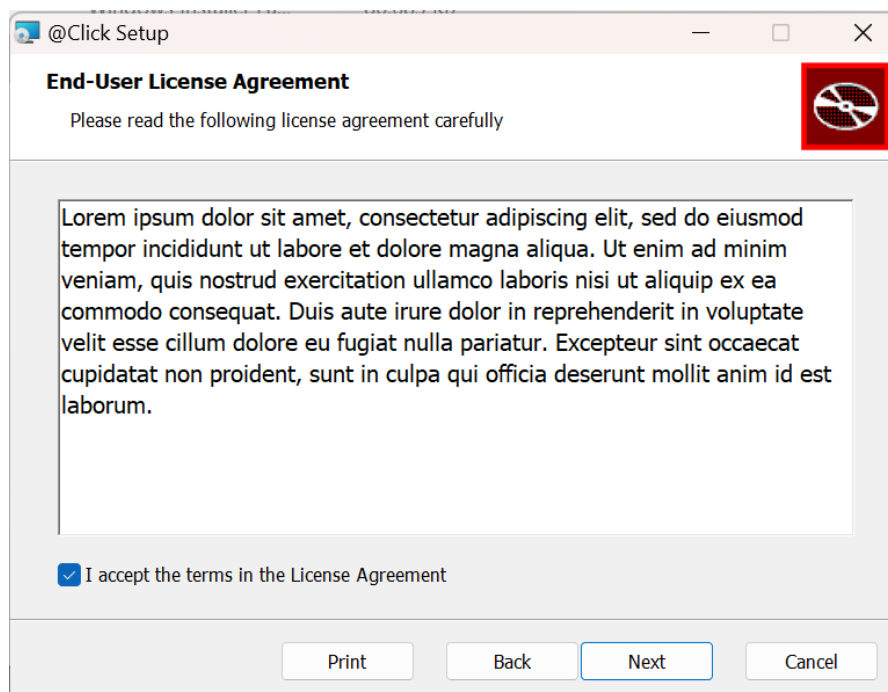
G.10.1.4 Install @Click

Run the installation file **Click Setup.msi**, then select **Next** to continue



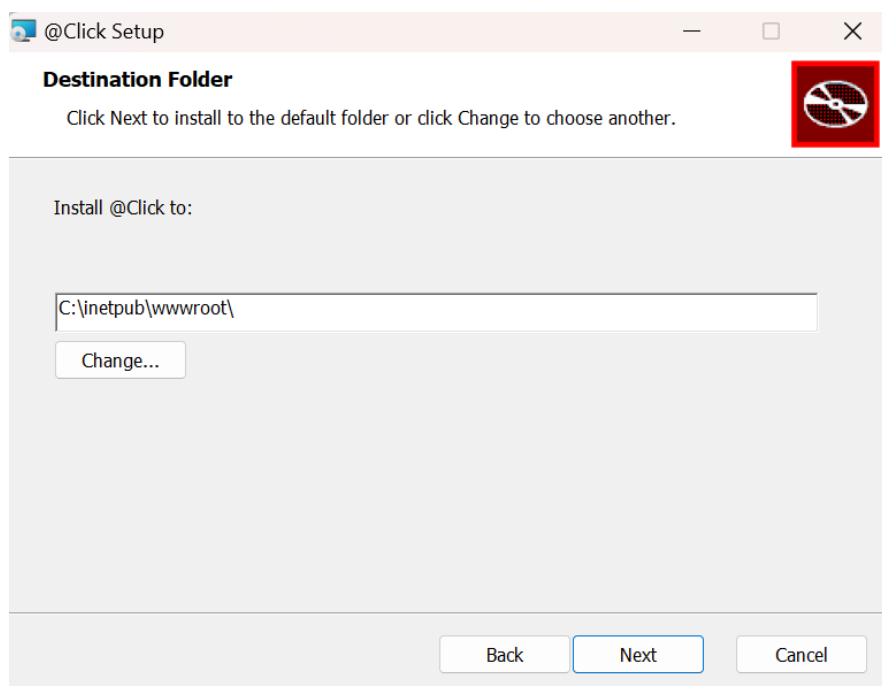
FigureG-110: @Click installation interface

Select **I accept the terms in the License Agreement**, then select **Next** to continue.



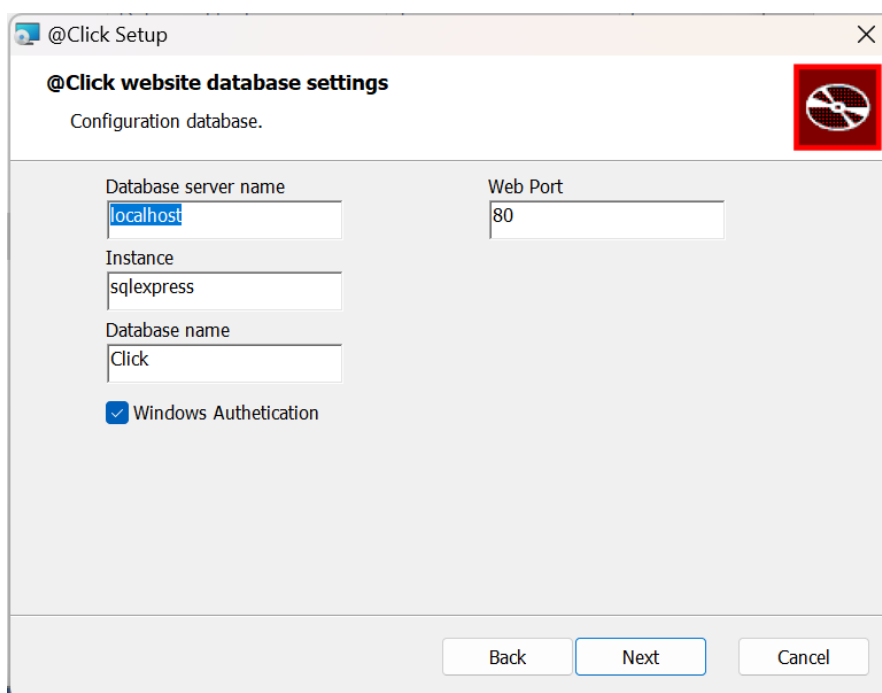
FigureG-111: License Agreement Interface

Select **Destination Folder**, then select **Next** to continue



FigureG-112: Destination Folder Configuration Interface

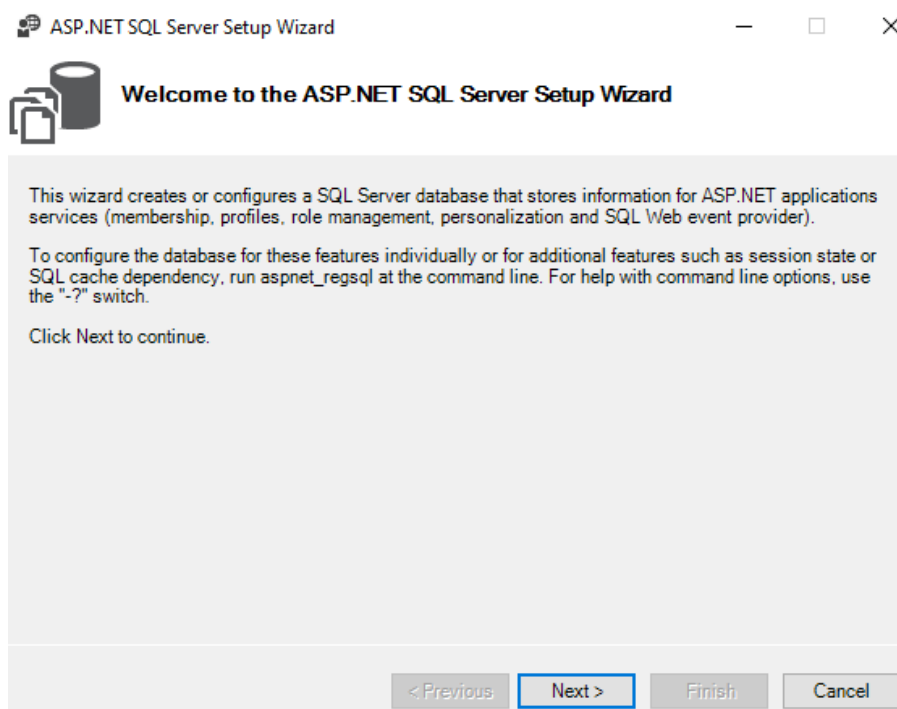
Configure the parameters as shown, then select **Next** to continue



FigureG-113: Database Installation Interface

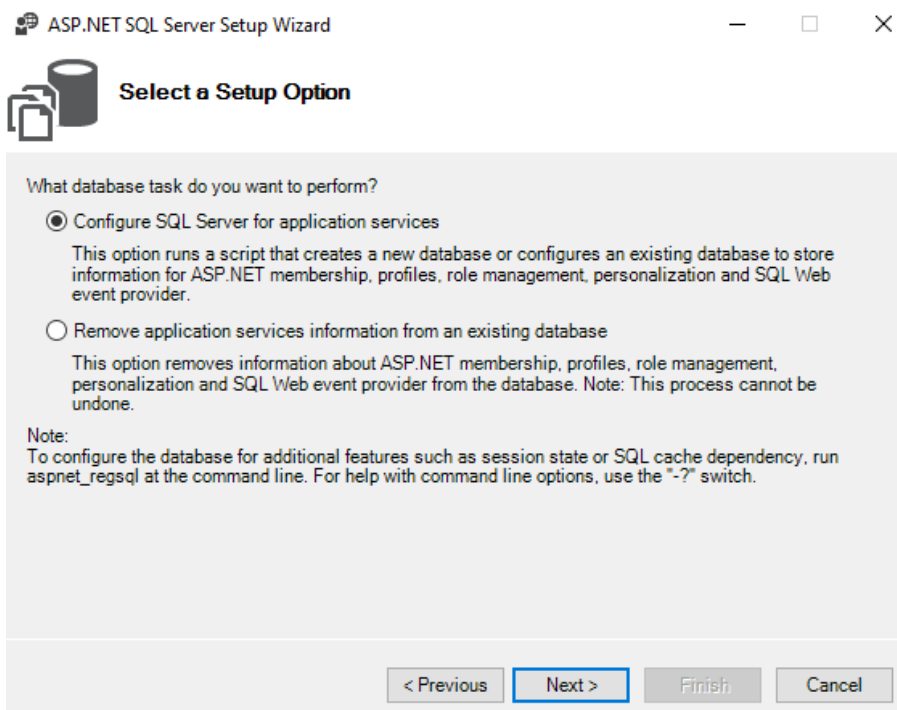
Select **Install** to continue the installation

Continue installing **ASP.NET SQL Server Setup Wizard**, select **Next** to continue



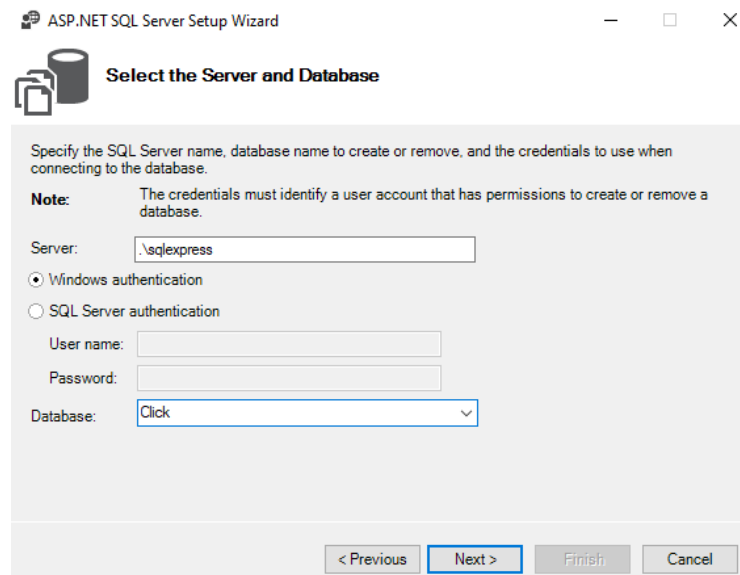
FigureG-114: Installation Interface ASP.NET SQL Server Setup Wizard

Select **Configure SQL Server for application services**, and then select **Next** to continue.



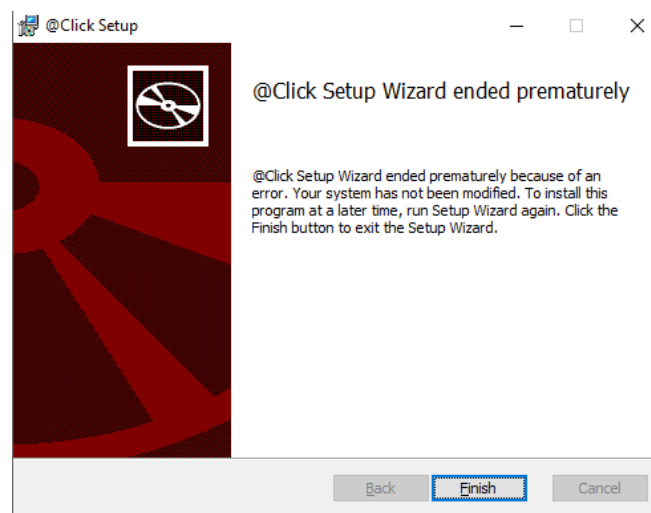
FigureG-115: Select a Setup Option

Configure the parameters as shown, then select **Next** to continue



FigureG-116: Server and Database Customization Interface

Select **Finish** to finish the installation process



FigureG-117: Interface ends the installation process

G.10.1.5 Installing a license for @Click

Open the **NLog file** at the @Click installation location, check the location where the LogClick file is saved, which is the file containing the Id. Open the Log file, get the ID to send to ATS, ATS will send back the @Click.lic file

```
<?xml version="1.0" encoding="utf-8" ?>
<nlog xmlns="http://www.nlog-project.org/schemas/NLog.xsd"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <targets>
    <target xsi:type="File" archiveAboveSize="1048576" maxArchiveFiles="10" name="logfile" fileName="d:\\Logs\\logClick.txt" />
  </targets>
  <rules>
    <logger name="*" writeTo="logfile" />
  </rules>
</nlog>
```

Debug 2022/11/03 11:48:18.374 Click.Global.Application_Start Application starting
Info 2022/11/03 11:48:18.596 Click.Global.ValidateLicense @Click is running in demo mode with id: BPJPNE038JE5200BFEBFBFF000806EC.

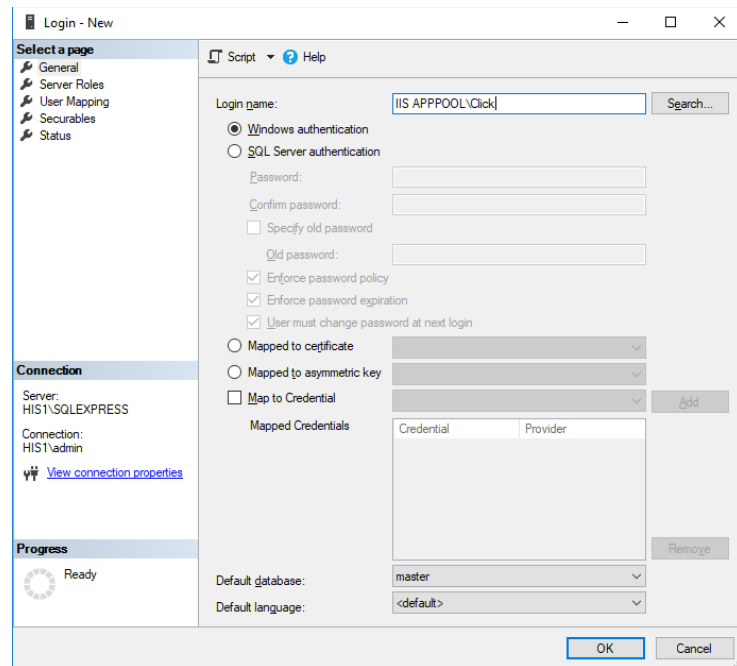
FigureG-118: LogClick Saving Location

Create the C:\Program Files\Common Files\@Click folder, copy the @Click.lic file to the folder you just created.

G.10.1.6 Configuration in SQL Managment Studio

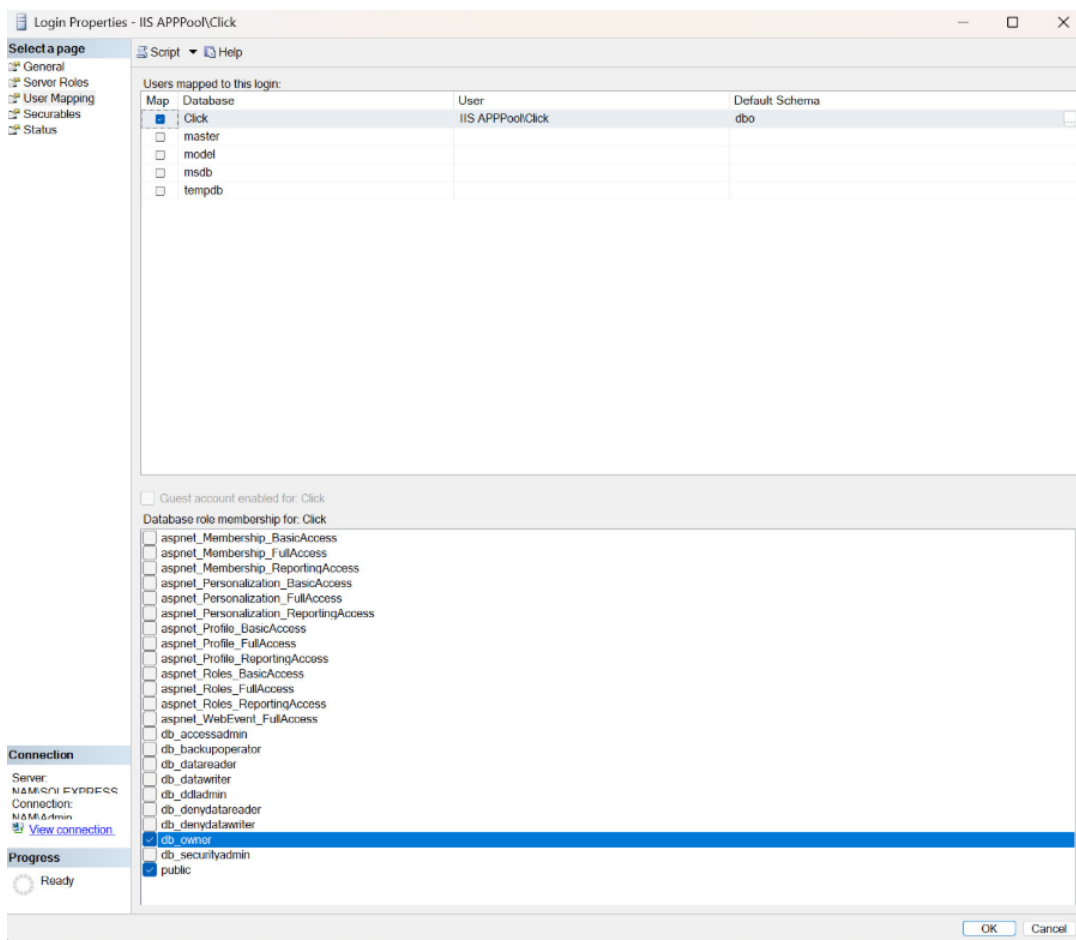
Open **SQL Server Management Studio** , select **Security** , select **IIS APPPool\Click**

- General Tab



FigureG-119: General Tab in IIS APPPool\Click

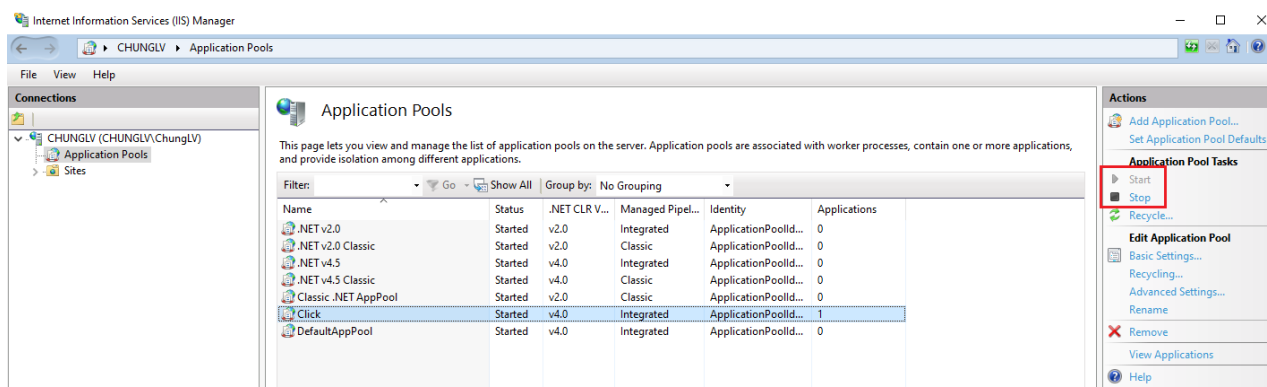
- User Mapping Tab



FigureG-120: User Mapping tab in IIS APPPool\Click

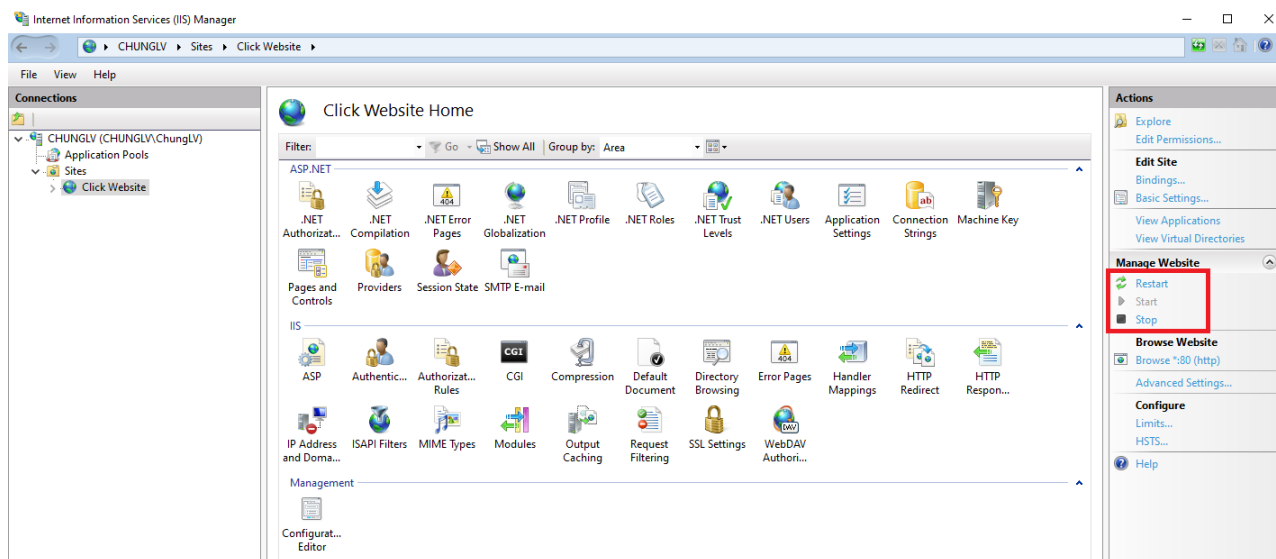
G.10.1.7 Configure enable ATS Web port

Open Internet Information Service (IIS) Manager, select Application Pools, select Click, select Stop and then Start to Restart Click



FigureG-121: Restart Click

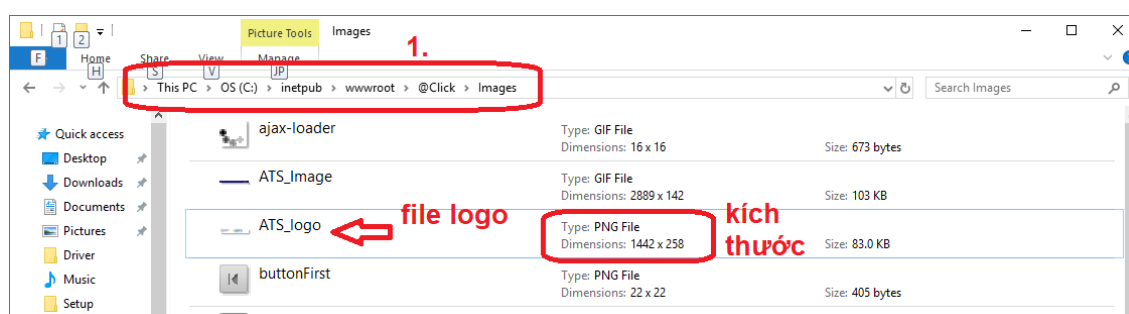
Open Internet Information Service (IIS) Manager, select Sites, select Click Website, select Stop then Start or Restart enable port ATS Web



FigureG-122: Enable ATS Web port

G.10.1.8 Logo Configuration on the Web Portal

Edit the **ATS_logo** file at the @Click installation location as shown below

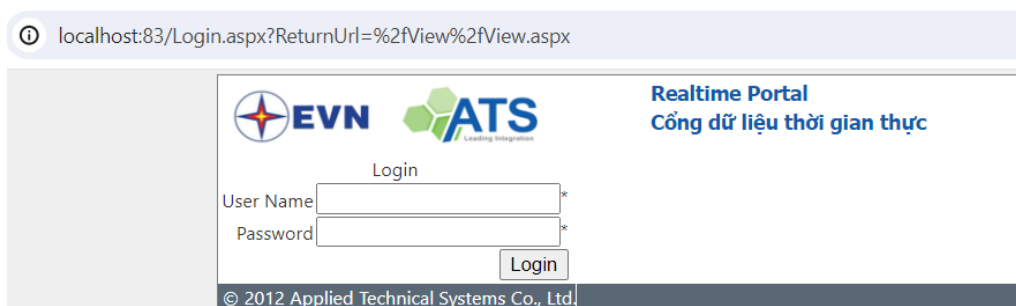


FigureG-123: Editing the logo on the Web Portal

G.10.2. Configuring @Click Web Server

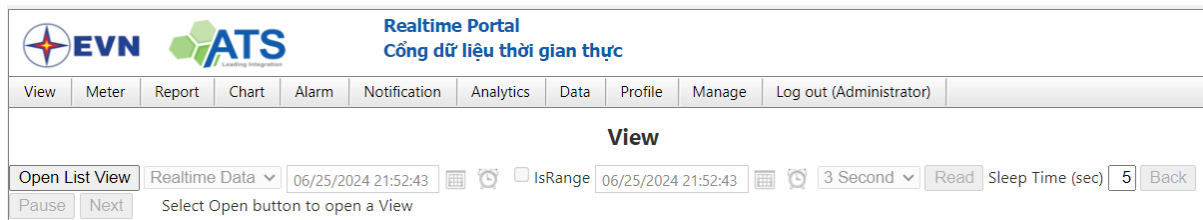
G.10.2.1 Log in to the Web Portal

The user using a Web browser on the machine where HIS Server is installed accesses the IP:Port address (with IP: IP address of the machine on which HIS Server is installed; Port: ATS Port Web, configured upon installation @Click)



FigureG-124: Web Portal Login Interface

After successfully logging in, the display interface is the default View screen of the system. In it, there is a menu bar of the main functions of the system:



FigureG-125: Main function menu of the system

Key functions include:

- **View:** Query Screen View.
- **Report:** Query the report form.
- **Chart:** Query the trend graph of data points over a specified period of time.
- **Alarm:** Queries the alarms of the SCADA system for a specified period of time.
- **Analytics:** Check the connection, exchange data to the HIS Server.
- **Data:** Upload reports by file, import reports manually.
- **Profile:** Manage the personal information of the currently logged in user.
- **Manage:** Manage users, reports, view... of the system.
- **Log out:** Log out of the system.

G.10.2.2 The main functions of the system

G.10.2.2.1 Manage function

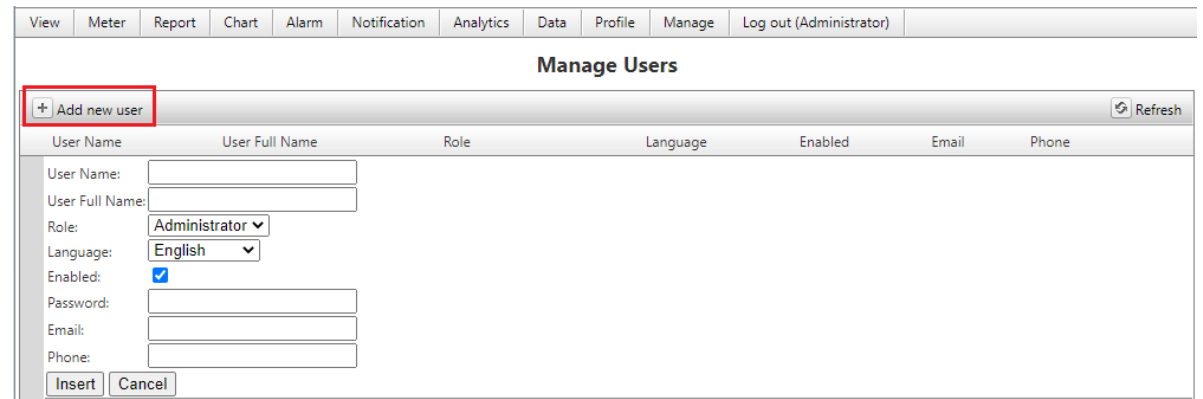
G.10.2.2.1.1. User

a. User Management

This function has the ability to add, edit, and delete users and groups.



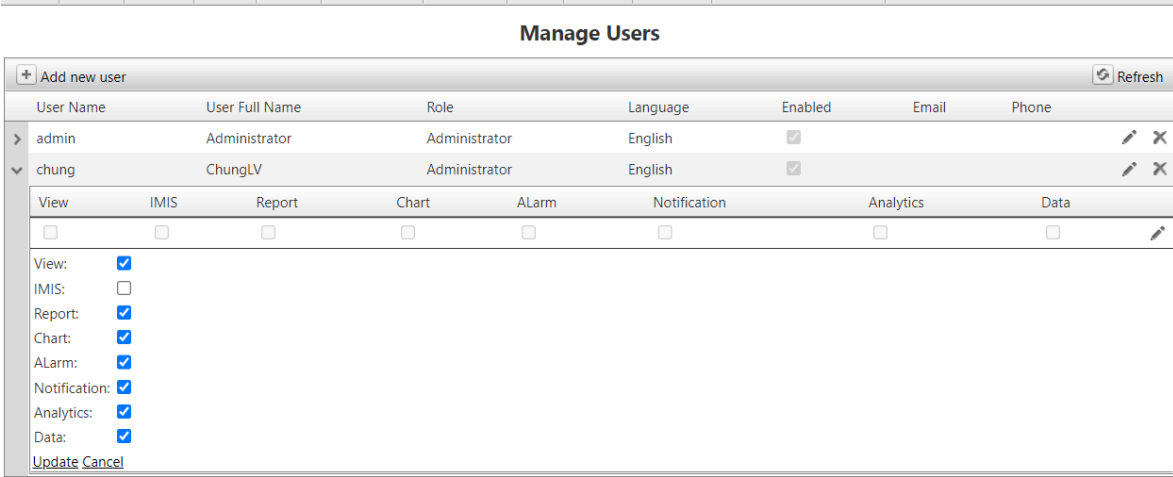
Users go to **the Manage > User > Manage Users** menu, select **Add new user** and enter the fields then select **Insert**



FigureG-126: User Configuration Interface

User Name	User Name
User Full Name	User's full name
Role	User Delegation: Administrator/Moderator/User
Language	Language: English/Vietnamese
Enabled	User's Enable/Disable Status
Password	Password
Email	Email Address
Phone	Phone Number

After creating a user, give the user permission to view the functions, click to select the **Expand**  icon in the user list, click to select the **Edit** icon 



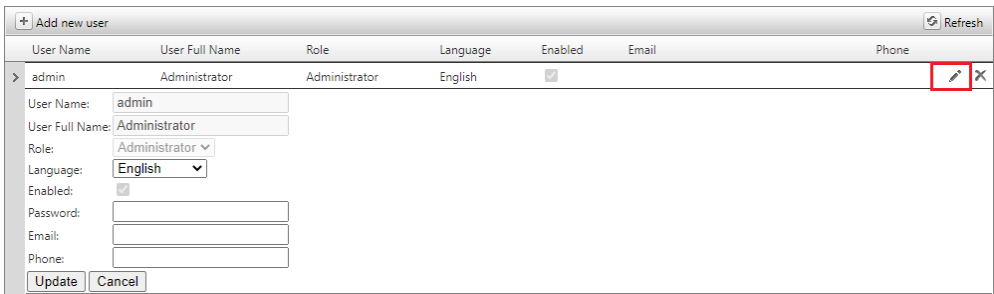
The screenshot shows the 'Manage Users' window. It has a table with columns: User Name, User Full Name, Role, Language, Enabled, Email, and Phone. Two users are listed: 'admin' (Administrator, English, Enabled) and 'chung' (ChungLV, Administrator, English, Enabled). Below the table, there are checkboxes for various functions: View, IMIS, Report, Chart, ALarm, Notification, Analytics, and Data. The 'View' checkbox is checked. At the bottom, there are 'Update' and 'Cancel' buttons.

FigureG-127: Option to grant user permissions

Click to select the functions for the user, then select **Update**.

 **Edit User**

Click the Manage > User > Manage Users menu, click the **Edit** icon , enter the information you want to change, then select **Update**.

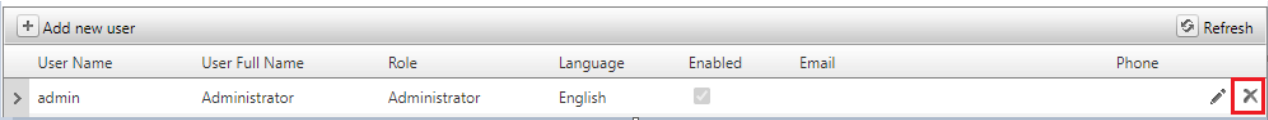


The screenshot shows the 'User Editing Interface' window. It has a table with columns: User Name, User Full Name, Role, Language, Enabled, Email, and Phone. One user is listed: 'admin' (Administrator, English, Enabled). Below the table, there are input fields for: User Name, User Full Name, Role, Language, Enabled, Password, Email, and Phone. At the bottom, there are 'Update' and 'Cancel' buttons.

FigureG-128: User Editing Interface

 Delete User

Click the **Manage > User > Manage Users** menu, click the **Delete icon**  , and then select **Ok**.

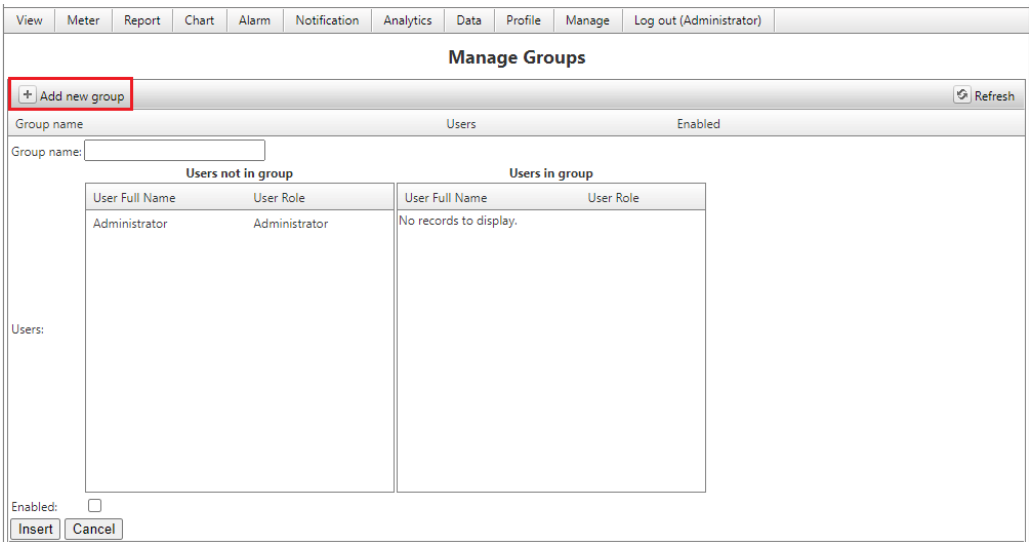


FigureG-129: User deletion interface

b. Group Management

 Add Group

Click the **Manage > User > Manage Groups** menu, select **Add new group** and enter the fields, select the users of the newly created group by left-clicking and dragging and dropping the user from **the Users not in group** column to the **Users in group** column then select **Insert**.

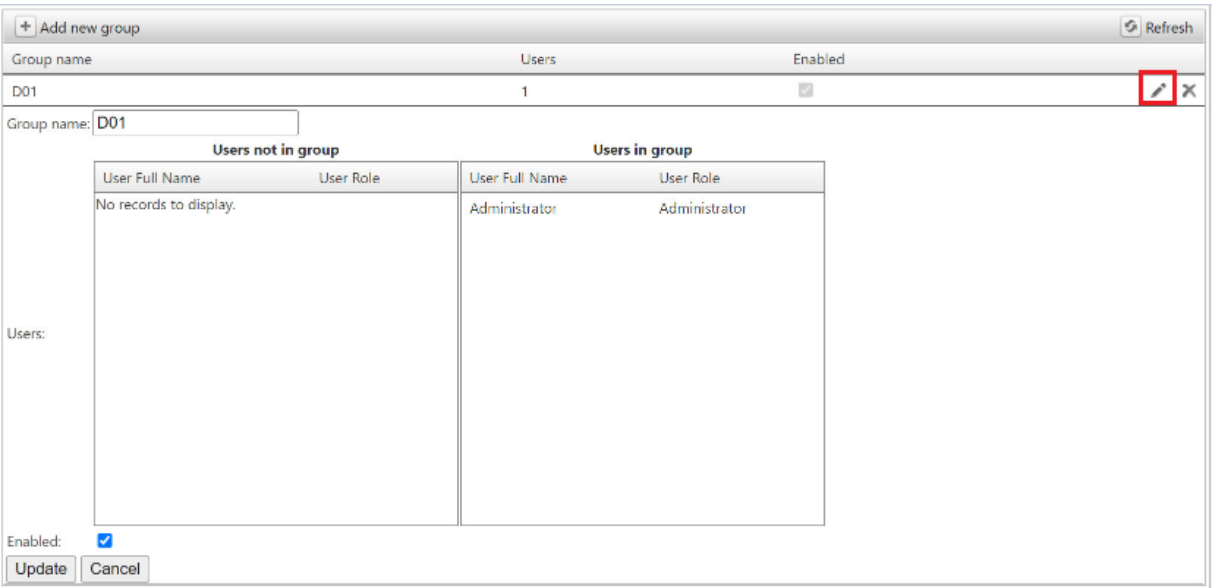


FigureG-130: Group Configuration Interface

Group name	Group Name
Enabled	Group Enable/Disable Status

 Edit Group

Click the **Manage > User > Manage Groups** menu, click the **Edit icon**  , enter the information you want to change, and then select **Update**.

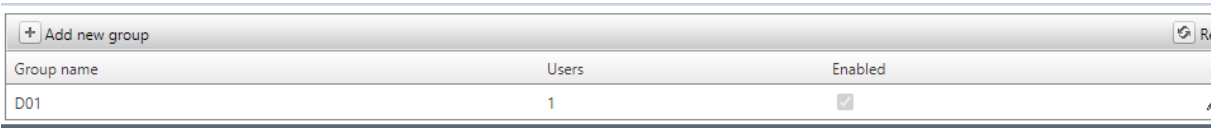


The Group Edit Interface shows a table with columns: Group name, Users, and Enabled. The first row shows 'D01', '1', and a checked 'Enabled' box. To the right of the 'Enabled' box are edit and delete icons. Below the table, the 'Group name' is 'D01'. There are two side-by-side tables: 'Users not in group' (empty) and 'Users in group' (containing one row: 'Administrator' with role 'Administrator'). At the bottom, there is an 'Enabled' checkbox (checked) and 'Update' and 'Cancel' buttons.

FigureG-131: Group Edit Interface

 Delete Group

Click the **Manage > User > Manage Groups** menu, click the **Delete** icon , and then select **Ok**.



The Delete group interface shows a table with columns: Group name, Users, and Enabled. The first row shows 'D01', '1', and a checked 'Enabled' box. To the right of the 'Enabled' box are edit and delete icons.

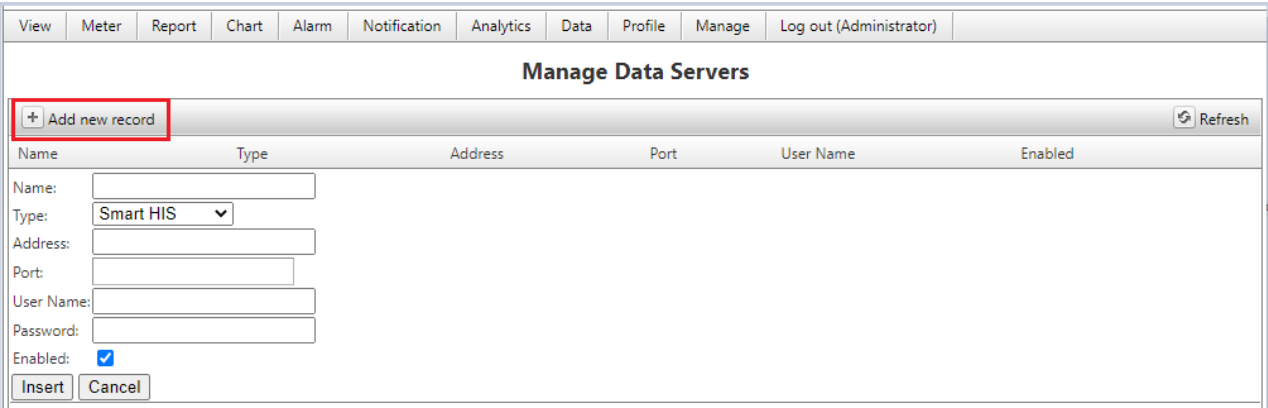
FigureG-132: Delete group interface

G.10.2.2.1.2. DataServer

This function has the ability to add, modify, and delete DataServers.

 Add a DataServer

Click the **Manage > DataServer menu> Manage DataServers**, select **Add new record**, enter the fields, then select **Insert**.



The DataServer Configuration Interface has a top menu bar with: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below is a 'Manage Data Servers' section with a table. The first row of the table has columns: Name, Type, Address, Port, User Name, and Enabled. To the left of the table are input fields for Name, Type (a dropdown menu with 'Smart HIS' selected), Address, Port, User Name, and Password. Below these fields is an 'Enabled' checkbox (checked) and 'Insert' and 'Cancel' buttons. The 'Add new record' button is highlighted with a red box.

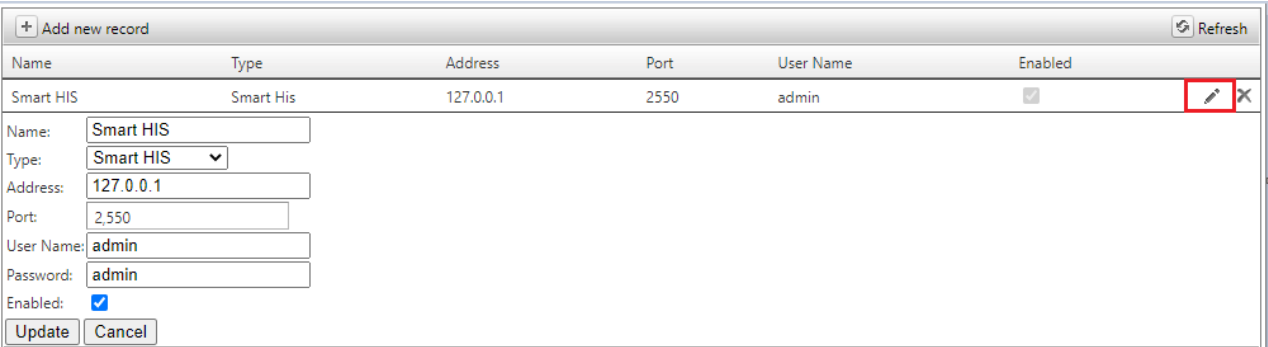
FigureG-133: DataServer Configuration Interface

In which:

- **Name:** The name of the DataServer
- **Address:** Address of HIS Server
- **Port:** The data port of the HIS Server
- **User Name/Password:** User/password logs in to HIS Server

 Editing DataServer

Click the **Manage > DataServer > Manage DataServers** menu, click the **Edit** icon , enter the information you want to change, and then select **Update**.



Name	Type	Address	Port	User Name	Enabled	
Smart HIS	Smart His	127.0.0.1	2550	admin	☒	 

Name:

Type:

Address:

Port:

User Name:

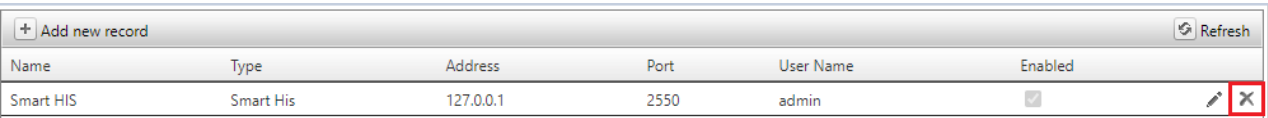
Password:

Enabled: ☒

FigureG-134: DataServer Editing Interface

 Deleting a DataServer

Click the **Manage > DataServer > Manage DataServers** menu, click the **Delete** icon , then select **Ok**

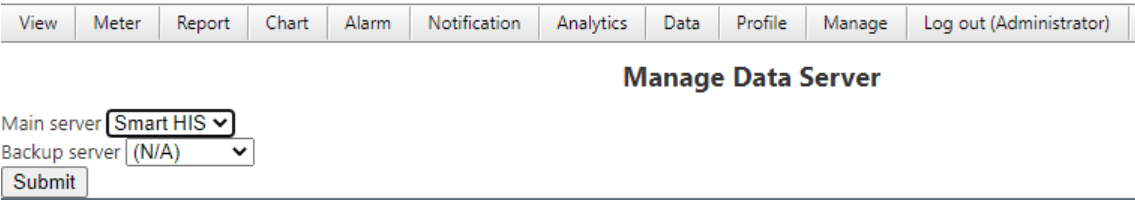


Name	Type	Address	Port	User Name	Enabled	
Smart HIS	Smart His	127.0.0.1	2550	admin	☒	 

FigureG-135: DataServer Deletion Interface

G.10.2.2.1.3. Alarms

To manage the server containing the alarm, Click the **Manage > Alarm** menu > **Manage DataServers**, select Main server, Backup server (if any) and **Submit**



View Meter Report Chart Alarm Notification Analytics Data Profile Manage Log out (Administrator)

Manage Data Server

Main server

Backup server

FigureG-136: Alarm Management Interface

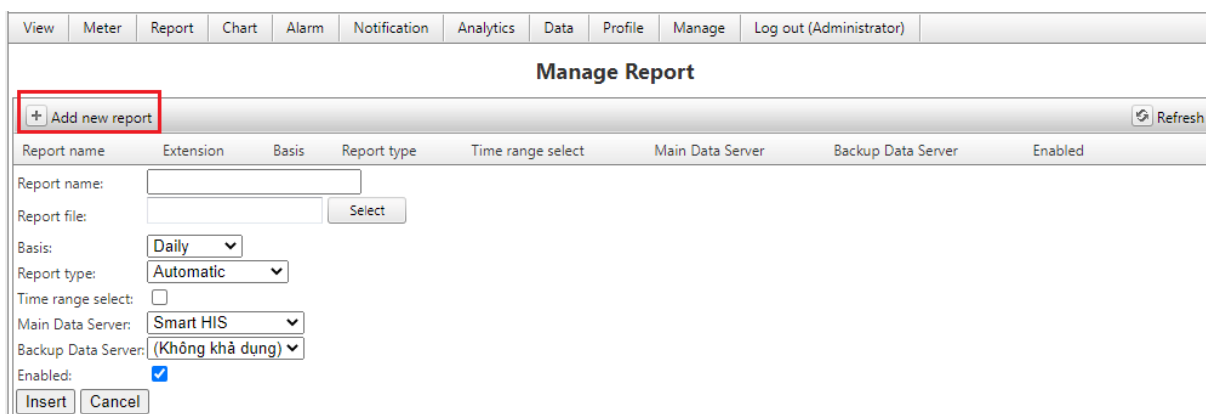
G.10.2.2.1.4. Report

a. Report Management

This function has the ability to add, edit, and delete reports.

Add Report

Click Manage > **Report menu** > **Manage Report**, select **Add new Report** and enter the fields then select **Insert**.



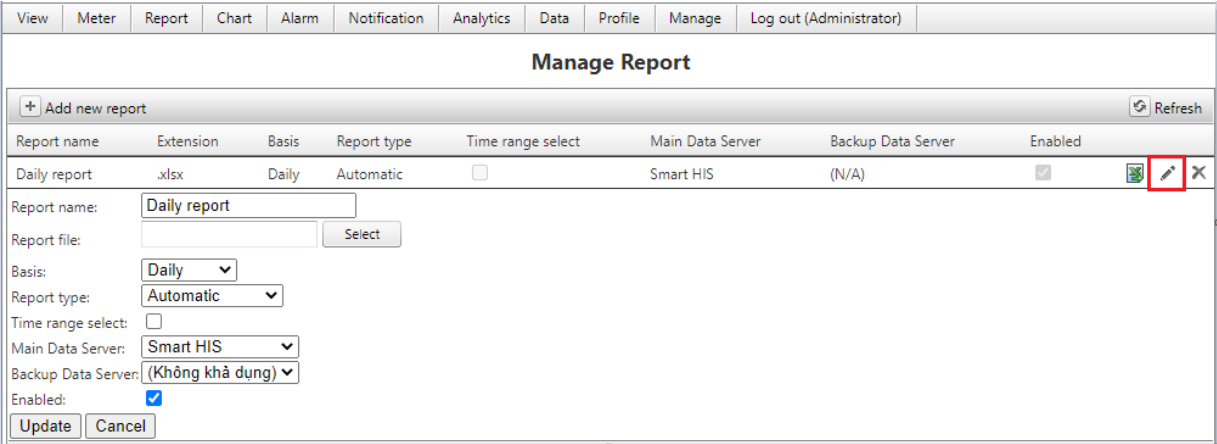
FigureG-137: Report Configuration Interface

In which:

- **Report name:** The display name of the report template.
- **Report file:** Select the report template file.
- **Basis:** The data extraction cycle corresponding to the report form includes:
 - **Hourly:** Hourly.
 - **Daily:** Daily.
 - **Weekly:** Weekly.
 - **Monthly:** Monthly.
 - **Quarterly:** Quarterly.
 - **Yearly:** Annual.
- **Report type:** The type of report includes:
 - **Automatic:** The report updates automatically.
 - **Upload:** Report the upload.
 - **Manual:** Manual update report.
 - **Semi-automatic:** Semi-automatic reporting.
- **Main DataServer:** Select the main server to query the data.
- **Backup DataServer:** Select a backup server to query data (in case the system is equipped with 2 past databases).
- **Enable:** Enable or Disable report.

 Edit a Report

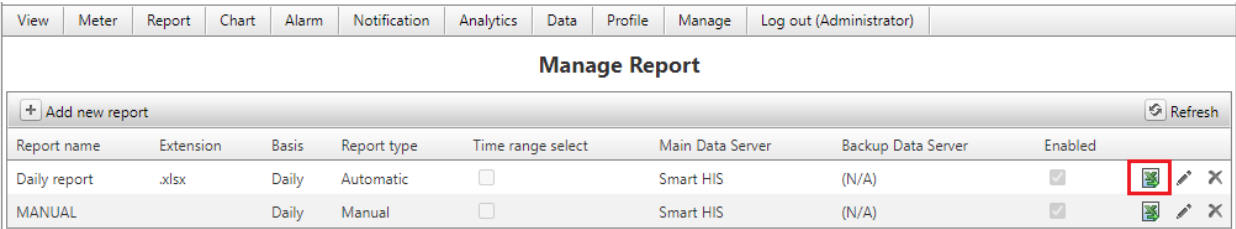
Click the **Manage > Report menu > Manage Report**, click the **Edit icon**  , enter the information you want to change, then select **Update**.



The screenshot shows the 'Manage Report' interface. At the top is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar is the 'Manage Report' section. It features a table with columns: Report name, Extension, Basis, Report type, Time range select, Main Data Server, Backup Data Server, and Enabled. The first row is 'Daily report' with extension '.xlsx', basis 'Daily', report type 'Automatic', and is enabled. To the right of this row are icons for download, edit (highlighted with a red box), and delete. Below the table is a form for editing the report. It includes fields for Report name (pre-filled with 'Daily report'), Report file (with a 'Select' button), Basis (dropdown menu set to 'Daily'), Report type (dropdown menu set to 'Automatic'), Time range select (checkbox), Main Data Server (dropdown menu set to 'Smart HIS'), Backup Data Server (dropdown menu set to '(Không khả dụng)'), and Enabled (checkbox checked). At the bottom of the form are 'Update' and 'Cancel' buttons.

FigureG-138: Report Editing Interface

Alternatively, you can download an existing report form by going to **the Manage > Report > Manage Report** menu, clicking on the **Download icon**  .

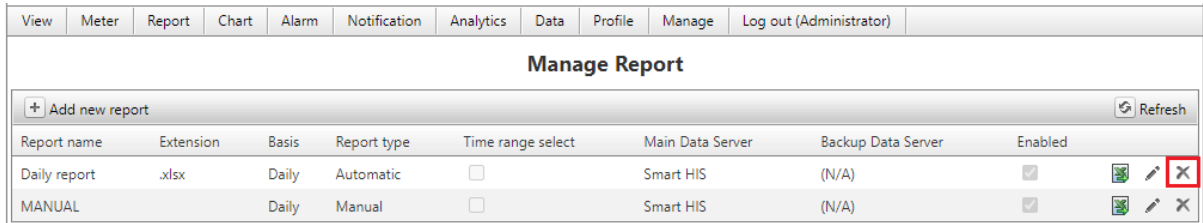


The screenshot shows the 'Manage Report' interface. It features a table with columns: Report name, Extension, Basis, Report type, Time range select, Main Data Server, Backup Data Server, and Enabled. The first row is 'Daily report' with extension '.xlsx', basis 'Daily', report type 'Automatic', and is enabled. The second row is 'MANUAL' with extension '.xlsx', basis 'Daily', report type 'Manual', and is enabled. To the right of each row are icons for download, edit, and delete. The download icon for the 'MANUAL' row is highlighted with a red box.

FigureG-139: Download existing report

 Delete a Report

Click the **Manage > Report > Manage Report** menu, click the **Delete icon**  , then select **Ok**.



The screenshot shows the 'Manage Report' interface. It features a table with columns: Report name, Extension, Basis, Report type, Time range select, Main Data Server, Backup Data Server, and Enabled. The first row is 'Daily report' with extension '.xlsx', basis 'Daily', report type 'Automatic', and is enabled. The second row is 'MANUAL' with extension '.xlsx', basis 'Daily', report type 'Manual', and is enabled. To the right of each row are icons for download, edit, and delete. The delete icon for the 'MANUAL' row is highlighted with a red box.

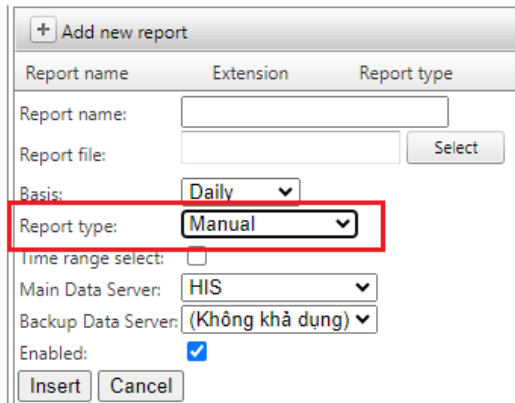
FigureG-140: Delete Report Interface

b. Hands-on Report Management

This function is capable of managing manual reports.

 Add a report

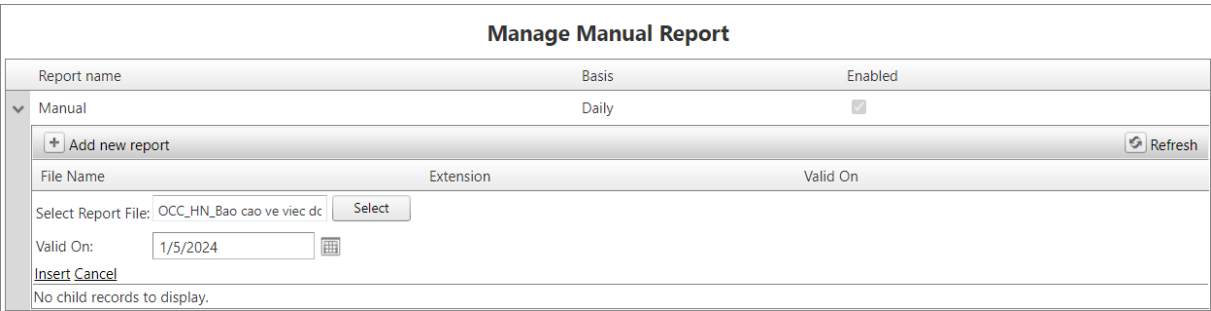
Adding a manual report is similar to adding a report, note that selecting the **Report type** is **Manual**.



The dialog box titled '+ Add new report' contains the following fields: 'Report name' (text input), 'Report file' (text input with a 'Select' button), 'Basis' (dropdown menu set to 'Daily'), 'Report type' (dropdown menu set to 'Manual', highlighted with a red box), 'Time range select' (checkbox), 'Main Data Server' (dropdown menu set to 'HIS'), 'Backup Data Server' (dropdown menu set to '(Không khả dụng)'), 'Enabled' (checkbox checked), and 'Insert'/'Cancel' buttons.

Click the Manage > **Report** > **Manage Manual Report** menu to open the list of managed manual reports.

Click the Expand icon , Select **Add new report** and select **Select Report File**, enter **Valid On** (the date the report starts working) and then Insert .

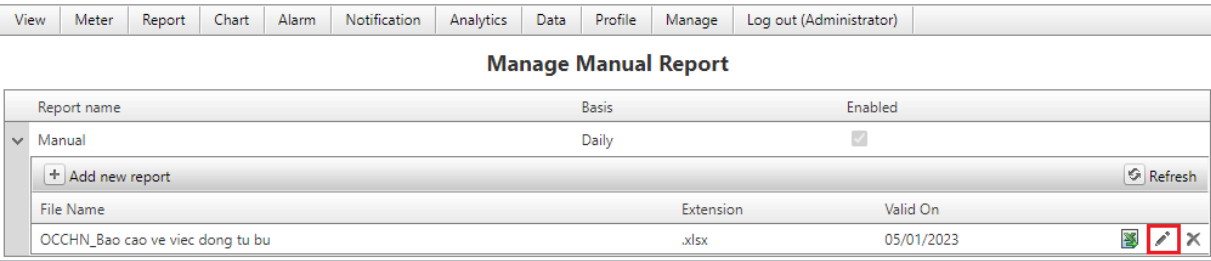


The 'Manage Manual Report' interface shows a table with columns: Report name, Basis, and Enabled. A dropdown menu is open for 'Manual', showing 'Daily' and a checked 'Enabled' checkbox. Below the table, there is a '+ Add new report' button and a 'Refresh' button. The 'File Name' field contains 'OCC_HN_Bao cao ve viec dc', and the 'Valid On' field contains '1/5/2024'. There are 'Insert' and 'Cancel' buttons at the bottom.

FigureG-141: Manual Report interface

 Edit a report

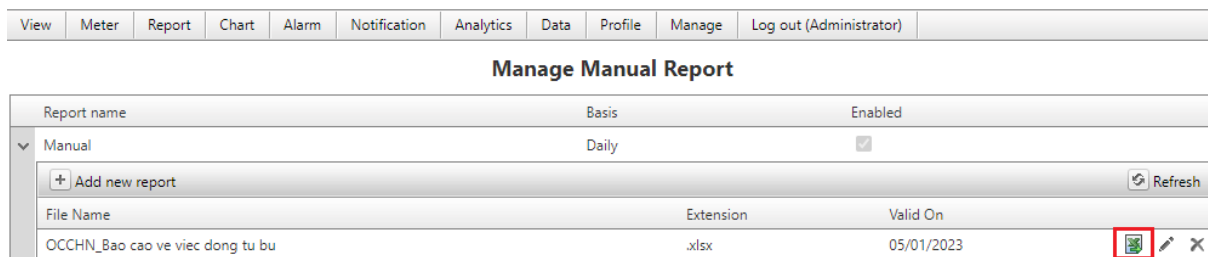
Click Manage > **Report** menu > **Manage Manual Report**, click the **Edit** icon , enter the information you want to change, then select **Update**.



The 'Manage Manual Report' interface shows a table with columns: Report name, Basis, and Enabled. A dropdown menu is open for 'Manual', showing 'Daily' and a checked 'Enabled' checkbox. Below the table, there is a '+ Add new report' button and a 'Refresh' button. The 'File Name' field contains 'OCC_HN_Bao cao ve viec dc', and the 'Valid On' field contains '05/01/2023'. There are 'Insert' and 'Cancel' buttons at the bottom. The 'Edit' icon (pencil) is highlighted with a red box.

FigureG-142: Manual Report Editing Interface

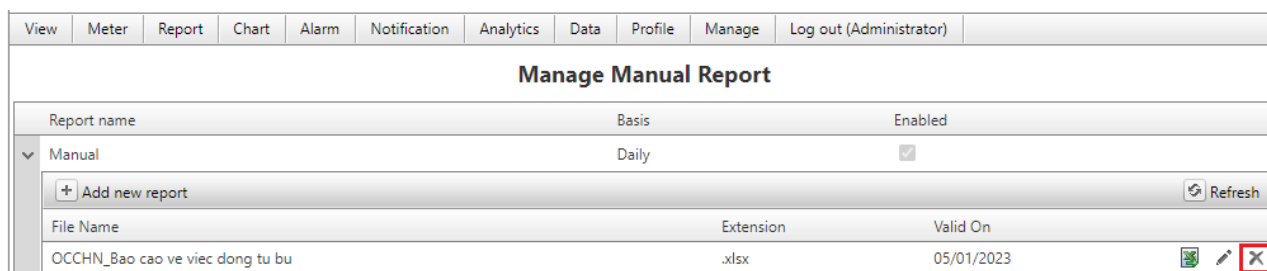
Alternatively, you can download an existing report form by, going to **the Manage > Report > Manage Manual Report** menu, selecting the report and clicking the Download icon  .



FigureG-143: Download Existing Manual Report

Delete a report

Click the **Manage > Report > Manage Manual Report** menu, click the **Delete** icon  , and then select **Ok**.

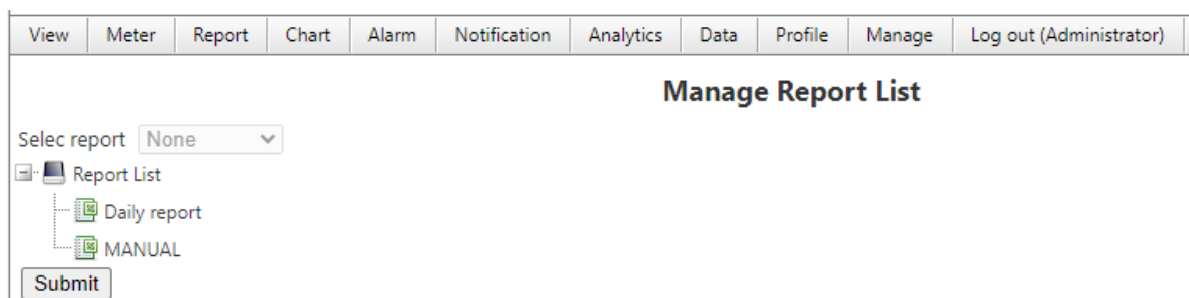


FigureG-144: Manual Report Deletion Interface

c. Manage Report Lists

This function has the ability to manage report lists.

Click the **Manage > Report > Manage Report List** menu.

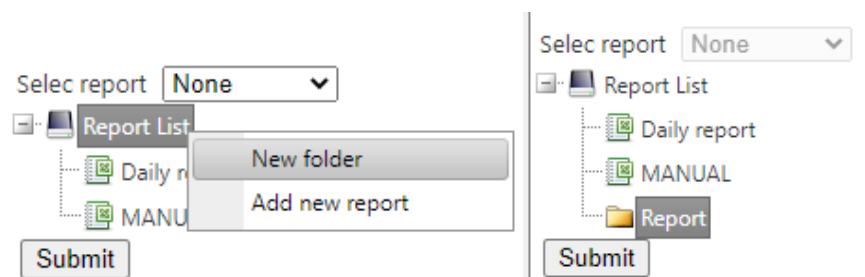


FigureG-145: Report List Management Interface

Users can add or delete folders or reports to the Report List.

Adding a Folder to the Report List

Click the **Manage > Report menu > Manage Report List**, Right Click on the **Report List** and select **New folder**.



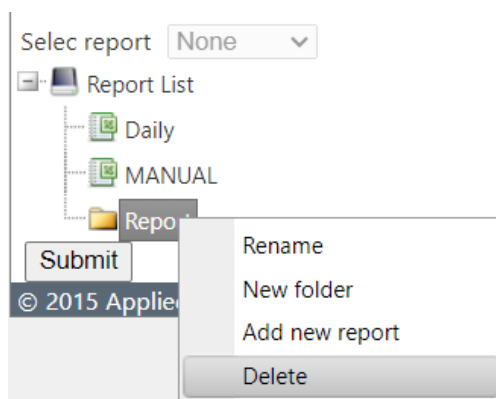
FigureG-146: The interface for adding folders in the Report List

Rename the folder by Double Click and rename the folder or Right Click select **Rename**, select **Submit**.

Add a folder to the folder by Right-clicking and selecting **Add new folder**, renaming the same as renaming the folder, selecting **Submit**.

Deleting a Folder in the Report List

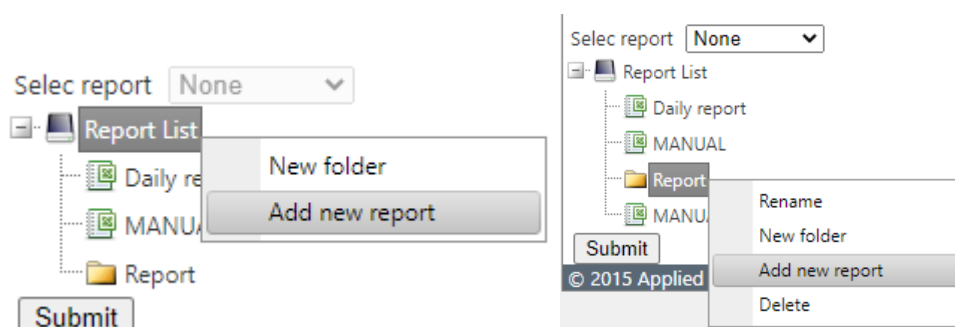
Click the **Manage > Report menu > Manage Report List**, Right Click on the **folder** you want to delete, select **Delete**, then select **Ok**.

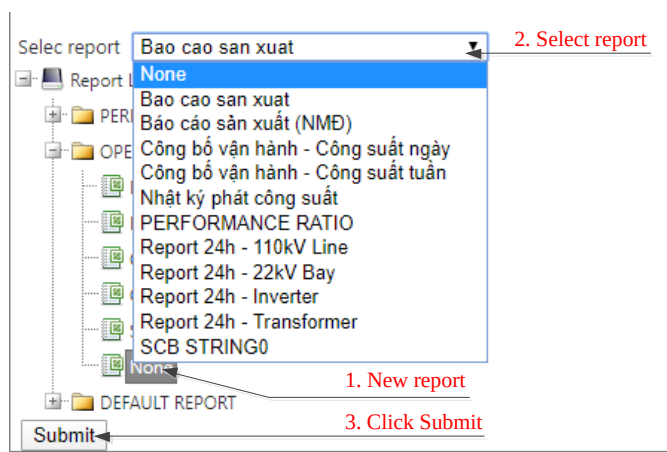


FigureG-147: The interface for deleting folders in the Report List

Add a Report to the Report List

Click the **Manage > Report menu > Manage Report List**, Right Click on the **Report List** and select **Add new report** or Right Click on the Folder to select **Add new report** to create a report, select **Select report** in the list of created reports and **Submit**.

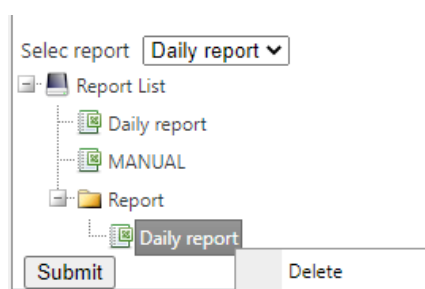




FigureG-148: Add Report interface

Deleting a Report in the Report List

Click the **Manage > Report menu > Manage Report List**, Right Click on the **Report** you want to delete, select **Delete**, then select **Ok** and **Submit**.



FigureG-149: The interface for deleting folders in the Report List

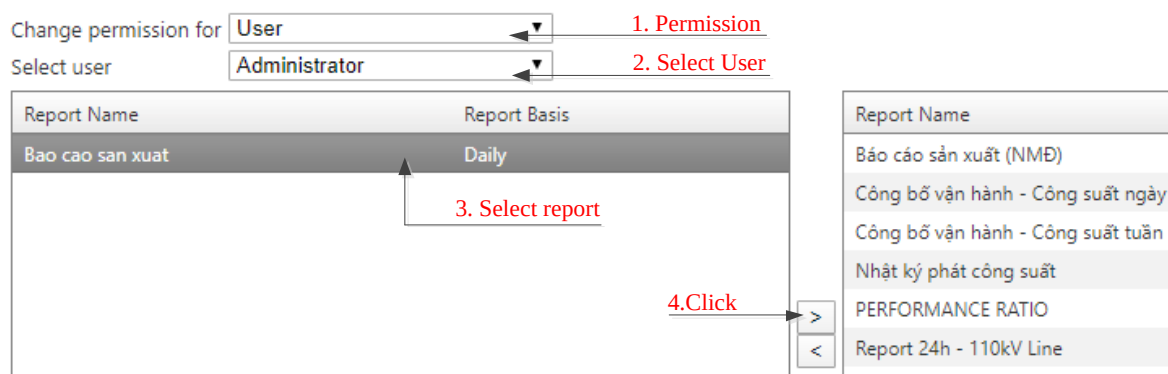
d. Manage permissions to view reports with users and groups

This function has the ability to manage the permissions to view reports for users and groups.

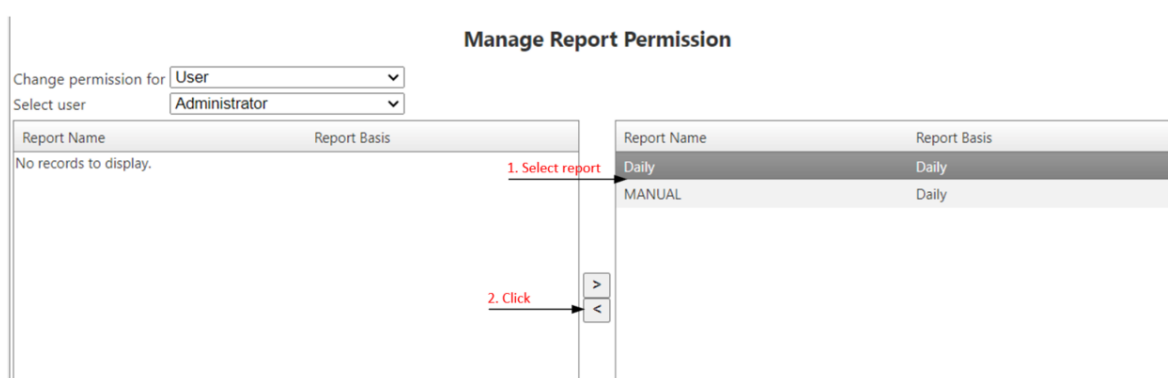
- Grant group/user permission to view reports:

Click **Manage > Report menu > Manage Report Permission**

1. Select **Group** or **User** in the **Change permission for box**.
2. Select **the Group** that needs to grant permission in the list of declared groups in the **Select group** box with **the Change permission for** case as Group. Select **the User** to grant permission from the list of declared users in the **Select user** box with **Change permission for** as User.
3. Select the report to which the group/user needs to be granted permissions.
4. Click **Add** .



FigureG-150: Report Permission Management Interface



- To revoke the permission to view reports for a group/user:

Click Manage > Report menu > Manage Report Permission

Select **Group** or **User** in the **Change permission for** box.

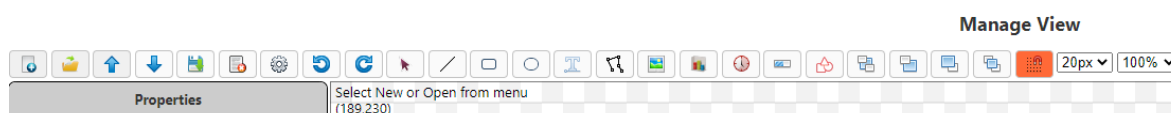
Select **the Group** that needs to be delicensed in the list of declared groups in the **Select group** box with the **Change permission for** case as Group. Select **the User** that needs to be ungranted permission in the list of declared users in the **Select user** box with **Change permission for** as User.

- Select the report that the group/user needs to delicense.
- Click **Remove** .

G.10.2.2.1.5. View

a. Managed View

Click Manage > View > Manage View menu



FigureG-151: Main Toolbar Management View

Functions on the toolbar:

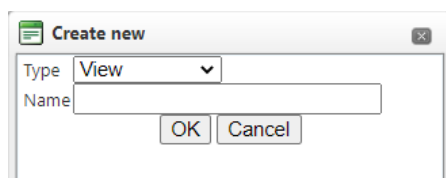
Function	Symbol	Describe
New		Create a new screen
Open		Open screen
Import		Import the screen
Export		Export the screen
Save		Save
Delete		Erase
Setting		Install
Undo		Undo the last action you took
Redo		Restore an action that was just canceled
Select		Audience Selection
Add Line		Add a straight line
Add Rectangle		Add a rectangle
Add Ellipse		Add an ellipse
Add Text		Add text
Add Image		Add Images
Add Chart		Add a chart
Add Gauge		Add a Gauge chart
Add Symbol		Add symbol
Bring Back		Move an audience below all other audiences
Bring Backward		Move the object below the object directly behind it
Bring Forward		Move the object that is on top of the object directly in front of it
Bring Front		Move an audience that is on top of all other objects
Snap To Grid		Moving the symbol on the View design screen will move according to the coordinates (5pixel)
Select grid size	20px	Grid Line Size
Select zoom ratio	100%	Zoom Ratio

 More screen view

Select the **New**  icon or select the **Import**  icon to add a screen view.

➤ Add a screen view using New:

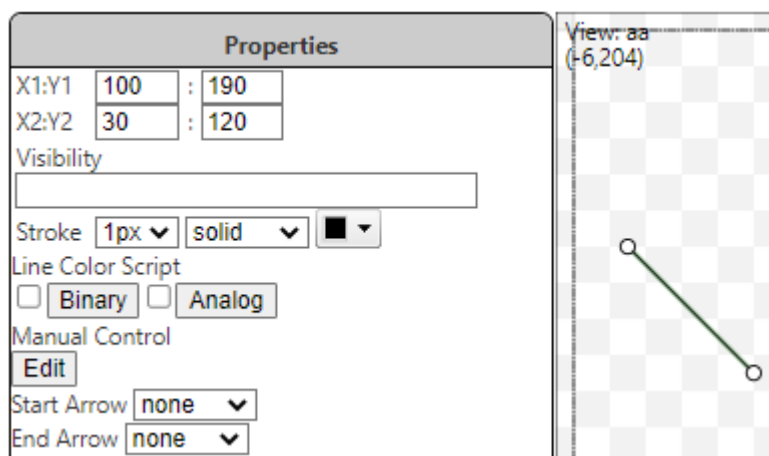
Select the **New**  icon, select Type **View**, and enter a screen name.



FigureG-152: Interface Add Screen View

Tools for drawing diagrams:

- Draw a line: Tap the **Line** button  on the toolbar, hold down the left mouse and draw on the background view.



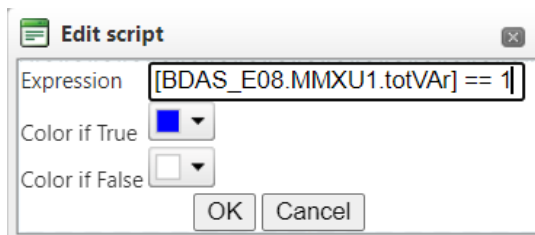
FigureG-153: Draw a straight line

In which:

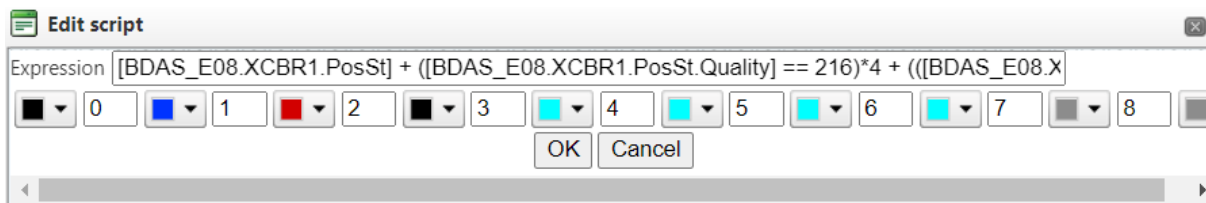
- X1:Y1: Coordinates of the starting point.
- X2:Y2: Endpoint coordinates.
- Visibility: Used to hide/show objects according to expressions. Example:

Visibility

- Stroke: Lines such as the width, style, and color of the line.
- Line Color Script: The color of the line according to the value of the Binary or Analog expression with the variables in the HIS Server.



FigureG-154: Script binary for the color of the line

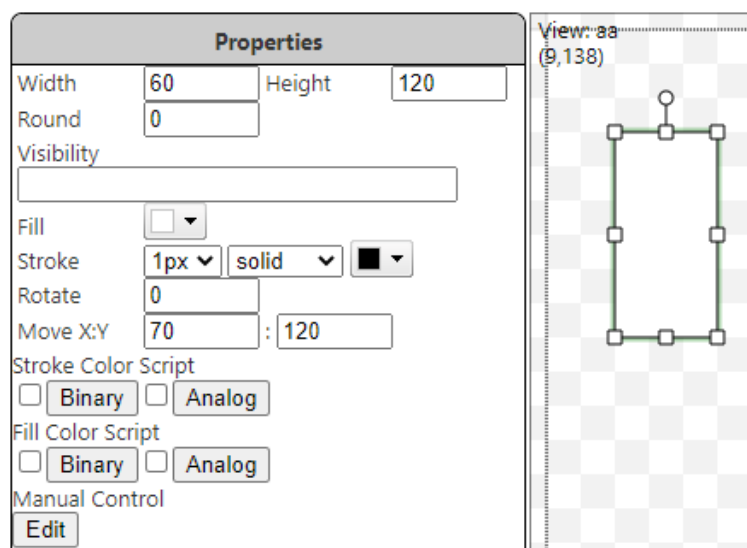


FigureG-155: Analog script for the color of the line

- Start Arrow: depending on whether the arrow at the beginning of the line is a triangle, a circle, a square or none.
- End Arrow: depending on whether the arrow at the end of the line is a triangle, circle, square or none.

The variable name in the Expression expression is placed in the [].

- Draw a rectangle: Tap the **Rectangle** button  on the toolbar, hold down the left mouse and draw on the background view.

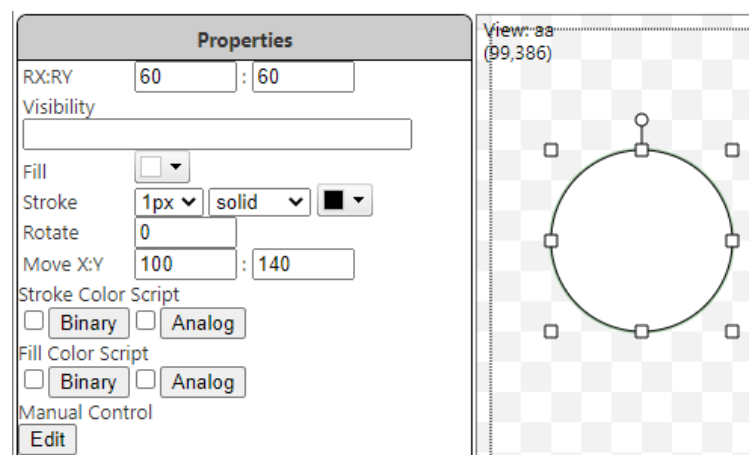


FigureG-156: Drawing a rectangle

In which:

- Width: The width of the rectangle.

- Height: The height of the rectangle.
 - Round: The curvature of the 4 rectangular corners.
 - Visibility: Used to hide/show objects according to expressions.
 - Fill: The background color of the rectangle.
 - Stroke: Lines such as the width, style, and color of the border.
 - Rotate: Rotation angle.
 - Move X:Y: The coordinates of the rectangle.
 - Stroke Color Script: The color of the outline according to the value of the Binary or Analog expression with variables in the system, similar to a line.
 - Fill Color Script: The color of the background according to the value of the Binary or Analog expression with the variables in the system, similar to the line.
- Draw an Ellipse (Ellipse): Press the **Ellipse button**  on the toolbar, hold down the left mouse, and draw on the background view.



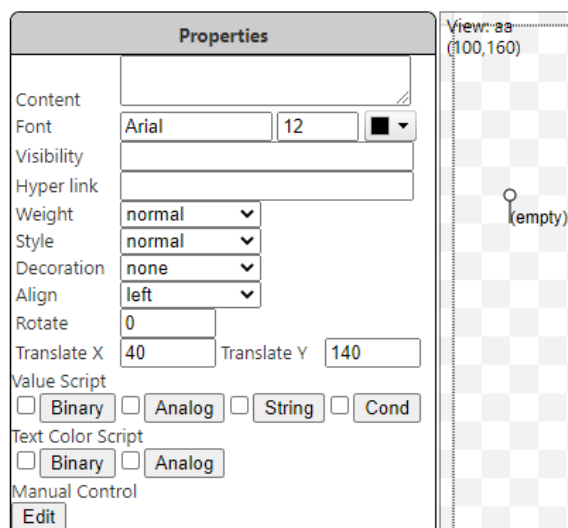
FigureG-157: Drawing an ellipse

In which:

- RX: Radius in the X-direction.
- RY: Radius in Y direction.
- Visibility: Used to hide/show objects according to expressions.
- Fill: The background color of the ellipse.
- Stroke: Lines such as the width, style, and color of the border.
- Rotate: Rotation angle.
- Move X:Y: The coordinates of the rectangle.
- Stroke Color Script: The color of the outline according to the value of the Binary or Analog expression with variables in the system, similar to a line.

- Fill Color Script: The color of the background according to the value of the Binary or Analog expression with the variables in the system, similar to the line.

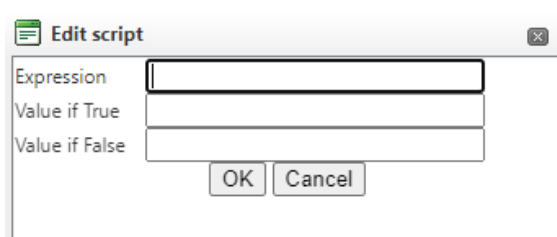
- Text: Click the **Text**  button on the toolbar, left-click on the background view.



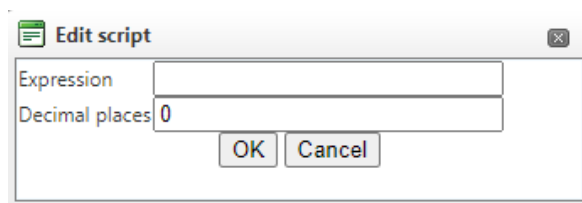
FigureG-158: Drawing a rectangle

In which:

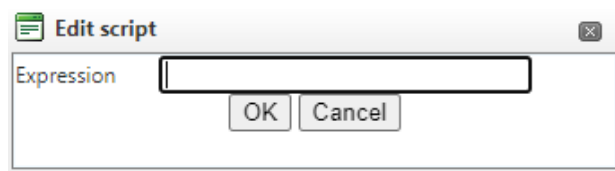
- Content: The content of the text.
- Font: The font type, font size, and color of the text.
- Visibility: Used to hide/show objects according to expressions.
- Hyper link: Create hyper links.
- Weight: The intensity of the text.
- Style: The tilt of the text.
- Decoration: An option to underline the text.
- Align: Align left, right, or center.
- Rotate: Rotation angle.
- Translate X, Translate Y: The X, Y coordinates of the text.
- Value Script: Displays content according to the value of the expression with variables in the system.



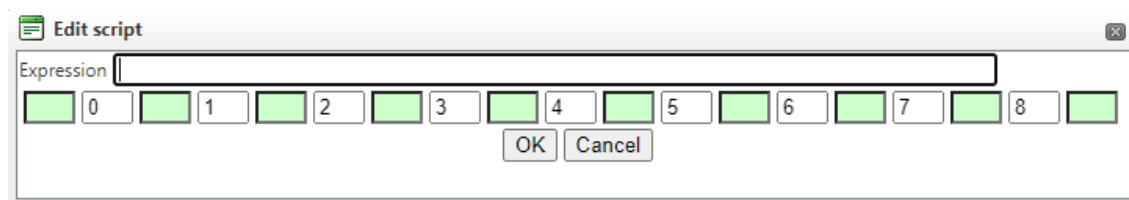
FigureG-159: Display content by value with Binary script



FigureG-160: Display content by value with Analog script

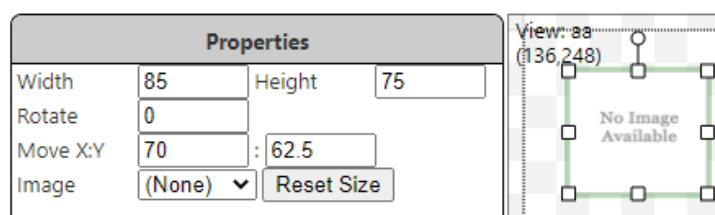


FigureG-161: Display content by value with String script



FigureG-162: Display content by value with Cond script

- Text Color Script: The color of the text according to the value of the Binary or Analog expression with variables in the system, similar to a line.
- Add Image: Tap the **Image button**  on the toolbar, hold down the left mouse, and draw on the background view.

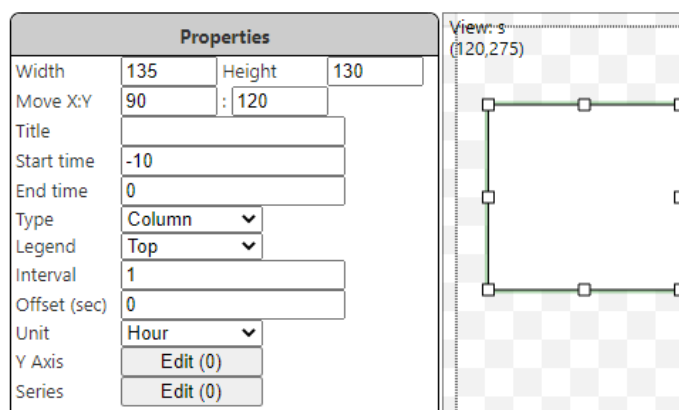


FigureG-163: Insert Image

In which:

- Width: The width of the image.
- Height: The height of the image.
- Rotate: Rotation angle.
- Move X:Y: The coordinates of the image.
- Image: Select an image that has been saved in the Image List. Refer to [Managing Image Lists](#)
- Reset Size: Automatically resize the photo to its original size in the Image List.

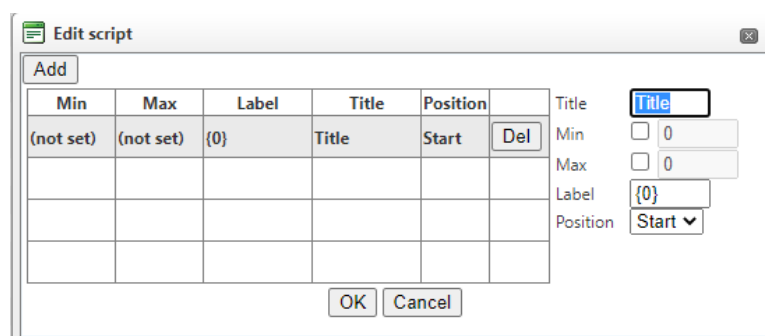
- Add Chart (Chart): Tap the **Chart** button  on the toolbar, hold down the left mouse and draw on the background view.



FigureG-164: Chart drawing

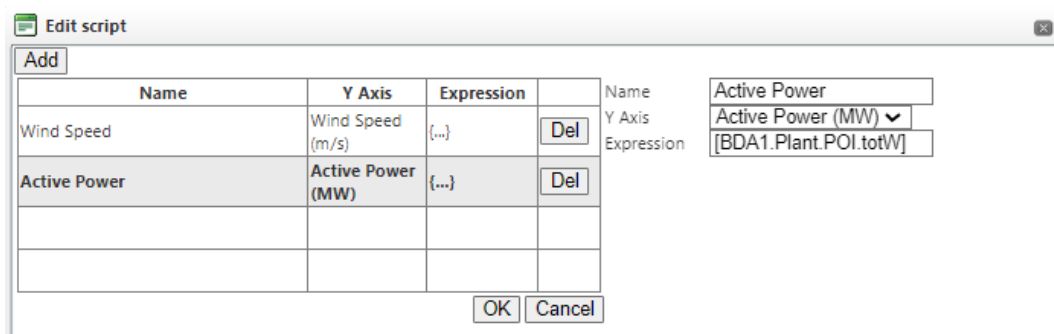
In which:

- Width: The width of the chart.
- Height: The height of the chart.
- Move X:Y: The coordinates of the chart.
- Title: The title of the chart.
- Start time: The time when the chart started compared to the current time.
- End time: The time when the chart is finished compared to the current time.
- Type: Chart type (Bar, Column, Line, Scatter Line, Pie).
- Legend: Top, Bottom, Left, Right options.
- Interval: The charting cycle.
- Offset (sec): Time Offset (unit: s), Data Query Cycle = Interval + Offset.
- Unit: Unit (Minute, Hour, Day, Month, Year).
- Y Axis: Y Axis.



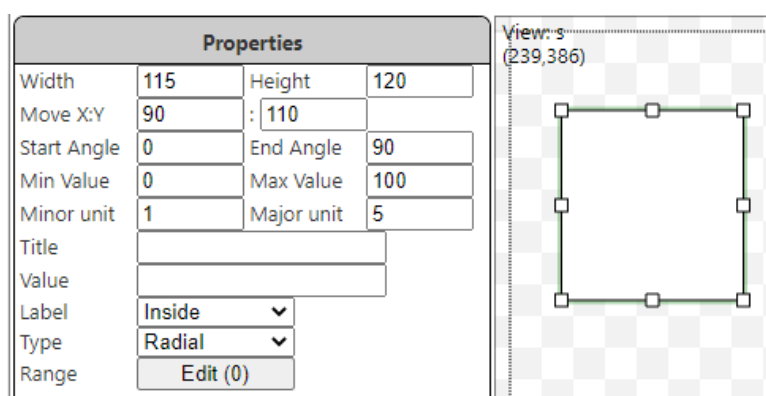
FigureG-165: Configure the display of the Y-axis in the chart

- Series: Configure the pens in the chart.



FigureG-166: Configure the pens in the chart

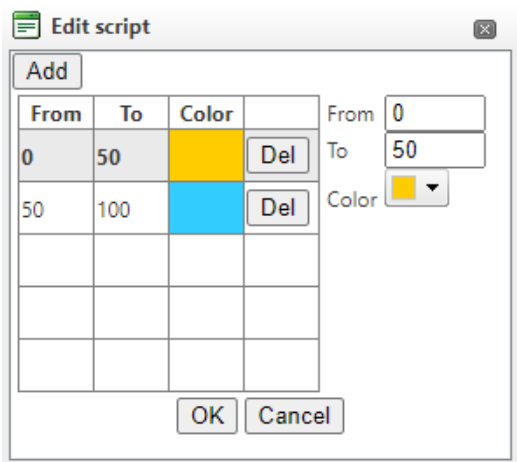
- Add Gauge (Gauge): Press the Gauge button  on the toolbar, hold down the left mouse and draw on the background view.



FigureG-167: Gauge drawing

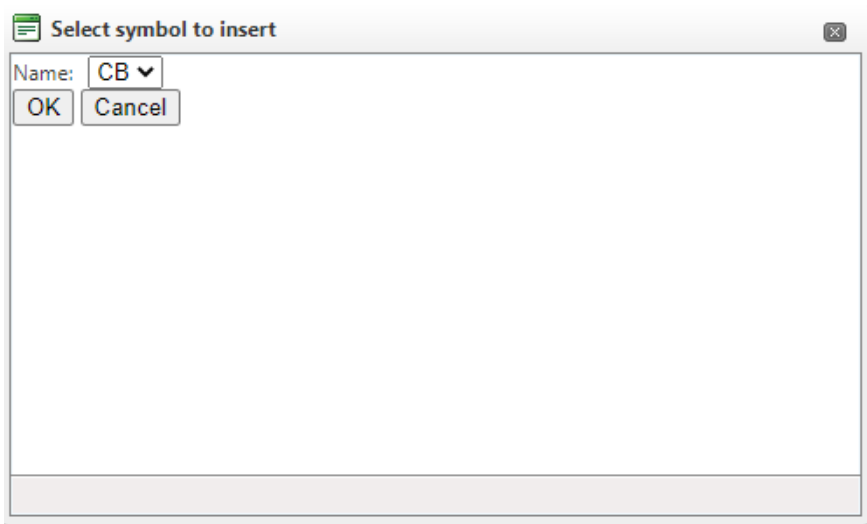
In which:

- Width: The width of the gauge.
- Height: The height of the gauge.
- Move X:Y: The coordinates of the gauge.
- Start Angle: How many degrees to start from.
- End Angle: How many degrees from the end angle.
- Min Value: The smallest value.
- Max Value: The greatest value.
- Minor Unit: The unit value between the minor dividing lines.
- Major Unit: The unit value between the major dividing lines
- Title: Title of the gauge.
- Value: Enter the tag name in the system.
- Label: Inside, outside options.
- Type: Gauge type (Radial, Linear).
- Range: Configure the color for the value ranges.



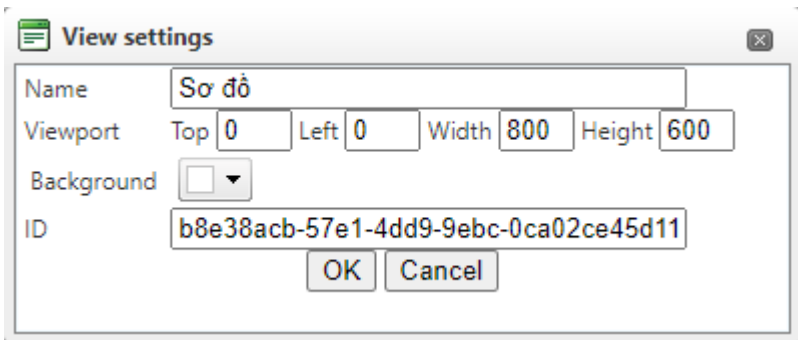
FigureG-168: Configure the color of the value band

- Add Symbol to the diagram: Click the **Symbol** button  on the toolbar, select the symbol in the symbol list, left-click on the background view to insert the symbol.



FigureG-169: Select a symbol in the symbol list

- Schematic Settings: Tap the **Settings**  button on the toolbar.



FigureG-170: Screen configuration interface view

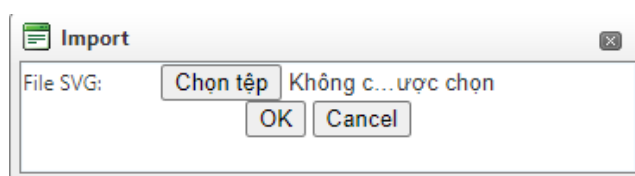
In which:

- Name: The name of the diagram.
- Top: The distance of the diagram to the top of the browser frame.
- Left: The distance of the diagram to the left side of the browser frame.
- Width: The design width of the screen view.
- Height: The design height of the screen view.
- Background: The background color when the display of the screen view.
- ID: The ID of the screen, used for the Hyper Link property on Text.

Once you're done drawing, click the **Save**  button on the toolbar to save it.

➤ Add a screen view using Import file:

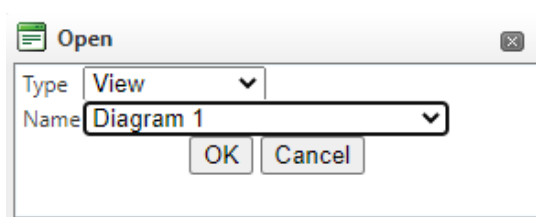
Select the **Import**  icon, select the file .xml. The .xml file can be exported from the HMI (refer to [the Export screen from HMI to Web Portal](#)).



FigureG-171: Interface adds View screen using Import file

Edit Screen View

Select the **Open** , select Type View, and select the screen you want to edit, click **OK** to open the screen.



FigureG-172: Open Screen View

After editing, users click the **Save**  to save it.

Users can also export the view screen to a .xml file by clicking the **Export**  icon.

Clear Screen View

Select the **Open** , select Type View, and select the screen you want to delete, click **OK** to open the screen.

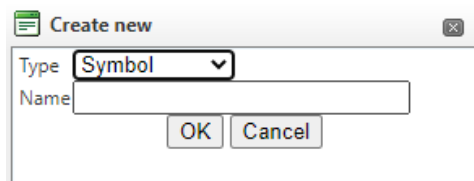
Users click the **Delete icon**  to delete the screen.

 *Export Screens from HMI to Web Portal*

Refer to the ChapterF.OneATS Grid Studio documentation.

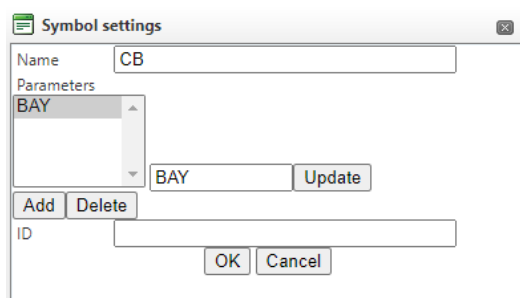
 *Add symbol*

Select the **New icon** , select Type View, and enter a screen name.



FigureG-173: Symbol Creation Interface

The operations of drawing 1 symbol are similar to drawing the view screen, only the difference is in the symbol setting section, click the **Setting icon**  to configure the parameters for the symbol.



FigureG-174: Symbol Configuration Interface

Click **Add** to add the Parameters.

Press **Delete** to delete the parameters.

To edit the name of the Parameters, select 1 Parameters in the list of Parameters, then edit the name and click **Update**.

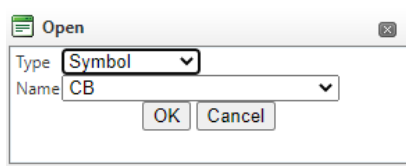
ID: The ID of the symbol generated by the system itself

Click **OK** to complete the symbol configuration.

After editing, users click the **Save icon**  to save it.

 *Editing Symbols*

Select the **Open icon** , select the Type symbol type, and select the symbol you want to edit, click **OK** to open it.



FigureG-175: Symbol opening interface

Click the **Setting icon**  to configure the parameters for the symbol.

After editing, users click the **Save icon**  to save it.

Removing symbols

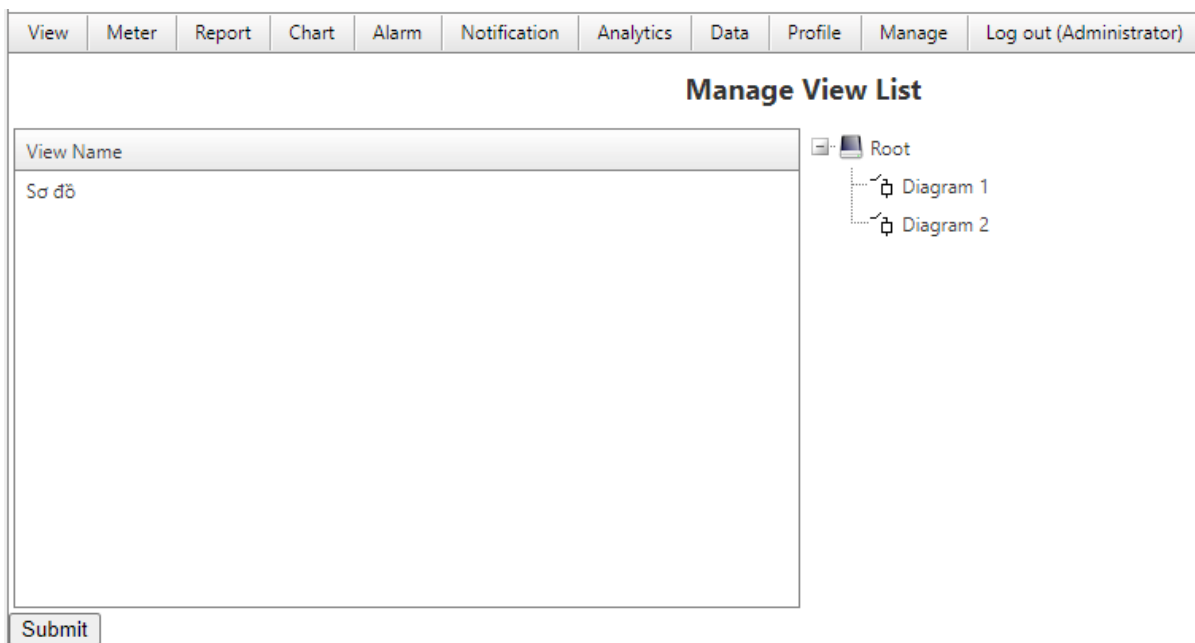
Select the **Open icon** , select the Type symbol type, and select the symbol you want to edit, click **OK** to open it.

The user clicks the **Delete icon**  to delete the symbol.

b. Manage list view

This function has the ability to manage the screen list.

Click Manage > **View menu** > **Manage View List**



FigureG-176: View List Management Interface

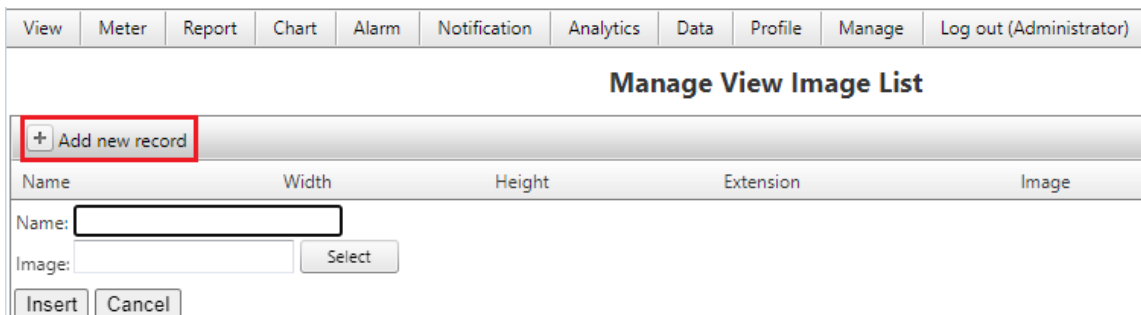
Select the screen, hold down the left mouse and drag the screen to the **Root node**, then click **Submit**.

c. Managing Image Lists

This function has the ability to manage image lists.

Add Image

Click the **Manage > View > Manage View Image List** menu, select **Add new record** and enter the fields then select **Insert**



The screenshot shows the 'Manage View Image List' interface. At the top is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar is the title 'Manage View Image List'. Underneath is a button labeled '+ Add new record' which is highlighted with a red box. Below this button is a form with fields for Name, Width, Height, Extension, and Image. The Name field is currently empty. Below the Name field is an 'Image:' label and a 'Select' button. At the bottom of the form are 'Insert' and 'Cancel' buttons.

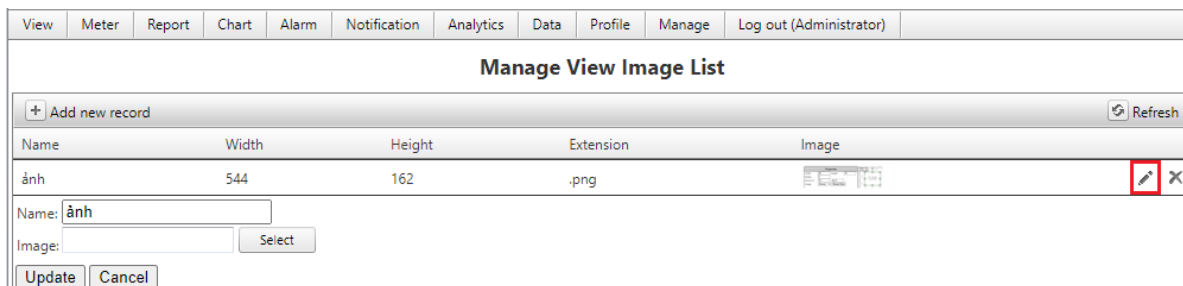
FigureG-177: Image Configuration Interface

In which:

- **Name:** The name of the Image.
- **Image:** Select the Image file.

Edit Image

Click the **Manage > View > Manage View Image List** menu, click the **Edit icon** , enter the information you want to change, and then select **Update**.

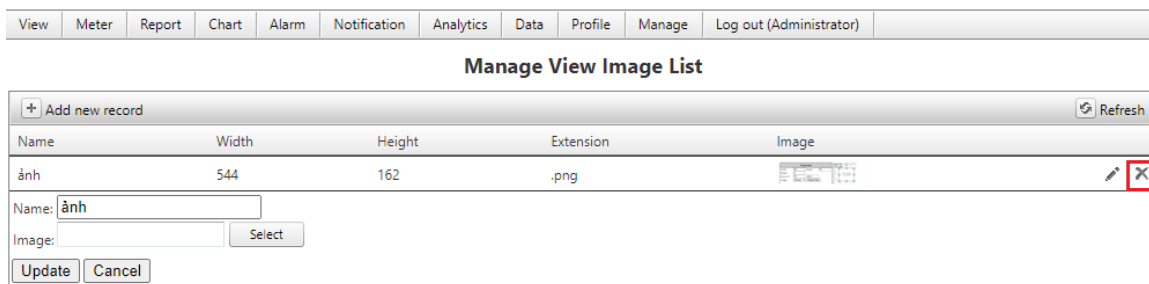


The screenshot shows the 'Manage View Image List' interface. At the top is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar is the title 'Manage View Image List'. Underneath is a button labeled '+ Add new record' and a 'Refresh' button. Below these is a table with columns: Name, Width, Height, Extension, and Image. The table contains one row with the following data: Name: ảnh, Width: 544, Height: 162, Extension: .png, and Image: . To the right of the table is an 'Edit' icon (pencil) and a 'Delete' icon (X), both highlighted with red boxes. Below the table is a form with fields for Name, Width, Height, Extension, and Image. The Name field is currently 'ảnh'. Below the Name field is an 'Image:' label and a 'Select' button. At the bottom of the form are 'Update' and 'Cancel' buttons.

FigureG-178: Image Editing Interface

Delete Image

Click the **Manage > View menu > Manage View Image List**, click the **Delete icon** , and then select **Ok**.



The screenshot shows the 'Manage View Image List' interface. At the top is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar is the title 'Manage View Image List'. Underneath is a button labeled '+ Add new record' and a 'Refresh' button. Below these is a table with columns: Name, Width, Height, Extension, and Image. The table contains one row with the following data: Name: ảnh, Width: 544, Height: 162, Extension: .png, and Image: . To the right of the table is an 'Edit' icon (pencil) and a 'Delete' icon (X), both highlighted with red boxes. Below the table is a form with fields for Name, Width, Height, Extension, and Image. The Name field is currently 'ảnh'. Below the Name field is an 'Image:' label and a 'Select' button. At the bottom of the form are 'Update' and 'Cancel' buttons.

FigureG-179: Image Deletion Interface

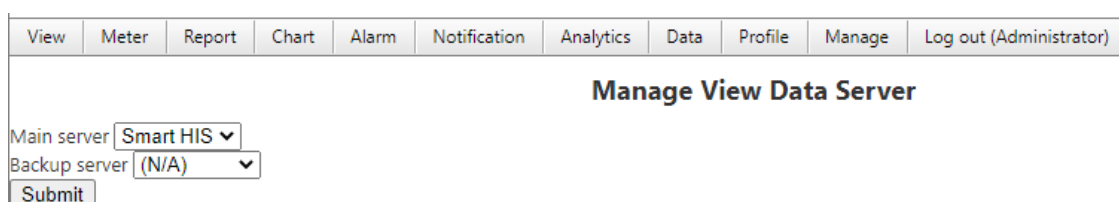
d. Manage screen view permissions with users and groups

This function has the ability to manage the right to view the screen for users and groups.

Click the Tab **menu Manage > View > Manage View Permission**, similar configurations to manage the grant of permission to view reports for users and groups.

e. View DataServer Management

Click **Manage > View menu > Manage View DataServers**, select Main server, Backup server (if any) and **Submit**.

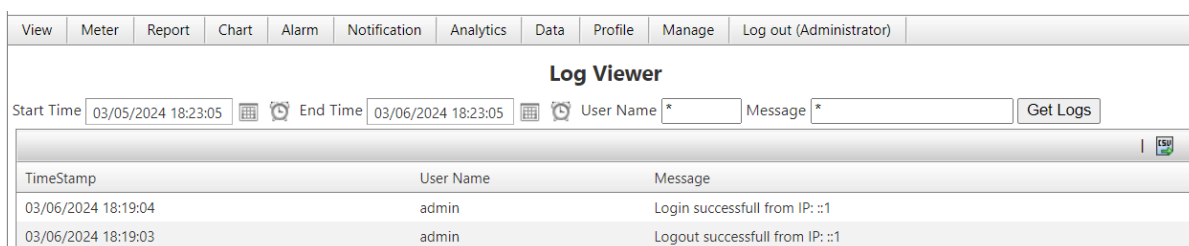


FigureG-180: View DataServer Management Interface

G.10.2.2.1.6. Log Viewer

Manage logs recorded on the system.

Click **Manage menu > Log Viewer**, select the query interval, select **Get Logs**



TimeStamp	User Name	Message
03/06/2024 18:19:04	admin	Login successfull from IP: ::1
03/06/2024 18:19:03	admin	Logout successfull from IP: ::1

FigureG-181: Log Viewer Manager

In which:

- **Start Time, End Time:** The duration of the system log query.
- **User Name:** Log query by web login username Click (each logged-in user will have different activities, resulting in different log information).
- **Message:** Query by Alarm message

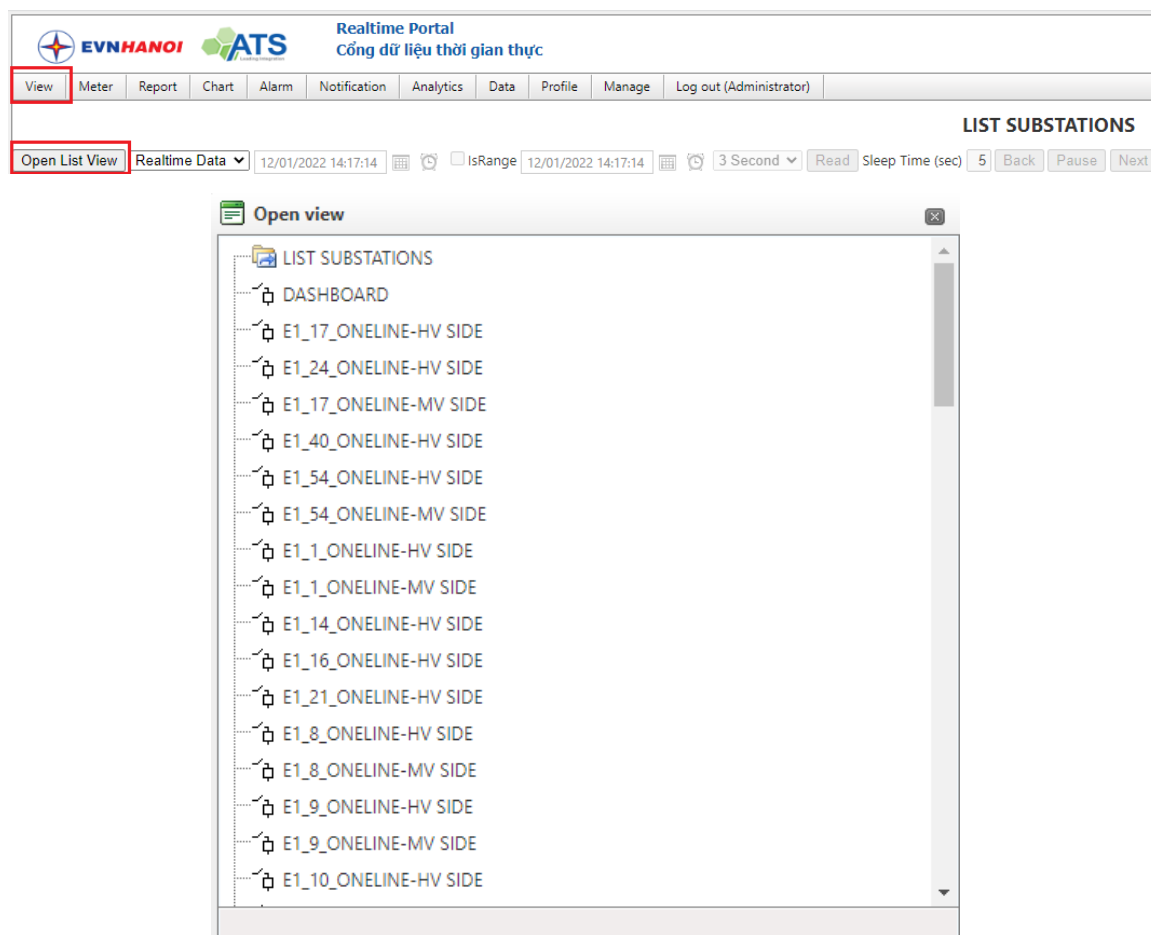
Download the file to your computer by clicking on the **Export to CSV**  icon to download the file in Excel format.

G.10.2.2.2 Function View

Users can view the screen in Realtime Data or Historical Data mode.

G.10.2.2.2.1. View the screen in Realtime Data mode

After successfully logging in, the display interface is the default View screen of the system, select **Open List View**, select Screen, select **Realtime Data mode**.



FigureG-182: Optional interface display screen view

G.10.2.2.2.2. View the screen in Historical Data mode

- a. View the screen in Historical Data mode at a time

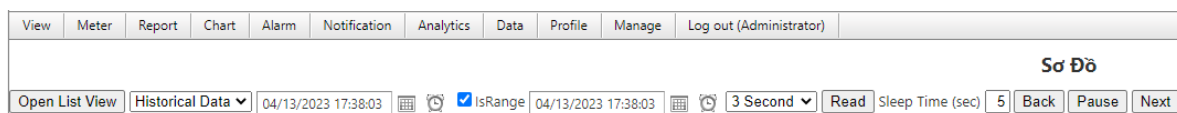
The user selects **Open List View**, selects the screen, selects the **Historical Data** mode, selects the query time, and clicks **Read** to query the data.



FigureG-183: View Historical Data at a Time

- b. View the screen in Historical Data mode for a period of time (Play Back)

The user selects **Open List View**, selects the screen, selects the **Historical Data** mode.



FigureG-184: Historical Data view over a period of time

- Check IsRange .
- Choose the start and end time.
- Select the sampling frequency (Interval time).
- Select the time for each display (Refresh time).
- Click Read to query the data.
- Click Back/Pause/Next to control the display.

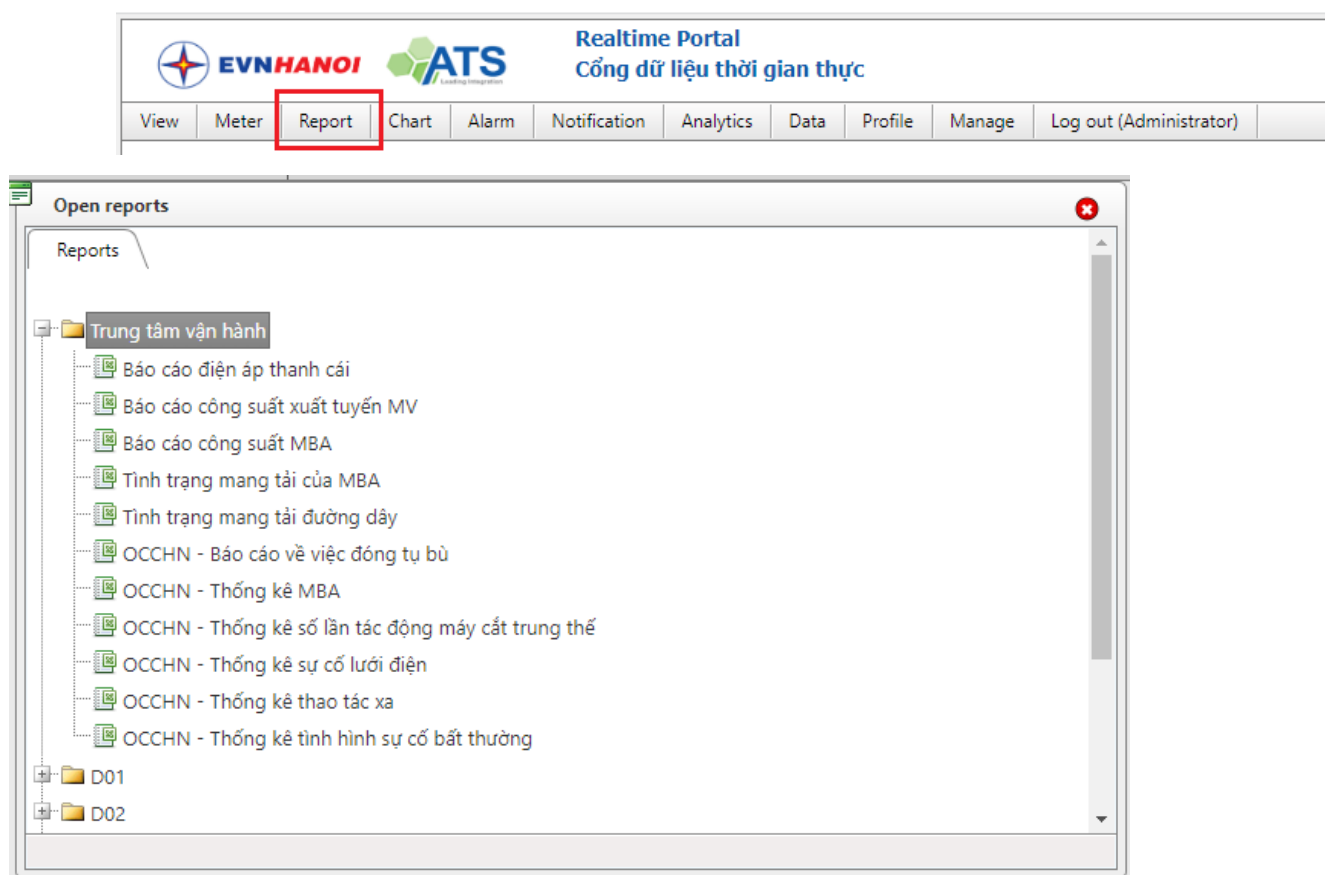
G.10.2.2.3 Report Function

Help users view the report templates that have been built, can view the data directly on a web browser, or download the report file to the machine as an excel or pdf file.

Similar to the view screen, users can also view reports in Realtime Data or Historical Data mode.

G.10.2.2.3.1. View reports in Realtime Data mode

Click the **Report** menu, select a report from the list of pre-built reports.



FigureG-185: List of reports

EVNHANOI

ATS

Realtime Portal

Cổng dữ liệu thời gian thực

View

Meter

Report

Chart

Alarm

Notification

Analytics

Data

Profile

Manage

Log out (Administrator)

Open

Realtime Data

12/01/2022

Reset

Báo Cáo Điện Áp Thanh Cái

CÔNG TY LƯỚI ĐIỆN CAO THẾ TP HÀ NỘI

BÁO CÁO ĐIỆN ÁP THANH CÁI CÁC TRẠM TRỰC THUỘC CTY LƯỚI ĐIỆN CAO THẾ HÀ NỘI

Ngày 01 tháng 12 năm 2022

STT	Tên trạm	Giờ	0:00	0:30	1:00	1:30	2:00	2:30	3:00	3:30	4:00	4:30	5:00	5:30	6:00	6:30	7:00	7:30	8:00	8:30	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30
1	E1.1 110kV Đông Anh	Busbar	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)	(kV)
		BB11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB19	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB41	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB42	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB43	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		BB31	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	E1.16 110kV Nội Bài	BB32	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB41	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
3	E1.17 110kV Bắc Thăng Long	BB42	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB11A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB12A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB41	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB42	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		BB43	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
4	E1.24 110kV Hải Bối	BB44	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
		BB45	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
		BB11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
		BB12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
		BB41	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
		BB42	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		

FigureG-186: Reporting in Realtime Data mode

Download the report to your computer as an excel file, select the **Download** icon

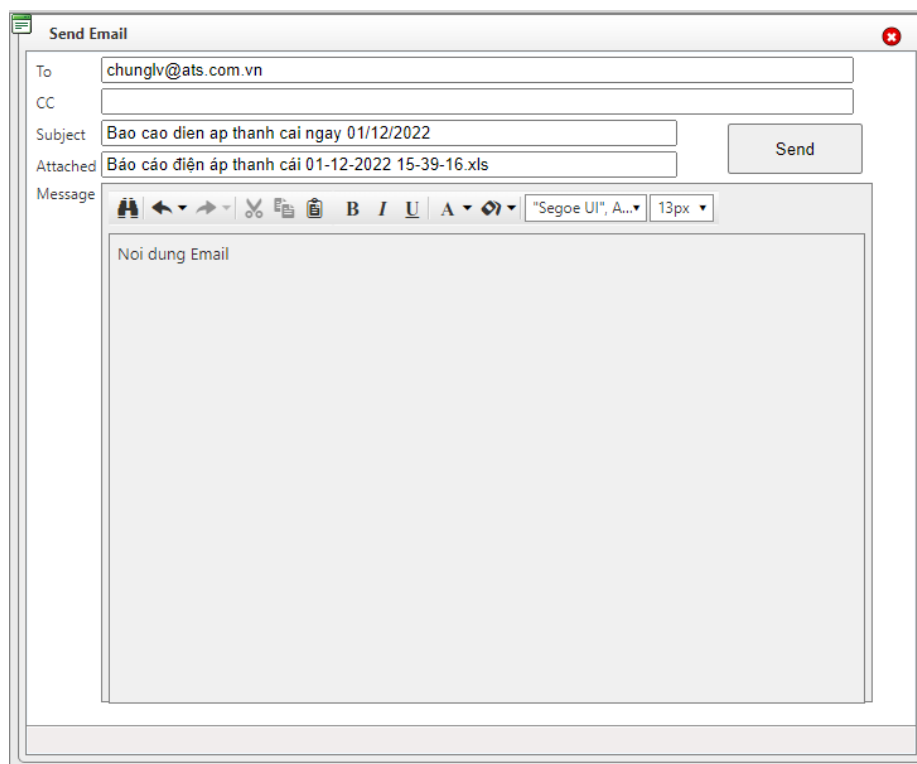


Download as a PDF file, select the **Export to PDF** icon.



Send an email with the report file attached, select the **Send Email** icon, and then enter the required email information to send.

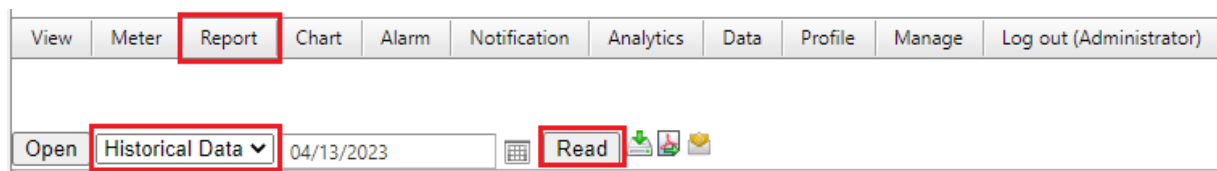




FigureG-187: Configuration interface to send reports by mail

G.10.2.2.3.2. View reports in Historical Data mode

Click the **Report** menu, select a report from the list of pre-built reports, select Historical Data mode, select Real to query the data.

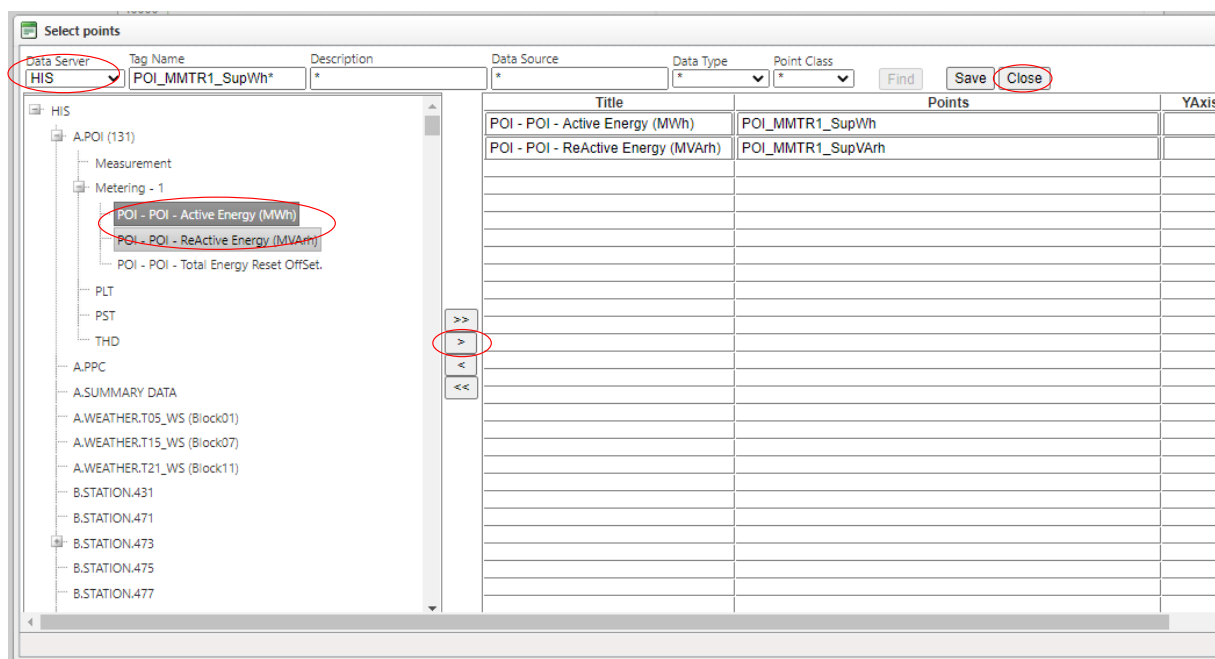


FigureG-188: Report in Historical Data mode

Performing a report load is similar to loading a report when in Realtime Data mode.

G.10.2.2.4 Graphing function

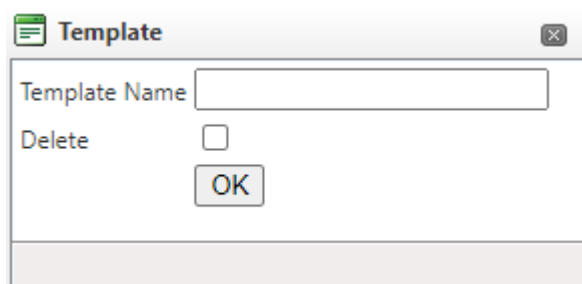
Select the **Chart** menu.



FigureG-189: Chart data point option interface

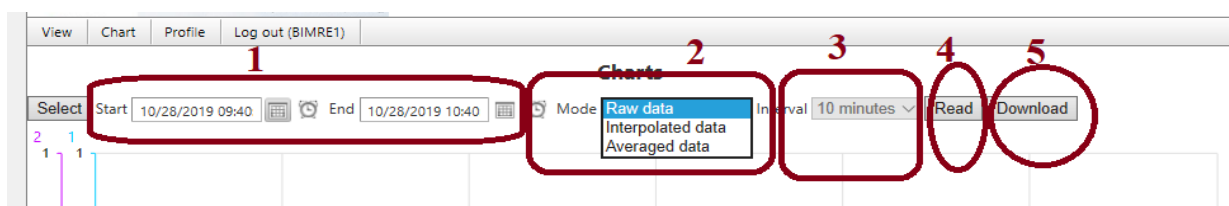
- Select **DataServer**:
 - HIS: A list of data stored in the HIS Server.
 - Template: Pre-created data groups.
 - User Define: Displays data according to the user's definition (configured user)
- Data point selection: After selecting the DataServer, from the data tree the user selects the data points. Hold *Shift* or *Ctrl* to select multiple data at once. Add the selection points by *clicking* on:
 - >> : Add all the first 50 points
 - > : Add selected points
 - < : Delete selected points
 - << : Delete all selected points

Color options, Y-axis for data points if needed.
- Select **Close** to complete the selection process.
- Select **Save** to save as a template.



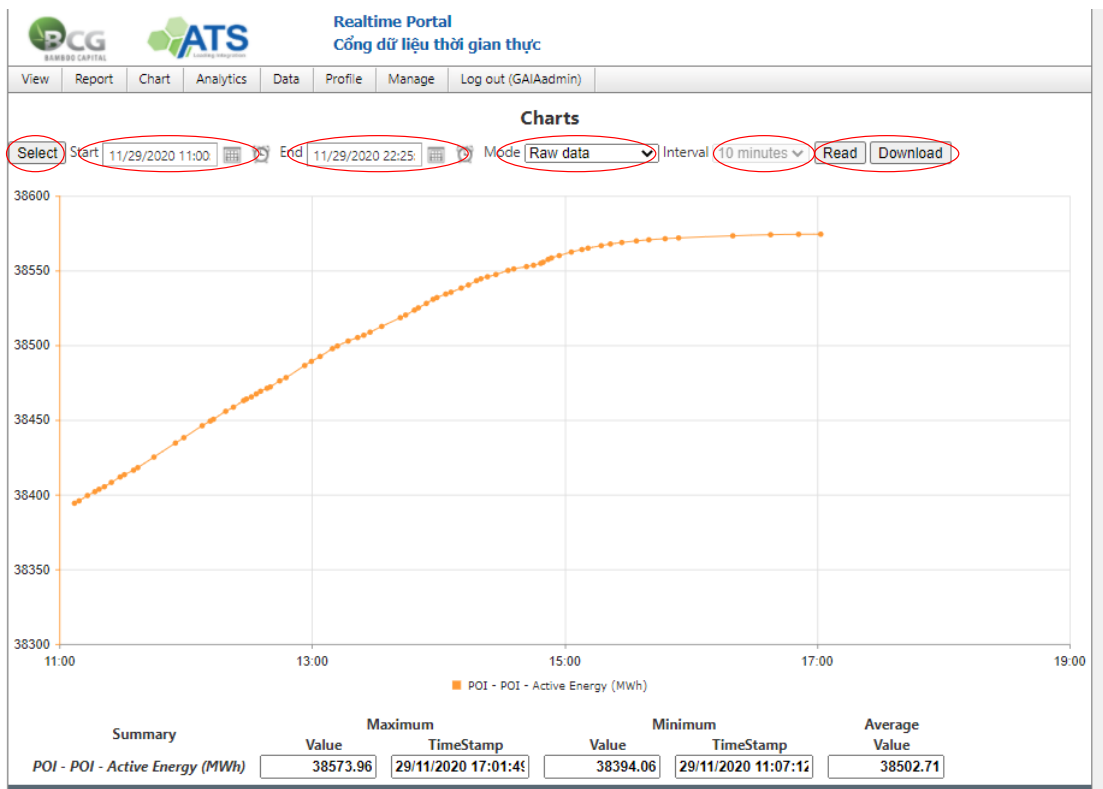
FigureG-190: Template Configuration Interface

- After selecting the data point, time options, data query type, and graphing.



FigureG-191: Chart Drawing Style Configuration Interface

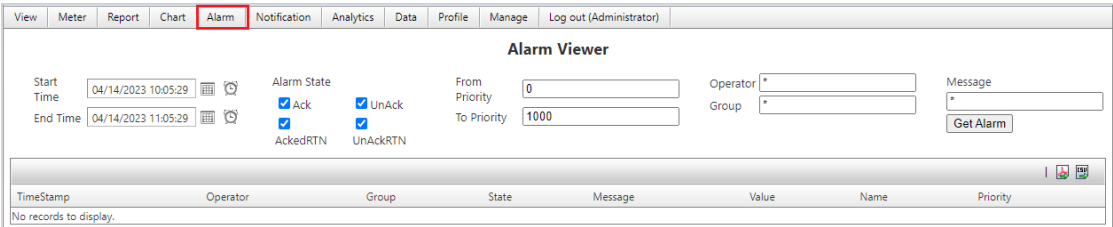
- (1) Select the time period you want to draw: *Start* and *End*
- (2) Select the data query type
 - Raw Data: The graph plots all the values stored in HIS.
 - Interpolated Data: The graph plots the calculated values interpolated from the values stored in the HIS Server over time.
 - Averaged Data: The graph plots the averaged calculated values from the values stored in the HIS Server.
- (3) Select the frequency of data queries.
- (4) Select *Read* to draw the graph.
- (5) Select *Download* to download the graph data as an Excel file.



FigureG-192: Illustration of data graphing

G.10.2.2.5 Alarm Query Function

Click the **Alarm** menu.



FigureG-193: Alarm Query Interface

The user selects the query interval, the status of the alarm, the Priority range, and selects **Get Alarm**.

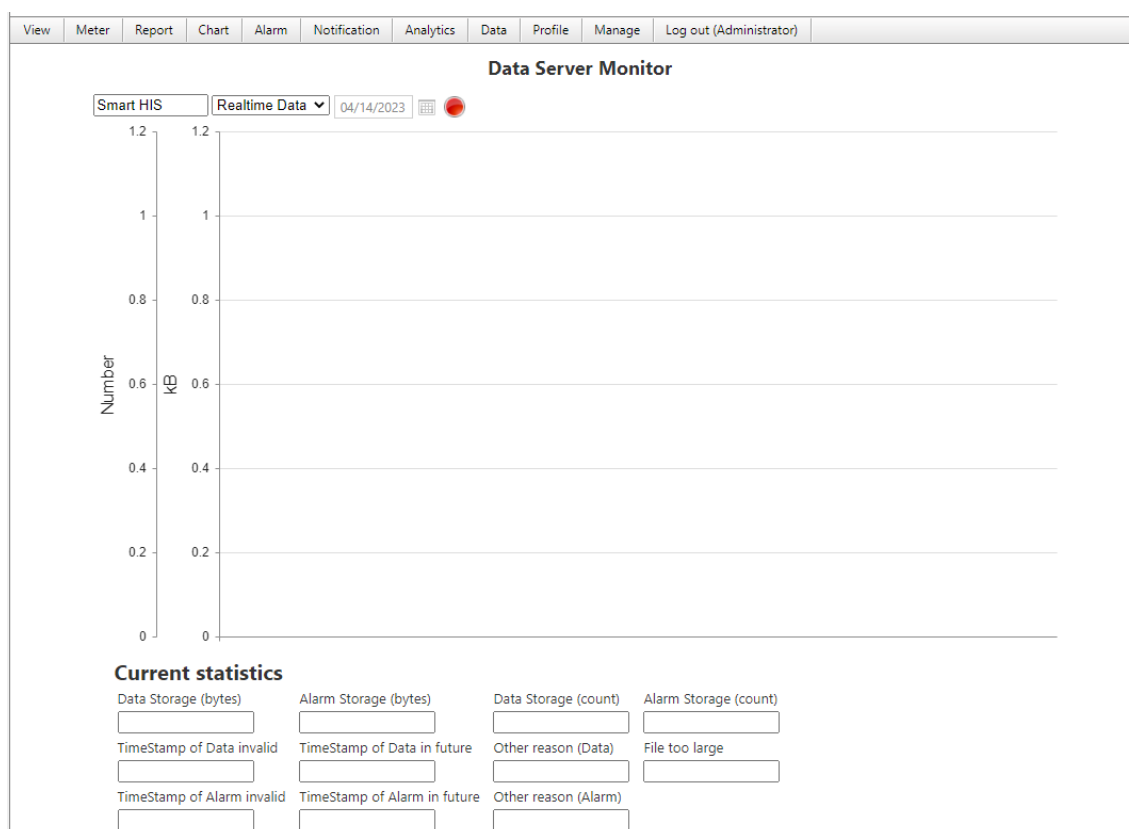
Start TimeEnd Time	Alarm Query Interval
Alarm State	Status of Alarm
From PriorityTo Priority	Alarm Priority Range
Operator	Alarm by User system
Group	Alarms by alarm group (refer to chapter E configuration)

Message	Query by Alarm message
---------	------------------------

Download the file to your computer by clicking the **Export to PDF**  icon to download the file as PDF, click the **Export to CSV**  icon to download the file in Excel format.

G.10.2.2.6 Connection test function, data exchange to HIS Server

Click the Analytics menu .



FigureG-194: Connection test interface, data exchange to HIS Server

In which:

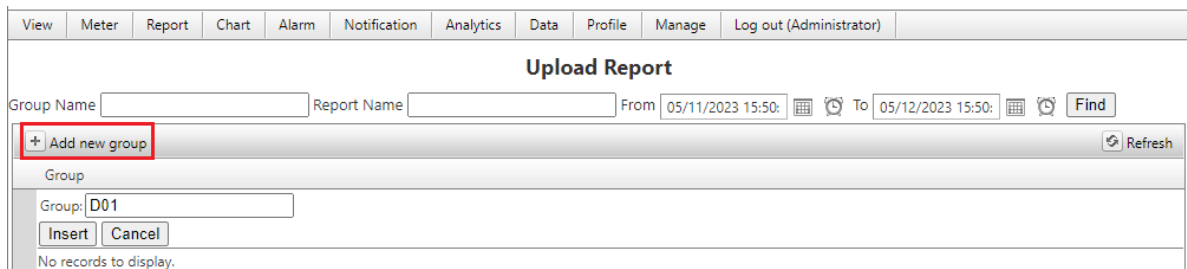
- Blue LED  : Good connection to HIS Server.
- Red LED  : Lost connection to HIS Server.

G.10.2.2.7 Upload Report, Manual Report function

G.10.2.2.7.1. Upload Report Function

a. Add Group

Click the **Data > Upload Report** menu, select **Add new group** and enter the group name in the **Group box**, select **Insert**.

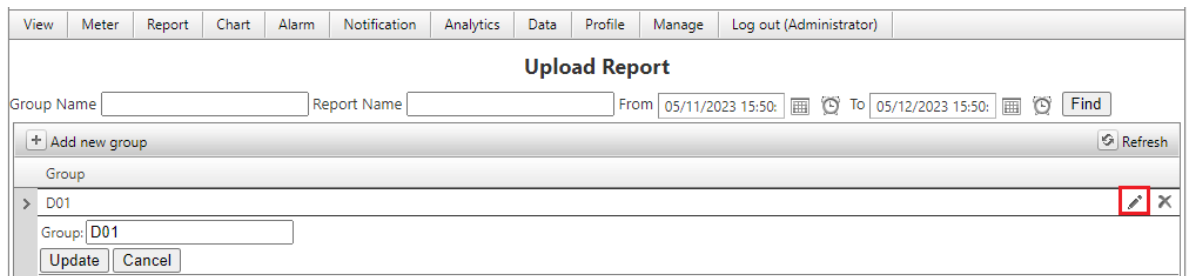


The screenshot shows the 'Upload Report' section of the OneATS interface. At the top, there is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar, the 'Upload Report' title is centered. Underneath, there are input fields for 'Group Name' and 'Report Name', followed by 'From' and 'To' date/time pickers (both set to 05/11/2023 15:50:00). A 'Find' button is to the right. Below these fields is a '+ Add new group' button, which is highlighted with a red box. To the right of this button is a 'Refresh' button. Below the '+ Add new group' button, there is a 'Group' section with a 'Group:' label and a text input field containing 'D01'. Below the input field are 'Insert' and 'Cancel' buttons. At the bottom, it says 'No records to display.'

FigureG-195: Add Group interface

b. Edit Group

Click the Data > **Upload Report** menu, click the **Edit icon** , enter the information you need to change, and select **Update**.

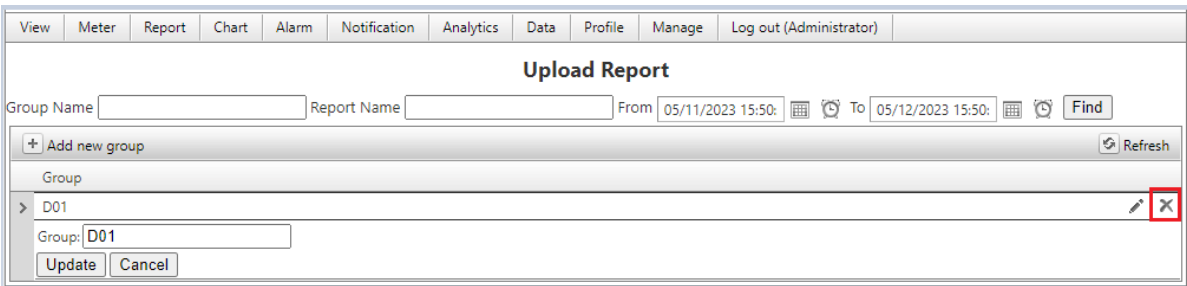


This screenshot shows the 'Upload Report' interface with a group 'D01' selected. The 'Group:' input field now contains 'D01'. Below the input field are 'Update' and 'Cancel' buttons. To the right of the 'Group:' input field, there is a red box highlighting the 'Edit' icon (pencil) and a 'Delete' icon (X).

FigureG-196: Report Editing Interface

c. Delete Group

Click the **Data > Upload Report** menu, click the **Delete icon** , and then select **Ok**.



This screenshot shows the 'Upload Report' interface with the 'Group:' input field containing 'D01'. Below the input field are 'Update' and 'Cancel' buttons. To the right of the 'Group:' input field, there is a red box highlighting the 'Delete' icon (X).

FigureG-197: Delete Group Interface

d. Add Report

Click the **Data > Upload Report** menu, select the Group you want to add a report to, select **Add new record** and enter the information fields, select **Insert**.

FigureG-198: Report Configuration Interface

In which:

- **Report name:** The display name of the report template.
- **Make time:** When the report was created.
- **Report file:** Select the report template file.

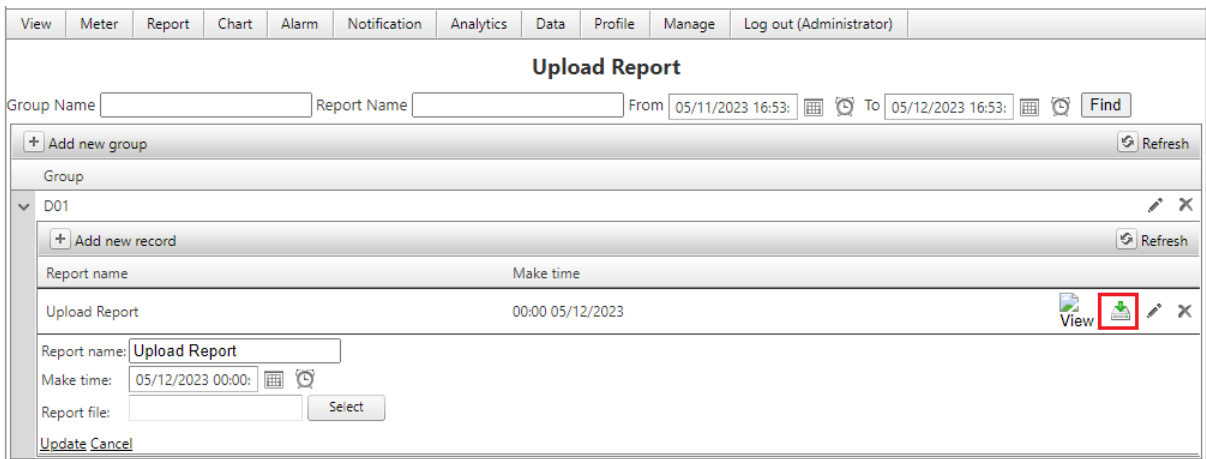
e. Edit a Report

Click the **Data > Upload Report** menu, click the **Edit icon**  , enter the information you want to change, then select **Update**.

FigureG-199: Report Editing Interface

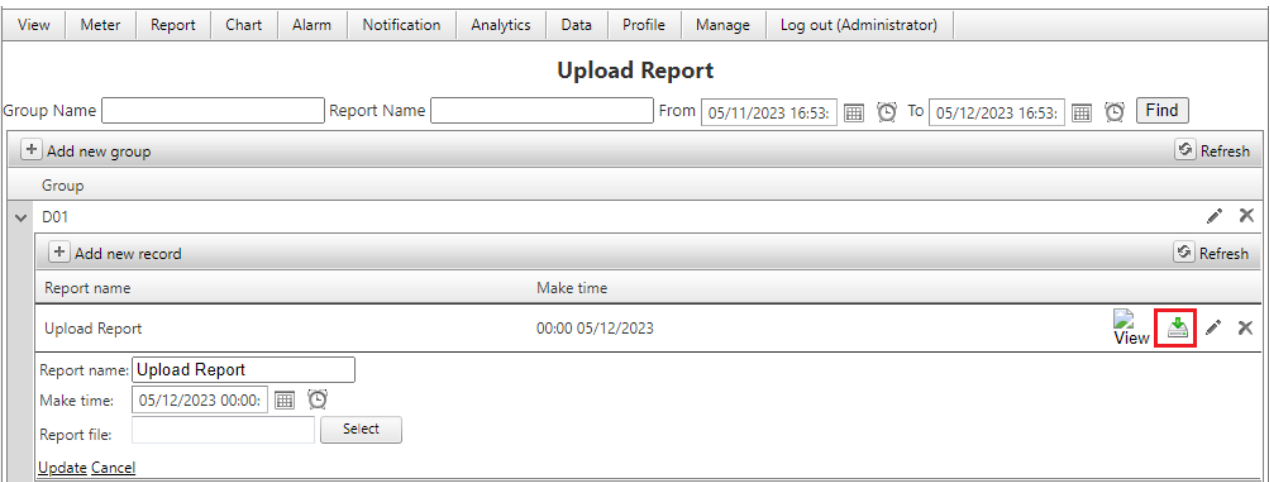
In addition, users can download and view existing reports by:

- Click the **Download icon**  to download the report:



FigureG-200: Download report

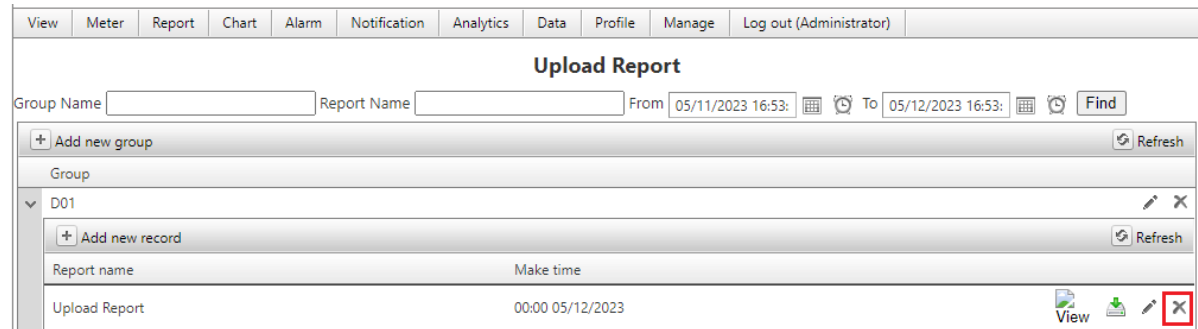
- Click on the **View** icon  to view the online report:



FigureG-201: View the report online

f. Delete a Report

Click the **Data > Upload Report** menu, click the **Delete** icon , and then select **Ok**.

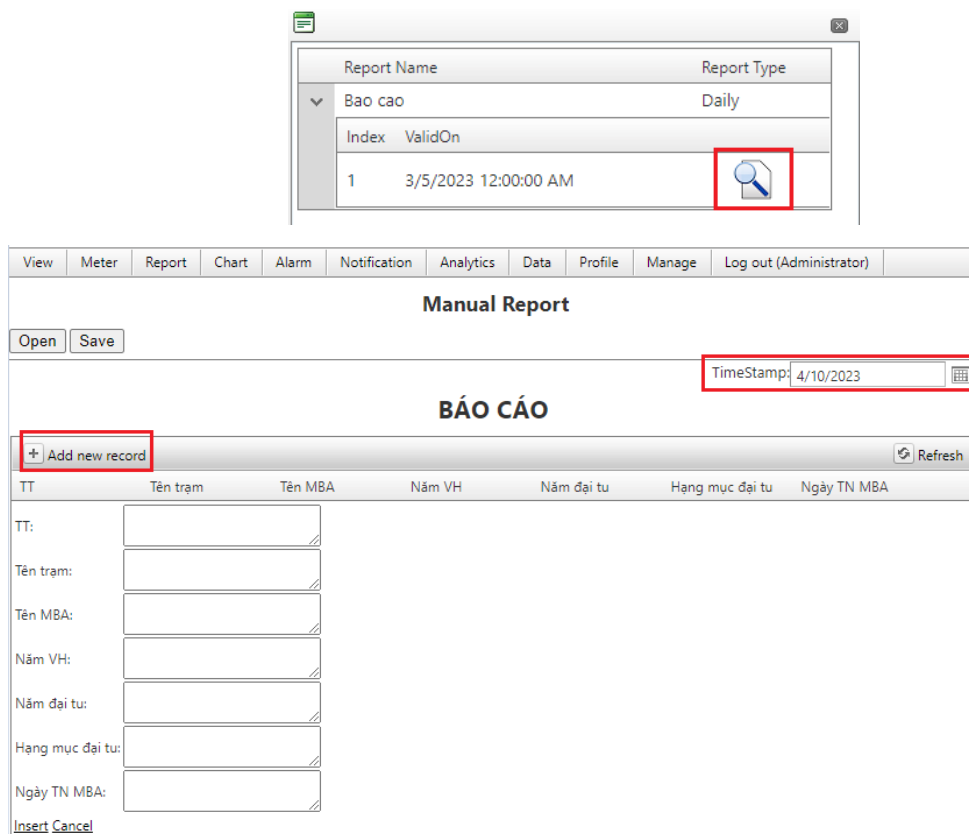


FigureG-202: Delete Report Interface

G.10.2.2.7.2. Manual Report Function

a. Add a report

Click the Data > **Manual Report** menu > **Open** to select the report, select the **View**  icon, select **TimeStamp**, select **Add new record**, enter the information of the report, and select **Insert**.

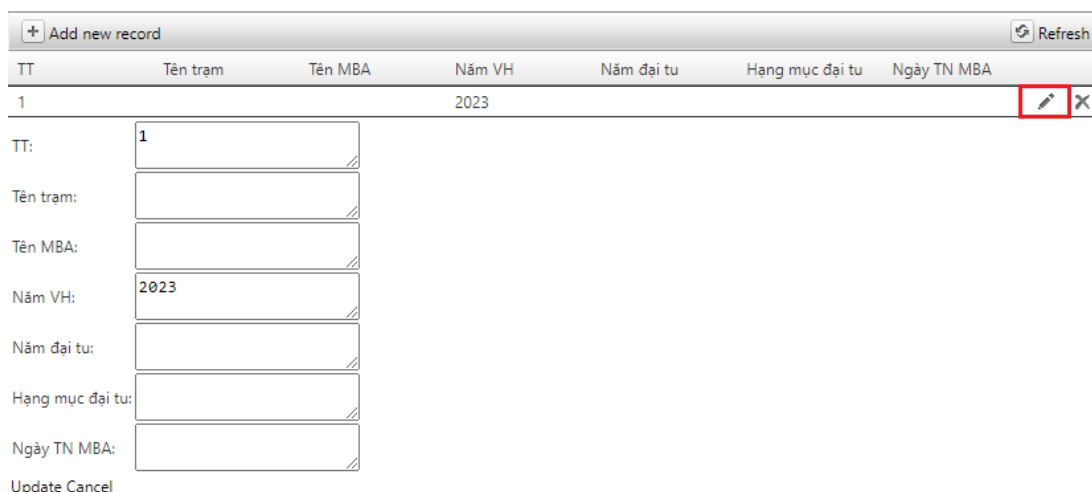


The screenshot shows the 'Manual Report' interface. At the top, there is a navigation bar with tabs: View, Meter, Report, Chart, Alarm, Notification, Analytics, Data, Profile, Manage, and Log out (Administrator). Below the navigation bar, there is a 'Manual Report' section with 'Open' and 'Save' buttons. A 'TimeStamp' field is highlighted with a red box, showing the value '4/10/2023'. Below this, there is a 'BÁO CÁO' section with an 'Add new record' button highlighted with a red box. The 'Add new record' button is located next to a table with columns: TT, Tên trạm, Tên MBA, Năm VH, Năm đại tu, Hạng mục đại tu, and Ngày TN MBA. Below the table, there are input fields for each column: TT, Tên trạm, Tên MBA, Năm VH, Năm đại tu, Hạng mục đại tu, and Ngày TN MBA. At the bottom, there are 'Insert' and 'Cancel' buttons.

FigureG-203: Interface for adding new reports in manual reports

b. Edit a report

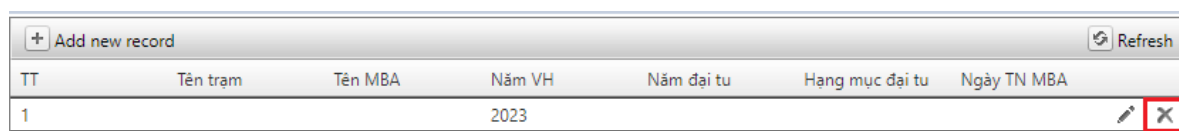
Click the Data > **Manual Report** > **Open** menu to select the report, select the **View**  icon, select **TimeStamp**, click the Edit icon , enter the information to be changed, and then select **Update**.



FigureG-204: Report editing interface in manual reports

c. Delete a report

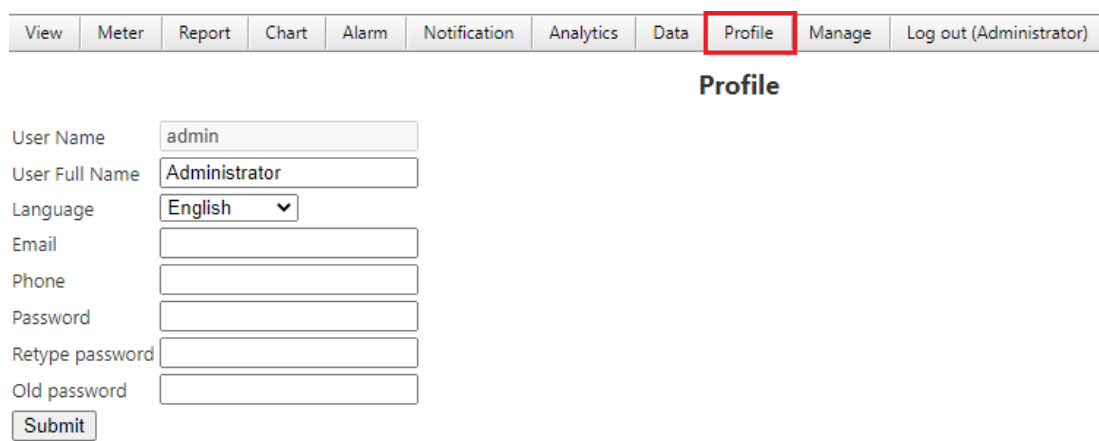
Click the **Data > Manual Report** menu > **Open** to select the report, select the **View** icon, select **TimeStamp**, click the Delete icon , and then select **Ok**.



FigureG-205: The interface for deleting reports in manual reports

G.10.2.2.8 User information management function

Click the **Profile** menu, enter the information you want to edit, then select **Submit**.



FigureG-206: User profile editing interface

In which:

- **Password:** The user's new password.
- **Retype password:** Retype the user's new password.
- **Old password:** The user's current password

G.10.2.2.9 Log out of the system

Click the Profile > **Log out** menu to log out of the system.

G.11. Appendix - Expression Syntax

Expressions are used in point equations, commands that query data from the client side. Expression Syntax:

- Use suffixes to query data components
 - .v: Used to get the value of the data
 - .q: Used to get the quality of the data
 - .t: Used to get the time-stamp of the data
- Point names with suffixes placed in [], e.g. [QTRI_LBAO_E01_MMXU1_AphsA.v]+10
- Types of operators that support the Lua language (refer to Lua Language 5.3). Common basic operators include

Operator Type	Operator symbols
Logical	▪ or, ▪ and, &&
Relational	▪ =, ==, !=, <> ▪ <, <=, >, >=
Additive	▪ +, -
Multiplicative	▪ *, /, %
Bitwise	▪ & (bitwise and), (bitwise or), ^ (bitwise xor) ▪ << (left shift), >>(right shift)
Unary	▪ !, not, -, ~ (bitwise not)
Primary	▪ (,)

- The calculation functions are supported in the Lua language (refer to the Lua Language 5.3 documentation). Common basic functions include

Function Name	Describe	Illustrated	
		Syntax	Result
math.abs	Return the absolute value of 1 number	math.abs(-2)	2
math.acos	Return the corner of some cosine	math.acos(1)	0

math.asin	Return the angle of a sine number	math.asin(0)	0
math.atan	Return the angle of 1 tangent number	math.atan(0)	0
math.ceil	Returns the smallest integer greater than or equal to the value of 1 number	math.ceil (1.5)	2
math.cos	Return the cosine value of 1 angle	math.cos(0)	1
math.exp	Return a power value with a definite exponent of e	math.exp(0)	1
math.floor	Returns the largest integer that is less than or equal to the value of 1 number	math.floor(1.5)	1
math.log	Return the logarithm value of 1 number with a definite base	math.log(1,10)	0
math.log10	Return the logarithm value of 1 number to base 10	math.log10(1)	0
math.max	Return the largest value of 2 numbers	math.max(1,2)	2
math.min	Returns the smallest value of 2 numbers	math.min(1,2)	1
math.pow	Return the power value of 1 number with a definite exponent	math.pow(3,2)	9
math.sqrt	Return the 2nd root value of 1 number	math.sqrt(4)	2
math.tan	Return the tangent value of 1 corner	math.tan(0)	0



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